Preliminary Remarks

Intervals of real numbers are written with brackets or parentheses according to whether the end points are included or not. Thus if $a < b$,

$$[a, b], \quad [a, b), \quad (a, b], \quad (a, b)$$

are sets of numbers x that satisfy the conditions

$$a \leq x \leq b, \quad a \leq x < b, \quad a < x \leq b, \quad a < x < b$$

respectively. $[a, b]$ is called a *closed interval*, and (a, b) an *open interval*; the other two may be called either half-closed or half-open. For intervals that are infinite in one direction (closed and open half-lines, or rays), we use the improper end points $+\infty$ and $-\infty$, which are not to be counted as belonging to the interval; thus

$$[a, +\infty), \quad (a, +\infty), \quad (-\infty, b], \quad (-\infty, b)$$

are the sets of numbers x that satisfy the respective conditions

$$x \geq a, \quad x > a, \quad x \leq b, \quad x < b.$$

$(-\infty, +\infty)$ is the set of all the real numbers (the entire straight line).

The largest and the smallest of a finite number of real numbers $x_1, x_2, \ldots, x_n$ are called their *maximum* and their *minimum* respectively and are denoted by

$$\max[x_1, x_2, \ldots, x_n], \quad \min[x_1, x_2, \ldots, x_n];$$

for example, $\max[2, -3] = 2$, $\min[2, -3] = -3$, $\max[2, 2] = \min[2, 2] = 2$. The same notation is also used for the largest and smallest numbers, if any, of an infinite set of numbers.

If a sequence of real numbers $x_1, x_2, \ldots$ is bounded from above — that is, if there are numbers v for which $v \geq x_n$ for every n —

then there is among these numbers v a smallest one, say v_0. It is variously called the *least upper bound* (the abbreviation l.u.b. is very common), the *supremum*, and sometimes the *upper boundary* of the sequence x_n; in the German-language editions of this book the term used (following Weierstrass) is *obere Grenze*; we shall write

$$v_0 = \sup[x_1, x_2, \dots] = \sup x_n.$$

For example, $\sup[0, 1/2, 2/3, 3/4, \dots] = \sup[(n-1)/n] = 1$. If the sequence has a maximum, this coincides with the supremum. The *greatest lower bound* (g.l.b.), or *infimum*, or *lower boundary* (or *untere Grenze*)

$$w_0 = \inf[x_1, x_2, \dots] = \inf x_n$$

of a sequence bounded from below is defined in an exactly similar way. The same terms are also applied to sets of numbers that are not given in the form of a sequence; for instance,

$$\sup_{a \le x \le b} f(x).$$

For a sequence bounded from above there exist all of the suprema

$$v_n = \sup[x_n, x_{n+1}, x_{n+2}, \dots]$$

with $v_1 \ge v_2 \ge \cdots$; if these v_n are bounded from below and hence have a (g.l.b.) v, this is called the *upper limit*, or *limit superior*, of the sequence; in symbols,

$$v = \limsup x_n = \overline{\lim}\, x_n.$$

The *lower limit*, or *limit inferior*

$$w = \liminf x_n = \underline{\lim}\, x_n,$$

is defined correspondingly.

If the assumptions concerning boundedness do not hold, the symbols $\pm\infty$ are used. For example, in the case of a sequence x_n not bounded from above, we write $\sup x_n = +\infty$ and $\overline{\lim}\, x_n = +\infty$; in the case of a sequence x_n bounded from above for which the suprema v_n are not bounded from below, we say that $\underline{\lim}\, x_n = -\infty$.

For the convergence of sequences and functions we will, as a rule, use the arrow as symbol, e.g.,

$$x_n \to x \quad \text{means} \quad \lim x_n = x;$$

likewise for proper divergence ($x_n \to +\infty$ and $x_n \to -\infty$).

A statement about the natural number n $(= 1, 2, 3, \dots)$ holds *ultimately*, or *for almost all* n (Kowalewski), if it holds from a particular n on ($n \ge n_0$), or for all values of n with at most a finite number of exceptions; it holds *infinitely often* if it holds for an infinite number of values of n (e.g., for $n = 2, 4, 6, 8, \dots$). When we speak of almost all or of infinitely many terms of a sequence x_n, we mean the terms x_n belonging to almost all or to infinitely many values of n, respectively, regardless of whether the terms themselves are different or not.

Chapter 1

SETS AND THE COMBINING OF SETS

§ 1.1. Sets

A set is formed by the grouping together of single objects into a whole. A set is a plurality thought of as a unit. If these or similar statements were set down as definitions, then it could be objected with good reason that they define *idem per idem*[1] or even *obscurum per obscurius*.[2] However, we can consider them as expository, as references to a primitive concept, familiar to us all, whose resolution into more fundamental concepts would perhaps be neither competent nor necessary. We will content ourselves with this construction and will assume as an axiom that an object M inherently determines certain other objects $a, b, c, \dots$ in some undefined way, and vice versa. We express this relation by the words: The set M *consists of* the objects $a, b, c, \dots$.

A set can consist of a natural number (positive whole number) of objects, or not; it is called *finite* or *infinite* accordingly. Examples of finite sets are: the set of the inhabitants of a city, the set of hydrogen atoms in the sun, and the set of natural numbers from 1 to 1000; examples of infinite sets are: the set of all natural numbers, the set of all the points of a line, and the set of all the circles in a plane. It is to the undying credit of Georg Cantor (1845–1918) that, in the face of conflict, both internal and external against apparent paradoxes, popular prejudices, and philosophical dicta (*infinitum actu non datur*[3]) and even in the face of doubts that had been raised by the very greatest mathematicians, he dared this step into the realm of the infinite.

[1] Something in terms of that same something.

[2] The obscure by the still more obscure.

[3] There is no actual infinite.

In so doing, he became the creator of a new science, Set Theory — the consideration of finite sets being nothing more than elementary arithmetic and combinatorial analysis — which today forms the basis of all mathematics. In our opinion, it does not detract from the merit of Cantor's ideas that some antinomies that arise from allowing excessively limitless construction of sets still await complete elucidation and removal.

Following G. Peano, we designate the fundamental relation between an object a and a set A to which it belongs in the following terms and using the following notation:

a is an element of A: $a \in A$,

The negation of this statement reads:

a is not an element of A: $a \notin A$.

Two sets are defined to be *equal*, in symbols

$$A = B,$$

if and only if each element of one is also an element of the other and vice versa (i.e. if they both contain the same elements). Hence a set is uniquely determined by its elements; we express this by writing the elements of the set, enclosed in braces, as a notation for the set, elements not listed explicitly being indicated by dots. Thus

$$A = \{a\}, \quad A = \{a, b\}, \quad A = \{a, b, c\}$$

are the sets consisting of the element a, the two elements a, b, and the three elements a, b, c, respectively;

$$A = \{a, b, c, \dots\}$$

is a set consisting of the elements a, b, c, and (possibly) others. Naturally, it must be specified in some way what the other elements are that have been indicated by dots; some examples are

the set of all natural numbers $\{1, 2, 3, \dots\}$,

the set of all even natural numbers $\{2, 4, 6, \dots\}$,

the set of all squares $\{1, 4, 9, \dots\}$,

the set of all powers of 2 $\{1, 2, 4, 8, \dots\}$,

and the set of all prime numbers $\{2, 3, 5, 7, \dots\}$.

A distinction must certainly be made, at least conceptually, between the object a and the set $\{a\}$ consisting of only this one element (even if the distinction is of no importance from a practical point of view), if for no other reason than that we will admit sets (or systems) whose elements are themselves sets. The set $A = \{1, 2\}$ consists of the two elements $1, 2$, while the set $\{a\}$ consists of the single element a.

For reasons of expedience, we also admit the *null set*, or *empty set*, denoted by 0, which contains no elements.[4] By the definition of equality of sets, there is only one null set. $A = 0$ means that the set A has no elements, is empty, "vanishes." If we were not to admit the null set as a set, then in countless cases where we speak of a set we would have to add "if this set exists." For, the definition of the elements of a set often does not tell us in the least whether such elements exist; for instance, it is not yet known whether the set of natural numbers n for which the equation $x^{n+1} + y^{n+1} = z^{n+1}$ has a solution in natural numbers x, y, and z is empty or non-empty (that is, whether Fermat's celebrated Last Theorem is true or false). Thus, the statement $A = 0$ may, in many cases, represent real knowledge but, in other cases, only a triviality; many mathematical statements — or, if one is not repelled by artificial devices, all mathematical statements — can be put in the form $A = 0$. The introduction of the null set, like that of the number zero, is dictated on grounds of expedience; on the other hand, it sometimes forces one to state explicitly, as one of the hypotheses of a theorem, that a set does not vanish (just as it is sometimes necessary to state explicitly that a number does not vanish).

If A and B are two sets, then the question arises whether the elements of one may not perhaps also belong to the other. If a and b denote elements of A and B respectively, then we first consider the following two pairs of alternatives:

$$\begin{array}{ll} \text{Every } a \in B, & \text{not every } a \in B, \\ \text{every } b \in A, & \text{not every } b \in A. \end{array}$$

By combining these we obtain four possible cases, the first three of

[4]Whether the symbol 0 denotes the null set or the number zero will always be apparent from the context.

which are denoted in symbols by the accompanying formulas:

$$\begin{array}{lll} (1) & \text{Every } a \in B \text{ and every } b \in A\text{: } A = B; & (1.1) \\ (2) & \text{every } a \in B \text{ and not every } b \in A\text{: } A \subset B; & (1.2) \\ (3) & \text{not every } a \in B \text{ and every } b \in A\text{: } A \supset B; & (1.3) \\ (4) & \text{not every } a \in B \text{ and not every } b \in A. & (1.4) \end{array}$$

In case (1), the two sets are equal, by the earlier explanation. In case (2), A contains only elements of B, but not all of them, which characterizes A as the smaller set and B as the larger set; this is expressed in symbols by $A \subset B$, which is supposed to be reminiscent of the notation $a < b$ in the case of numbers. In case (3), we have the opposite situation, so that the statement $A \supset B$ amounts to the same as $B \subset A$. In general, what we will have will be none of these three cases, but rather case (4); but there is no reason to have a special notation for this case.

The relation "smaller than" is *transitive*; that is, from $A \subset B$ and $B \subset C$ it follows that $A \subset C$. (This also holds, of course, for the relations "greater than" and "equal to.")

If every $a \in B$, so that either case (1) or case (2) holds, then we denote this combination[5] by

$$A \subseteq B, \quad A \text{ is a subset of } B$$

(also, A is a part of, or contained in, B); when we have the more restrictive $A \subset B$, then A is sometimes called a *proper subset* of B. Among the subsets of B we therefore include B itself as well as the null set; for when $A = 0$, then the relation "every $a \in B$" is satisfied, trivially to be sure, by virtue of the fact that there is no a at all.[6] Thus $0 \subseteq B$ or $B \supseteq 0$ means that B is not empty. The usefulness of this convention becomes apparent, for instance, in counting the subsets of a finite set. The subsets of $\{1, 2, 3\}$ are:

$$0,\ \{1\},\ \{2\},\ \{3\},\ \{1,2\},\ \{1,3\},\ \{2,3\},\ \{1,2,3\};$$

[5]The equality sign is dispensed with in some texts; in place of our rounded less than sign, other variants of the symbol are in use.

[6]More explicitly: The statement "If $a \in A$, then $a \in B$ also" is true because the hypothesis $a \in A$ is never satisfied; a statement of this kind is said to be *vacuously true.* If p and q are statements, then the statement "If p is true, then q is true" (p implies q) is certainly true when p is false. A false statement implies any statement; if $2 \times 2 = 5$, then ghosts exist.

there are $8 = 2^3$ of them. A set of n elements contains $\binom{n}{m}$ subsets that contain m elements each, where $\binom{n}{m} = \frac{n!}{m!\,(n-m)!}$ is the binomial coefficient and where $\binom{n}{0} = \binom{n}{n} = 1$. The total number of subsets is

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{m} + \cdots + \binom{n}{n} = 2^n.$$

This simple result speaks in favor of the concept of subset as introduced above.

If $A \subset B$, then we let

$$B - A$$

denote the set of elements of B that are not elements of A; this difference of the two sets is also called the *complement* of A in B. Clearly,

$$B - 0 = B, \quad B - B = 0, \quad B - (B - A) = A.$$

Since some other authors follow a different usage, we wish to emphasize explicitly that in forming a difference we always assume that the subtrahend is a subset of the minuend. For instance, when we write $C - (B - A)$, we assume first of all that $A \subseteq B$, and second that $B - A \subseteq C$. Example: Let $A = \{5, 6, 7, \dots\}$ be the set of all the natural numbers from five on, $B = \{1, 2, 3, \dots\}$ the set of all the natural numbers, and $C = \{1, 2, 3, 4, 5, 6\}$ the set of the first six natural numbers. Then we have $B - A = \{1, 2, 3, 4\}$, $C - (B - A) = \{5, 6\}$.

§ 1.2. Functions

The concept of function is almost as fundamental and primitive as the concept of set. A functional relation is formed from pairs of elements just as a set is formed from individual elements.

We consider, instead of individual elements, a grouping together of two elements in a definite order — or, as we shall call it, an *ordered pair* (a, b) of elements — where a is the first element and b the second. Two such ordered pairs will be considered to be the same if and only if they have the same first element and the same second element:

$$(a^*, b^*) = (a, b) \quad \text{means} \quad a^* = a \text{ and } b^* = b.$$

Accordingly, the pairs (a, b) and (b, a) are different if $a \neq b$; on the other hand, nothing prevents us from forming an ordered pair (a, a)

of two equal elements. For instance, by combining natural numbers, we obtain the ordered pairs

$$(1,1),\ (1,2),\ (2,1),\ (1,3),\ (2,2),\ (3,1),\ \ldots\,;$$

the double subscripts of the elements of a matrix or a determinant are ordered pairs of this kind. By combining real numbers, we obtain ordered pairs of numbers (x, y); the use of such pairs of cartesian coordinates, where the abscissa x and the ordinate y may not be interchanged, makes it possible to represent the points of the plane.

The ordered pair (a, b) is a different concept from the set $\{a, b\}$; in the latter, a and b are assumed to be distinct, and it does not matter in what order they occur.

Ordered pairs make possible the introduction of the concept of *function*, and they will also serve to define multiplication (§ 4) and the ordering (§ 9) of sets. Let P be a set of ordered pairs $p = (a, b)$; for every pair p that occurs in P $(p \in P)$, we call b an *image* of a and a a *pre-image* (or inverse image) of b. Let A be the set of all pre-images a (i.e., all the first elements of pairs $p \in P$), B the set of all images b (i.e., all the second elements of pairs $p \in P$). Each a determines its images, each b its pre-images; and this, then, is the connection that the set of pairs P establishes between the sets A and B; we say that we have a *mapping* from one set onto the other.

In the particular case in which each a has only a single image b, we denote this element b, which is determined by a and depends on a, by

$$b = f(a),$$

and we say that this is a *single-valued* (or one-valued) or *univalent function* of a defined on the set A. For instance, the set of pairs $(1,2), (2,1), (3,2)$ defines a mapping of the set $A = \{1,2,3\}$ onto the set $B = \{1,2\}$; in fact, it defines a function that is single-valued on the set A, namely the function

$$f(1) = 2, \quad f(2) = 1, \quad f(3) = 2.$$

If in addition to this, each b has only a single pre-image a, then this element a determined by b will be denoted by

$$a = g(b),$$

and we thus have a single-valued function of b defined in B. Each of these functions is said to be the *inverse* of the other; both are said to be *one-to-one*; we call the mapping between A and B *one-to-one* (or one-one, or *schlicht*, or bi-unique), sometimes referring to it as a *one-to-one correspondence*, and we say that two sets A and B that can have this relation to each other are *equivalent*: in symbols,

$$A \sim B, \quad B \sim A.$$

This basic concept of equivalence will be fundamental for the second chapter; let us content ourselves here with the example of the equivalence between the set A of all natural numbers and the set B of all (positive) even numbers. The set of ordered pairs

$$(1, 2),\ (2, 4),\ (3, 6),\ \ldots$$

represents the one-to-one correspondence, since it assigns to every natural number a the image $b = 2a$ and to every even number b the pre-image $a = \frac{1}{2}b$.

If the element a has several images, then the notation $b = f(a)$ can be retained with the understanding that $f(a)$ represents not one but several (perhaps infinitely many) elements b; we then have a *many-valued function* $f(a)$. The corresponding statement holds for $g(b)$. These two functions, which are in general many-valued functions, are still called inverse to each other. A case which will come up often is the one in which $f(a)$ is single-valued while the inverse $g(b)$ is many-valued. For instance, the set of ordered pairs $(a, \sin a)$, where a runs through all the real numbers, defines a mapping of the set A of all real numbers onto the interval $B = \{-1 \leq y \leq 1\}$; the function $y = \sin a$ is single-valued, while the inverse $a = \arcsin y$ is many-valued.

We note, for future use, a formula concerning the pre-images of subsets: If $B_0 \subseteq B$, then the pre-image set A_0 of all pre-images of elements $b \in B_0$ is a subset of A, and we have

$$A_0 = \{a \colon f(a) \in B_0\}.$$

If $B_0 \subseteq B_1 \subseteq B$, then the corresponding pre-image sets satisfy $A_0 \subseteq A_1 \subseteq A$. The pre-image of the sum of sets is the sum of the pre-image sets, and the pre-image of the intersection is the intersection of the pre-images; i.e., if $B = B' + B''$, then $A = A' + A''$, where A', A'' are the pre-images of B', B''.

§ 1.3. Sum and Intersection

If A, B, are two sets, then by their *sum*,[7] or *union*,

$$S = A \cup B$$

we mean the set of all elements that belongs to A or B (or both); by their *intersection*

$$D = A \cap B$$

we mean the set of all elements that belong to both A and B. If $D = 0$, that is, if A and B have no element in common, then they are called *disjoint* (or mutually exclusive or non-overlapping); in this case, and in this case only, we also use the notation

$$S = A + B$$

for the sum, and we note that in this case we clearly have $S - A = B$, $S - B = A$.

Example: Let A be the interval[8] $[1, 3]$, i.e., the set of all real numbers x for which $1 \leq x \leq 3$, and let B be the interval $[2, 4]$. Then S is the interval $[1, 4]$; D, the interval $[2, 3]$.

If A, B, are disjoint finite sets and if A consists of m elements and B of n elements, then $A + B$ consists of $m + n$ elements.

We have

$$S - A = B - D, \quad S - B = A - D,$$

the first of these being the set of all elements that belong to B and not to A. Thus the intersection

$$D = B - (S - A) = A - (S - B)$$

may be written in terms of the sum and the difference.

The formation of sums and intersections may be immediately extended to any number of sets, finitely many or infinitely many. As a symbol for the sum, we use the German letter $\mathfrak{S}$; in the case of disjoint summands we may use the Greek Σ instead; and as a symbol

[7] The more common notation for the + sign is $\cup$; this notation, however, fails to take into account the distinction between + and $\cap$. On the other hand, in many places one can find the sum denoted by a simple plus sign (without the dot). The symbol $\cup$ was introduced by Carathéodory.

[8] For the notation for intervals, see p. 9.

for the intersection we use the German $\mathfrak{D}$. Let us assume that there correspond to the natural numbers $1, 2, \dots, n$, or for that matter to all of the natural numbers $1, 2, \dots,$ sets $A_1, A_2, \dots$ (so that we are already making use of the concept of single-valued function given in § 2) which, moreover, need by no means all be distinct; then their sum

$$\mathfrak{S} = A_1 \cup A_2 \cup \cdots = \bigcup_n A_n$$

is the set of all the elements that belong to at least one of the A_n, and their intersection

$$\mathfrak{D} = A_1 \cap A_2 \cap \cdots = \bigcap_n A_n$$

is the set of all elements that belong to all the A_n simultaneously. In case the summands are disjoint, i.e., pairwise non-overlapping, or, in symbols,

$$A_m \cap A_n = 0 \quad \text{for } m \neq n,$$

and in this case only, we may write the sum also in the form

$$\mathfrak{S} = A_1 + A_2 + \cdots = \sum_n A_n.$$

Finally, we have the general case: To each element m of a set $M = \{m, n, p, \dots\}$ let there correspond a set A_m; then the sum of these sets,

$$\mathfrak{S} = A_m \cup A_n \cup A_p \cup \cdots = \bigcup^M A_m,$$

is the set of all the elements that belong to at least one A_m, and the intersection of these sets,

$$\mathfrak{D} = A_m \cap A_n \cap A_p \cap \cdots = \bigcap^M A_m,$$

is the set of all the elements that belong to all of the A_m simultaneously; in case the summands are disjoint (pairwise non-overlapping), we also write

$$\mathfrak{S} = A_m + A_n + A_p + \cdots = \sum^M A_m.$$

Examples: Let $M = \{0, 1, 2, \dots\}$ be the set of integers ≥ 0, and let A_m be the set of natural numbers that are divisible by precisely the m-th power of 2 (i.e., that are divisible by 2^m but not by any higher power of 2), so that

$$\begin{aligned}
A_0 &= \{1, 3, 5, 7, \ldots\}, \\
A_1 &= \{2, 6, 10, 14, \ldots\}, \\
A_2 &= \{4, 12, 20, 28, \ldots\}, \\
A_3 &= \{8, 24, 40, 56, \ldots\}, \\
&\text{etc.}
\end{aligned}$$

Then

$$\mathfrak{S} = A_0 \cup A_1 \cup A_2 \cup \cdots = \sum_n A_n$$

is the set of all natural numbers.

Let M be the set of real numbers $m > 1$, and let $A_m = [0, m)$ be the interval $0 \leq x < m$. Then

$$\mathfrak{D} = \bigcap A_m = [0, 1]$$

is the interval $0 \leq x \leq 1$.

These two operations, the forming of the sum and the forming of the intersection, are commutative, associative, and distributive with respect to each other; that is, to give only the simplest cases, we have

$$\begin{gathered}
A \cup B = B \cup A, \quad A \cap B = B \cap A, \\
(A \cup B) \cup C = A \cup (B \cup C) = A \cup B \cup C, \quad (A \cap B) \cap C = A \cap (B \cap C) = A \cap \\
(A \cup B) \cap C = A \cap C \cup B \cap C, \\
A \cap B \cup C = (A \cup C) \cap (B \cup C).
\end{gathered}$$

Of the last two formulas — the distributive ones — the first is obvious and the second is also easy to see; we also note the next most general case

$$\begin{aligned}
C \cap \bigcup A_n &= \bigcup (C \cap A_n), \\
C \cup \bigcap A_n &= \bigcap (C \cup A_n).
\end{aligned}$$

If all the A_n are subsets of an encompassing set E, and if $B_n = E - A_n$ are their respective complements, then

$$E = \bigcup A_n \cup \bigcap B_n = \bigcap A_n \cup \bigcup B_n.$$

For, every element of E belongs either to at least one A_m (and so to $\bigcup A_m$), or to none, in which case it belongs to all the B_m (and so to $\bigcap B_m$). This important formula can be expressed in concise form as follows: The complement of a sum is the intersection of the complements, and the complement of an intersection is the sum of the complements. Thus, if a set P is formed from the sets A_n by the repeated taking of sums and intersections, then its complement $Q = E - P$ can be obtained by replacing the A_n by their complements B_n, and at the same time interchanging the operations $\mathfrak{S}$ and $\mathfrak{D}$. Thus from one of the formulas that constitute the distributive law,

$$A \cap \bigcup A_n = \bigcup A \cap A_n,$$

there immediately follows the other,

$$A \cup \bigcap B_n = \bigcap (A \cup B_n).$$

If this process, the transition to complements, is applied to an inequality between sets (set-theoretic inequality), then the $<$ and $>$ signs must also be interchanged (because $P \subset P^*$ implies $Q \supset Q^*$ for the complements).

Let $A_1, A_2, A_3, \ldots,$ be a sequence of sets, i.e., to every natural number n let there correspond a set A_n. By its *upper limit* (or *limes superior*), denoted by

$$\overline{A} = \overline{\lim}\, A_n$$

(or by lim sup), we shall mean the set of those elements x that belong to infinitely many A_n ($x \in A_n$ for infinitely many n); by its *lower limit* (*limes inferior*), denoted by

$$\underline{A} = \underline{\lim}\, A_n$$

(or by lim inf), we shall mean the set of those elements x that belong to almost all A_n ($x \in A_n$ for almost all n). Since the second requirement is more stringent than the first, we always have $\overline{A} \supseteq \underline{A}$; if, in particular the equality sign holds, then the set $A = \overline{A} = \underline{A}$ is called the *limit* of the sequence A_n of sets, and the sequence is called *convergent*; we write

$$A = \lim A_n.$$

Examples: The sequence $M, N, M, N, \ldots$ has $M \cup N$ as its upper limit and $M \cap N$ as its lower limit; it converges only if $M = N$.

An increasing sequence $A_1 \subseteq A_2 \subseteq \cdots$ converges to the sum $\mathfrak{S}A_n$, a decreasing sequence $A_1 \supseteq A_2 \supseteq \cdots$ converges to the intersection $\mathfrak{D}A_n$, and a sequence of disjoint sets converges to the null set. Let A_1 denote the interval $[0,1]$; A_2 and A_3, the intervals $[0,\frac{1}{2}]$ and $[\frac{1}{2},1]$, respectively; A_4, A_5, and A_6, the intervals $[0,\frac{1}{3}]$, $[\frac{1}{3},\frac{2}{3}]$, and $[\frac{2}{3},1]$, respectively; and so on. Then the upper limit of the sequence A_n is the interval $[0,1]$, and the lower limit is the null set. [9]

If E is once again a set containing all the A_n, and if $B_n = E - A_n$, then we have

$$\overline{E} = A \cup B = \underline{A} \cup \overline{B}.$$

For, an $x \in E$ belongs either to almost all A_n, i.e., to only finitely many B_n, or to infinitely many B_n.

The sets $\overline{A}$ and $\underline{A}$ may be formed from the A_n by the repeated use of the sum and intersection; in fact

$$\begin{aligned} \overline{A} &= \mathfrak{D}_1 \cup \mathfrak{D}_2 \cup \mathfrak{D}_3 \cup \cdots, \quad & \mathfrak{D}_n &= A_n \cap A_{n+1} \cap A_{n+2} \cap \cdots, \\ \underline{A} &= \mathfrak{S}_1 \cap \mathfrak{S}_2 \cap \mathfrak{S}_3 \cap \cdots, \quad & \mathfrak{S}_n &= A_n \cup A_{n+1} \cup A_{n+2} \cup \cdots. \end{aligned}$$

The first formula follows at once from the definition, and the simplest way of proving the second is by taking complements.

The sets $\overline{A}$ and $\underline{A}$ remain unchanged if we add, remove, or change a finite number of sets of the sequence A_n. Furthermore, if A_{p_n} is a subsequence (where p_n runs through a sequence of natural numbers), then we clearly have

$$\underline{\lim}\, A_n \subseteq \underline{\lim}\, A_{p_n} \subseteq \overline{\lim}\, A_{p_n} \subseteq \overline{\lim}\, A_n.$$

If the whole sequence converges to A, so does every subsequence.

Characteristic functions (C. de la Vallée Poussin). There can be associated with every subset A of a fixed set E a function $f(x)$ defined on E which takes on only the values 0 and 1, and which uniquely determines and is uniquely determined by A; namely the function

$$f(x) = 1 \text{ for } x \in A, \quad f(x) = 0 \text{ for } x \in E - A.$$

This is called the *characteristic function* of the set A; we denote it simply by $[A]$, omitting the argument x and thus emphasizing only its dependence on A. To the whole set E there corresponds the constant function $[E] = 1$; to the null set 0, the constant function $[0] = 0$.

[9] For the meaning of *almost all*, see the Introduction.

To the combinations of sets there correspond simple combinations of the characteristic functions. Thus

$$[B - A] = [B] - [A] \cap (B)$$

is an equation which, like those that follow, holds for every x. In fact, for the three possible cases, we have

	$x \in E - B$	$B - A$	A
$[A]$	0	0	1
$[B]$	0	1	1
$[B - A]$	0	1	0

In particular, we have

$$[E - A] = 1 - [A].$$

Furthermore,

$$[A \cap B] = [A] \cdot [B],$$

and by forming the complements, we obtain

$$[A \cup B] = 1 - (1 - [A])(1 - [B]),$$
$$[A \cup B] + [A \cap B] = [A] + [B],$$

and in particular, for disjoint summands,

$$[A + B] = [A] + [B].$$

On the other hand, we also have

$$[A \cup B] = \max\{[A], [B]\}, \quad [A \cap B] = \min\{[A], [B]\};$$

and more generally, for the sum $\mathfrak{S} = \bigcup A_m$ and the intersection $\mathfrak{D} = \bigcap A_m$ of an arbitrary number of sets, we have

$$[\mathfrak{S}] = \max\{[A_m]\}, \quad [\mathfrak{D}] = \min\{[A_m]\}.$$

For the upper limit $\overline{A} = \overline{\lim}\, A_n$ and the lower limit $\underline{A} = \underline{\lim}\, A_n$ of a sequence of sets, we have

$$[\overline{A}] = \overline{\lim}[A_n], \quad [\underline{A}] = \underline{\lim}[A_n];$$

for, $\overline{\lim}[A_n]$ is $= 1$ if and only if $[A_n] = 1$ (i.e. $x \in A_n$) infinitely often, so that $x \in \overline{A}$ (otherwise $\overline{\lim}[A_n] = 0$); and $\underline{\lim}[A_n]$ is $= 1$ if and only if $x \in A_n$ ultimately, that is, for all n greater than some fixed N, so that $x \in \underline{A}$. In the case of the limit $A = \lim A_n$ of a convergent sequence of sets, we have $[A] = \lim[A_n]$.

§ 1.4. Products and Powers

It remains to define the product of sets. Let us form from two sets A and B the set P of ordered pairs (§ 2) $p = (a, b)$, where a runs through all the elements of A and b runs through all the elements of B, so that

$$p \in P \quad \text{means that} \quad a \in A \text{ and } b \in B.$$

If the sets[10] are finite, A consisting of m elements and B of n elements, then P consists of mn elements, so that this set of pairs has in fact the character of a product. Note that this complete set of pairs, which contains all the pairs for which $a \in A$ and $b \in B$, has as subsets all those sets of pairs that, by § 2, define a functional relation between A and B; it itself defines the "most-valued" mapping, the mapping that is the most strongly many-valued — in the sense that every b corresponds under the mapping to each a, and every a to each b. The simple notation AB, already having been used for the intersection, is no longer available as a symbol for the product. But we can form an ordered pair (A, B) from the sets A and B themselves, and we are free to let this new symbol denote a set also: we define the ordered pair of sets to be the set of ordered pairs of elements. That is, for ordered pairs of elements and of sets we shall define the $\in$-relation

$$(a, b) \in (A, B) \quad \text{to mean} \quad a \in A \text{ and } b \in B,$$

while objects other than pairs of elements are not to be admitted as elements of a set (A, B). Thus

$$P = (A, B)$$

is the product of the two sets.

Example: Let A and B be sets of real numbers. Let the pair of numbers (a, b) be represented by the point of the plane with cartesian coordinates a and b or, for short, by the point (a, b); through every point $(a, 0)$ $(a \in A)$ of the x-axis draw a line parallel to the y-axis, and through every point $(0, b)$ $(b \in B)$ of the y-axis draw a line parallel to the x-axis; then the intersections (a, b) of these lines form the product (A, B).

[10] They do not have to be disjoint and could even be identical.

The product may be thought of as the sum of equivalent summands (§ 6).

There is no difficulty in generalizing the product to more than two factors. Ordered triples (a, b, c) may be defined analogously to ordered pairs (a, b); they are groupings together of three elements in a definite order, so that equality is defined thus:

$$(a^*, b^*, c^*) = (a, b, c) \quad \text{means that} \quad a^* = a,\ b^* = b, \text{ and } c^* = c;$$

moreover, these three elements do not have to be distinct. The ordered triples (a, b, c), where a runs through all the elements of A, b runs through all the elements of B, and c runs through all the elements of C, by definition form the set (A, B, C) — the product of the three sets A, B, and C. We may proceed similarly for any finite number of factors.

The commutative and associative laws hold for the product — not, to be sure, in the sense of equality, but in the sense of equivalence (§ 2). Thus, although the elements (a, b) and (b, a) of the sets (A, B) and (B, A) respectively are not the same, they are nevertheless in one-to-one correspondence, so that $(A, B) \sim (B, A)$. Similarly, the elements $((a, b), c)$ of $((A, B), C)$, i.e., the ordered pairs whose first elements are ordered pairs (a, b) and whose second elements are elements c, are not identical with the triples (a, b, c) — the members of (A, B, C) — but are in one-to-one correspondence with them, and we have

$$((A, B), C) \sim (A, (B, C)) \sim (A, B, C).$$

In order to define products with an arbitrary set of factors, we first generalize the concept of ordered pair or ordered triple by assigning to each element m of a set $M = \{m, n, \dots\}$ some quite arbitrary element a_m; in other words, we define a single-valued function $f(m) = a_m$ on M. Grouping these elements together yields what is called a *complex*, or *element complex*

$$p = (a_m, a_n, a_q, \dots)$$

or (following Cantor) a *covering* of the elements m by the elements a_m. Neither name implies a new construction; both are just synonyms for the function $f(m)$ defined on M. Two complexes are said to be equal if and only if there corresponds to every m the same element a_m (as its image), that is,

$$(a^*_m, a^*_n, a^*_q, \dots) = (a_m, a_n, a_q, \dots) \quad \text{means that} \quad a^*_m = a_m,\ a^*_n = a_n,\ a^*_q = a_q,\ \dots$$

or, in other words, two functions are considered to be equal if and only if they have the same "value" for every m: $f^*(m) = f(m)$; this rule replaces the requirement of a definite order in which the elements must appear in the ordered pairs, triples, etc. Moreover, the complexes depend on the basic set M, a fact that is not fully expressed in the above notation; a complex of three elements that assigns to the elements $1, 2, 3$ the elements a, b, c as images is not the same as one that assigns to the elements $4, 5, 6$ the same elements a, b, c, and therefore it is also not the same as an ordered triple (a, b, c), which assigns the elements a, b, c to three definite places in a written or printed array. However — and, after all, this is what counts — the triples (a, b, c) can certainly be brought into one-to-one correspondence with the complexes that assign to the members of an arbitrary set of three elements the three elements a, b, c as images.

If we now assign a set A_m to every $m \in M$, then we obtain a *set complex*

$$P = (A_m, A_n, A_q, \dots)$$

and, at the same time, we define this to be a set of complexes, namely, the set of element complexes $p = (a_m, a_n, a_q, \dots)$, in which a_m runs through all the elements of A_m, a_n runs through all the elements of A_n, etc. P is defined to be the *product* of the sets $A_m, A_n, A_q, \dots$. Using the Greek Π as an abbreviation for the product, we also write

$$P = \prod^{M} A_m.$$

The correspondence between the m and the A_m amounts to a definition of a single-valued function $F(m) = A_m$ on M, a function whose "values" are not elements, but sets. And the product P is the set of all single-valued functions $f(m)$ for which

$$f(m) \in F(m) \quad \text{for every } m \in M.$$

Finally, if we let all the A_m coincide — $A_m = A$ for all m — then we define the *power*

$$P = A^M$$

(with base A and exponent M) as a product of factors that are all equal. This, therefore, is the set of all element complexes $p = (a_m, a_n, \dots)$ whose elements belong to A ($a_m \in A$, $a_n \in A$, $\dots$) or the set of all single-valued functions $a = f(m)$ that assign an image $a \in A$ to every $m \in M$.

An important example: Let $A = \{a, b\}$ consist of two elements; A^M is the set of functions $f(m)$ for which $f(m) = a$ or b. Each such function $f(m)$ determines the set M_1 of those m for which $f(m) = a$, and the set M_2 of those m for which $f(m) = b$, where

$$M = M_1 + M_2$$

is a decomposition of M into two complementary subsets. Conversely, if M_1 is an arbitrary subset of M and $M_2 = M - M_1$ is its complement in M, and if we define

$$f(m) = a \text{ for } m \in M_1, \quad f(m) = b \text{ for } m \in M_2,$$

then we have one of our functions $f(m)$. The functions $f(m)$ and the sets $M_1 \subseteq M$ are thus in one-to-one correspondence, that is, A^M is equivalent to the set of all subsets of M. [11]

It can be seen similarly that, for a set $A = \{a, b, c\}$ with three elements, A^M is equivalent to the set of all partitions

$$M = M_1 + M_2 + M_3$$

into three disjoint summands, where the order of the summands counts; that is, the partition $M = M_1' + M_2' + M_3'$ is to be considered the same as the above if and only if $M_1' = M_1$, $M_2' = M_2$, $M_3' = M_3$.

What has been said in this chapter about the basic concepts so far introduced should only be regarded as preliminary orientation. In particular, the construction of sets by means of sum and intersection will have to be dealt with more thoroughly later on (Chapter V). In the next three chapters, the intersection will play a minor role as compared with the other combinatorial operations on sets; in the later chapters, on the other hand, the product will play the minor role.

[11] For $a = 1$, $b = 0$, $f(m)$ is the characteristic function (§ 3) of M_1.

Chapter 2

CARDINAL NUMBERS

§ 2.1. Comparison of Sets

We called two sets equivalent (§ 2), in symbols

$$A \sim B,$$

if there exists a one-to-one correspondence between their elements such that to each a there corresponds a unique $b = f(a)$ and to each b a unique $a = g(b)$. Clearly, we have

$$A \sim A;$$
$$\text{if } A \sim B, \text{ then } B \sim A;$$
$$\text{if } A \sim B, \ B \sim C, \text{ then } A \sim C;$$

we say that the relation of equivalence is *reflexive*, *symmetric*, and *transitive*.

Finite sets[1] are obviously equivalent if and only if they have the same number of elements. We shall therefore say in general that equivalent sets have the same *cardinal number* or *cardinality* (or *power*). That is, we assign to each set A an object $\mathfrak{a}$ in such a way that equivalent sets, and equivalent sets only, have the same object corresponding to them:

$$\mathfrak{a} = \mathfrak{b} \quad \text{means that} \quad A \sim B.$$

We call these new objects *cardinal numbers* or *powers* and say: A has the cardinality $\mathfrak{a}$, $\mathfrak{a}$ is the cardinality of A, A has cardinal number $\mathfrak{a}$,

[1] A complete theory of sets should also formulate precisely the theory of finite sets and of the natural numbers; we shall assume here that these preliminaries have been taken care of.

$\mathfrak{a}$ is the cardinal number of A, and also (using $\mathfrak{a}$ linguistically as a number) A has $\mathfrak{a}$ elements.

This formal explanation says what the cardinal numbers are supposed to do, not what they are. More precise definitions have been attempted, but they are unsatisfactory and unnecessary. Relations between cardinal numbers are merely a more convenient way of expressing relations between sets; we must leave the determination of the "essence" of the cardinal number to philosophy.

A finite set consisting of n elements $(n = 1, 2, 3, \dots)$ will be assigned the number n as its cardinality; the null set, the number 0.

The cardinality of the set $\{1, 2, 3, \dots\}$ of natural numbers is called $\aleph_0$ (read: aleph-null).[2] Sets of this cardinality, which can be written in the form of a sequence,

$$\{a_1, a_2, a_3, \dots\} \quad (a_m \neq a_n \text{ for } m \neq n)$$

are all *countable*[3] (or enumerable or denumerable).

We assign to the set of all real numbers and to the equivalent set of all the points of a straight line the cardinality $\aleph$ (Alef); it is called the cardinality of the *continuum.*

The cardinality of countable sets is often denoted by $\mathfrak{a}$; that of the continuum, by $\mathfrak{c}$.

When we first introduced the concept of equivalence, we mentioned that the set of all natural numbers is equivalent with the set of all even numbers. As is shown by the following diagram, in which the numbers in the same column correspond, the sets of all natural numbers, of all even numbers, of all odd numbers, of all squares, of all powers of 10, are all equivalent to each other, and therefore all have the same cardinality:

$$\begin{array}{cccccccc}
1 & 2 & 3 & \dots & n & & \dots & \\
2 & 4 & 6 & \dots & 2n & & \dots & \\
1 & 3 & 5 & \dots & 2n-1 & & \dots & \\
1 & 4 & 9 & \dots & n^2 & & \dots & \\
10 & 100 & 1000 & \dots & 10^n & & \dots & \\
2 & 3 & 5 & \dots & p_n & \dots & & (p_n \text{ is the } n\text{-th prime}).
\end{array}$$

[2] $\aleph$ (aleph) is the first letter of the Hebrew alphabet.

[3] Its elements can be "counted" (enumerated) by means of all the natural numbers. Sets that are finite (including the null set) or countable will be called *at most countable*; infinite sets that are not countable are called *uncountable* (or nonenumerable or nondenumerable).

One can extend the list and form sequences of numbers that have ever faster rates of growth (consider, for example, 10, 10^{10}, $10^{10^{10}}$, ...), which nevertheless, no matter how sparsely they are sown in the set of all natural numbers, have a cardinality no smaller than that of the whole set. This violation of the axiom "*totum parte majus*" (the whole is greater than any of its parts) is one of those so-called paradoxes of the infinite which one must get used to; there are, of course, differences between the laws that pertain to finite sets and those that pertain to infinite sets; these differences do not, of course, in the least justify any objections to infinite sets.

The sets of points (or of numbers) constituting the intervals $0 \leq x \leq 1$ and $0 \leq y \leq 1000$ have the same cardinality even though the second is a thousand times as long as the first; the one-to-one correspondence is given by $y = 1000x$ or by any other projection. The interval $-\pi/2 < x < \pi/2$ and the set of all real numbers have the same cardinality by virtue of the correspondence $y = \tan x$.

We shall see in a moment that this relation — the equivalence between a set and a proper subset thereof — is characteristic of infinite sets.

I. *Every infinite set contains a countable subset.*

Let a_1 be an element of the infinite set A, a_2 an element of the (still) infinite set $A - \{a_1\}$, a_3 an element of the (still) infinite set $A - \{a_1, a_2\}$, etc. The pairwise distinct elements $a_1, a_2, a_3, \ldots$ constitute a countable subset of A.

II. *Every infinite set is equivalent to a proper subset.*

By removing from the infinite set A a countable subset and denoting its complement in A by B, we obtain

$$A = \{a_1, a_2, \ldots, a_n, \ldots\} + B;$$

this is equivalent with, say,

$$\{a_2, a_3, \ldots, a_n, \ldots\} + B$$

(let a_n in the first set correspond to a_{n+1} in the second set, and let each $b \in B$ correspond to b itself).

Conversely, no set that is equivalent to one of its proper subsets can be finite. This property therefore *characterizes* infinite sets (and was used by Dedekind as the definition of infinite sets).

THEOREM 2.1.1 (Equivalence Theorem of F. Bernstein)**.** *If each of two given sets is equivalent to a subset of the other, then the two given sets are themselves equivalent.*

Let $A \sim B_1$, $B \sim A_1$, where A_1 is a subset of A, and moreover a proper subset, as otherwise there is nothing to prove; thus $A_1 \subset A$, $B_1 \subset B$. By the one-to-one mapping of B onto A_1, B_1 is also mapped onto a proper subset A_2 of A_1, so that

$$A \supset A_1 \supset A_2, \quad A \sim B_1 \sim A_2.$$

Thus the theorem is reduced to the following:

If $A \supset A_1 \supset A_2$, $A \sim A_2$, then $A \sim A_1$,

i.e., if a set lies between two equivalent sets, it is equivalent to them.

Under the one-to-one mapping from A onto A_2, let A_1 be mapped onto A_3, A_2 onto A_4, A_3 onto A_5, etc., so that

$$A \supset A_1 \supset A_2 \supset A_3 \supset A_4 \supset A_5 \supset \cdots .$$

If

$$D = A \cap A_1 \cap A_2 \cdots$$

is the intersection of the sets A_n, then

$$\begin{aligned} A &= D + (A - A_1) + (A_1 - A_2) + (A_2 - A_3) + (A_3 - A_4) + \cdots , \\ A_1 &= D + (A_1 - A_2) + (A_2 - A_3) + (A_3 - A_4) + \cdots . \end{aligned}$$

For to prove, say, the first formula, we note that an element $a \in A$ either belongs to all the A_n, and thus to D, or there exists a first A_n to which it does not belong while it still belongs to A_{n-1}, so that $a \in A_{n-1} - A_n$ (where we have set $A_0 = A$). From

$$A - A_1 \sim A_1 - A_2 \sim A_2 - A_3 \sim \cdots$$

it follows that

$$(A-A_1)+(A_2-A_3)+(A_4-A_5)+\cdots \sim (A_1-A_2)+(A_3-A_4)+(A_5-A_6)+\cdots ,$$

the set on the left-hand side being mapped onto the set on the right. By mapping the remaining elements of A onto themselves, we obtain a one-to-one mapping of A onto A_1, so that $A \sim A_1$.

Until now we have only discussed equivalence, that is, the *equality* of two cardinal numbers. The next question would be: If two sets are not equivalent, so that their cardinal numbers are unequal, then can one of these cardinal numbers be defined in a natural way as being the larger and the other as the smaller? In short, do cardinal numbers have the character of magnitude? Can they be compared with one another?

It seems at first that the answer would have to be in the negative. If A and B denote sets, and A_1 and B_1 any subsets of A and B respectively, then there are four possibilities:

(1) There is an $A_1 \sim B$, and a $B_1 \sim A$.

(2) There is no $A_1 \sim B$, but there is a $B_1 \sim A$.

(3) There is an $A_1 \sim B$, but no $B_1 \sim A$.

(4) There is no $A_1 \sim B$, and no $B_1 \sim A$.

In case (1) we have, by the Equivalence Theorem, $\mathfrak{a} = \mathfrak{b}$; in case (2) it is natural to define $\mathfrak{a} < \mathfrak{b}$, and in case (3) to define $\mathfrak{a} > \mathfrak{b}$. In case (4), because of its symmetry with respect to the two sets, it is obviously impossible to define either $\mathfrak{a} < \mathfrak{b}$ or $\mathfrak{a} > \mathfrak{b}$, for both would have to hold; equally inadmissible is $\mathfrak{a} = \mathfrak{b}$, which would contradict the natural definition of equality that we have been using so far. We thus have a fourth relation, which we write as $\mathfrak{a} \parallel \mathfrak{b}$ and call *incomparability* between $\mathfrak{a}$ and $\mathfrak{b}$, while in the first three cases $\mathfrak{a}$ and $\mathfrak{b}$ will be called *comparable*. Thus:

$$\begin{array}{ll} (1)\ \mathfrak{a} = \mathfrak{b} & \\ (2)\ \mathfrak{a} < \mathfrak{b} & \mathfrak{a}, \mathfrak{b} \text{ comparable} \\ (3)\ \mathfrak{a} > \mathfrak{b} & \\ (4)\ \mathfrak{a} \parallel \mathfrak{b} & \mathfrak{a}, \mathfrak{b} \text{ incomparable.} \end{array}$$

Comparability could therefore only be saved if we could show that the fourth case can never actually occur.

§ 2.2. Sum, Product, and Power

Let A and B be disjoint. If both are finite, and A consists of m elements and B of n elements, then $A+B$ consists of $m+n$ elements. Furthermore, if $A \sim A_1$, $B \sim B_1$, and A_1 and B_1 are also disjoint, then $A+B \sim A_1+B_1$. These remarks justify the following definition:

The sum $\mathfrak{a} + \mathfrak{b}$ of two cardinal numbers is the cardinality of the set-theoretic sum $A \cup B$, where A and B are any two disjoint sets having the cardinalities $\mathfrak{a}$ and $\mathfrak{b}$ respectively.

More generally, if the cardinalities $\mathfrak{a}_m$ correspond to the elements m of a set

$$M = \{m, n, p, \dots\},$$

then

$$\sum^M \mathfrak{a}_m = \mathfrak{a}_m + \mathfrak{a}_n + \mathfrak{a}_p + \cdots$$

is the cardinality of the set-theoretic sum

$$\sum^M A_m = A_m \cup A_n \cup A_p \cup \cdots,$$

where the A_m are disjoint (pairwise non-overlapping) sets of cardinality $\mathfrak{a}_m$.

Examples: The simplest examples of countable sets are: a set consisting of countably many finite sets (e.g., the set of all prime numbers, since it is a subset of the set of all natural numbers); a finite sum of countable sets (e.g., the set of all integers, positive, negative, and zero). We have

$$\aleph_0 + n = \aleph_0, \quad \aleph_0 + \aleph_0 = \aleph_0.$$

The set of all natural numbers can moreover be split into countably many countable sets, as the following arrays, among others, show:

$$\begin{array}{cccc|cccc}
1 & 3 & 5 & 7\dots & 1 & 2 & 4 & 7\dots \\
2 & 6 & 10 & 14\dots & 3 & 5 & 8 & \dots \\
4 & 12 & 20 & 28\dots & 6 & 9 & \dots & \\
8 & 24 & 40 & 56\dots & 10 & \dots & & \\
\multicolumn{4}{c|}{\text{(dyadic array)}} & \multicolumn{4}{c}{\text{(diagonal array)}}
\end{array}$$

In the first, the m-th row consists of those numbers that are divisible by 2^{m-1} and by no higher power of 2; in the second, the numbers are arranged diagonally (from upper right to lower left). We thus obtain the equation

$$\aleph_0 + \aleph_0 + \aleph_0 + \cdots = \aleph_0$$

or, more precisely,

$$\mathfrak{a}_1 + \mathfrak{a}_2 + \mathfrak{a}_3 + \cdots = \aleph_0 \quad \text{for } \mathfrak{a}_1 = \mathfrak{a}_2 = \mathfrak{a}_3 = \cdots = \aleph_0.$$

But

$$1 + 1 + 1 + \cdots = \aleph_0$$

also, as can be seen from the decomposition of a countable set into its individual elements, and therefore, by the Equivalence Theorem,

$$\mathfrak{a}_1 + \mathfrak{a}_2 + \mathfrak{a}_3 + \cdots = \aleph_0 \quad \text{for } 1 \leq \mathfrak{a}_n \leq \aleph_0,$$

e.g., $2 + 2 + 2 + \cdots = \aleph_0$,

$$1 + 2 + 3 + 4 + \cdots = \aleph_0.$$

More precisely, $\mathfrak{a}_1 + \mathfrak{a}_2 + \mathfrak{a}_3 + \cdots = \aleph_0$ for $\mathfrak{a}_1 = \mathfrak{a}_2 = \mathfrak{a}_3 = \cdots = \aleph_0$.

For every infinite cardinal number $\mathfrak{a}$, we have $\mathfrak{a} + \aleph_0 = \mathfrak{a}$. For by § 5, I, we may write $\mathfrak{a} = \mathfrak{b} + \aleph_0$, so that

$$\mathfrak{a} + \aleph_0 = (\mathfrak{b} + \aleph_0) + \aleph_0 = \mathfrak{b} + (\aleph_0 + \aleph_0) = \mathfrak{b} + \aleph_0 = \mathfrak{a}.$$

The cardinality of an infinite set is not reduced by the elimination of a finite number of elements; that is, if $\mathfrak{a} = \mathfrak{b} + n$, where $\mathfrak{a}$ is infinite and n finite, then $\mathfrak{a} = \mathfrak{b}$. In fact,

$$\mathfrak{a} = \mathfrak{a} + n = \mathfrak{b} + (\aleph_0 + n) = \mathfrak{b} + \aleph_0 = \mathfrak{b},$$

since $\mathfrak{b}$ is also infinite.

If A consists of m and B of n elements, then the product (A, B) has mn elements (§ 4). Furthermore, we have in general: If $A_1 \sim A$ and $B_1 \sim B$, then $(A_1, B_1) \sim (A, B)$. This justifies the following definition:

The product $\mathfrak{ab}$ of two cardinal numbers is the cardinality of the set-theoretic product (A, B), where A and B are any two sets having the cardinalities $\mathfrak{a}$ and $\mathfrak{b}$ respectively.

The product may also be formed for arbitrarily many factors. If $\mathfrak{a}_m$ are the cardinalities corresponding to the elements m of a set M, then

$$\prod^{M} \mathfrak{a}_m = \mathfrak{a}_m \cdot \mathfrak{a}_n \cdot \mathfrak{a}_p \cdots$$

is the cardinality of the product

$$\prod^{M} A_m = (A_m, A_n, A_p, \dots).$$

For powers we set

$$\mathfrak{a}^m = \prod^{M} \mathfrak{a}, \quad \text{where } m = |M|;$$

thus in the realm of the infinite also, the addition of equal summands leads to multiplication, and the multiplication of equal factors leads to the forming of powers. To prove the first formula, let us consider those members of the set (A, M) of pairs (a, m) with $a \in A$ and $m \in M$ whose second members are all equal to some given m; these form a set A_m equivalent to A, and the equation between cardinal numbers then follows from the set-theoretic equation

$$(A, M) = \sum^{M} A_m.$$

To prove the second formula, we can simply set $A_m = A$ in $\prod A_m$ and obtain A^M.

The commutative, associative, and distributive laws hold for the processes just defined, but we will not plague the reader with a complete exposition of them; we will content ourselves with noting that

$$\mathfrak{a}(\mathfrak{b}_1 + \mathfrak{b}_2 + \cdots) = \mathfrak{a}\mathfrak{b}_1 + \mathfrak{a}\mathfrak{b}_2 + \cdots\,,$$

$$(\mathfrak{a}_1 + \mathfrak{a}_2 + \cdots)\mathfrak{b} = \mathfrak{a}_1\mathfrak{b} + \mathfrak{a}_2\mathfrak{b} + \cdots\,,$$

$$\mathfrak{a}^{\mathfrak{b}_1+\mathfrak{b}_2+\cdots} = \mathfrak{a}^{\mathfrak{b}_1} \cdot \mathfrak{a}^{\mathfrak{b}_2} \cdots\,, \quad (\mathfrak{a}^{\mathfrak{b}})^{\mathfrak{c}} = \mathfrak{a}^{\mathfrak{b}\mathfrak{c}}, \quad (\mathfrak{a}_1 \cdot \mathfrak{a}_2 \cdots)^{\mathfrak{b}} = \mathfrak{a}_1^{\mathfrak{b}} \cdot \mathfrak{a}_2^{\mathfrak{b}} \cdots\,.$$

Examples of product and power:

$$2\aleph_0 = \aleph_0 + \aleph_0 = \aleph_0,$$

$$n\aleph_0 = \aleph_0 + \cdots + \aleph_0 = \aleph_0,$$

and from the sum of countably many summands each of which is $\aleph_0$ we obtain

$$\aleph_0 \cdot \aleph_0 = \aleph_0 + \aleph_0 + \cdots = \aleph_0,$$

$$n\aleph_0 = \aleph_0 \cdot \aleph_0 = \cdots = \aleph_0, \quad \aleph_0^n = \aleph_0 \cdot \aleph_0 \cdots = \aleph_0.$$

$\mathfrak{a}^{\aleph_0}$ is the cardinality of the set of all functions $a = f(n)$ (n a natural number, $a \in A$), or, alternatively, of all sequences

$$(a_1, a_2, \ldots) \quad (a_n \in A),$$

where A denotes a set of cardinality $\mathfrak{a}$; for instance, $2^{\aleph_0}$ is the cardinality of the set of all dyadic sequences, i.e., of those sequences formed from the digits 0 and 1; while $10^{\aleph_0}$ is the cardinality of the set of all decimal sequences, i.e., of those sequences formed from the digits $0, 1, \ldots, 9$.

§ 2.3. The Scale of Cardinal Numbers

I. *For every cardinal number* $\mathfrak{m}$*, we have* $2^{\mathfrak{m}} > \mathfrak{m}$.

First of all, there exists a system of subsets equivalent to M, namely the subsets consisting of the single elements $\{m\}$, $m \in M$. Hence $2^{\mathfrak{m}} \geq \mathfrak{m}$.

On the other hand, let us suppose that there exists an arbitrary system of subsets equivalent to M; that is, we allow a subset $M_m \subseteq M$ to correspond one-to-one to each $m \in M$; we wish to show that there then exists still another $N \subseteq M$, different from all the M_m, so that the equation $2^{\mathfrak{m}} = \mathfrak{m}$ is then impossible. The element m may either belong to the set M_m corresponding to it, or not:

$$\text{either } m \in M_m \quad \text{or} \quad m \notin M_m;$$

the set N consisting of those elements of the second kind is then certainly different from all the M_m. For since

$$m \in N \quad \text{means} \quad m \notin M_m,$$

if we were to have $N = M_m$ for some m, we would come out with the contradiction $m \in N$ and $m \notin N$. Expressed differently: If m belongs to M_m, then it does not belong to N; if m does not belong to M_m, then it belongs to N; M_m and N therefore always differ, since m belongs to one set but not to the other.

I. ensures the existence of distinct infinite cardinal numbers, and in fact of an infinite number of distinct ones; starting with one infinite $\mathfrak{m}$ (for instance $\aleph_0$ or $\aleph$), we can form

$$\begin{aligned} \mathfrak{m}_1 &= 2^{\mathfrak{m}} > \mathfrak{m}, \\ \mathfrak{m}_2 &= 2^{\mathfrak{m}_1} > \mathfrak{m}_1, \\ \mathfrak{m}_3 &= 2^{\mathfrak{m}_2} > \mathfrak{m}_2, \end{aligned}$$

(I. also holds, by the way, for finite $\mathfrak{m}$: $1 = 2^0 > 0$, $2 = 2^1 > 1$, $\ldots$). Furthermore $\mathfrak{m} + \mathfrak{m}_1 + \mathfrak{m}_2 + \cdots$ is then an even greater cardinal number, and one can continue the process by this means; in general, we have:

II. *If a cardinal number* $\mathfrak{a}_m$ *corresponds to every* $m \in M$*, and if there is no greatest among them, then the sum*

$$\mathfrak{a} = \sum^{M} \mathfrak{a}_m$$

exceeds every $\mathfrak{a}_m$.

For, first of all, we certainly have $\mathfrak{a} \geq \mathfrak{a}_m$ for each m; but equality is impossible, for then $\mathfrak{a}$ would be the largest of the given cardinal numbers. For example, if there existed incomparable cardinals $\mathfrak{a}$ and $\mathfrak{b}$, then $\mathfrak{a} + \mathfrak{b}$ would exceed both.

I and II make possible an unboundedly ascending series of cardinal numbers. Admittedly they also present an opportunity for an antinomy (p. 11): Since to every set of cardinal numbers there corresponds a still larger one, no such set can encompass all cardinal numbers, and the "set of all cardinal numbers" is therefore unconceivable. We are thus confronted with the fact that the condition that "all" things of a certain kind be made into a collection may not always be capable of fulfillment; when we think we have them all, they are not all in the collection after all. The unsettling thing about this antinomy is not that we obtain a contradiction, but that we were not prepared for one: the set of all cardinal numbers seems a priori to be as innocuous as the set of all natural numbers. This creates an uncertainty as to whether other infinite sets — perhaps even all of them — may not be such sham sets, pseudo-sets, sets that are not really sets, and creates the problem of removing this uncertainty. Set Theory must be built on a new basis — an axiomatic basis — in such a way that antinomies are impossible. In this book, we cannot go into the investigations, begun by Zermelo, whose purpose is to do this, investigations that give every promise of success; and we must adhere to our "naive" concept of set.[4]

III. (*König's Theorem*) *If to every* $m \in M$ *there correspond two cardinal numbers* $\mathfrak{a}_m$ *and* $\mathfrak{b}_m$, *and if we always have* $\mathfrak{a}_m < \mathfrak{b}_m$, *then*

$$\sum^{M} \mathfrak{a}_m < \prod^{M} \mathfrak{b}_m.$$

This is a generalization of Theorem I, which may be obtained from it by setting $\mathfrak{a}_m = 1$ *and* $\mathfrak{b}_m = 2$.

Proof. For the sake of convenience, we denote some of the elements of M by $1, 2, 3, \ldots$. Let the A_m be disjoint sets of cardinality $\mathfrak{a}_m$; the B_m, disjoint sets of cardinality $\mathfrak{b}_m$. Since the B_m can be replaced by equivalent sets, A_m may be assumed to be a subset of B_m, so that

$$A_m \subset B_m, \quad B_m = A_m \cup C_m, \quad C_m > 0.$$

[4]Compare Appendix E.

If we set

$$A = \sum^{M} A_m = A_1 + A_2 + A_3 + \cdots\ , \quad B = \prod^{M} B_m = (B_1, B_2, B_3, \dots),$$

then we must prove that $\mathfrak{a} < \mathfrak{b}$. B is the set of complexes

$$p = (b_1, b_2, b_3, \dots, b_m, \dots) \quad (b_m \in B_m).$$

First of all, $\mathfrak{a} \leq \mathfrak{b}$. For if c_m is a fixed element of $C_m = B_m - A_m$, then the complexes

$$\begin{array}{ll} (a_1, c_2, c_3, \dots, c_m, \dots) & (a_1 \in A_1), \\ (c_1, a_2, c_3, \dots, c_m, \dots) & (a_2 \in A_2), \\ (c_1, c_2, a_3, \dots, c_m, \dots) & (a_3 \in A_3), \\ \quad \cdots\cdots\cdots & \end{array}$$

each of which contains only one of the elements a_n (which runs through the set A_n) and otherwise contains only c's, are subsets of B that are pairwise non-overlapping and equivalent to $A_1, A_2, \dots, A_m, \dots$, respectively; A is thus equivalent to a subset of B.

On the other hand, let

$$P = \sum^{M} P_m = P_1 \cup P_2 \cup P_3 \cup \cdots$$

be a subset of B equivalent to A ($P_m \sim A_m$). We will show that it cannot be equivalent to the whole set B; this makes the equation $\mathfrak{a} = \mathfrak{b}$ impossible, leaving only $\mathfrak{a} < \mathfrak{b}$.

Let us consider the complexes

$$p_m = (b_{m1}, b_{m2}, \dots, b_{mm}, \dots)$$

belonging to P_m, and in particular, the elements b_{mm} belonging to them; they form a subset D_m of B_m, of cardinality $\leq \mathfrak{a}_m$. For, there are only $\mathfrak{a}_m$ complexes p_m, and since the b_{mm} that correspond to distinct p_m do not even have to be distinct, it follows that there are at most $\mathfrak{a}_m$ distinct b_{mm}. Hence we have $D_m < B_m$ or, in other words, $B_m = D_m \cup E_m$, $E_m > 0$. If we now choose an arbitrary element e_m of E_m for every $m \in M$, then the complex

$$p = (e_1, e_2, e_3, \dots, e_m, \dots)$$

differs from all the P_m (since $e_m \neq b_{mm}$), and in fact it differs from all the P_m for every m. Hence it does not belong to P, and $P = B$ was not possible. This proves König's Theorem.

If, in particular $0 < \mathfrak{a}_1 < \mathfrak{a}_2 < \mathfrak{a}_3 < \cdots$ is a sequence of increasing cardinal numbers, then

$$\mathfrak{a}_1 + \mathfrak{a}_2 + \mathfrak{a}_3 + \cdots < \mathfrak{a}_1\mathfrak{a}_2\mathfrak{a}_3 \cdots ;$$

if we set $\mathfrak{a} = \mathfrak{a}_1 + \mathfrak{a}_2 + \mathfrak{a}_3 + \cdots$ and $\mathfrak{b} = \mathfrak{a}_1\mathfrak{a}_2\mathfrak{a}_3 \cdots$, then

$$\mathfrak{a} = \mathfrak{a}_1+\mathfrak{a}_2+\mathfrak{a}_3+\cdots < 1+\mathfrak{a}_1\mathfrak{a}_2\mathfrak{a}_3 \cdots = \mathfrak{b} \leq \mathfrak{a}_1\mathfrak{a}_2\mathfrak{a}_3 \cdots = \mathfrak{a}^{\aleph_0}, \quad \mathfrak{a} < \mathfrak{b} \leq \mathfrak{a}^{\aleph_0}.$$

Thus there exist cardinal numbers $\mathfrak{a}$ for which $\mathfrak{a} < \mathfrak{a}^{\aleph_0}$, and in fact there are an infinite number of such cardinal numbers, since one can start with an arbitrarily large $\mathfrak{a}_1$. However, in the same way there exist cardinal numbers $\mathfrak{c}$ for which $\mathfrak{c} = \mathfrak{c}^{\aleph_0}$; these are the cardinal numbers of the form $\mathfrak{b}^{\aleph_0}$, inasmuch as $(\mathfrak{b}^{\aleph_0})^{\aleph_0} = \mathfrak{b}^{\aleph_0 \cdot \aleph_0} = \mathfrak{b}^{\aleph_0}$.

Under the conditions of III we of course also have

$$\sum^M \mathfrak{a}_m \leq \sum^M \mathfrak{b}_m, \quad \prod^M \mathfrak{a}_m \leq \prod^M \mathfrak{b}_m;$$

however, we might have equality in this case. Thus, for instance,

$$1 \cdot 2 \cdot 3 \cdot 4 \cdots = 2 \cdot 3 \cdot 4 \cdot 5 \cdots = \aleph_0!,$$
$$1 + 2 + 3 + \cdots = 2 + 3 + 4 + \cdots = \aleph_0;$$

for, the product is $\geq 2 \cdot 2 \cdot 2 \cdots = 2^{\aleph_0}$ and

$$\aleph_0! \leq \aleph_0 \cdot \aleph_0 \cdot \aleph_0 \cdots = \aleph_0^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}.$$

§ 2.4. The Elementary Cardinal Numbers

We shall call the following three cardinal numbers *elementary*:

1. The cardinality $\aleph_0$ of countable sets;

2. The cardinality $\aleph$ of the continuum, which we note is identical with $2^{\aleph_0}$;

3. The cardinal number $2^{\aleph}$.

The cardinal number $\aleph_0$ (the smallest infinite cardinal number)

We already know that the sum of a finite or countably infinite number of summands $\aleph_0$ or the product of a finite number of factors $\aleph_0$ is itself equal to $\aleph_0$. Let us list a few countable sets in addition to the ones we have already given.

(1) The set of all integers ≤ 0. The negative integers, the number 0, and the positive integers together form a set of cardinality $\aleph_0 + 1 + \aleph_0 = \aleph_0$. The integers can be put into the form of a simple sequence $\{a_1, a_2, \dots\}$ by ordering them as follows:

$$0, 1, -1, 2, -2, 3, -3, \dots .$$

(2) The set of all pairs of natural numbers. Its cardinality is $\aleph_0 \cdot \aleph_0 = \aleph_0$. One can construct a one-to-one correspondence between the pairs of natural numbers and the natural numbers themselves, or convert a double sequence into a simple one, by combining the pairs of numbers in a matrix (where (p, q) appears in the p-th row and q-th column)

$$\begin{matrix} (1,1) & (1,2) & (1,3) & \dots \\ (2,1) & (2,2) & (2,3) & \dots \\ (3,1) & (3,2) & (3,3) & \dots \\ \vdots & \vdots & \vdots & \ddots \end{matrix}$$

at the same time arranging all the natural numbers in a similar matrix (two examples may be found on p. 34) and letting the elements having the same position correspond to each other. The diagonal array yields the correspondence

$$\begin{matrix} n: & 1 & 2 & 3 & 4 & 5 & 6 & \dots \\ (p,q): & (1,1) & (1,2) & (2,1) & (1,3) & (2,2) & (3,1) & \dots, \end{matrix}$$

where the pairs of numbers are arranged according to increasing value of the sum $p + q$ and, if the sum is the same, according to increasing p. The dyadic array yields the correspondence

$$n = 2^{p-1}(2q - 1),$$

which is uniquely solvable for p and q.

(3) The set of all finite complexes, or n-tuples, of natural numbers

$$(p_1),\ (p_1, p_2),\ (p_1, p_2, p_3),\ \ldots,\ (p_1, p_2, \ldots, p_k),\ \ldots,$$

where k and p_λ run through all natural numbers.

Its cardinality is

$$\aleph_0 + \aleph_0^2 + \aleph_0^3 + \cdots = \sum_n \aleph_0^n = \aleph_0 \cdot \aleph_0 = \aleph_0.$$

The correspondence between these k-tuples and the natural numbers n can be given, for instance, by developing the numbers n dyadically as follows:

$$n = 2^{p_1 - 1} + 2^{p_1 + p_2 - 1} + 2^{p_1 + p_2 + p_3 - 1} + \cdots + 2^{p_1 + \cdots + p_k - 1},$$

for example, there corresponds to the number $27 = 1 + 2 + 2^3 + 2^4$ the n-tuple $(1, 1, 2, 1)$.

(4) The set of the rational numbers. If we allow the positive rational number p/q, where p and q are relatively prime natural numbers, to correspond to the pair of numbers (p, q) (Example (2)), then the set of numbers p/q becomes equivalent to a part of the set of the (p, q), and therefore it has a cardinality $\leq \aleph_0$; however, since it is infinite, the cardinality must be $\aleph_0$. The diagonal array yields an arrangement of the p/q according to increasing $p + q$; and if the sum $p + q$ is the same for two fractions, they are arranged with numerators in ascending order, the reducible fractions being thus omitted.

The set of all rational numbers also is countable, and so is the set of all the rational numbers in an interval. The rational numbers in the interval $(0, 1)$ may be put in the form of a simple sequence by arranging them, say, with denominators in ascending order and, in case the denominators are equal, with numerators in ascending order,

$$\tfrac{1}{2},\ \tfrac{1}{3},\ \tfrac{2}{3},\ \tfrac{1}{4},\ \tfrac{3}{4},\ \tfrac{1}{5},\ \tfrac{2}{5},\ \tfrac{3}{5},\ \tfrac{4}{5},\ \tfrac{1}{6},\ \cdots$$

That the set of all rational numbers — which, after all, is everywhere dense on the straight line (geometrically speaking) — does not have a larger cardinality than the set of integers, is just another of the many facts of set theory that appear surprising and even paradoxical.

(5) The set of (real) algebraic numbers. This set also, itself much finer and more dense than that of the rational numbers, is only countable. We prove this for real and for complex algebraic numbers simultaneously. An algebraic number of the k-th degree is a root of an irreducible equation

$$a_0x^k + a_1x^{k-1} + \cdots + a_{k-1}x + a_k = 0$$

with rational coefficients (which is uniquely determined by it). There are infinitely many such equations, but surely no more than there are $(k+1)$-tuples $(a_0, a_1, \ldots, a_k)$ of rational numbers (since not all $(k+1)$-tuples yield irreducible equations); so there are at most $\aleph_0^{k+1} = \aleph_0$ such equations. Thus there are $\aleph_0$ irreducible equations of the k-th degree, each one of which has k roots; so there are $\aleph_0 \cdot k = \aleph_0 \cdot \aleph_0 = \aleph_0$ algebraic numbers of the k-th degree and $\sum \aleph_0 = \aleph_0 \cdot \aleph_0 = \aleph_0$ algebraic numbers altogether.

The set of all pairs, triples, etc., of algebraic or rational numbers is, of course, also countable. The set of all elements each of which can be represented by a finite number of symbols of a finite or countable system of symbols is countable if, in the case of a finite system of symbols, we admit n-tuples with arbitrarily large n. For we have

$$\aleph_0 + \aleph_0^2 + \aleph_0^3 + \cdots = \aleph_0,$$
$$n + n^2 + n^3 + \cdots = \aleph_0,$$
$$n + n + n + \cdots = \aleph_0.$$

For instance, the set of all arbitrarily long "words" (i.e., n-tuples of letters, which may or may not make sense) constructible from a finite alphabet is countable, as is the set of all possible books, symphonies, and the like.

The cardinality $\aleph$ of the continuum

We have already found (p. 34) that the sum of finitely or countably many summands $\aleph$ is equal to $\aleph$:

$$n\aleph = \aleph \cdot \aleph_0 = \aleph.$$

By means of the decimal representation $0.a_1a_2a_3\ldots$, every number of the interval $[0, 1]$ is represented at least once and at most twice. Hence

$$\aleph \leq 10^{\aleph_0} \leq 2^{\aleph_0 \cdot \aleph_0} = \aleph,$$

so that $\aleph = 10^{\aleph_0}$. The same of course holds if, instead of the decimal representation, we make use of any other representation to a base > 1, so that

$$\aleph = n^{\aleph_0} = b^{\aleph_0} \cdots$$

(but $1^{\aleph_0} = 1$). It then follows that

$$\aleph^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0} = 2^{\aleph_0} = \aleph,$$

that is, $\aleph$ belongs to that category of cardinal numbers, already considered, that equal their $\aleph_0$-th power, and is certainly not the sum of an increasing sequence of cardinal numbers. By the Equivalence Theorem we then certainly have

$$\aleph = \aleph^2 = \aleph^3 = \cdots = \aleph^{\aleph_0}$$

(or directly: $\aleph^2 = \aleph \cdot \aleph = 2^{\aleph_0} \cdot 2^{\aleph_0} = 2^{\aleph_0 + \aleph_0} = 2^{\aleph_0} = \aleph$) as well as $\aleph = n\aleph = \aleph\aleph_0 = \aleph^2 = \aleph^3 = \cdots = \aleph\aleph_0\aleph_0 \cdots$.

In these formulas there is again a surprise in store for us: for $\aleph^2 = \aleph \cdot \aleph$ is, after all, the cardinality of the set of all pairs (x, y) of real numbers, or of all points of the plane; $\aleph^3$ is the cardinality of three-dimensional space, or of the set of all number triples (x, y, z), etc. until we get to $\aleph^{\aleph_0}$, which is the cardinality of $\aleph_0$-dimensional space, that is, of the set of all sequences $(x_1, x_2, x_3, \ldots)$ of real numbers. So we have: *All spaces that have a finite or a countable number of dimensions have the same cardinality* $\aleph$. This, too, sounds paradoxical enough, and it seems to overthrow the concept of dimension, which will later be reinstated only by means of quite a different order of ideas (namely, the requirement of not only a one-to-one but a bi-continuous mapping); and yet the fact that the line and the plane have, in a manner of speaking, the same number of points is no more puzzling than the fact that the natural numbers may be divided into odd and even ones; that is, $\aleph^2 = \aleph$ is basically nothing but $\aleph_0 + \aleph_0 = \aleph_0$. For, by splitting the decimal representation of a number, we can make two; and from two representations we can make one, by interweaving the two; that is the entire secret. To make the equivalence between line and surface perfectly clear, we represent every real number of the interval $J = (0, 1]$ (i.e. $0 < x \leq 1$) by a dyadic fraction with infinitely many ones (with a modification of the usual expansion to avoid its two-valuedness), that is we can put it in the form

$$x = \frac{1}{2^{\xi_1}} + \frac{1}{2^{\xi_1 + \xi_2}} + \frac{1}{2^{\xi_1 + \xi_2 + \xi_3}} + \cdots,$$

where the ξ_n are natural numbers; this can always be done and can be done in only one way, and it incidentally illustrates the equation $\aleph = \aleph_0^{\aleph_0}$. Let us write this representation for short as

$$x = [\xi_1, \xi_2, \xi_3, \dots].$$

From two such numbers x and $y = [\eta_1, \eta_2, \eta_3, \dots]$ we then obtain a single one

$$t = [\xi_1, \eta_1, \xi_2, \eta_2, \dots],$$

and conversely from

$$t = [t_1, t_2, t_3, t_4, \dots]$$

we again obtain

$$x = [t_1, t_3, t_5, \dots], \quad y = [t_2, t_4, t_6, \dots],$$

so that the ordered pairs (x, y) of numbers and the numbers t are brought into one-to-one correspondence; that is, the square $0 < x \leq 1$, $0 < y \leq 1$ is mapped biuniquely onto the line-segment $0 < t \leq 1$, and the product of intervals (J, J) is mapped biuniquely onto the interval J. We deal similarly with number triples (x, y, z) and with finite n-tuples; for sequences of numbers $(x, y, z, \dots)$ we represent the mapping by means of the diagonal array:

$$t = [t_1, t_2, t_3, t_4, t_5, t_6, \dots]$$

and conversely,

$$\begin{aligned} x &= [t_1, t_3, t_5, \dots], \\ y &= [t_2, t_5, t_9, \dots], \\ z &= [t_4, t_8, t_{13}, \dots], \\ &\dots \end{aligned}$$

From $\aleph = 2^{\aleph_0}$ it follows that $\aleph > \aleph_0$; the continuum is not countable. A conjecture that was made at the beginning of Cantor's investigations, and that remains unproved to this day, is that $\aleph$ is the cardinal number next larger than $\aleph_0$; this conjecture is known as the *continuum hypothesis*, and the question as to whether it is true or not is known as the *problem of the continuum.* The uncountability of the continuum may also be seen as follows: No countable set of real numbers (of the interval J)

$$\begin{aligned} a &= [a_1, a_2, a_3, \dots], \\ b &= [b_1, b_2, b_3, \dots], \\ c &= [c_1, c_2, c_3, \dots], \\ &\cdots \end{aligned}$$

can include all these numbers, as there are others

$$e = [e_1, e_2, e_3, \dots] \quad (e_1 \neq a_1, e_2 \neq b_2, e_3 \neq c_3, \dots),$$

This "diagonal process" is the simplest model for the proof of Theorem I, § 7.

As there are only $\aleph_0$ rational and algebraic numbers, there certainly must exist irrational and transcendental numbers, and in fact as many ($\aleph$) of them as of real numbers. For, the cardinality of a non-denumerable (i.e., infinite and not countable) set is not diminished by the removal of a countable number of elements. (The proof is like that of the corresponding remark on p. 35.) The real numbers $0 < x < 1$ can be mapped onto the irrational numbers $0 < y < 1$ by means of, say,

$$x = \frac{1}{\xi_1 +} \frac{1}{\xi_2 +} \frac{1}{\xi_3 +} \cdots ,$$

where x is given by the above dyadic representation and y by the continued fraction formed from the sequence of natural numbers $\xi_1, \xi_2, \dots$.

The cardinal number $2^{\aleph}$

This cardinal number, which is in turn $> \aleph$, belongs to the set of all subsets of the continuum (that is, to the set of all sets of real numbers) or to the set of all linear, planar, or spatial point sets; it is also the cardinality of the set of all single-valued functions $f(x)$ of a real variable where $f(x)$ can take on two values; however, from

$$\aleph^{\aleph} = (2^{\aleph_0})^{\aleph} = 2^{\aleph_0 \cdot \aleph} = 2^{\aleph}$$

it follows at once that the set of all the functions $f(x)$ that can run through all the real numbers is also only of the cardinality $2^{\aleph}$. The cardinality of special classes of functions may of course be smaller;

the set of all continuous functions $f(x)$, for instance, only has the cardinality $\aleph$. For, $f(x)$ is then determined by its values $f(r)$ at the rational points r; the set of all $f(r)$ has the cardinality $\aleph^{\aleph_0} = \aleph$, and the set of all continuous $f(x)$ has a cardinality which is $\leq \aleph$ (since not every $f(r)$ gives rise to a continuous $f(x)$), and also $\geq \aleph$ (the functions $f(x) =$ constant already form a set of cardinality $\aleph$, and hence the cardinality of this set is exactly equal to $\aleph$).

We close with a collection of formulas for $\mathfrak{a} + \mathfrak{b}$, $\mathfrak{a}\mathfrak{b}$, and $\mathfrak{a}^{\mathfrak{b}}$, where the cardinal numbers involved are $\aleph_0$, $\aleph$, and the natural numbers n; to the extent that they are not proved above, they follow from the Equivalence Theorem.

$$
\begin{array}{rlll}
\text{(a)} & n + \aleph_0 = \aleph_0, & n + \aleph = \aleph, & \aleph_0 + \aleph = \aleph, \\
\text{(b)} & n \cdot \aleph_0 = \aleph_0, & n \cdot \aleph = \aleph, & \aleph_0 \cdot \aleph_0 = \aleph_0, \\
\text{(c)} & \aleph_0 \cdot \aleph = \aleph, & \aleph \cdot \aleph = \aleph, & \aleph_0^n = \aleph_0, \\
\text{(d)} & n + n = n \cdot 2, & \aleph_0 + \aleph_0 = \aleph_0, & \aleph + \aleph = \aleph, \\
\text{(e)} & n\aleph_0 = \aleph_0, & \aleph_0\aleph_0 = \aleph_0, & \aleph\aleph = \aleph, \\
\text{(f)} & \aleph^{\aleph_0} = \aleph, & \aleph_0^{\aleph} = 2^{\aleph}, & \aleph^{\aleph} = 2^{\aleph}.
\end{array}
$$

Chapter 3

ORDER TYPES

§ 3.1. Order

Many sets present themselves at the outset in a natural order, in which for each two distinct elements, one precedes and one follows. We express such an ordering by the usual way in which a set is written, the element that precedes being to the left of the one that follows. Thus letters appear in alphabetical order and the natural numbers in their order of magnitude, as follows:

$$1, 2, 3, \ldots ;$$

similarly for the real numbers. However, we can also prescribe such orderings as we please: for instance, we can order the natural numbers in decreasing order of magnitude

$$\ldots, 3, 2, 1,$$

or we can put the odd numbers before the even numbers, ordering each class according to increasing or decreasing order of magnitude:

$$\begin{gathered} 1, 3, 5, \ldots, 2, 4, 6, \ldots \\ 1, 3, 5, \ldots, \ldots, 6, 4, 2 \\ \ldots, 5, 3, 1, 2, 4, 6, \ldots \\ \ldots, 5, 3, 1, \ldots, 6, 4, 2. \end{gathered}$$

We can order a set of people according to their weight, their height, their age, the alphabetical order of their names, or the numbers on the stubs of their theatre tickets.

Thus, a set is *ordered* by assigning an order of precedence to any two distinct elements a and b, that is, by a rule that states of two distinct elements a and b that one precedes and the other follows. If a is to come before b, and b after a, then we write[1]

$$a < b, \quad b > a.$$

The relation $<$ is to be *transitive*, that is,

$$\text{if } a < b \text{ and } b < c, \text{ then } a < c.$$

Of course, the space-like or time-like characteristics which seem to attach to this explanation because of the use of the prepositions "before" and "after" are of no consequence, and we will convince ourselves by the use of other definitions that what we have here is nothing but an application of the concept of function.

Let us form the set of ordered pairs $p = (a, b)$ from distinct elements of the set A (which must of course contain at least two elements); the pairs p and $p^* = (b, a)$ are distinct and will be called *inverse* to each other. We then split the set of all ordered pairs into two complementary subsets $P + P^*$ by means of the following rule:

(a) Of two inverse pairs, one belongs to P, the other to P^*;

(b) If $p = (a, b)$ and $q = (b, c)$ belong to P, then so does $r = (a, c)$.

If we then write

$$a < b \quad \text{or} \quad b > a$$

for $(a, b) = p \in P$, then an ordering in the above sense is defined. (And conversely: Every ordering yields a partition $P + P^*$, which we define by considering those pairs $p = (a, b)$ as elements of P for which $a < b$). Here it is clear that the idea of "preceding" and "following" introduces nothing new or mysterious, but that it is only a question of distinguishing one pair of two inverse pairs, or one element of two distinct elements. We call P the *ordering set*.

Even simpler is the following explanation: With every element a of A we associate a unique set $M(a)$ such that the a are in one-to-one correspondence with the $M(a)$ and such that the $M(a)$ are comparable (in the sense of p. 13); by virtue of the one-to-one correspondence, it follows that

$$M(a) \subset M(b) \quad \text{for } a < b.$$

[1] It seems to us unnecessary to use other forms of the symbols $<$ and $>$.

If we then write $a < b$ instead of $M(a) < M(b)$, an ordering of the set A is defined; conversely, from any ordering we can form a system of sets $M(a)$ of the kind indicated.

For instance: For each a, let

$$M(a) = \text{the set of all elements of } A \text{ that are } < a.$$

The actual existence of orderings of A or of sets of pairs P or of systems of sets $M(a)$ of the kind indicated does not as yet follow from the above considerations; later on (§ 12) we shall see that every set can not only be ordered, but can also even be ordered in a special way (well-ordered). An upper bound on the number of distinct orderings, that is, on the cardinality of the set of distinct orderings of a set A is $2^{\mathfrak{a}^2}$, where $\mathfrak{a}$ is the cardinality of A. For, (A, A), which has the cardinality $\mathfrak{a} \cdot \mathfrak{a} = \mathfrak{a}^2$, is the set of all pairs (a, b) of (equal or distinct) elements of A; and the ordering set P is a subset of (A, A). There are $2^{\mathfrak{a}^2}$ subsets of (A, A), and hence at most $2^{\mathfrak{a}^2}$ ordering sets, or distinct orderings, of A. For instance, for a finite set of n elements, the number of distinct orderings is

$$n! = 1 \cdot 2 \cdots n < 2^{n^2}.$$

Until further notice, capital italic letters shall denote ordered sets, and the equation $A = B$ shall mean that A and B have not only the same elements but also the same ordering when individual elements are given, the order in which they are written from left to right will denote their order in the set as well; thus

$$A = \{\dots, a, \dots, b, \dots, c, \dots\}$$

is a set in which $a < b < c$, where the dots indicate the presence of other elements before a, between a and b, between b and c, and after c. Accordingly,

$$A = \{a, \dots, b, \dots\}$$

denotes a set in which there is no element before a, so that a is the *first* element;

$$A = \{\dots, b, \dots, c\}$$

denotes a set in which there is no element after c, so that c is the *last* element; and

$$A = \{\dots, a, b, \dots\}$$

denotes a set in which there is no element between a and b, so that a and b are *neighboring elements*.

If we interchange the symbols $<$ and $>$ throughout A, then we obtain the *inverse* ordered set

$$A^* = \{\dots, c, \dots, b, \dots, a, \dots\},$$

whose ordering set is P^*.

Two ordered sets are called *similar*, written

$$A \simeq B,$$

if there exists a one-to-one correspondence $b = f(a)$, $a = g(b)$ between them that preserves the order; that is, $a < a_1$ when and only when $b < b_1$.

For instance, the set $\{1, 2, 3, 4, \dots\}$ is similar to $\{2, 3, 4, \dots\}$, but not to $\{2, 3, 4, \dots, 1\}$.

Like equivalence, similarity is reflexive, symmetric, and transitive. Two similar sets are said to have the same *order type*; that is, with every ordered set A we associate a symbol α, called its *order type* (type, for short), in such a way that similar sets, and only similar sets, have the same order type:

$$\alpha = \beta \quad \text{means} \quad A \simeq B.$$

The type of the set A^* inverse to A is called α^*.

Similarity implies equivalence, but the converse is in general false: $A \simeq B$ implies $A \sim B$, and $\alpha = \beta$ implies $\mathfrak{a} = \mathfrak{b}$. Hence we may say that a type α has a definite cardinal number $\mathfrak{a}$.

A finite set of n elements can be ordered (permuted) in $n!$ ways, but the resulting ordered sets are always similar to the set $\{1, 2, \dots, n\}$; again we call this type n, since any confusion between this and the cardinal number n is quite innocuous. A set having one element shall have type 1, the null set type 0.

The set $\{1, 2, 3, \dots\}$ of the natural numbers in their natural order shall have type ω; thus the inversely ordered set $\dots, 3, 2, 1$ has the type ω^*.

§ 3.2. Sum and Product

Let A and B be disjoint ordered sets. Then

$$S = A + B$$

shall denote the sum of the two sets in the following order: The order of the elements a among themselves, and that of the elements b among themselves is retained, and every a precedes every b ($a < b$). This sum is therefore to be distinguished from $B + A$, which, though it contains the same elements, does not contain them in the same order: The addition of ordered sets is *not* commutative.

SET THEORY

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III. Order Types

2.10 Sum and Product

§ 10. Sum and Product

Example: $A = \{1, 3, 5, \ldots\}$, $B = \{2, 4, 6, \ldots\}$.

$$A + B = \{1, 3, 5, \ldots,\ 2, 4, 6, \ldots\}$$
$$B + A = \{2, 4, 6, \ldots,\ 1, 3, 5, \ldots\}.$$

It can also be said that in $S = A + B$ we have $A < B$; the entire set A precedes the entire set B. In general, if A and B are subsets of an ordered set, then $A < B$ shall mean

$$a < b \quad \text{for all } a \in A \text{ and } b \in B;$$

the relations $a < B$ and $A < b$ shall have analogous meanings.

If A_1 and B_1 are also disjoint ordered sets and if $A_1 \simeq A$, $B_1 \simeq B$, then $A_1 + B_1 \simeq A + B$. This justifies the following definition:

The sum (or *type-sum*) $\alpha + \beta$ is the type of $A + B$, where A and B are arbitrary disjoint sets whose types are α and β respectively.

In general, let us associate with every element m of an ordered set

$$M = \{\ldots, m, \ldots, n, \ldots, p, \ldots\}$$

an ordered set A_m, and let these sets be disjoint (pairwise non-overlapping). Then the sum

$$S = \sum_M A_m = \cdots + A_m + \cdots + A_n + \cdots + A_p + \cdots$$

is defined to be the set formed from all the elements of all the A_m in the following order: for each m the order of the elements $a \in A_m$

among themselves remains unchanged, while for $m < n$, the entire set A_m precedes the entire set A_n: $A_m < A_n$.

If we replace the summands by similar ones (once more, of course, these have to be disjoint), then the sum is also transformed into a similar one, and this justifies the following definition: if to the elements $m \in M$ there correspond the types α_m, then by the sum

$$\sum_M \alpha_m = \cdots + \alpha_m + \cdots + \alpha_n + \cdots + \alpha_p + \cdots$$

we shall mean the type of the above set S.

A general associative law holds, of which we will adduce only the simplest case

$$(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma) = \alpha + \beta + \gamma;$$

on the other hand, there is no commutative law: if the order of M is changed, then S, and in general the type σ, is changed as well.

Examples: The partition of the sequence of natural numbers (of type ω) into

$$\{1, 2, \ldots, n\} + \{n + 1, n + 2, \ldots\},$$

where the second summand is of type ω, yields

$$n + \omega = \omega.$$

On the other hand $\omega + n$ is the type of

$$\{n + 1, n + 2, \ldots, 1, 2, \ldots, n\},$$

and this set is certainly not of type ω, as it has a last element; hence

$$\omega + n \neq n + \omega;$$

and the types ω, $\omega + 1$, $\omega + 2$, $\ldots$ are clearly pairwise distinct. The four orderings (p. 48) of the natural numbers obtained by placing the odd numbers before the even ones have the types

$$\omega + \omega,\ \omega + \omega^*,\ \omega^* + \omega,\ \omega^* + \omega^*,$$

respectively, which, as can easily be seen, are distinct from one another and from the types $\omega + n$ and their inverses $n + \omega^*$. $\omega^* + \omega$ is also the type of the set of all integers

$$\{\ldots, -2, -1, 0, 1, 2, \ldots\}$$

in their natural order.

If with every natural number m we associate a natural number a_m, then the partition of the sequence of natural numbers into groups of a_m members each yields the equation (type-equation)

$$\sum a_m = a_1 + a_2 + a_3 + \cdots = \omega;$$

for instance,[1]

$$\begin{aligned} 1 + 1 + 1 + \cdots &= \omega \\ 1 + 2 + 3 + \cdots &= \omega. \end{aligned}$$

On the other hand, if we divide the natural numbers into countably many sequences, for instance as follows (diagonal array),

$$\{1, 2, 4, \ldots\} + \{3, 5, 8, \ldots\} + \{6, 9, 13, \ldots\} + \cdots ,$$

then we once again obtain a new type

$$\sum \omega = \omega + \omega + \omega + \cdots .$$

When inverting a sum, one must invert each summand and the order of the summands; that is,

$$\text{if } S = \sum_M A_m, \text{ then } S^* = \sum_{M^*} A_m^*,$$

and the corresponding result holds for types. For instance

$$(\alpha + \beta)^* = \beta^* + \alpha^*.$$

Products of finitely many factors. The ordered pairs (a, b), $a \in A$, $b \in B$, where A and B are ordered (not necessarily disjoint) sets, can be arranged in an order that is called, quite appropriately, the *lexicographic order*, or *order by first difference.* To be specific, we define

$$(a, b) < (a_1, b_1) \quad \text{if} \quad \begin{cases} \text{either } a < a_1 \\ \text{or } a = a_1, \; b < b_1. \end{cases}$$

[1]Since we have also written $1+1+1+\cdots = \aleph_0$, the undistinguishing use of the finite numbers as cardinal numbers and as order-types might arouse misgivings; on the other hand, it can be seen at once from any equation containing infinite symbols whether it is a number equation or a type equation.

The set ordered in this way will again be called (A, B); it is the ordered product of the ordered pairs, and is to be distinguished from (B, A), to which it is equivalent but not, in general, similar. If $A_1 \simeq A$, $B_1 \simeq B$, then $(A_1, B_1) \simeq (A, B)$, and this once again justifies defining the product of α and β as the type of (A, B). Unfortunately, we cannot escape an inconvenience of historical origin: The type of (A, B) is called $\beta\alpha$, not $\alpha\beta$.[2]

Example:

$$A = \{1, 2\},\ B = \{1, 2, 3, \ldots\},\ \alpha = 2,\ \beta = \omega.$$

(A, B) is the lexicographically ordered set of pairs (a, b), which therefore has the order

$$(1,1),\ (1,2),\ (1,3),\ \ldots,\ (2,1),\ (2,2),\ (2,3),\ \ldots;$$

it is of type $\beta\alpha = \omega \cdot 2 = \omega + \omega$. (B, A) is the set of all pairs (b, a) in the order

$$(1,1),\ (1,2),\ (2,1),\ (2,2),\ (3,1),\ (3,2),\ \ldots;$$

it is of type $\alpha\beta = 2\omega = \omega$.

The addition of equal summands here again leads to multiplication; that is, if all the $\alpha_m = \alpha$ and if μ denotes the type of M, then

$$\sum_M \alpha_m = \mu\alpha.$$

In fact, $\mu\alpha$ is the type of (M, A), that is, of the set of lexicographically ordered pairs (m, a). If the set of these pairs for a given m is A_m, then

$$(M, A) = \sum_M A_m,$$

from which, because $A_m \simeq A$, the type equation follows. $\mu\alpha$ is obtained by "the insertion of α in μ," that is, by the insertion of a set of type α for every element of a set of type μ.

[2] Cantor originally wrote $\alpha\beta$, then changed this to $\beta\alpha$, and the latter notation became the overwhelmingly predominant one; one further change would make the confusion permanent. One could do away with the lack of agreement by ordering anti-lexicographically, i.e., by last differences; but this again would be uncomfortable.

Examples. $\omega+\omega+\omega=\omega\cdot 3$, $3+3+3+\cdots=3\omega=\omega\omega$. In general $n\omega=n+n+n+\cdots=\omega$, $\omega n=\omega+\omega+\cdots+\omega$, the latter having n summands.

The distributive law holds only with respect to the second factor; that is, we have

$$\beta\sum_M \alpha_m=\sum_M \beta\alpha_m,$$

whereas if the factor β is placed at the right, the equation need not hold. In fact, if $A=\sum_M A_m$, then for every set B we have

$$(A,B)=\sum_M (A_m,B),$$

as can be seen at once from the lexicographical ordering of the pairs (a,b), and the type equation follows from this; it can also be deduced easily from the "insertion" of β in $\alpha=\sum_M \alpha_m$. In particular,

$$\beta(\alpha+\alpha')=\beta\alpha+\beta\alpha',$$

but $(\alpha+\alpha')\beta=\alpha\beta+\alpha'\beta$ does not necessarily hold.

Example. $2(\omega+1)=2\omega+2=\omega+2$, since (by actual insertion)

$$2(\omega+1)=\omega+2+\omega+2+\cdots=\omega+2.$$

On the other hand,

$$(\omega+1)\cdot 2=(\omega+1)+(\omega+1)=\omega+1+\omega+1=\omega\cdot 2+1.$$

Inversion of (A,B) yields the lexicographic ordering of the pairs (a,b) of (A^*,B^*), so that

$$(A,B)^*=(A^*,B^*)\quad\text{and}\quad(\beta\alpha)^*=\beta^*\alpha^*;$$

in the inversion of a product, the factors are to be inverted but not their order (unlike the case of addition).

The extension of multiplication to three or more factors is obvious. Thus, let (A,B,C) be the lexicographically ordered set of triples (a,b,c), in which we then have $(a,b,c)<(a_1,b_1,c_1)$ if

$$\begin{cases}\text{either } a<a_1\\ \text{or } a=a_1,\ b<b_1\\ \text{or } a=a_1,\ b=b_1,\ c<c_1;\end{cases}$$

its type is called $\gamma\beta\alpha$. Obviously the associative law holds:

$$\gamma(\beta\alpha) = (\gamma\beta)\alpha = \gamma\beta\alpha.$$

We can deal in a similar way with any finite number of factors, and even more: If $M = \{1, 2, 3, \ldots\}$ is the set of all the natural numbers, then the complexes (sequences)

$$p = (a_1, a_2, a_3, \ldots) \quad (a_n \in A_n)$$

can also be ordered lexicographically, for two distinct complexes p and $q = (b_1, b_2, b_3, \ldots)$ have a first place m at which they differ; that is,

$$a_1 = b_1,\ a_2 = b_2,\ \ldots,\ a_{m-1} = b_{m-1},\ a_m \neq b_m;$$

and we then define $p < q$ if $a_m < b_m$. The product ordered in this way will be called $(A_1, A_2, A_3, \ldots)$, and its type will be called $\cdots \alpha_3\alpha_2\alpha_1$.

Example: If every A_n is the set of all natural numbers, then we obtain $\cdots \omega\omega\omega$ as the type of the lexicographically ordered set of sequences $p = (a_1, a_2, a_3, \ldots)$ of natural numbers. If we let the real number

$$x = \left(\frac{1}{2}\right)^{a_1} + \left(\frac{1}{2}\right)^{a_1 + a_2} + \cdots$$

correspond to each p (biuniquely), then we see that to the lexicographic ordering of the p there corresponds the inverse order by magnitude of the x; that is, $p < q$ means $x > y$. Since x runs through the interval $(0, 1)$, it follows that $\cdots \omega\omega$ is the type of the numbers $1 - x$ in their natural order, that is, the type of the interval $[0, 1)$.

2.11 The Types η and λ

§ 11. The Types η and λ

Obviously we would be able to extend the lexicographic ordering to any product $\prod_M A_m$ if we could only be certain that any two of its complexes have a first place at which they differ, or in general, if every subset of M has a first element (that is, if M is well-ordered (Chap. IV)). Later (§ 16), we shall go into these and further extensions of the concept of product. A generalized concept of powers can also only be explained there: in the meantime we shall, of course, write $\alpha\alpha = \alpha^2$, $\alpha\alpha\alpha = \alpha^3$, ... (The previously mentioned product $\cdots\omega\omega\omega$ would be written ω^ω.) Thus

$$\omega^2 = \omega\omega = \omega + \omega + \omega + \cdots$$

is the type of a sequence of sequences or of a double sequence, ω^3 that of a sequence of double sequences, or of a triple sequence; etc. We have

$$\omega + \omega^2 = \omega(1 + \omega) = \omega\omega = \omega^2;$$

$\omega^2 + \omega = \omega(\omega + 1)$ is a type differing from the first; and

$$(\omega + \omega)\omega = (\omega \cdot 2)\omega = \omega(2\omega) = \omega\omega = \omega^2$$

is different from

$$\omega(\omega\omega + \omega) = \omega(\omega \cdot 2) = \omega^2 \cdot 2 = \omega^2 + \omega^2.$$

The types that are obtained from ω and finite types by finitely many additions and multiplications may be called entire rational functions, or *polynomials*, in ω; they can be put, and (as we shall also see) put uniquely, in the form

$$\omega^n a + \omega^m a_1 + \cdots + \omega^{n_k} a_k,$$

$(n > m > \cdots > n_k \geq 0;\ a, a_1, \ldots, a_k$ natural numbers).

§ 11. The Types η and λ

Every type α has a definite cardinal number $\mathfrak{a}$. The various types of cardinality $\mathfrak{a}$ form a class of types $\mathfrak{T}(\mathfrak{a})$; to obtain this class one need only order a fixed set A of cardinality $\mathfrak{a}$ in all the ways possible, where different orderings of course do not necessarily yield distinct types.

The classes of types of finite cardinality $\mathfrak{T}(n)$ ($n = 0, 1, 2, \ldots$) always contain only one type n. But of the types belonging to $\mathfrak{T}(\aleph_0)$ we already know infinitely many: ω, $\omega+1$, $\omega+2$, $\ldots$, $\omega+\omega$, and ω^2, among others.

Theorem 2.11.1. *I. The set of countable types has the cardinality of the continuum.*

That is, $\mathfrak{T}(\aleph_0)$ has the cardinality $\mathfrak{c}$. Half of this theorem we already know: There are (p. 50) at most $2^{\aleph_0} \cdot 2^{\aleph_0} = \mathfrak{c}$ distinct countable types.

On the other hand, let $\iota = \omega^* + \omega$ be the type of the set $\{\ldots, -2, -1, 0, 1, 2, \ldots\}$ of the integers in their natural order, let

$$a = (a_1, a_2, a_3, \ldots)$$

be a sequence of natural numbers, and let

$$\iota a_1 + \omega + \iota a_2 + \omega + \iota a_3 + \omega + \cdots$$

be a type, obviously countable, determined by a. If we can show that a in turn determines the sequence a, that is, that there exists a one-to-one correspondence between the a and the α, then we have proved that there are also at least $\aleph_0^{\aleph_0} = \mathfrak{c}$ distinct countable types and, by the Equivalence Theorem, that there are precisely $\mathfrak{c}$ of them.

Thus, we have to prove that if $\alpha = \iota a_1 + \omega + \iota a_2 + \omega + \iota a_3 + \omega + \cdots$ and $\beta = \iota b_1 + \omega + \iota b_2 + \omega + \cdots$, then $a_1 = b_1$, $a_2 = b_2$, $\ldots$. This follows from the following statements:

(a) If $A_1 + A_2 \simeq B_1 + B_2$, and if A_1 and B_1 are finite and A_2 and B_2 have no first element, then $A_1 \simeq B_1$ and $A_2 \simeq B_2$.

For, in the similarity mapping, an element $b_1 \in B_1$ cannot be the image of an element $a_2 \in A_2$, because b_1 has only finitely many elements preceding it (and possibly none), while infinitely many elements precede a_2. Similarly no b_2 can correspond to an a_1. Thus, the a_1 must correspond to the b_1 and the a_2 to the b_2.

(b) If $A_1 + A_2 \simeq B_1 + B_2$, and A_2 and B_2 are of type ι, then $A_2 \simeq B_2$.

Here again, no b_1 can correspond to an a_2, since the set of elements preceding a_2 contains a subset (A_1) without a last element, while the set of elements preceding b_1 is of type ω^* and has no subset without a last element (except, of course, for the null set). Again, therefore, A_1 must be mapped onto B_1 and A_2 onto B_2.

By (a), therefore, it follows from

$$\iota a_1 + \omega + \cdots \simeq \iota b_1 + \omega + \cdots$$

that $\iota a_1 \simeq \iota b_1$, $\omega + \iota a_2 + \omega + \cdots \simeq \omega + \iota b_2 + \omega + \cdots$, and then, by (b), that $\iota a_2 + \omega + \cdots \simeq \iota b_2 + \omega + \cdots$, then again that $a_1 = b_1$, $a_2 = b_2$, and so on.

We will call an (infinite) set without a first and a last element *unbounded*, and an (infinite) set without neighboring elements *dense*; we will extend these adjectives to the types of these sets as well. We therefore require that there exist elements of the set preceding any given element and following any given element, and in the second case, between any two elements. The set of all the rational numbers and the set of all the real numbers in their natural order are sets of this kind; their types will be called η and λ (λ is the type of the continuum).

Theorem 2.11.2. *II. If A is countable and B is unbounded and dense, then A is similar to a subset of B.*

Theorem 2.11.3. *III. Any two unbounded dense countable sets are similar.*

Proof of II. Let $A = \{a_1, a_2, \ldots\}$; we maintain that A can be mapped onto a subset of B in an order-preserving way. The proof is by induction. First, we associate with a_1 an arbitrary element of B. Next, let us suppose that we already have images of the elements of $A_n = \{a_1, a_2, \ldots, a_n\}$ of the required kind; we wish to find such an image for a_{n+1}. Either a_{n+1} lies between two elements of A_n or $a_{n+1} < A_n$ or, finally, $a_{n+1} > A_n$. Thus, we need only find an element of B between two elements of B, or before such an element (the image of the first element of A_n), or after such an element, and this is possible because B is unbounded and dense. There is therefore a suitable image of a_{n+1}, and thus, one by one, we can associate images with each of the a_n with preservation of order.

Proof of III. Let $A = \{a_1, a_2, \ldots\}$ and $B = \{b_1, b_2, \ldots\}$ be unbounded and dense; by II we can map A onto a subset of B and B onto a subset of A,[3] and by setting up the individual correspon-

[3]The similarity analogue of the Equivalence Theorem (namely, that if each of two sets has a subset similar to the other then the two sets are similar) is false. Example: Two intervals, one with and one without its endpoints.

dences alternately in one direction and the other, we can also map A onto B. In fact, let us associate a_1 with b_1 and write $a^1 = a_1$, $b^1 = b_1$. We now proceed by induction: let us assume that the pairs $(a^1, b^1), \ldots, (a^n, b^n)$ have already been formed in an order-preserving way, that is, that the sets

$$A^n = \{a^1, \ldots, a^n\} \quad \text{and} \quad B^n = \{b^1, \ldots, b^n\}$$

are, as subsets of A and B respectively, similar as they stand; we wish to add the pair (a^{n+1}, b^{n+1}) to them. For even n let a^{n+1} be the a_k with lowest index k that does not belong to A^n, and b^{n+1} the b_k with lowest index k that has the same position with respect to B^n that a^{n+1} has with respect to A^n. For odd n, the other way around: let b^{n+1} be the lowest b_k as yet unmapped and a^{n+1} the lowest a_k consistent with the order relationships. The existence of suitable images is guaranteed by the unboundedness and density of the two sets, while at the same time no element can be left out in the mapping; thus $A \simeq B$.

From II (in which one can choose A as well as B to be the set of all rational numbers, of type η) and III we therefore obtain:

Theorem 2.11.4. *IV. Every unbounded dense set contains a subset of type η. A set of type λ contains countable subsets of every type. Every unbounded dense countable set is of type η.*

Examples. If M (of type μ) is finite or countable, then $\mu\eta = \eta\mu$ is always an unbounded dense countable type, so that $\mu\eta = \eta$; for example

$$n + \eta = \eta = \eta n = \eta\omega = \omega\eta = \eta.$$

Dense countable types can only be distinguished from each other by the existence or non-existence of first or last elements; there are therefore precisely four such sets:

$$\eta, \quad 1 + \eta, \quad \eta + 1, \quad 1 + \eta + 1.$$

$\mu(1+\eta) = (1+\eta)\mu$ (M finite or countable) is either $= 1+\eta$ or $= \eta$, according as M does or does not have a first element; for example,

$$\lambda + \eta = \lambda + \eta\eta = (1 + \eta)\eta = 1 + \eta, \quad (1 + \eta)\omega^* = (1 + \eta)\eta = \eta.$$

The set of all rational numbers $> a$ is of type η. Thus there exists, for instance, a function $s = f(r)$ that preserves order, i.e., increases monotonically with r, which assigns a rational number $s > a$ to every rational number $r > 0$, and vice versa. Clearly this can be extended to a continuous monotonically increasing function $y = f(x)$ that maps the half-line $x > 0$ onto the half-line $y > a$ and assigns to every rational x a rational y, and to every irrational x an irrational y. For rational a this is trivial, of course, as $y = x + a$ induces such a correspondence.

The set of all rational numbers in the interval $(0, 1)$ and that of the dyadic rationals (fractions whose denominator is a power of 2) in the same interval are similar, both having type η. The correspondence (which we will extend at once to a correspondence

$$y = f(x), \quad (0 < x \leq 1,\ 0 < y \leq 1)$$

applying to the whole interval) which assigns to a dyadic rational x a rational y and vice versa, may be constructed as follows. We divide the interval $0 < x \leq 1$ at $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$ and the interval $0 < y \leq 1$ at $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots$, into the respective subintervals

$$\left(\frac{1}{2^m}, \frac{1}{2^{m-1}}\right], \quad \left(\frac{1}{m+1}, \frac{1}{m}\right],$$

where m is a natural number. We can write

$$x = \left(\frac{1}{2}\right)^{m_1} \cdot x_1, \quad 0 < x_1 \leq 1,$$

$$y = \frac{1}{m_1 + 1 - y_1}, \quad 0 < y_1 \leq 1$$

for this.

By repeating this process for x_1 and y_1, and continuing in this way indefinitely, we obtain for x the dyadic expansion

$$x = \left(\frac{1}{2}\right)^{m_1} + \left(\frac{1}{2}\right)^{m_1+m_2} + \cdots$$

and for y the continued fraction

$$y = \cfrac{1}{m_1 + \cfrac{1}{m_2 + \cfrac{1}{m_3 + \cdots}}}$$

where $m = (m_1, m_2, m_3, \ldots)$ is a sequence of natural numbers. The correspondence between x and y defined by equal m has the desired property, namely that of associating a rational y with a dyadic rational x and vice versa; furthermore it associates a rational (but not a dyadic rational) x with a quadratic irrational y, and vice versa (Minkowski). A more detailed investigation must be left to the reader versed in the elements of the theory of continued fractions.

The type λ of the set of all real numbers is also that of an interval without end points or that of a half-line without end point ($x > a$, $x < a$).

The four intervals

$$(a, b), \quad [a, b), \quad (a, b], \quad [a, b]$$

have the types λ, $1 + \lambda$, $\lambda + 1$, $1 + \lambda + 1$.

From $(a, b) + [b, c) \simeq (a, c)$ $(a < b < c)$ we obtain $\lambda + 1 + \lambda = \lambda$, whereas $\lambda + \lambda \neq \lambda$; similarly, stringing together suitable intervals yields

$$1 + \lambda \leq \lambda, \quad 1 + \lambda \cdot 2 = (1 + \lambda)\lambda = \lambda, \quad \lambda = (\lambda + 1)\lambda = \lambda,$$

among others.

To characterize λ more precisely we employ the following considerations. A partitioning

$$A = P + Q \quad (P < Q)$$

of the ordered set A into a "lower section" P and an "upper section" Q can present the following four cases:

P has a last element and Q has a first element:	Jump
P has a last element and Q has no first element:	} Cut
P has no last element and Q has a first element:	
P has no last element and Q has no first element:	Gap.

A set without jumps is dense — the set of all rational numbers, for example, which however has gaps. A set without jumps or gaps is called *continuous* in the sense of Dedekind — the set of all real numbers, for example (hence the expression "continuum" and "cardinality of the continuum"). We form the continuum from the set of all rational numbers by filling in the gaps (by means of the irrational numbers); the well-known way in which this was done by Dedekind may be carried over to an arbitrary dense set, which we will assume in addition to be unbounded. Let us consider the partitions

$$A = P + Q$$

where P has no last element; Q may have a first element or not. These lower sections themselves form an ordered set $\mathfrak{P}$, where we define $P < P_1$ by $P < P_1$, and we assert that $\mathfrak{P}$ is continuous. For, in the first place, $\mathfrak{P}$ is dense, that is, between P_1 and $P_2 > P_1$ there always lies another P; in fact, since $P_2 - P_1$ has no last element and therefore has infinitely many elements, if we choose an element a of $P_2 - P_1$ which is not the first element of $P_2 - P_1$ and allow P to be the set of all elements of A that are $< a$, then $P_1 < P < P_2$. In the second place, $\mathfrak{P}$ also has no gaps. For let $\mathfrak{P} = \mathfrak{P}_1 + \mathfrak{P}_2$ be a partition of $\mathfrak{P}$ into a lower class and an upper class, that is, into two classes of lower sections P_1 and P_2, where we always have $P_1 < P_2$. Let us form the sum $P = \mathfrak{S}\, P_1$ of the P_1 and the intersection $Q = \mathfrak{D}\, Q_1$ of their complements ($A = P_1 + Q_1$); P and Q again form a partition $A = P + Q$, where P is thus a lower section and clearly has no last element. From $P_1 \leq P$ it then follows that $P \leq P_2$; that is, we always have $P_1 \leq P \leq P_2$, and thus P is either the last element of

$\mathfrak{P}_1$ or the first of $\mathfrak{P}_2$; thus $\mathfrak{P}_1 + \mathfrak{P}_2$ is neither a jump nor a gap, but a cut. $\mathfrak{P}$ is thus continuous.

Let us use the following terminology:

$B \subseteq A$ is said to be *dense in* A if between any two elements of A there always lies an element of B. (If B is dense in A, then it of course follows that both A and B are themselves dense.)

We can now characterize the type λ of the continuum by means of the following theorem:

Theorem 2.11.5. *V. Every continuous set has a subset of type λ. Every unbounded continuous set for which there exists a countable set that is dense in it is of type λ.*

Proof. Let A be continuous; we may assume that it is unbounded. It contains (by IV) a subset B of type η. If $B = P + Q$ is a gap of B, then we must have at least one element of A lying between P and Q; otherwise, as is easy to see, A also has a gap. So A has a subset C obtainable from B by filling in its gaps, that is, a subset of type λ. On the other hand, if B is also dense in A (every set dense in A is itself unbounded and dense, and hence, if countable, is of type η), then only a single element of A can lie between P and Q, so that $A \simeq C$. This completes the proof of the theorem.

There exist infinitely many distinct order types that have the cardinality of the continuum. If $\vartheta = 1 + \lambda + 1$ is the type of the closed interval $J = [0, 1]$, then the powers ϑ, $\vartheta\vartheta$, $\ldots$ are continuous and distinct from each other. For, let $J_2 = (J, J)$, $J_3 = (J, J, J)$, $\ldots$, J_m be the lexicographically ordered set of the complexes $x = (x_1, x_2)$, $\ldots$, $x = (x_1, x_2, \ldots, x_m)$ where each x_i runs through the interval J. That J_m is dense is obvious; that J_m has no gaps may be reduced to the corresponding property of J_{m-1} in the following way. Let $H_m(a)$ be the set of complexes $(a, x_2, \ldots, x_m)$, similar to J_{m-1}, obtained from J_m by letting x_1 have the fixed value $x_1 = a$; we have

$$J_m = \sum_J H_m(a).$$

In a partition $J_m = P_m + Q_m$ there are two possibilities: either (i) one of the summands $H_m(a)$ is also partitioned, so that this case is reduced to a partition $J_{m-1} = P_{m-1} + Q_{m-1}$ or (ii) the partition is of the form

$$P_m = \sum_P H_m(a), \quad Q_m = \sum_Q H_m(a), \quad J = P + Q,$$

and, if a_0 is the largest a, then P_m has the last element $(a_0, 1, \ldots, 1)$ and, if b_0 is the smallest b, then Q_m has the first element $(b_0, 0, \ldots, 0)$.

To prove that the ϑ^m are distinct, we show that if $m > 1$ and J_m is similar to a subset of J_n, then $n > 1$ and J_{m-1} is similar to a subset of J_{n-1}. Let us assume that the set J_m of complexes $x = (x_1, \ldots, x_m)$ is mapped by a similarity mapping onto a subset of J_n, the set of complexes $y = (y_1, \ldots, y_n)$. Let us assume that the images of the two complexes $(x_1, 0, \ldots, 0)$ and $(x_1, 1, \ldots, 1)$ are $\eta = (\eta_1, \eta_2, \ldots, \eta_n)$ and $\overline{\eta} = (\overline{\eta}_1, \overline{\eta}_2, \ldots, \overline{\eta}_n)$ respectively; it follows that $\underline{\eta} < \overline{\eta}$, and so $\underline{\eta}_1 \leq \overline{\eta}_1$. It is then impossible that $\eta_1 < x_1$ throughout — that is, for every x_1 — for, the open intervals $(\underline{\eta}_1, \overline{\eta}_1)$ would then be disjoint and the set of all these open intervals would, like the set of the x_1, have the cardinality $\mathfrak{c}$, whereas since each of these intervals must contain a rational number, there are in fact at most countably many of them. Hence for some x_1, we have $\eta_1 = \overline{\eta}_1$, so that, as one consequence, $n > 1$.

Let α be a value of x_1 for which the images of the complexes $(\alpha, 0, \ldots, 0)$ and $(\alpha, 1, \ldots, 1)$ are $(\beta, \eta_2, \ldots, \eta_n)$ and $(\beta, \overline{\eta}_2, \ldots, \overline{\eta}_n)$ respectively, with the same $\eta_1 = \overline{\eta}_1 = \beta$; the previously introduced set $H_m(\alpha)$ is then similar to a subset of $H_n(\beta)$, and hence J_{m-1} is similar to a subset of J_{n-1}. This proves the above statement.

From this it follows, finally, that when $m > n$, J_m cannot be similar to a subset of J_n (and, in particular, not to J_n itself). For J_{m-1} would then be similar to a subset of J_{n-1} and, continuing in this way, we would obtain that J_{m-n+1} would be similar to a subset of J_1, in contradiction to the above.

Chapter 3

Ordinal Numbers

CHAPTER IV
ORDINAL NUMBERS

3.1 The Well-Ordering Theorem

§ 12. The Well-Ordering Theorem

We define: An ordered set is said to be *well ordered* if every (non-empty) subset has a first element. The order type of a well-ordered set is called an *ordinal number*.

In a well-ordered set there is no subset of type ω^*, and every decreasing sequence of elements $a > b > c \cdots$ has only finitely many members. This property could also serve as definition.

After every element, unless it is the last one, there always follows a next one; for every partition $A = P + Q$, the upper section Q has a first element, whether the lower section P has a last element or not. Conversely, if for every partition (including the trivial one $0 + A$) the upper section Q has a first element, then A is well ordered; for if $B > 0$ is an arbitrary subset of A, and P the set of all elements $< B$, and if $A = P + Q$, then the first element of Q is also the first element of B.

The finite sets $\{1, 2, \ldots, n\}$, the set $\{1, 2, 3, \ldots\}$ of all natural numbers, and the set $\{1, 3, 5, \ldots, 2, 4, 6, \ldots\}$ are well ordered, and their types, n, ω, and $\omega + \omega$ are ordinal numbers. The order types ω^*, η, and λ are not ordinal numbers.

An infinite well-ordered set A has a first element a_0, then a second a_1, a third a_2, etc.; if, in addition to this sequence, it contains other elements, then there is among these a first a_ω, and then a next one $a_{\omega+1}$, etc.

We therefore have

$$A = \{a_0, a_1, a_2, \ldots, a_\omega, a_{\omega+1}, \ldots, a_\xi, \ldots\}. \tag{3.1}$$

The notation indicated here, which we will use in the sequel, is that the index of every element is the type of the set of elements preceding it. In order that this should hold true for finite indices as well, we have started with 0; the index of a_n is the type n of the set $\{a_0, \ldots, a_{n-1}\}$, that of a_0 the type 0 of the null set.

After this preliminary orientation we prove

ZERMELO'S WELL-ORDERING THEOREM: Every set can be well ordered.

If we wish to arrive at a well ordering of the form (1), then we must, say, assign to every set $P_\alpha = \{a_0, a_1, \ldots, a_{\alpha-1}\}$ a definite element a_α as a next, or immediately following, element; that is, we must think of a definite element a_α as having been chosen from the set $Q_\alpha = A - P_\alpha$ of the as yet unordered elements. If we proceed in this way, then the very acts of choosing appear in a certain order: a_α may, and must, be chosen only when its predecessors $a_0, \ldots, a_{\alpha-1}$ have been chosen. The Well-Ordering Theorem can also be proved by means of these successive acts of choosing, but only later on, after a more thorough investigation of ordinal numbers. In order to prove the theorem at once, we use a system of simultaneous, mutually independent, acts of choosing.[1] We assign to every subset P of A other than A itself an element $a = f(P) \in A - P$ not belonging to it or, we choose from every subset Q of A other than 0, an element $a = \varphi(Q) \in Q$ belonging to it. Both these statements mean the same thing: we have $f(P) = \varphi(A - P)$, or $\varphi(Q) = f(A - Q)$. We prefer the first form, and we shall call $a = f(P)$ the *differential element* of P, and the set formed from P by the addition of the differential element we shall call the *successor* to P. This process involves more acts of choosing than are absolutely necessary, since in the well ordering (1) the set $P = \{a_0, a_2\}$, for instance, and its differential element would not be used at all. On the other hand, as we have mentioned, this process makes it possible for the acts of choosing to be mutually independent or, expressed differently, the function $a = f(P)$ has a domain of definition which is fixed from the very beginning, namely the set of all $P < A$.

The method by which a well ordering of A necessarily results from the correspondence $a = f(P)$ is basically very simple, although it places something of a burden on the abstract thinking of the reader. We consider a system of sets $< A$ that

(a) contains the null set,

(b) contains the sum of arbitrarily many sets if it contains each of them, and

(c) contains the successor P_1 to any set $P < A$ if it contains P.

Such a system is called a *chain*. There exist chains; for instance, the largest one: the system of all sets $\subseteq A$. The intersection of

[1] Compare Appendix F.

arbitrarily many chains is clearly also a chain; therefore there exists a smallest chain $\mathfrak{K}$, which is the intersection of all chains. We shall now investigate this chain, and we shall stipulate that all the sets occurring in the sequel, such as P and X, will always belong to $\mathfrak{K}$.

The fundamental problem is to show that all the sets of $\mathfrak{K}$ are comparable (in the sense of p. 13), that is, that for two sets P and X we always have one of the three relations $X \subseteq P$ or $P \subseteq X$. Calling a set P *normal* if it is comparable with every set X, we must prove that all sets are normal. The first step consists in proving the following theorem:

Theorem 3.1.1. *I. If $P < A$ is normal, then all sets are either $\subseteq P$ or $\supseteq P_1$.*

We show that these X ($\subseteq P$ or $\supseteq P_1$) form a chain; this chain must be identical with $\mathfrak{K}$, since $\mathfrak{K}$ is the smallest chain. We must therefore establish that these sets satisfy the chain properties:

(a) The empty set is an X.

Obvious, since $0 \subseteq P$.

(b) The sum of arbitrarily many X is an X.

Let $S = \mathfrak{S}\, X_\alpha$; either every $X_\alpha \subseteq P$ and so $S \subseteq P$, or at least one $X_\alpha \supseteq P_1$ and so $S \supseteq P_1$.

(c) The successor to every $X < A$ is an X.

For $X \supseteq P_1$, we have $X_1 \supseteq P_1$; for $X = P$ we have $X_1 = P_1$. For $X \subset P$ we must have $X_1 \subseteq P$; for, in any case, X_1 is comparable with P, which is normal; but if $X_1 \supset P$, then $X_1 - X = (X_1 - P) + (P - X)$ would have to contain at least two elements, whereas we know that this set contains only the one element $f(X)$. We can now conclude that:

Theorem 3.1.2. *II. All sets are normal.*

Again, we show that the normal sets form a chain, which must therefore be identical with $\mathfrak{K}$.

(a) The empty set is normal.

(b) The sum of arbitrarily many normal sets is normal.

Let $P = \mathfrak{S}\, P_\alpha$ be a sum of normal sets and X an arbitrary set, so that $P_\alpha \gtreqqless X$. Either every $P_\alpha \subseteq X$ and therefore $P \subseteq X$, or at least one $P_\alpha \supset X$ and therefore $P \supset X$. Thus P is comparable with every X.

(c) The successor P_1 to any normal set $P < A$ is normal.

This is proved by Theorem I.

Because of the comparability of all sets, $\mathfrak{K}$ can now be ordered by putting the smaller of two distinct sets before the larger ($P_1 < P_2$ shall mean $P_1 \subset P_2$). This ordering is a well ordering. Since $\mathfrak{K}$ itself has a first element (the null set), it need only be shown that, in a partition $\mathfrak{K} = \mathfrak{K}_1 + \mathfrak{K}_2$ into a lower section and an upper section, the latter has a first element. Let $P_1 \in \mathfrak{K}_1$, $P_2 \in \mathfrak{K}_2$, $P_1 < P_2$, and let P be the sum of all the P_1. Then $P_1 = P \subseteq P_2$, so that P is either the first P_2 or the last P_1. But in the second case, by the dichotomy of Theorem I, P_1 must be the first P_2. In either case there is a first P_2.

Finally the sets $P < A$ can be put into a one-to-one correspondence with the elements $a \in A$, and thus A will also be well ordered. This is accomplished by means of the relation $a = f(P)$, which determines the differential element of P. Distinct sets $P_1 < P_2$ have distinct differential elements a_1 and a_2; for, P_2 contains the successor to P_1, so that $a_1 \in P_2$, $a_2 \notin P_2$. On the other hand, every a is the differential element of one, and thus only one, set P. For let $P = F(a)$ be the sum of all the sets P_α that do not contain a (the null set, for example, is one such set). Then we must have $a = f(P)$, for otherwise $P_1 \supset P$ would also not contain a. The elements $a \in A$ and the sets $P < A$ (i.e., all the sets of $\mathfrak{K}$ except for the last, A) can be put into one-to-one correspondence by means of

$$a = f(P), \quad P = F(a);$$

the ordering of the a induced by that of the P ($a_1 < a_2$ means $P_1 < P_2$) thus well-orders A. Since $P_1 < P_2$ means the same as $a_1 \in P_2$, it follows that $P = F(a)$ is the set of all the elements $a_1 < a$ and, conversely, $a = f(P)$ is the next element after P in the well ordering (but this only holds for those sets $P < A$ that appear in the chain $\mathfrak{K}$).

An example of this process of well ordering is the following: From every set Q of natural numbers, that number $a = \varphi(Q)$ is chosen which has the least number of prime factors; if there is more than one number in Q with a minimal number of prime factors, the smallest is chosen. This induces the following well ordering of the natural numbers: First comes the number 1, then the prime numbers in order of magnitude, then the numbers containing exactly two prime factors, again in order of magnitude, and so on; this well ordering is of the type

$$1 + \omega + \omega^2 + \omega^3 + \cdots = \omega^\omega.$$

3.2 The Comparability of Ordinal Numbers

§13. THE COMPARABILITY OF ORDINAL NUMBERS

Every element a of the well-ordered set A determines

$$\text{the segment } P_a = \text{the set of all elements} < a \text{ and}$$
$$\text{the remainder } Q_a = \text{the set of all elements} \geq a,$$

and hence determines the partition $A = P_a + Q_a$ $(P_a < a, Q_a > 0)$. Conversely, every partition of A into a lower section P and an upper section Q is of this form, as Q has a first element a. If a is the first element of A, then we set $P = 0$, $Q = A$.

We now have:

Theorem 3.2.1. *I. If $b = f(a)$ is a similarity mapping of the well-ordered set A onto a subset B, then we always have $f(a) \geq a$.*

That is, in such a transformation the image of an element can never be an earlier element. In fact, if there were elements a for which $f(a) < a$, then there would be a first among these; let this be a and let $b = f(a)$ be its image, so that $b < a$ and, because of similarity, $f(b) < f(a)$, that is, $f(b) < b$; then b would after all be an earlier element having this property.

In particular:

Theorem 3.2.2. *II. A well-ordered set is never similar to one of its segments.*

For, if A were similar to the segment B determined by a, then we would have $f(a) \in B$ and hence $f(a) < a$.

The relation between two well-ordered sets A and B characterized by the statement that A is similar to a segment of B remains unchanged when these sets are replaced by similar ones. This justifies the following definition:

If α and β are two ordinal numbers and if A and B are well-ordered sets of these types, then

$$\alpha < \beta \text{ or } \beta > \alpha$$

(α *smaller* than β, β *larger* than α) shall mean that A is similar to a segment of B.

Clearly the transitive law holds:

$$\text{if } \alpha < \beta \text{ and } \beta < \gamma, \text{ then } \alpha < \gamma$$

(A is similar to a segment of B, B to a segment of C, and therefore A to a segment of C).

Let

$$A = \{\ldots, a, \ldots, b, \ldots\}$$

be a well-ordered set of type α; then by definition the numbers $<$ α are in one-to-one correspondence with the segments of A, and this correspondence is a similarity mapping: Every a determines its segment P_a of type α_a, and if $a < b$, then P_a is a segment of P_b and $\alpha_a < \alpha_b$. Thus

$$W(\alpha) = \{\alpha_0, \alpha_1, \alpha_2, \ldots, \alpha_\omega, \ldots\}$$

which proves the above statement. Conversely, this enables us to carry out in general the notation indicated in § 12, (1), i.e., to index the elements of a well-ordered set of type α by means of the correspondence with the numbers of

$$W(\alpha) = \{0, 1, \ldots, \xi, \ldots\} \quad (\xi < \alpha)$$

in such a way so that in

$$A = \{a_0, a_1, \ldots, a_\xi, \ldots\} \quad (\xi < \alpha)$$

the index of every element is the type of the segment belonging to it. We need only be careful to include the ordinal number 0 as the type of the null set, which is the segment of the first element. Thus, for example,

$$W(1) = \{0\}, \quad W(2) = \{0, 1\}, \quad W(n) = \{0, 1, \ldots, n-1\}$$

for finite $n > 0$, while $W(0) = 0$ is the null set.

Now let α and β be two ordinal numbers, $A = W(\alpha)$, $B = W(\beta)$, and $D = AB$ their intersection, that is, the set of those ordinal numbers γ that are at the same time $< \alpha$ and $< \beta$. D is well ordered and its type δ is an ordinal number. Either $D = A$ or D is a segment of A (determined by $a \in A$); that is, either $\delta = \alpha$ or $\delta < \alpha$; analogously, either $\delta = \beta$ or $\delta < \beta$. Of the four cases, the case $\delta < \alpha$, $\delta < \beta$ is impossible. For, if D is a segment of A determined by α_1 and a segment of B determined by β_1, then $\alpha_1 \in A$, $\beta_1 \in B$, and because $D = A \cdot B$, $\alpha_1 \notin D$ means $\alpha_1 \geq \beta_1$, and $\beta_1 \notin D$ means $\beta_1 \geq \alpha_1$; therefore $\alpha_1 = \beta_1$, which is a contradiction since $\alpha_1 = \beta_1$ would have to belong to D. It follows that:

Theorem 3.2.3. *III. (Comparability Theorem). Two ordinal numbers α and β are always comparable; that is, one and only one of the three relations*

$$\alpha < \beta, \quad \alpha = \beta, \quad \alpha > \beta$$

holds.

If in particular $A \subseteq B$, then $\alpha \leq \beta$. For if $\alpha > \beta$, then B would be similar to a segment P of A (determined by $a \in A$), and in the mapping from B onto P, the image of a would be in P and thus would be $< a$, contradicting Theorem I. Nevertheless, even when $A < B$ we may have $\alpha = \beta$, but of course only if A is not a segment of B; for example, all the infinite subsets of the sequence of natural numbers are of type ω.

With the Well-Ordering Theorem and the Comparability Theorem proved, the gap in the theory of cardinal numbers is now filled in: Two cardinal numbers are always comparable. For, $\mathfrak{a}$ and $\mathfrak{b}$ can be considered to be the cardinalities of the well-ordered sets A and B of type α and β respectively, and then either

$$\begin{array}{rl} \alpha = \beta, & \qquad \mathfrak{a} = \mathfrak{b} \\ \text{or } \alpha < \beta, & \qquad \mathfrak{a} \leq \mathfrak{b} \\ \text{or } \alpha > \beta, & \qquad \mathfrak{a} \geq \mathfrak{b}; \end{array}$$

for, $\alpha < \beta$ means that A is similar to a segment of B, so that A is equivalent to a subset of B. Reversing the implication, we obtain

$$\begin{array}{rl} \text{either } \mathfrak{a} = \mathfrak{b}, & \qquad \alpha \lesseqqgtr \beta \\ \text{or } \mathfrak{a} < \mathfrak{b}, & \qquad \alpha < \beta \\ \text{or } \mathfrak{a} > \mathfrak{b}, & \qquad \alpha > \beta, \end{array}$$

where the first equation says that, in the case of equal cardinalities, various well orderings are still possible (for instance, when $\mathfrak{a} = \aleph_0$ we may have $\alpha = \omega, \omega + 1, \ldots$).

Theorem 3.2.4. *IV. In every (non-empty) set of ordinal numbers there is a smallest; that is, every set of ordinal numbers ordered by magnitude is well ordered.*

For, if W is such a set, let α be an arbitrary number in it; if α is not already the smallest, then among the numbers $< \alpha$ that belong to W there is a first, and this is the smallest number of W.

Theorem 3.2.5. *V. For every set of ordinal numbers there are larger ordinal numbers; in particular, there is a next larger ordinal number.*

For we can choose (p. 38) a cardinal number $\mathfrak{a}$ larger than the cardinalities of all the ordinal numbers in W; if α is an ordinal number of cardinality $\mathfrak{a}$, then α is larger than all the ordinal numbers in W; in short, $\alpha > W$. The smallest number $> W$ is then either α itself or a certain number of $W(\alpha)$.

According to V, the concept of the "set of all ordinal numbers" is inconceivable (cf. p. 39).

The numbers $> \alpha$ are the numbers $\alpha + \xi$ $(\xi > 0)$, and conversely (A is similar to a segment of $A + B$); the smallest number $> \alpha$ is $\alpha + 1$.

A number $\lambda > 0$ that has no immediate predecessor, that is, for which $W(\lambda)$ has no last element, is called a *limit number*; the smallest limit numbers are ω, $\omega + \omega = \omega \cdot 2$, $\omega \cdot 3$, $\ldots$. A number that is not a limit number is called *isolated*; aside from 0, isolated numbers are numbers of the form $\alpha + 1$. If the set

$$W = \{\xi_1, \xi_2, \ldots\}$$

has a last element or no last element, then the corresponding set of numbers $> W$ either has a first element $\alpha + 1$ or no first element α, and conversely.

3.3 The Combining of Ordinal Numbers

§ 14. The Combining of Ordinal Numbers

We have already defined the sum and the product for order types. It is easily seen that a sum of ordered sets

$$S = \sum_M A_m$$

is well ordered if M and the summands A_m are well ordered.

We thus have: A well-ordered sum of ordinal numbers, and a product of finitely many ordinal numbers, is itself an ordinal number.

For instance, ω^2, ω^3, $\omega + \omega^3 + \omega^5 + \cdots$ are ordinal numbers.

In this situation we can to a certain extent also assign a meaning to subtraction and division. We first take note of the following inequalities:

From $\alpha < \beta$ it follows that

$$\begin{cases} \xi + \alpha < \xi + \beta, & \omega + \alpha \leq \omega + \beta \\ \alpha\omega < \beta\omega & (\omega > 0), \quad \alpha\xi \leq \beta\xi. \end{cases} \tag{3.2}$$

For, $\alpha < \beta$ means (we may assume that A is a segment of B) that $B = A + C$ $(C > 0)$, and vice versa. Hence

$$\begin{aligned} \xi + \beta = \xi + A + C = (\xi + \alpha) + C > \xi + \alpha, \\ \xi\beta = \xi(A + C) = \xi\alpha + \xi C > \xi\alpha \quad \text{for } \xi > 0. \end{aligned}$$

If the summand or factor ξ comes after the α and the β, then equality can occur. For instance,

$$\begin{aligned} &\omega + 1 < \omega + 2, \qquad && 1 + \omega = 2 + \omega = \omega, \\ &\omega \cdot 1 < \omega \cdot 2, && 1 \cdot \omega = 2 \cdot \omega = \omega. \end{aligned}$$

The converse of (1) yields the following:

$$\begin{cases} \text{From } \xi + \alpha < \xi + \beta \text{ or } \alpha + \xi \leq \beta + \xi & \alpha < \beta \text{ follows} \\ \text{From } \xi\alpha < \xi\beta \text{ or } \alpha\xi \leq \beta\xi & \alpha < \beta \text{ follows} \\ \text{From } \alpha\omega = \beta\omega \text{ and } \omega > 0 & \alpha = \beta \text{ follows.} \end{cases} \tag{3.3}$$

It does not follow from $\alpha + \omega = \beta + \omega$ or from $\alpha\omega = \beta\omega$ that $\alpha = \beta$.

Subtraction. As we have just seen, α and $\beta > \alpha$ always uniquely define a number ξ satisfying the equation

$$\alpha + \xi = \beta,$$

and we denote it by

$$\xi = -\alpha + \beta,$$

so that $\alpha + (-\alpha + \beta) = \beta$. ξ is the type of $W(\beta) - W(\alpha)$, that is, of a remainder (p. 68) of $W(\beta)$ or, for short, a *remainder type* of β. What is more, for fixed β and $\alpha < \alpha_1 < \beta$ we clearly have $\xi > \xi_1$; the distinct remainder types therefore form an inversely well-ordered set, which however is also well ordered, since the remainder types are ordinal numbers; it follows that there are only finitely many distinct remainder types for a given ordinal number.

For instance, the only remainder type of ω is ω; $\omega + 3$ has the remainder types $\omega + 3$, 3, 2, and 1, respectively, corresponding to the partitions (ν finite)

$$\omega + 3 = \nu + (\omega + 3) = 0 + 3 = (0 + 1) + 2 = (0 + 2) + 1.$$

On the other hand, when $\xi > \alpha$ the equation

$$\eta + \alpha = \beta$$

is not always solvable for η (in order that the equation be meaningful, α must be a remainder type of β); for instance, $\eta + \omega = \omega + 1$ is not solvable.

Division. Let $\alpha > 0$ and ξ be given. Then for every β there exist numbers η and ζ satisfying

$$\xi = \alpha\eta + \zeta \quad (\zeta < \alpha), \tag{3.4}$$

and η and ζ are uniquely determined by α and ξ.

For, if A and B are well ordered and of type α and β respectively, then $\alpha\beta$ is the type of the product (B, A), that is, of the lexicographically ordered set of pairs (b, a). ξ is the type of a segment of (B, A), which we suppose to be determined by (b, a); this segment consists of the pairs (y, x) for which $y < b$ and $x \in A$, and of the pairs (b, x) for which $x < a$. Thus if ξ and η are the types of the segments of A and B determined by a and b, then the above formula follows; at the same time, it can be seen that (b, a) and ξ, η are determined by α and ξ.

Letting β be arbitrary, we can say:

If $\alpha > 0$, then every number ξ can be represented in the form

$$\xi = \alpha\eta + \zeta \quad (\zeta < \alpha), \tag{3.5}$$

where ζ and η are uniquely determined by α and ξ.

For, one can choose β so large (for instance $\beta = \xi + 1$) that $\xi < \alpha\beta$, and one can then apply (3); what is more, the representations (4) corresponding to two distinct β cannot differ, as then there would be two distinct representations

$$\xi = \alpha\eta_1 + \zeta_1 = \alpha\eta_2 + \zeta_2$$

with the same α and $\zeta_1, \zeta_2 < \alpha$; but since of two numbers $\alpha\eta_1 + \zeta_1$ and $\alpha\eta_2 + \zeta_2$ one must be $\leq$ the other, and since a remainder type of a remainder type is itself a remainder type, the smaller one would have to be representable in both of the above forms, which can happen only if $\eta_1 = \eta_2$ and $\zeta_1 = \zeta_2$.

Let an ordinal number α_ξ be associated with every ordinal number ξ. We define the sum

$$f(\alpha) = \sum_{\xi < \alpha} \alpha_\xi = \alpha_0 + \alpha_1 + \cdots + \alpha_\xi + \cdots$$

— which is to be regarded as a function of α only when the summands are fixed — as an ordinal number by means of the following inductive rule:

$$f(0) = 0;$$

for $\alpha > 0$, $f(\alpha)$ is the smallest number $\geq f(\xi) + \alpha_\xi$ (for all $\xi < \alpha$). (3.6)

If for the sake of simplicity we take all $\alpha_\xi > 0$ (summands $= 0$ are to be left out), then for $\alpha > \xi$ we have

$$f(\xi) + \alpha_\xi \geq f(\alpha) = f(\xi) + \alpha_\xi \geq f(\xi),$$

the numbers $f(\alpha)$ as well as the numbers $f(\xi) + \alpha_\xi$ having the same order as their arguments ξ. In particular, therefore,

$$f(\xi + 1) = f(\xi) + \alpha_\xi. \tag{3.7}$$

If furthermore α is a limit number, then

$$f(\alpha) = \lim f(\xi) \quad (\xi < \alpha); \tag{3.8}$$

for in that case $\xi < \alpha$ implies $\xi{+}1 < \alpha$ and the first number $\geq f(\xi{+}1)$ is identical with the first number $> f(\xi)$, so that $f(\alpha)$ is identical with $\lim f(\xi)$ (compare (5) and (6)).

It is easy to see that this present definition of sum is identical with the earlier one, that is, that the earlier conditions remain in force here.

We also define the product

$$f(\alpha) = \prod_{\xi<\alpha} \alpha_\xi = \alpha_0 \cdot \alpha_1 \cdots \alpha_\xi \cdots$$

in the same way:

$$\begin{gathered} f(0) = 1; \\ \text{for } \alpha > 0,\ f(\alpha) \text{ is the smallest number } \geq f(\xi) \cdot \alpha_\xi \text{ (for all } \xi < \alpha). \end{gathered} \tag{3.9}$$

If for the sake of simplicity we take all the factors to be > 1 (ones are to be omitted), then first of all it is clear that $f(\alpha) > 0$, and that

$$f(\xi) \cdot \alpha_\xi \geq f(\alpha) \geq f(\xi) \cdot \alpha_\xi > f(\xi),$$

and we obtain as before

$$f(\xi + 1) = f(\xi) \cdot \alpha_\xi \tag{3.10}$$

and for a limit number α

$$f(\alpha) = \lim f(\xi) \quad (\xi < \alpha). \tag{3.11}$$

Products with finite α agree with our earlier products,

$$f(1) = \alpha_0, \quad f(2) = \alpha_0\alpha_1, \quad \ldots$$

and hence

$$f(\omega) = \lim f(n), \quad \alpha_0\alpha_1\alpha_2 \cdots = \lim \alpha_0\alpha_1 \cdots \alpha_n;$$

for instance[2]

$$2 \cdot 3 \cdot 4 \cdot \dots = \lim\{2, 6, 24, \dots\} = \omega.$$

In particular, if all the factors $\alpha_\xi = \alpha > 1$, we define the product to be the power $f(\alpha) = \mu^\alpha$. Thus

$$\mu^{\xi+1} = \mu^\xi \cdot \mu, \tag{3.12}$$

and for a limit number α,

$$\mu^\alpha = \lim \mu^\xi \quad (\xi < \alpha); \tag{3.13}$$

for instance

$$2^\omega = \lim 2^n = \lim\{2, 4, 8, \dots\} = \omega,$$

and in general

$$2^\omega = 3^\omega = \dots = \aleph_0^\omega = \mathfrak{c}^\omega = \omega,$$

whereas

$$\omega\omega^\omega = \lim \omega^n = \lim\{\omega, \omega^2, \omega^3, \dots\} = \omega^\omega,$$

for which ω^ω does not equal ω.

The powers thus defined are not, as might be assumed, the powers in the sense of set products; the latter are in general not the types of set products. Hence μ^α, for instance, does not necessarily have cardinality $\mathfrak{a}^\alpha$ (while $\alpha + \beta$ and $\alpha\beta$ have the cardinalities $\mathfrak{a} + \mathfrak{b}$ and $\mathfrak{a}\mathfrak{b}$ respectively); $2^\omega = \omega$ has only the cardinality $\aleph_0$, not $2^{\aleph_0}$.

Just as a natural number can be expressed by means of the powers of 10, so every ordinal number can be expressed by means of the powers of an arbitrary basis $\beta > 1$. Let $\xi > 0$ be an ordinal number and β^γ the smallest power of β that is $> \xi$ (that there are such powers can be easily seen from the inequality, easily proved by induction, $\mu^\xi \geq \xi$, from which it follows that $\beta^{\xi+1} > \xi$). γ is not a limit number, because otherwise for every $\xi < \gamma$ we would also have $\xi + 1 < \gamma$, so that

$$\beta^\xi \leq \xi, \quad \beta^{\xi+1} \leq \xi, \quad \beta^\gamma = \lim \beta^{\xi+1} \leq \xi,$$

whereas $\beta^\gamma > \xi$. Therefore γ (> 0) must have an immediate predecessor α, and so

$$\beta^\alpha \leq \xi < \beta^{\alpha+1} = \beta^\alpha \cdot \beta.$$

[2]Note that the "factors" 2, 3, 4 are the ordinal numbers 2, 3, 4, not cardinal numbers.

By the division theorem, there exists a unique representation

$$\xi = \beta^\alpha \cdot \gamma_\alpha + \zeta \quad (\zeta < \beta^\alpha). \tag{3.14}$$

Now $1 \le \gamma_\alpha < \beta$, and if $\zeta > 0$, there is a largest power β^{α_1} $(\alpha_1 < \alpha)$ that is $\le \zeta$, and for this there exists a representation

$$\zeta = \beta^{\alpha_1} \cdot \gamma_1 + \zeta_1 \quad (\zeta_1 < \beta^{\alpha_1}),$$

and so on. In this way we obtain the representation

$$\xi = \beta^\alpha \gamma_\alpha + \beta^{\alpha_1}\gamma_1 + \cdots + \beta^{\alpha_n}\gamma_n, \tag{3.15}$$

where

$$\alpha > \alpha_1 > \cdots > \alpha_n \ge 0, \quad 1 \le \gamma_i < \beta,$$

and this representation is unique. The proof is analogous to that for finite ξ.

Examples. $\beta = 2$:

$$\xi = 2^\alpha + 2^{\alpha_1} + \cdots + 2^{\alpha_n}.$$

$\beta = \omega$:

$$\xi = \omega^\alpha \cdot c_\alpha + \omega^{\alpha_1} \cdot c_1 + \cdots + \omega^{\alpha_n} \cdot c_n, \tag{3.16}$$

(c_i natural numbers).

In particular, every number $\xi < \beta^\omega$ can be represented as a polynomial in β (with finite exponents), and in fact it has a unique representation of the form (14).

In the representation (14) it can happen, moreover, that ξ is not expressed at all by means of smaller numbers, but that the equation

$$\xi = \beta^\xi \tag{3.17}$$

holds (which is impossible for finite numbers when $\beta > 1$); thus, as we have found, $\omega = 2^\omega$. If the numbers ε_ν are defined for $\nu = 0, 1, 2, 3, \ldots$ by $\varepsilon_0 = 1$, $\varepsilon_{\nu+1} = \beta^{\varepsilon_\nu}$, then $(\beta > 1)$

$$\begin{aligned}
&\varepsilon_0 < \varepsilon_1, \text{ so that } \beta^{\varepsilon_0} < \beta^{\varepsilon_1},\\
&\varepsilon_1 < \varepsilon_2, \text{ so that } \beta^{\varepsilon_1} < \beta^{\varepsilon_2},\\
&\varepsilon_2 < \varepsilon_3, \text{ etc.}
\end{aligned}$$

When $\varepsilon = \lim \varepsilon_\nu$,

$$\varepsilon = \lim \varepsilon_{\nu+1} = \lim \beta^{\varepsilon_\nu} = \beta^\varepsilon,$$

so that ε is a number with the property (16); it is the limit of

$$1,\ \beta,\ \beta^\beta,\ \beta^{\beta^\beta},\ \ldots$$

and is the smallest number with the property (16) (by induction, $\varepsilon_\nu \leq \xi$ for every ν, if $\xi = \beta^\xi$). The numbers with the property (16) form a well-ordered set; its elements, arranged in order of magnitude, form a sequence

$$\varepsilon_0,\ \varepsilon_1,\ \varepsilon_2,\ \ldots,\ \varepsilon_\omega,\ \ldots$$

whose type is a limit number. The smallest of these numbers is ε_0 (for $\beta = 2$):

$$\varepsilon_0 = \lim\{2, 4, 16, 65536, \ldots\} = \omega.$$

The next is

$$\varepsilon_1 = \lim\{\omega, \omega^\omega, \omega^{\omega^\omega}, \ldots\},$$

and so on. The numbers ε are limit numbers; for, if $\xi < \varepsilon$, then also $\beta^\xi < \beta^\varepsilon = \varepsilon$; if ξ were the last number $< \varepsilon$, then β^ξ would be the last number $< \varepsilon$, which contradicts $\xi < \beta^\xi$.

Since a remainder type of a remainder type of ξ is itself a remainder type of ξ, it follows that the smallest remainder type of a number ξ is always a power of ω (one possibility being $\omega^0 = 1$); in the expansion (15) ω^{α_n} is the smallest remainder type of ξ.

Natural Sums and Products. The expansion (15) exhibits an ordinal number as a kind of polynomial in ω, in general with infinite exponents. If we compute with these polynomials as with ordinary polynomials, then we obtain, following Hessenberg, *natural sums* $\sigma(\xi, \eta)$ and *natural products* $\pi(\xi, \eta)$ of ordinal numbers, constructs that are considerably closer to the corresponding ones for finite numbers than $\xi + \eta$ and $\xi\eta$ are.

Let us write the expansion in powers of ω in the simple form

$$\xi = \sum_\alpha \omega^\alpha x_\alpha, \quad \eta = \sum_\alpha \omega^\alpha y_\alpha,$$

where for every α at most one of the coefficients x_α, y_α is $\neq 0$ and all the coefficients are finite. Then we define

$$\sigma(\xi, \eta) = \sum_\alpha \omega^\alpha (x_\alpha + y_\alpha), \quad \pi(\xi, \eta) = \sum_\alpha \omega^{\alpha_\beta} x_\alpha y_\beta.$$

Natural sums, then, behave completely like finite sums; in the sequel, we shall often make use of this fact.

We obtain the natural product by multiplying

$$\xi = \sum_{\alpha} \omega^{\alpha} x_{\alpha}, \quad \eta = \sum_{\beta} \omega^{\beta} y_{\beta}$$

like polynomials, adding the exponents naturally, so that

$$\pi(\xi, \eta) = \sum_{\gamma} \omega^{\gamma} \cdot z_{\gamma}, \quad z_{\gamma} = \sum_{\sigma(\alpha,\beta)=\gamma} x_{\alpha} y_{\beta} = \pi(x_{\alpha}, y_{\beta}),$$

or

$$\pi(\xi, \eta) = \sum_{\alpha,\beta} \omega^{\alpha+\beta} x_{\alpha} y_{\beta} = \sum_{\gamma} \omega^{\gamma} z_{\gamma},$$

where γ is extended over the finitely many pairs for which $\sigma(\alpha, \beta) = \gamma$. We do no more than mention this, and leave it to the reader to establish the analogy with finite products that holds here also.

3.4 The Alefs

§ 15. THE ALEFS

All cardinal numbers can be looked upon as being the cardinalities of well-ordered sets (by the Well-Ordering Theorem) and therefore are comparable. By a *number class* $Z(\mathfrak{a})$ we shall mean the set of all ordinal numbers α that have cardinal number $\mathfrak{a}$; it consists of all the representatives of the single ordinal number $\mathfrak{a}$; we are already acquainted with infinitely many representatives of $Z(\aleph_0)$, namely

$$\omega,\ \omega+1,\ \omega+2,\ \dots,\ \omega^2,\ \dots,\ \omega^3,\ \dots,\ \omega^n,\ \dots,\ \omega^\omega,\ \dots$$

If $\mathfrak{a} < \mathfrak{b}$ and if α and β are any two ordinal numbers of the number classes $Z(\mathfrak{a})$ and $Z(\mathfrak{b})$ respectively, then $\alpha < \beta$.

Theorem 3.4.1. *I. Every set of cardinal numbers when ordered according to size is well ordered.*

For, if we associate with every cardinal number an ordinal number from the corresponding number class, then the set of cardinal numbers is similar to a set of ordinal numbers and is therefore well ordered (§ 13, Chap. IV).

Theorem 3.4.2. *II. For every set of cardinal numbers there exist larger ones, and in particular, there exists a unique next larger one.*

The first statement follows from § 7 and the second follows from I; for the set of cardinal numbers larger than the given ones is, according to I, well ordered, and therefore has a smallest element.

According to I and II, the cardinal numbers (as ordered sets) are similar to the ordinal numbers, and to no others; so every cardinal number $\mathfrak{a}$ is assigned an index α, the type of the set of all cardinal numbers $< \mathfrak{a}$, and we write

$$\mathfrak{a} = \aleph_\alpha.$$

Conversely, to every ordinal number α there corresponds an alef $\aleph_\alpha$, so that the alefs, ordered according to size, form a sequence, similar to the sequence of ordinal numbers,

$$\aleph_0,\ \aleph_1,\ \aleph_2,\ \dots,\ \aleph_\omega,\ \aleph_{\omega+1},\ \dots,\ \aleph_\alpha,\ \dots$$

The cardinal numbers $< \aleph_\alpha$ form a set of cardinality $\aleph_\alpha$, and so the set of all ordinal numbers $< \omega_\alpha$ has the same cardinality; we can also express this as follows:

The set $W(\omega_\alpha)$ has the cardinality $\aleph_\alpha$.

For $\alpha = 0$ this is obvious. If it holds for all $\beta < \alpha$, then it also holds for α: for if α has a predecessor $\alpha - 1$, then the set of cardinal numbers $< \aleph_\alpha$ is obtained from the set of cardinal numbers $\leq \aleph_{\alpha-1}$ by the addition of one element, and therefore has the same cardinality; if α is a limit number, then the set of cardinal numbers $< \aleph_\alpha$ is a well-ordered sum of sets of cardinalities $< \aleph_\alpha$, and therefore has cardinality $\leq \aleph_\alpha$, while on the other hand it certainly has cardinality $\geq \aleph_\alpha$.

Theorem 3.4.3. *III. The cardinality of $Z(\aleph_\alpha)$ is $\aleph_{\alpha+1}$.*

For, the set of all ordinal numbers of cardinality $\aleph_\alpha$ has cardinality $\leq \aleph_{\alpha+1}$, since it is a subset of $W(\omega_{\alpha+1})$, which has cardinality $\aleph_{\alpha+1}$. But if its cardinality were $< \aleph_{\alpha+1}$, then the set $W(\omega_{\alpha+1})$ would be the sum of at most $\aleph_{\alpha+1}$ sets, each of cardinality $\leq \aleph_\alpha$, and would therefore itself have cardinality $\leq \aleph_{\alpha+1}$, whereas its cardinality is actually $\aleph_{\alpha+1}$.

The well-ordered set of all ordinal numbers of cardinality $\aleph_\alpha$ has, by III, a type $\omega_{\alpha+1}$, so that

$$Z(\aleph_\alpha) = W(\omega_{\alpha+1}) - W(\omega_\alpha),$$

or, in the sense of the addition of ordered sets,

$$W(\omega_{\alpha+1}) = W(\omega_\alpha) + Z(\aleph_\alpha), \tag{3.18}$$

$$W(\omega_\alpha) = \sum_{\beta<\alpha} Z(\aleph_\beta) \quad (\aleph_\alpha), \tag{3.19}$$

where for $\alpha = 0$ we set the last sum $= 0$, and $W(\alpha)$ denotes, as before, the set of all ordinal numbers $< \alpha$. $W(\omega_0) = W(\omega)$ represents the union of the finite classes $Z(0)$, $Z(1)$, $\ldots$; moreover (following Cantor), $W(\omega_\alpha)$ is often called the first class, $Z(\aleph_0)$ the second class, $Z(\aleph_1)$ the third class.

Every ordinal number is of the form $\omega\nu + \mu$ ($\mu < \omega$, i.e. μ finite; see § 14, (4)); in particular, a limit number is of the form $\lambda = \omega\nu$, so that we have for its cardinal number

$$\mathfrak{l} = \aleph_0 \cdot \nu = \aleph_0 \cdot \mathfrak{m} = \aleph_0\mathfrak{m} = \mathfrak{l}.$$

Since every beginning number is clearly a limit number (an infinite cardinal number cannot be changed by the addition of a finite number), it follows that:

Theorem 3.4.4. *IV. ω_α is the smallest ordinal number of cardinality $\aleph_\alpha$.*

Theorems III and IV need not hold for beginning numbers whose subscript is a limit number; for example, the numbers $\omega_0, \omega_1, \omega_2, \ldots$ belong to $W(\omega_\omega)$, but the limit ω_ω of this ω-sequence does not. The beginning numbers for which IV holds are called *regular*; $\omega_{\alpha+1}$ and $\omega_0 = \omega$ are thus regular. No regular beginning numbers whose index is a limit number are known; they would have to be tremendously large.

3.5 The General Concept of Product

§ 16. The General Concept of Product

Let $M = \{\ldots, m, \ldots, n, \ldots, p, \ldots\}$ be an ordered set whose elements m correspond to ordered sets A_m; from this set we obtain the as yet unordered product,

$$A = \prod A_m = (\cdots \times A_m \times \cdots \times A_n \times \cdots \times A_p \times \cdots)$$

as the set of complexes

$$a = (\cdots, a_m, \cdots, a_n, \cdots, a_p, \cdots) \quad (a_m \in A_m).$$

Two such complexes a and

$$b = (\cdots, b_m, \cdots, b_n, \cdots, b_p, \cdots)$$

are called *distinct* if they differ at at least one place (i.e., if there exists an m for which $a_m \neq b_m$); we will say that they are *different at m*. For two distinct complexes, the set of places of differences, as a non-empty subset of M, may or may not have a first element. If it has a first element, and if $a_m \lesseqgtr b_m$ at this first place m, then we accordingly say $a \lesseqgtr b$ and thereby obtain an ordering of the set of all complexes.

If M is well-ordered, then every non-empty subset of M has a first element, and therefore any two distinct complexes can be compared; the ordering then becomes the lexicographic ordering, in which $a < b$ if, at the first place m where they differ, $a_m < b_m$. If α_m is the type of A_m and μ the type of M, then the type of the lexicographically ordered product is called the *product*

$$\prod_M \alpha_m = \cdots \alpha_m \cdots \alpha_n \cdots \alpha_p \cdots ,$$

where the order of the factors (α_m is the type of A_m) is the opposite of the order of the factors in the set product and in M. The inverted argument $M^* = N$ may appropriately be called the *exponent*. When the factors are equal, $A_m \simeq B$, then we obtain from the product the power B^N, having the type $\beta\mu^* = \mu^\nu$ (β, μ, and ν are the respective types of B, M, and N). We have already found among others, the power $\omega^* = 1 + \lambda$ (p. 56).

In the case of an arbitrary M, which is not necessarily well-ordered, we must define the lexicographic order insofar as it is definable. That is, if $M(a,b)$ has a first element m, which we will then denote by $m(a,b)$, and if $a_m \lesseqqgtr b_m$, then let $a \lesseqqgtr b$. $a < b$ then implies $b > a$, and the transitive law holds: if $a < b$ and $b < c$, then $a < c$. For, let $m_1 = m(a,b)$ and $m_2 = m(b,c)$. If $m_1 = m_2$, then $a_{m_1} < b_{m_1} < c_{m_1}$, so that $a_{m_1} < c_{m_1}$, and since either m_1 is also the first element of $M(a,c)$ or there is an earlier element m_0, in which case $a_{m_0} = b_{m_0} = c_{m_0}$, so m_0 cannot be a place of difference of a and c, it follows that $a < c$. If $m_1 \neq m_2$, say $m_1 < m_2$, then $a_{m_1} < b_{m_1}$ and $b_{m_1} = c_{m_1}$, so $a_{m_1} < c_{m_1}$, and m_1 is the first place where a and c differ; therefore $a < c$. The case $m_2 < m_1$ is analogous.

If $M(a,b)$ has no first element, then a and b are called *incomparable*. The transitive law then also holds for incomparable complexes: if a and b are incomparable and b and c are incomparable, then so are a and c. For, if $m \in M(a,b) \cdot M(b,c)$, then a_m, b_m, c_m are all distinct, and two of them must coincide; but this is impossible, since from $a_m = b_m$ it would follow that $m \notin M(a,b)$, etc. Therefore $M(a,c) \supseteq M(a,b) \cup M(b,c)$, and a union of two sets without first elements also has no first element.

The set of complexes can therefore be divided into classes of mutually comparable complexes, and we will look for a way of ordering these classes and which retain as many of the characteristics of a set product as possible.

For this purpose, we again avail ourselves of the theory of well-ordering. If, for two complexes, the set $M(a,b)$ of places of differences of a and b is well-ordered, the complexes will be called *congruent*, a term borrowed from number theory:

$$a \equiv b \text{ or } b \equiv a.$$

The case $M(a,b) = 0$ is also included here: $a \equiv a$. By (1), the transitive law holds: $a \equiv b$ and $b \equiv c$ implies $a \equiv c$. Hence A can be divided into classes in such a way that congruent complexes belong to the same class and incongruent complexes to distinct classes; such a class $A(a)$ — namely the set of complexes congruent to a, and hence to each other — is either identical with $A(b)$ (when $b \equiv a$) or has no element in common with $A(b)$ (when $b \not\equiv a$). The set of all these classes will again be called the (generalized) product, in the hope of being able to define a suitable order for it. To this end, let

$$a \equiv (\ldots, a_m, a_n, \ldots, a_p, \ldots), \quad a_m \in A_m.$$

To investigate the class $A(a)$ and its type, we let

$$A_m = B_m + \{a_m\} + C_m, \quad \alpha_m = \beta_m + 1 + \gamma_m$$

be respectively the partition of A_m determined by a_m, and the type of that partition; furthermore, let X_m, Y_m, Z_m run through the sets A_m, B_m, C_m. The complexes congruent to a then appear in the following order:

$$\begin{aligned}
&(\ldots, X_m, a_n, \ldots, a_p, \ldots)\\
+&(\ldots, B_m, X_n, \ldots, a_p, \ldots)\\
+&(\ldots, B_m, a_n, X_p, \ldots)\\
+&(\ldots, B_m, B_n, X_p, \ldots)\\
+&(\ldots, B_m, a_n, C_p, \ldots)\\
+&(\ldots, a_m, X_n, \ldots, a_p, \ldots)\\
+&(\ldots, a_m, B_n, X_p, \ldots)\\
+&(\ldots, a_m, a_n, X_p, \ldots)\\
+&(\ldots, a_m, B_n, C_p, \ldots)
\end{aligned}$$

and so on, where we must keep in mind that of the sets $M(a,x)$ corresponding to the complexes x in the first row the first element is m, in the second row the first element is n, etc. This arrangement, which may be continued indefinitely, is thus well-ordered. For, each of the rows corresponds to a complex formed from the elements B_m, a_m, C_m, where the first and last of these elements each occur at most once and where at least one B_m occurs; these complexes of the symbols B_m, a_m, C_m, ordered lexicographically (in the order $B_m < a_m < C_m$), form a well-ordered set (since the set of indices m is well-ordered). Each row, however, has the type of the product of the sets X_m occurring in it, where the first elements of the corresponding $M(a,x)$ all coincide. If therefore $\nu(a)$ denotes the order type of the set of all complexes congruent to a, then we have

$$\nu(a) = \prod \alpha_m^{\varepsilon_m} = \cdots \alpha_m^{\varepsilon_m} \alpha_n^{\varepsilon_n} \alpha_p^{\varepsilon_p} \cdots,$$

where the exponents ε_m are the numbers $-1, 0, +1$; namely, $\varepsilon_m = -1$ means that the row in which the first element of $M(a,x)$ is m has complexes of the form $(\ldots, B_m, \ldots)$, and for this row the product of the X_m has the type $\cdots \beta_m \cdots$; analogously, $\varepsilon_m = +1$ means a row with C_m and the type $\cdots \gamma_m \cdots$; finally, $\varepsilon_m = 0$ means that a_m remains fixed throughout the row. The factors in this product, in which the exponents are $= 0$, are to be omitted.

The order types $\nu(a)$ of the classes $A(a)$ are, by the way, not necessarily distinct. It can certainly happen that different classes $A(a)$ and $A(b)$ are similar; in that case, however, the elements a_m and b_m for which $\beta_m \neq \beta'_m$ or $\gamma_m \neq \gamma'_m$ must form a well-ordered subset of M, and conversely.

The generalized product $\prod A_m$ is the set of classes $A(a)$, ordered according to magnitude of their types $\nu(a)$. If this ordering is possible at all — that is, if the types $\nu(a)$ are mutually distinct — then it is a well ordering. For, the well-ordering of the types follows from the well-ordering of the set of all complexes of exponents ε_m, ordered lexicographically.

Chapter 4

Systems of Sets

CHAPTER V
SYSTEMS OF SETS

4.1 Rings and Fields

§ 17. Rings and Fields

For the sake of clarity, we shall call a set of sets a *system of sets*; we shall denote systems of sets by capital German letters. $M \in \mathfrak{M}$ thus means that the set M is an element of the system $\mathfrak{M}$. The sets M under consideration are pure sets, without relations (order) between their elements, so that in this respect we have returned to the point of view of the first two chapters; nevertheless the knowledge of ordinal numbers that we have since acquired will be useful to us and, on occasion, indispensable. We shall direct our attention, in particular, to certain largest and smallest systems of sets which, in accordance with rules to be described, can have no elements adjoined to or removed from them.

1. Rings. A system of sets will be called a *ring*[1] if the sum and the intersection of any two sets of the system are themselves sets of the system. The same is then true for any finite number of sets of the system. A ring is thus in a sense a largest system of sets — one that cannot be extended by the operations of taking sums and intersections (of a finite number of elements).

Examples. The system of all subsets of a given set is a ring. If S is an arbitrary set, then the system of all finite subsets of S is a ring; likewise, the system of all subsets of S that are either finite or have a finite complement. If S is the plane and M consists of the points and the straight lines, then the finite sums of points and straight lines form a ring; if we add the empty set and the whole plane, we obtain the system of all algebraic curves. If S is the set of all real numbers, then the half-open intervals

$$I = [\alpha, \beta) \quad (\alpha \leq \beta)$$

form a system of sets that is not a ring; but the sets

$$S = I_1 + I_2 + \cdots + I_n$$

(finite sums of intervals) do form a ring.

[1] The justification for this term is that in the algebra of these systems of sets (cf. § 19) addition and multiplication can be introduced (the latter, to be sure, in a way different from that of ordinary algebra) which have almost all the properties of the corresponding operations in the domain of integers; the difference of two sets then behaves like the difference of two integers.

In the above examples, we could define this ring — that is to say, the sets belonging to it — very easily. But, in anticipation of less elementary cases that will arise later, we give a general existence proof. There certainly do exist rings $\supseteq \mathfrak{M}$; for if S is the sum of all the sets $\in \mathfrak{M}$, then the system $\mathfrak{S}$ of all subsets of S is a ring containing $\mathfrak{M}$. Since the intersection of any number of rings is obviously itself a ring, it follows that the intersection $\mathfrak{R}$ of all rings $\mathfrak{R}_\alpha$ for which $\mathfrak{M} \subseteq \mathfrak{R}_\alpha$ is a ring containing $\mathfrak{M}$. This ring is the smallest ring possible; that is to say, it is contained in every ring containing $\mathfrak{M}$, since $\mathfrak{M} \subseteq \mathfrak{R}_\alpha \subseteq \mathfrak{S}$ and hence $\mathfrak{R} \subseteq \mathfrak{R}_\alpha \subseteq \mathfrak{S}$.

This smallest of the rings containing $\mathfrak{M}$, referred to henceforth as $\mathfrak{R}$, can obviously be constructed as follows: It consists of the finite sums

$$R = D_1 + D_2 + \cdots + D_n,$$

whose summands, in turn, are of the form

$$D = M_1 M_2 \cdots M_n,$$

that is, are intersections of finitely many sets $M \in \mathfrak{M}$. It is clear that the sets R so defined belong to $\mathfrak{R}$; but they themselves already constitute a ring (which is therefore $\mathfrak{R}$), since it follows from the associative law that the sum of two R is itself an R and from the distributive law that the intersection of two R is itself an R.

2. Fields. A ring is called a *field* if it contains the difference of any two of its sets; it then also contains the null set and, with every set M, its complement $M' = S - M$ (S is the sum of all the sets of the system). A field is thus a largest system of sets — one that cannot be extended by the operations of taking sums, intersections, and differences (of a finite number of elements). (In the algebra of logic, the term *system of domains* is also used.)

Examples. The system of all subsets of a set is a field. The finite subsets of an infinite set S form a ring but not a field. The finite sums of half-open intervals are a ring but not a field; if, however, we had used instead of half-open intervals $I = [\alpha, \beta)$ either open intervals (α, β) or closed intervals $[\alpha, \beta]$, then the sets S would still form a ring but would no longer form a field.

Exactly as in the case of a ring, we can see that there exists a smallest field $\mathfrak{F}$ containing any given system of sets $\mathfrak{M}$; but the construction of its elements is, in this case, not quite so trivial. Since $\mathfrak{F}$ contains the null set if it contains any sets whatever, we begin by assuming that the null set is already contained in $\mathfrak{M}$; furthermore, $\mathfrak{F}$ contains the smallest ring $\mathfrak{R} \supseteq \mathfrak{M}$ and is also the smallest field containing $\mathfrak{R}$. Hence we may as well suppose $\mathfrak{M}$ to be a ring that contains the null set. We then consider a finite number of sets $M_1 \supseteq M_2 \supseteq \cdots \supseteq M_n$ or, to simplify our notation, a decreasing sequence of sets M

$$M_1 \supseteq M_2 \supseteq M_3 \supseteq \cdots \tag{4.1}$$

with ultimately vanishing terms. The differences $M_1 - M_2$, $M_2 - M_3$, ... are disjoint. The set

$$A = (M_1 - M_2) + (M_2 - M_3) + (M_3 - M_4) + \cdots \tag{4.2}$$

will be called a (finite) *chain of differences* of the system $\mathfrak{M}$. Clearly, all these sets A will be elements of $\mathfrak{F}$; we shall show that as they stand they constitute a field, which is therefore the field $\mathfrak{F}$ we have been looking for.

The sum of two A's is an A. Let

$$\begin{aligned} A &= \sum_i (M_i - M_i'), \\ B &= \sum_k (N_k - N_k') \end{aligned}$$

with ultimately vanishing M_i, M_i', N_k, N_k' which belong to the system $\mathfrak{M}$. If we now define

$$\begin{aligned} P_i &= M_i N_k, \\ P_i' &= \sum_{i+k=l} (M_i N_k + M_i' N_k'), \\ \text{that is,} \quad P_0 &= M_0 N_0,\ P_1' = M_0 N_1 + M_1 N_0, \ldots \\ P_l &= M_0 N_l + M_1 N_{l-1} + \cdots + M_l N_0, \ldots \end{aligned} \tag{4.3}$$

(where the subscripts i, k, l range over the integers $0, 1, 2, \ldots$), then the P_l, P_l' belong to the ring $\mathfrak{M}$ and ultimately vanish. Furthermore, $P_l \supseteq P_l'$, and if we take into account that

$$P_{l+1}' = \sum_{i+k=l+1} (M_i N_k + M_{i+1}' N_{k+1}'),$$

we have $P_l \supseteq P'_{l+1}$, so that

$$P_0 \supseteq P'_0 \supseteq P_1 \supseteq P'_1 \supseteq P_2 \supseteq P'_2 \supseteq \cdots.$$

We shall now show that

$$AB = (P_0 - P'_0) + (P_1 - P'_1) + \cdots \tag{4.4}$$

and even that, term by term,

$$\sum_{i+k=l} (M_i - M'_i)(N_k - N'_k) = P_l - P'_l. \tag{4.5}$$

Let us denote the sum on the left-hand side of (8) by C_l. If $x \in C_l$, then x belongs to one of the summands, say $x \in (M_i - M'_i)(N_k - N'_k)$; so that $x \in M_i N_k \subseteq P_l$, while, at the same time, $x \notin P'_l$. For if $i + k = i_0 + k_0 = l$, then either $i = i_0$ and $M_i N_k \subseteq M'_{i_0}$ or $i < i_0$, $k > k_0$, and $M_i N_k \subseteq N_{k_0} \subseteq N'_{k_0}$; thus $x \notin M'_{i_0} N_{k_0}$, and similarly $x \notin M_{i_0} N'_{k_0}$. Hence $C_l \subseteq P_l - P'_l$. Conversely, if $x \in P_l - P'_l \subseteq P_l$ and, say, $x \in M_i N_k$, then $x \notin M'_i$, since otherwise we should have $x \in M'_i N_k \subseteq P'_l$; and similarly $x \notin N'_k$, so $x \in (M_i - M'_i)(N_k - N'_k) \subseteq C_l$. Thus (8) and (7) are proved; and it follows that AB is an A.

The difference of two A's is an A. For, if A is given by (2), then $S - A$ is again an A, namely

$$S - A = (0 - M_1) + (M_1 - M_2) + (M_2 - M_3) + \cdots$$

and therefore

$$A - B = A(S - B)$$

is an A by what has just been proved.

3. Extended fields. In anticipation of a later application (§ 30, II and III), we wish to extend our chains of differences to infinity. We let Greek letters range over the ordinal numbers $< \omega_\alpha$, where ω_α is a fixed beginning number, the first ordinal number of cardinality $\aleph_\alpha$. From the system $\mathfrak{M}$, which is again assumed to be a ring containing the null set, we choose sets M to form a decreasing well-ordered sequence of type ω_α:

$$M_0 \supseteq M_1 \supseteq \cdots \supseteq M_\xi \supseteq M_{\xi+1} \supseteq \cdots \tag{4.6}$$

whose components are thus denoted by $M_{\xi+1}$ and use them to form the set

$$A = (M_0 - M_1) + (M_2 - M_3) + \cdots + (M_{2\xi} - M_{2\xi+1}) + \cdots = \sum_\xi (M_{2\xi} - M_{2\xi+1}), \tag{4.7}$$

where it must be remembered that the ordinal numbers are either even (2ξ) or odd ($2\xi + 1$), the limit ordinals, in particular, being even. We again call A a chain of differences of the system $\mathfrak{M}$, and specifically, a chain of type ω_α, if all the sets (9) are non-empty, and of type $\tau < \omega_\alpha$, if $M_{\tau+1}$ is the first vanishing set occurring in (9). Of course, the ordinal type depends not only on the set itself but also on the representation chosen. We now assert:

I. If the system $\mathfrak{M}$ (which is a ring containing the null set) has, in addition, the property that the intersection

$$M_0 M_1 M_2 \cdots = \mathfrak{D}\, M_\xi$$

of every decreasing well-ordered sequence (9) of type ω_α is a set of the system, then the chains of differences (10) constitute a field.

In this case A is called a *chain of type* $\leq \omega_\alpha$, since the intersection $M^* = \mathfrak{D}\, M_\xi$ can be added to the sequence (9) as a last element (which, of course, need not vanish); in (10) the terms from a certain point on are then $= 0$.

The proof is a generalization of that given above for finite chains. The sum of two A's is again an A: If A and B are given by (10), we form the sums

$$P_\gamma = \sum_{\sigma(\xi,\eta)=\gamma} M_\xi N_\eta, \quad P'_\gamma = \sum_{\sigma(\xi,\eta)=\gamma} (M'_\xi + N'_\eta),$$

and then

$$AB = \sum_\gamma (P_\gamma - P'_\gamma). \tag{4.8}$$

The difference $S - A$ is again an A:

$$S - A = \sum_\xi (M_{2\xi+1} - M_{2\xi+2}) \tag{4.9}$$

$$= (M_1 - M_2) + (M_3 - M_4) + \cdots + (M_{2\xi+1} - M_{2\xi+2}) + \cdots$$

The intersection of two A's is itself an A.[2] Once more we write as in (13):

$$A = \sum_\xi (M_\xi - M'_\xi), \tag{4.10}$$

$$B = \sum_\eta (N_\eta - N'_\eta).$$

[2]The difference of two A's is also an A, since $A - B = A(S - B)$.

In order to generalize (6) we have to make use of the natural sum. We define

$$P_\zeta = \sum_{\sigma(\xi,\eta)=\zeta} M_\xi N_\eta, \tag{4.11}$$

$$P'_\zeta = \sum_{\sigma(\xi,\eta)=\zeta} (M'_\xi + N'_\eta),$$

where $\sum^\zeta$ means that we are to sum over the pairs ξ, η with $\sigma(\xi,\eta) = \zeta$. Since there are at most a finite number of these, the sets P_ζ and P'_ζ belong to the ring $\mathfrak{M}$. From $\xi < \xi_1$ and $\sigma(\xi,\eta) = \zeta$ it follows also that $\zeta < \sigma(\xi_1,\eta)$ (p. 84, footnote), and it is easily seen that P_ζ and P'_ζ must ultimately vanish provided that M_ξ, M'_ξ and N_η, N'_η do so. Furthermore

$$P_0 \supseteq P'_0 \supseteq P_1 \supseteq P'_1 \supseteq P_2 \supseteq P'_2 \supseteq \cdots \supseteq P_\zeta \supseteq P'_\zeta \supseteq \cdots .$$

In fact, $P_\zeta \supseteq P'_\zeta$ follows immediately; in addition $P'_\zeta \supseteq P_{\zeta+1}$ for $\xi < \xi_1$. For if $x \in P_\zeta$ and if x is an element of, say, $M_\xi N_\eta$ with $\sigma(\xi,\eta) = \zeta$, then there exists (p. 81) a pair of numbers ξ_1, η_1 with $\sigma(\xi_1,\eta_1) = \zeta + 1$ and $\xi_1 \leq \xi$, $\eta_1 \leq \eta$; if $\xi_1 < \xi$, then we have $M_{\xi_1} \supseteq M_\xi$, $N_{\eta_1} \supseteq N_\eta$ and $x \in P'_{\zeta+1}$, and similarly for $\eta_1 < \eta$.

Now once again we have

$$AB = \sum_\zeta (P_\zeta - P'_\zeta) \tag{4.12}$$

and even, term by term,

$$\sum_{\sigma(\xi,\eta)=\zeta} (M_\xi - M'_\xi)(N_\eta - N'_\eta) = P_\zeta - P'_\zeta; \tag{4.13}$$

the proof has the same form as the proof of (8), because the natural sum has all the properties of ordinary sums (in particular, the inequalities $\xi \leq \sigma(\xi,\eta)$, $\eta \leq \sigma(\xi,\eta)$).

For $\omega_\alpha = \omega_0 = \omega$, where $\mathfrak{M}$ is only required to be a ring, we again obtain the result of the preceding subsection: The finite chains of differences constitute a field (the smallest field containing $\mathfrak{M}$). For $\omega_\alpha = \omega_1 = \Omega$ it follows that chains of differences which are at most countable (of type $< \Omega$) constitute a field, provided that the intersection of countably many $M \in \mathfrak{M}$ is an M.

If the intersection of any number of $M \in \mathfrak{M}$ is an M, then all the chains of differences over $\mathfrak{M}$ form a field. For, all these chains of differences (each having a fixed representation) are of type $< \omega_\alpha$, provided that ω_α is taken sufficiently large.

4.2 Borel Systems

§ 18. Borel Systems

1. The σ and δ processes. We have called a system of sets a ring if it contains the sum and intersection of two (or any finite number of) sets of the system. If we now extend this to the sum and intersection of a countable number of sets, we arrive at the concept of a Borel System. But the simple method that enabled us to construct the smallest ring containing a given system of sets $\mathfrak{M}$ does not allow of generalization to the present case. For if we were to try, as in the earlier situation (p. 90), to form the intersections

$$D = M_1 M_2 \cdots M_n \cdots$$

of countably many sets of $\mathfrak{M}$, and then the finite or countable sums of such D, and then again the countable intersections of such sums, etc., then we would obtain an uncountable sequence of systems of sets

$$\mathfrak{M},\ \mathfrak{M}_\sigma,\ \mathfrak{M}_{\sigma\delta},\ \mathfrak{M}_{\sigma\delta\sigma},\ \ldots$$

that would not lead us to our goal, as we shall see later. That is to say, the Borel system is not obtained by the iteration of the two processes of forming countable sums and intersections, each performed a countable number of times.

A system of sets will be called a *σ-system* if it contains the sum of every sequence of sets of the system; a *δ-system* if it contains the intersection of every such sequence. Every σ-system also contains the intersection of a finite number of sets of the system; every δ-system also contains the sum of two as well as of a finite number of sets of the system also belongs to the system.

The smallest σ-system over a given system of sets $\mathfrak{M}$ (whose existence we deduce in the same way as we did that of the smallest ring) will be called $\mathfrak{M}_\sigma$. It is formed by the sets

$$M_\sigma = M_1 + M_2 + M_3 + \cdots = \mathfrak{S}\, M_n \quad (M_n \in \mathfrak{M}), \tag{4.14}$$

that is, by the sums of sequences of sets M_n. For the M_σ must belong to $\mathfrak{M}_\sigma$, while at the same time they themselves already constitute a σ-system (transformation of a double sequence into a simple sequence).

If $\mathfrak{M}$ is a ring, then $\mathfrak{M}_\sigma$ is also a ring; for, by the distributive law, the intersection of two M_σ is an M_σ. Furthermore, in this case the

M_σ can be written as sums of increasing sequences of sets M, for it follows from (1) that

$$M_\sigma = \mathfrak{S}\, S_n, \quad S_n = M_1 + M_2 + \cdots + M_n,$$

where the S_n are themselves sets M_σ and $S_1 \subseteq S_2 \subseteq \cdots$.

In a δ-system, the intersection of a finite number of sets of the system also belongs to the system. The smallest δ-system $\mathfrak{M}_\delta$ over $\mathfrak{M}$ is formed by the sets

$$M_\delta = M_1 M_2 M_3 \cdots = \mathfrak{D}\, M_n \quad (M_n \in \mathfrak{M}); \tag{4.15}$$

the M_δ are intersections of sequences of sets M_n. If $\mathfrak{M}$ is a ring, then $\mathfrak{M}_\delta$ is a ring, and the M_δ can be represented as intersections of decreasing sequences of sets M:

$$M_\delta = \mathfrak{D}\, D_n, \quad D_n = M_1 M_2 \cdots M_n.$$

A system of sets which is at one and the same time a σ-system and a δ-system will be called a *$(\sigma\delta)$-system*, or a *Borel system.*

For a given system of sets there exists once more a smallest Borel system $\mathfrak{B} = \mathfrak{M}_{\sigma,\delta}$ containing it. The sets B belonging to it are called the *Borel sets* generated by $\mathfrak{M}$, or by the sets $M \in \mathfrak{M}$.

We can represent these sets in the following way. Let i, k, l, m, ... range over the natural numbers, and define

$$S = \mathfrak{S}\, D_i, \quad D_i = \mathfrak{D}\, S_{ik}, \quad S_{ikl} = \mathfrak{S}\, D_{ikl}, \quad D_{iklm} = \mathfrak{D}\, S_{iklm}, \quad \ldots$$

with the proviso that for every sequence of natural numbers i, k, l, m, ... the corresponding sequence of sets D_i, S_{ik}, D_{ikl}, S_{iklm}, ... is ultimately to contain only sets M.

Similarly, we begin the alternate formation of sums and intersections with the latter, and we let

$$D = \mathfrak{D}\, S_i, \quad S_i = \mathfrak{S}\, D_{ik}, \quad D_{ikl} = \mathfrak{D}\, S_{ikl}, \quad S_{iklm} = \mathfrak{S}\, D_{iklm}, \quad \ldots$$

and require that every sequence S_i, D_{ik}, S_{ikl}, D_{iklm}, ... ultimately contain only sets M. Then the sets S, as well as the sets D, are the Borel sets generated by $\mathfrak{M}$.

For since the D_i of the first group of formulas are arbitrary sets D of the second group, it follows that every D_i (the sum of a sequence of sets D) is an S and vice versa and that, in particular, every $D =$

$D_{ikl\cdots}$ is itself an S. Similarly, the S are identical with the D, and every S is a D. Hence the S are identical with the D and form a Borel system that contains the sets M $(S = D_i = S_{ik} = \cdots = M)$; the smallest Borel system $\mathfrak{B}$ is contained in it; that is, every set B is an S.

Conversely, every S is also a B. For if the S were not a B, there would have to be at least one D_i which is not a B, then at least one S_{ik}, one D_{ikl}, and so forth, so that there would have to be at least one sequence of integers i, k, l, ... for which no B occurred in the sequence D_i, S_{ik}, D_{ikl}, ... But this would contradict the requirement that the sequence is ultimately to contain only sets M.

The representation thus obtained, which yields all the Borel sets at one stroke, does not, however, give us as clear an idea of these sets as we should like, and we shall therefore try to give a stepwise construction of the Borel system $\mathfrak{B}$. If the notation of (1) and (2) is to be extended in a natural way, then $\mathfrak{B}$ must in any case contain the following sets.

the sets M $(\in \mathfrak{M})$,

the sums M_σ of sequences of sets M,
the intersections $M_{\sigma\delta}$ of sequences of sets M_σ,
the sums $M_{\sigma\delta\sigma}$ of sequences of sets $M_{\sigma\delta}$,
etc., (4.16)

where we alternate the formation of sums and intersections, since — to take one example — the sets $M_{\sigma\sigma}$, which are sums of sequences of sets M_σ, are identical with the sets M_σ. Similarly, if we begin with intersections instead of sums, we must include in our Borel System the following sets:

the sets M $(\in \mathfrak{M})$,

the intersections M_δ of sequences of sets M,
the sums $M_{\delta\sigma}$ of sequences of sets M_δ,
the intersections $M_{\delta\sigma\delta}$ of sequences of sets $M_{\delta\sigma}$,
etc. (4.17)

The systems formed by these sets ought to be denoted by

$$\mathfrak{M}_\sigma,\ \mathfrak{M}_{\sigma\delta},\ \mathfrak{M}_{\sigma\delta\sigma},\ \mathfrak{M}_{\sigma\delta\sigma\delta},\ \ldots \quad \text{or} \quad \mathfrak{M}_\delta,\ \mathfrak{M}_{\delta\sigma},\ \mathfrak{M}_{\delta\sigma\delta},\ \mathfrak{M}_{\delta\sigma\delta\sigma},\ \ldots$$

(hence the parentheses in the notation $\mathfrak{B} = \mathfrak{M}_{\sigma,\delta}$ may not be omitted).

By way of example, we remark that the upper and lower limits (p. 22)

$$\overline{M} = \overline{\lim}\, M_n = S_1 S_2 S_3 \cdots = M_\sigma + M_{\sigma+1} + M_{\sigma+2} + \cdots ,$$
$$\underline{M} = \underline{\lim}\, M_n = D_1 + D_2 + D_3 + \cdots = M_\sigma \cdot M_{\sigma+1} \cdot M_{\sigma+2} \cdots$$

are Borel sets generated by the sets M_n ($\in \mathfrak{M}$); the sets S_n and D_n are sets M_σ and M_δ, the upper limit $\overline{M}$ is an $M_{\sigma\delta}$, the lower $\underline{M}$ an $M_{\delta\sigma}$; the limit of a convergent sequence is both of these simultaneously.

Even though we have now formed all the sets (3) or (4) having a finite number of indices, σ and δ, the Borel system is by no means yet complete. In continuation of the procedure of, say, (3) we must next adjoin the intersections

$$N = M_\sigma M_{\sigma\delta} M_{\sigma\delta\sigma} \cdots ,$$

then the N_σ, $N_{\sigma\delta}$, $N_{\sigma\delta\sigma}$, ..., and finally again,

$$P = N N_\sigma N_{\sigma\delta} N_{\sigma\delta\sigma} \cdots ,$$

and so on. To give this "and so on" a precise meaning we avail ourselves of the ordinal numbers.

2. The construction of the Borel System $\mathfrak{B}$. Let ξ and η be ordinal numbers $< \Omega$, where $\Omega = \omega_1$ is the beginning number of the class $Z(\aleph_1)$. We again distinguish between even ordinals 2ξ and odd ordinals $2\xi + 1$. We now assign to each η a system $\mathfrak{A}^\eta$ of sets A^η by means of the following inductive rule:

The sets A^0 are the sets M ($\mathfrak{A}^0 = \mathfrak{M}$);

The sets A^η are the sums of sequences of sets A^ξ ($\xi < \eta$) if $\eta > 0$ is odd and th
(4.18)

Then the systems of all sets A^η so defined is the smallest Borel system $\mathfrak{B}$ containing $\mathfrak{M}$.

It is clear that all the sets A^η must belong to $\mathfrak{B}$. It only remains to show that these sets do indeed constitute a Borel System. If a sequence of sets A^{ξ_1}, A^{ξ_2}, ... is now given, let ξ be the ordinal number immediately following ξ_1, ξ_2, ... (because of § 15, IV, we have $\xi < \Omega$). One of the numbers ξ, $\xi + 1$ is even and the other odd, and so both the sum and the intersection of the given sequence occur among the sets A^ξ, $A^{\xi+1}$. From this we see that there would have been no point in extending the definition to $\eta = \Omega$ (for then we should have found $\mathfrak{A}^\Omega = \mathfrak{A}^{\Omega+1} = \cdots = \mathfrak{B}$).

The systems $\mathfrak{A}^\eta$ increase with increasing index; that is, $\mathfrak{A}^\xi \subseteq \mathfrak{A}^\eta$ for $\xi < \eta$.[3] From this it follows that the A^η could also have been defined as sums or intersections (depending on whether η is even or odd) of sequences of sets A^ξ for $\xi < \eta$; for a limit ordinal η the A^η are intersections of sequences of sets A^ξ $(\xi < \eta)$.

Thus, the sets (3) begin the induction; the

$$A^1,\ A^2,\ A^3,\ A^4,\ \ldots$$

are identical with the M_σ, $M_{\sigma\delta}$, $M_{\sigma\delta\sigma}$, $M_{\sigma\delta\sigma\delta}$, ...; then we get the A^η as intersections of sequences of preceding sets, the $A^{\xi+1}$ as sums of sequences of sets A^ξ, and so forth. We call the $\mathfrak{A}^\eta$ *Borel classes* and say that a set belongs exactly to the class $\mathfrak{A}^\eta$ if it belongs to that class but not to any class preceding it — that is, if it is an A^η but not an A^ξ (for $\xi < \eta$). (It is generally customary, although for us somewhat inconvenient, to take the classes as disjoint, that is, to call $A^\eta - \sum_{\xi<\eta} A^\xi$ a class.)

A slightly different way of generating $\mathfrak{B}$ is derived from the above by interchanging the roles played by sum and intersection, so that the system $\mathfrak{B}^\eta$ of sets B^η are defined as follows:

The sets B^0 are the sets M $(\mathfrak{B}^0 = \mathfrak{M})$;

The sets B^η are the intersections, if η is odd, and the sums, if $\eta > 0$ is even, of (4.19)

All the sets B^η also form the smallest Borel system $\mathfrak{B}$ containing $\mathfrak{M}$. The $B^{\xi+1}$ are the intersections or sums (depending on whether ξ is even or odd) of sequences of sets B^ξ. The induction this time starts with the sets (4); that is, the sets

$$B^1,\ B^2,\ B^3,\ B^4,\ \ldots$$

are identical with the sets M_δ, $M_{\delta\sigma}$, $M_{\delta\sigma\delta}$, $M_{\delta\sigma\delta\sigma}$, ...

These systems of sets, too, are called Borel classes. It is easy to see that every A^ξ is a $B^{\xi+1}$ and every B^ξ an $A^{\xi+1}$; the $A^1 = M_\sigma$, for example, occur among the $B^2 = M_{\delta\sigma}$, and the $B^1 = M_\delta$ among the $A^2 = M_{\sigma\delta}$.

It may happen, of course, that not all the sets A^ξ or B^ξ are really needed to complete the Borel System; that is, it may happen that some $\mathfrak{A}^\xi$ is, as it stands, the whole Borel system. This depends

[3] For since $A^\xi = A^\xi A^{\xi+1} \cdots = A^\xi + A^{\xi+1} + \cdots$, every A^ξ is also an $A^{\xi+1}$, $A^{\xi+2}$, ...

on the given system $\mathfrak{M}$: if, for instance, $\mathfrak{M}$ itself happens to be a Borel system, then $\mathfrak{A}^0 = \mathfrak{B}$ to begin with, and every extension is superfluous. We shall see later (§ 33, I), however, that it is just in the most important cases that every step gives rise to new sets and all the $\mathfrak{A}^\xi$ are required for the formation of $\mathfrak{B}$.

One additional remark: If for any $\xi \geq 1$ it should happen that $\mathfrak{A}^\xi = \mathfrak{A}^{\xi+1}$, then $\mathfrak{A}^\xi = \mathfrak{B}$. For of the two systems $\mathfrak{A}^\xi$ and $\mathfrak{A}^{\xi+1}$, the one is a σ-system, and the other a δ-system; when they coincide, we have a Borel system.

3. Representation of Borel sets. We have seen (to continue with the second form) that every $B^{\xi+1}$ is an intersection, for even ξ, and a sum for odd ξ, of a sequence of sets B^ξ. If η is a limit ordinal, then

$$B^\eta = C_1 C_2 \cdots \quad (C_n \text{ a } B^{\xi_n} \text{ with } \xi_n < \eta); \tag{4.20}$$

but this form suffers from a certain lack of definiteness in that the ξ_n may depend on the set B^η which is to be represented, and we should like to transform it in such a way as to make the ξ_n a fixed sequence depending only on η (as was the case for $\eta = \xi + 1$, where we were able to choose $\xi_n = \xi = \eta - 1$).

§ 18. Borel Systems

Let us now suppose that a unique set

$$X = \Phi(M_1, M_2, M_3, \ldots) \tag{7}$$

is made to correspond, by some rule or other, to every sequence of sets $M_1, M_2, \ldots$ (say, for example, $X = M_1 M_2 M_3 \cdots$ or $X = M_1 + M_2 + M_3 + \cdots$), so that Φ denotes a single-valued function of the sets M_n. If the sets M_n are chosen from a given system of sets $\mathfrak{M}$ in every possible way, then X ranges over some system of sets $\overline{\mathfrak{M}}$. Then we assert:

Theorem IV.1 (I). *For every $\xi \geq 1$ a function Φ_ξ, depending only on ξ, can be determined in such a way that*

$$X = \Phi_\xi(M_1, M_2, M_3, \ldots) \qquad (M_n \in \mathfrak{M}) \tag{8}$$

represents precisely the Borel sets $\overline{\mathfrak{B}}_\xi$ generated by $\mathfrak{M}$.

The statement that X represents precisely the $\overline{\mathfrak{B}}_\xi$ is of course supposed to mean that X represents all the $\overline{\mathfrak{B}}_\xi$ and no other sets, or that X ranges over the system of sets $\overline{\mathfrak{B}}_\xi$.

The proof is very simple. If we set

$$\Phi_1(M_1, M_2, M_3, \ldots) = M_1 M_2 M_3 \cdots , \tag{9}$$

then $X = \Phi_1$ represents precisely the $\mathfrak{M}_\sigma = \overline{\mathfrak{B}}_1$.

If, further, the function Φ_ξ is already defined, then corresponding to the formation of the $\overline{\mathfrak{B}}_{\xi+1}$ from the $\overline{\mathfrak{B}}_\xi$ we obtain a suitable function $\Phi_{\xi+1}$, in the following way. We divide the set of natural numbers into a countable number of subsets by means, say, of the dyadic scheme of p. 34. For odd ξ we then set

$$\begin{aligned} &\Phi_{\xi+1}(M_1, M_2, M_3, \ldots) \\ &\quad = \Phi_\xi(M_1, M_3, M_5, \ldots) + \Phi_\xi(M_2, M_6, M_{10}, \ldots) \\ &\qquad + \Phi_\xi(M_4, M_{12}, M_{20}, \ldots) + \cdots \end{aligned} \tag{10}$$

and for even ξ

$$\begin{aligned} &\Phi_{\xi+1}(M_1, M_2, M_3, \ldots) \\ &\quad = \Phi_\xi(M_1, M_3, M_5, \ldots) \cdot \Phi_\xi(M_2, M_6, M_{10}, \ldots) \\ &\qquad \cdot \Phi_\xi(M_4, M_{12}, M_{20}, \ldots) \cdots . \end{aligned} \tag{11}$$

For a limit ordinal γ, finally, we set

$$\begin{aligned}\Phi_\gamma&(M_1, M_2, M_3, \ldots)\\ &= \Phi_{\gamma_1}(M_1, M_2, M_3, \ldots) + \Phi_{\gamma_2}(M_1, M_2, M_3, \ldots)\\ &\quad + \Phi_{\gamma_3}(M_1, M_2, M_3, \ldots) + \cdots\end{aligned} \tag{12}$$

in conformity with the representation (6).

The functions Φ_ξ have now been defined by induction, and it is clear that they do the required job, for the separation of the indices on the right-hand side of the last three formulas — for example, in (10) and (11) — permits of each Φ_ξ being made into a pre-assigned $\overline{\mathfrak{B}}_\xi$ independently of the rest, so that $\Phi_{\xi+1}$ does in fact represent all the sets $\overline{\mathfrak{B}}_{\xi+1}$; the same is true of (12). For instance

$$\Phi_2(M_1, M_2, M_3, \ldots) = (M_1 M_3 M_5 \cdots) + (M_2 M_6 M_{10} \cdots) + (M_4 M_{12} M_{20} \cdots) + \cdots$$

gives precisely the sets $\mathfrak{M}_{\sigma\delta} = \overline{\mathfrak{B}}_2$ for $M_n \in \mathfrak{M}$.

Now the functions, inductively defined in (9) through (12), that occur here are all of a special form. They are all sums of intersections of subsequences of the sequence of sets $M_1, M_2, \ldots$.[1] These special functions Φ are thus formed as follows: We let the intersection

$$M_\nu = M_{n_1} M_{n_2} M_{n_3} \cdots$$

correspond to the sequence of increasing natural numbers

$$\nu = (n_1, n_2, n_3, \ldots);$$

then we let N be a set of such sequences and let

$$X = \Phi_N(M) = \sum_{\nu \in N} M_\nu = \sum_{\nu \in N} M_{n_1} M_{n_2} M_{n_3} \cdots = \Phi(M_1, M_2, M_3, \ldots), \tag{13}$$

where the function symbol Φ and the set N correspond to one another. We call these frequently occurring functions $\mathfrak{S}\mathfrak{d}$-functions. If, as before, we now denote the formation of sums and intersections of countably many sets by σ and δ and of arbitrarily many sets by s and d, then in (13) every summand M_ν is some $\mathfrak{M}_\delta$, the sum itself is an $\mathfrak{M}_{s\delta}$, (in case N is countable, an $\mathfrak{M}_{\sigma\delta}$; in case N contains only a single element, an $\mathfrak{M}_\delta$). By interchanging the roles of sum and intersection we get, analogously, the $\mathfrak{S}\mathfrak{d}$-functions, the intersections of sums of subsequences of a sequence of sets; the complements of the sets (13) are an example of such functions.

Since we have made the claim that the functions Φ_ξ are $\mathfrak{S}\mathfrak{d}$-functions as in (13), we have to prove the following: For every $\xi \geq 1$ there exists a set N_ξ depending only on ξ such that

$$\Phi_\xi(M_1, M_2, M_3, \ldots) = \sum_{\nu \in N_\xi} M_\nu. \tag{14}$$

For example, N_1 consists of the single sequence $(1, 2, 3, \ldots)$; N_2 of the sequences $(1, 3, 5, \ldots)$, $(2, 6, 10, \ldots)$, $(4, 12, 20, \ldots)$, $\ldots$; and so forth. Let us now assume that (14) is true for some particular ξ; then

$$\Phi_\xi(M_1, M_3, M_5, \ldots) = \sum_{\alpha \in A_\xi} M_\alpha, \quad \Phi_\xi(M_2, M_6, M_{10}, \ldots) = \sum_{\beta \in B_\xi} M_\beta, \quad \Phi_\xi(M_4, M$$

[1] The sets $\overline{\mathfrak{B}}_0$ could have been included in Theorem I using $\Phi_0(M_1, M_2, M_3, \ldots) = M_1$, but what follows below would not apply to them.

where a sequence $\alpha = (2n_1 - 1, 2n_2 - 1, 2n_3 - 1, \ldots)$ consisting entirely of odd numbers corresponds to each $\nu = (n_1, n_2, n_3, \ldots)$, and thus a fixed set A_ξ to the set N_ξ; B_ξ, Γ_ξ, ... are to be understood analogously.

If, then, (10) is valid, we have

$$\Phi_{\xi+1}(M_1, M_2, M_3, \ldots) = \sum_{\nu \in N_{\xi+1}} M_\nu, \quad N_{\xi+1} = A_\xi + B_\xi + \Gamma_\xi + \cdots ,$$

and the set N_γ is determined analogously in the case of formula (12). In the formation of the intersections (11) there appears first, by the distributive law, the sum[2]

$$\sum M_{n_1} M_{n_2} M_{n_3} \cdots ,$$

and the summation extends over $\alpha \in A_\xi$, $\beta \in B_\xi$, $\gamma \in \Gamma_\xi$, ...; that is, the sequence $(\alpha, \beta, \gamma, \ldots)$ of sequences ranges over the set product $(A_\xi, B_\xi, \Gamma_\xi, \ldots)$. Any sequence of sequences determines a single sequence ν simply by the uniting of the disjoint sequences $\alpha, \beta, \gamma, \ldots$, and the product determines in this sense a set $N_{\xi+1}$ of sequences ν, whence it then follows that

$$\Phi_{\xi+1}(M_1, M_2, M_3, \ldots) = \sum_{\nu \in N_{\xi+1}} M_\nu .$$

We have thus proved the following, more precise, version of Theorem I:

Theorem IV.2 (II). *For every $\xi \geq 1$, a set N_ξ depending only on ξ, whose elements $\nu = (n_1, n_2, n_3, \ldots)$ are increasing sequences of natural numbers, can be found for which the $\mathfrak{S}\mathfrak{d}$-function*

$$X = \sum_{\nu \in N_\xi} M_\nu = \sum_{\nu \in N_\xi} M_{n_1} M_{n_2} M_{n_3} \cdots = \Phi_\xi(M_1, M_2, M_3, \ldots) \qquad (M_n \in \mathfrak{M}) \tag{15}$$

represents precisely the Borel sets $\overline{\mathfrak{B}}_\xi$ generated by $\mathfrak{M}$.

Of course, the Borel sets $\overline{\mathfrak{B}}^\xi$ (p. 99) permit of analogous treatment; the Φ_ξ would then be intersections of sums — rather than sums of intersections — of subsequences of the sequence M_n, that is, $\mathfrak{S}\mathfrak{d}$-functions.

[2]The set of summands is already uncountable in the case of the $\overline{\mathfrak{B}}^3$.

§ 19. Suslin Sets

Theorem II of the preceding section immediately suggests the question whether there is not perhaps a fixed $\mathfrak{S}\mathfrak{d}$-function or a fixed set N of sequences of increasing natural numbers $\nu = (n_1, n_2, \ldots)$ such that the set

$$X = \sum_{\nu \in N} M_\nu = \sum_{\nu \in N} M_{n_1} M_{n_2} M_{n_3} \cdots = \Phi(M_1, M_2, M_3, \ldots) \qquad (M_n \in \mathfrak{M}) \tag{1}$$

ranges precisely over all the Borel sets generated by $\mathfrak{M}$. This question has to be answered in the negative. If, however, we are willing to strike out the word "precisely," an affirmative answer becomes possible: there is a representation (1) that produces all the Borel sets, but in general it gives other sets as well. We shall return to this point toward the end of the section and shall first discuss the sets in question in terms of a different symbolism in which, instead of the sets M_n, there appear sets with multiple indices — indices which, however, vary independently of each other.

This time we assign sets M_{n_1}, $M_{n_1 n_2}$, $M_{n_1 n_2 n_3}$, ..., not to the natural numbers, but to the finite complexes (n_1), (n_1, n_2), (n_1, n_2, n_3), ... of natural numbers (these, too, are a countable collection); except for external form, this is nothing but a sequence of sets. We indicate some of these sets (omitting the dots that denote additional sets, in order not to clutter up the scheme):

$$\begin{array}{ccccc} M_1 & M_2 & M_3 & M_4 & \cdots \\ M_{11} & M_{12} & M_{13} & M_{14} & \cdots \\ M_{21} & M_{22} & M_{23} & M_{24} & \cdots \\ M_{31} & M_{32} & M_{33} & M_{34} & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \end{array} \tag{2}$$

After this, we assign to each sequence

$$\nu = (n_1, n_2, n_3, \ldots) \tag{3}$$

of natural numbers, which this time do not need to be pairwise distinct, the intersection

$$M_\nu = M_{n_1} M_{n_1 n_2} M_{n_1 n_2 n_3} \cdots \tag{4}$$

constructed from the sets which correspond to the finite sections of the sequence ν; we then form the sum

$$X = \sum_{\nu} M_\nu \tag{5}$$

taken over all the sequences ν.

This, then, is a function $\Phi(M_1, M_2, M_3, \ldots)$ of the generating sets (2). If these sets are taken in all possible ways from a system of sets $\mathfrak{M}$, then X ranges over a system of sets $\mathfrak{A}$, and these sets X are called the *Suslin sets* generated by $\mathfrak{M}$ (or, generated by the sets $M \in \mathfrak{M}$). The summands M_ν are then indeed sets $\mathfrak{M}_\delta$, but there are uncountably many, just as was the case for the Borel sets $\overline{\mathfrak{B}}_\xi$ for $\xi \geq 3$. In addition, it should be observed that X does not coincide with the intersection

$$\sum_{n_1} M_{n_1} \cdot \sum_{n_2} M_{n_1 n_2} \cdot \sum_{n_3} M_{n_1 n_2 n_3} \cdots$$

(which is an $\mathfrak{M}_{\sigma\delta\sigma}$), for this latter, developed by means of the distributive law, is the same as

$$\sum_{n_1} M_{n_1} M_{n_2} M_{n_3} \cdots ,$$

where all the indices range over all the natural numbers independently of each other, while in our case we ought to have $n_1 = n_2 = n_3 = \cdots$ etc.

Let us call the Suslin sets generated by $\mathfrak{M}$ the sets $\mathfrak{M}_\sigma$ and let us call the system formed by them $\mathfrak{M}_\S$.

Every $\mathfrak{M}_\sigma$ and $\mathfrak{M}_\delta$ is an $\mathfrak{M}_\S$; thus the Suslin formula embraces the formation of sums and intersections of sequences. For if a sequence of sets $M^1, M^2, \ldots$ is given and if we set $M_{n_1 n_2 \cdots n_k} = M^{n_1}$, then $M_\nu = M^{n_1}$ and $X = \sum M^{n_1}$; if, on the other hand, we set $M_{n_1 n_2 \cdots n_k} = M^k$, then every M_ν and consequently X itself equals $\prod M^k$.

The truly fundamental property of the Suslin sets, however, is given by the following theorem:

Theorem IV.3 (I). *The Suslin sets generated by Suslin sets $\mathfrak{M}_\S$ are themselves sets $\mathfrak{M}_\S$.*

Or in brief: Every set $\mathfrak{M}_{\S\S}$ is merely a set $\mathfrak{M}_\S$ (just as every $\mathfrak{M}_{\sigma\sigma}$ is an $\mathfrak{M}_\sigma$). The Suslin process is so comprehensive that its iteration produces nothing new.

The proof looks difficult only because there are so many indices, but it is really quite simple. Let

$$P = \sum_\nu M_\nu, \quad M_\nu = M^{(1)}_{n_1} M^{(2)}_{n_1 n_2} M^{(3)}_{n_1 n_2 n_3} \cdots$$

be a Suslin set whose generating sets are themselves also Suslin sets:

$$M_{n_1}^{(1)} = \sum_{\alpha} M_{n_1,a_1,a_2,a_3,\cdots}^{(1)},$$
$$M_{n_1n_2}^{(2)} = \sum_{\beta} M_{n_1n_2,b_1,b_2,b_3,\cdots}^{(2)},$$
$$M_{n_1n_2n_3}^{(3)} = \sum_{\gamma} M_{n_1n_2n_3,c_1,c_2,c_3,\cdots}^{(3)},$$
$$\cdots\cdots\cdots$$

where $\nu = (n_1, n_2, n_3, \ldots)$, $\alpha = (a_1, a_2, a_3, \ldots)$, $\beta = (b_1, b_2, b_3, \ldots)$, $\ldots$ range independently of each other through all the sequences of natural numbers.

It is to be shown that P can be formed as a Suslin set from the sets M themselves. By the distributive law, P is the sum of all intersections

$$M_{n_1,a_1,a_2,\ldots}^{(1)} M_{n_1n_2,b_1,b_2,\ldots}^{(2)} M_{n_1n_2n_3,c_1,c_2,\ldots}^{(3)} \cdots \tag{6}$$

where the summation extends over all the sequences $\nu, \alpha, \beta, \gamma, \ldots$ or all the natural numbers $n_1, a_1, b_1, c_1, \ldots$. Again using, say, the dyadic scheme, we transform the sequence of sequences $\nu, \alpha, \beta, \gamma, \ldots$ into a single sequence $\mu = (m_1, m_2, m_3, \ldots)$; that is, we do so in such a way that the matrices

$$\begin{array}{cccc|cccc}
n_1 & n_2 & n_3 & \cdots & m_1 & m_2 & m_3 & \cdots \\
a_1 & a_2 & a_3 & \cdots & m_2 & m_4 & m_6 & \cdots \\
b_1 & b_2 & b_3 & \cdots & m_3 & m_7 & m_{11} & \cdots \\
c_1 & c_2 & c_3 & \cdots & m_5 & m_9 & m_{13} & \cdots \\
\cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots
\end{array}$$

become equal to one another. In this way (6) goes over into

$$M_{m_1}^{(1)} M_{m_1m_2}^{(2)} M_{m_1m_2m_3}^{(3)} M_{m_1m_2m_3m_4}^{(4)} M_{m_1m_2m_3m_4m_5}^{(5)} \cdots = M(1)M(2)M(3)\cdots M(k) \tag{7}$$

where $M(k)$ denotes a set depending only on $m_1, m_2, \ldots, m_{2k}$ (though not, in general, on all these numbers) and where k runs through all the natural numbers once; for it is easy to see that the superscripts $1, n_2, \ldots, n_k, \ldots, n_{2k-1}$ on which a set in (7) depends have smaller indices than the subscripts, which begin with n_{2k}, and that thus the largest n_j on which the set depends is of even index. If, finally, we set up a one-to-one correspondence between the pairs of numbers (k, n) and the natural numbers l, then the set depending on $m_1, m_2, \ldots, m_{2k}$ and the superscript n becomes a set M_l depending on $m_1, m_2, \ldots, m_{2l}$, and (7) becomes a Suslin set formed from the sets M_l.

This proves I. The sets $\mathfrak{A} = \mathfrak{M}_{\S}$ generate no new Suslin sets; every $\mathfrak{M}_{\S\S}$ is an $\mathfrak{M}_{\S}$. Thus, in particular, every $\mathfrak{M}_\sigma$ and $\mathfrak{M}_\delta$ is an $\mathfrak{M}_{\S}$, and the sets $\mathfrak{M}_{\S}$ form a Borel system $\overline{\mathfrak{B}}$ containing $\mathfrak{M}$; the smallest Borel system $\overline{\mathfrak{B}}$ is therefore contained in $\mathfrak{M}_{\S}$: all the Borel sets generated by $\mathfrak{M}$ are also Suslin sets. The converse is not true in general, as we shall see later (§ 33, I).

Finally, we can return easily from the Suslin formula (5), (4) to a representation of the form (1) and in so doing answer the question raised at the beginning of this section. All we have to do is to produce a one-to-one correspondence between the natural numbers l and the finite complexes of natural numbers $(n_1, n_2, \ldots, n_k)$, for instance the one arising from the dyadic representation

$$l = 2^{n_1-1} + 2^{n_1+n_2-1} + \cdots + 2^{n_1+n_2+\cdots+n_k-1}. \tag{8}$$

Then we set

$$M^l = M_{n_1, n_2, \ldots, n_k}.$$

To every sequence $\nu = (n_1, n_2, n_3, \ldots)$ of arbitrary natural numbers there corresponds, then, a sequence $\lambda = (l_1, l_2, l_3, \ldots)$ of increasing natural numbers of the form

$$l_1 = 2^{n_1-1}, \quad l_2 = 2^{n_1-1} + 2^{n_1+n_2-1}, \quad l_3 = 2^{n_1-1} + 2^{n_1+n_2-1} + 2^{n_1+n_2+n_3-1}, \quad \ldots$$

(that is, the numbers $l_1, l_2 - 1, l_3 - l_2, \ldots$ are a subsequence of $1, 2, 4, 8, \ldots$). If Λ is the set of all these sequences λ, then (5) goes over into

$$X = \sum_{\lambda \in \Lambda} M^{l_1} M^{l_2} M^{l_3} \cdots .$$

If we return to the notation of (1), this proves the theorem:

Theorem IV.4 (II). *There exists a fixed $\mathfrak{S}\mathfrak{d}$-function*

$$X = \Phi(M_1, M_2, M_3, \ldots)$$

which, for an arbitrary system of sets $\mathfrak{M}$, produces all the Borel sets $\overline{\mathfrak{B}}$ generated by $\mathfrak{M}$; that is, every Borel set is representable in the form (1), where, however, the function Φ also gives (for every $\mathfrak{M}$) non-Borelian sets X of the system $\mathfrak{M}_{\S}$.

Chapter V

Point Sets

§ 20. Distance

We have until now dealt in part with pure sets, in part with sets in which there are relations among the elements (ordered sets). It is to the second of these classes that the sets we are about to study belong: sets in which any two elements have a distance from each other.

1. Metric spaces

Let E be a set, whose elements we now refer to as *points*. To each pair of points (x, y) we assign a real number xy, the distance between the two points,[1] that is, a real function $xy = f(x, y)$ defined in (E, E). We require of this function that the following *distance axioms*, or postulates, be satisfied:

(α_0) $xx = 0$;

(β) $xy = yx > 0$ for $x \neq y$;

(γ) $xy + yz \geq xz$.

The axiom (γ), the most important among them, is called the *triangle inequality* or *triangle axiom* (the sum of two sides of a triangle is at least as great as the third side). The set E is called a *metric set*, a *point set*, or a *metric space*. In the case of the last of these terms we have in mind the relation of E and its subsets and points, in the first two, of the inclusion of E in some containing space.

[1] Also denoted by $|x - y|$ or $d(x, y)$.

The example closest to hand is the set of real numbers E_1, where distance is defined as the absolute value $|x - y|$ of the difference of the two numbers. Next comes n-dimensional Euclidean space, E_n; its elements are the complexes

$$x = (x_1, x_2, \dots, x_n)$$

of n real numbers; equality has the usual meaning ($x = y$ means $x_1 = y_1, \dots, x_n = y_n$), and distance is defined by

$$xy = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2},$$

where we take the positive root. The postulates (α_0) and (β) are satisfied, and we shall return to the proof of (γ) later on.

If two spaces can be mapped on each other one-to-one and with preservation of distance (that is, if the points x correspond one-to-one to the points ξ in such a way that $xy = \xi\eta$), then they are said to be *isometric*. For example, E_1 is isometric with the set of points $(x, 0)$ of E_2. Isometry, fundamentally nothing but the congruence of elementary geometry, is a concept for metric sets analogous to equivalence for pure sets and to similarity for ordered sets (another, more important, analogue is that of homeomorphism, § 38); it is unnecessary, however, to introduce names corresponding to cardinal number and ordinal type. Two isometric spaces each thought of in relation to its subsets and its points may be regarded simply as identical (two isometric sets in a containing space, however, may not).

A space isometric with E_n is called an n-dimensional Euclidean space; $x_1, \dots, x_n$ are called the rectangular Cartesian coordinates of the point that corresponds to the complex x. The isometric mapping can be done in infinitely many ways (coordinate transformation, orthogonal substitution). Ordinary space, idealized from our everyday experience and observation, is a three-dimensional Euclidean space, whose planes and lines are two-dimensional and one-dimensional Euclidean spaces, respectively. It is here that the familiar geometrical configurations arise: spheres (including their interiors), cones, cylinders, and so on.

2. Linear spaces

Let us define, in a metric space, the concepts of straight line, plane, and higher-dimensional linear space. We form finite linear combinations

$$\alpha x + \beta y + \gamma z + \cdots$$

of points, with real coefficients, under the assumption that

$$\alpha + \beta + \gamma + \cdots = 1,$$

and let

$$0x = 0, \quad 1x = x, \quad (-1)x = -x,$$

so that it is clear what we mean by such symbols as $-x$, $x - y$, $\alpha x + \beta y$, and $\alpha x + \beta y + \gamma z$.

We say that three points x, y, z lie on a straight line, or that they are *collinear*, if there exist three numbers α, β, γ, not all zero, such that

$$\alpha x + \beta y + \gamma z = 0, \qquad \alpha + \beta + \gamma = 0.$$

Here γ cannot be zero except when $x = y$ (for then every point z lies on a straight line with x and y); for $x \neq y$, we can choose $\gamma = -1$, since only the ratios $\alpha : \beta : \gamma$ matter, and thus from

$$z = \alpha x + \beta y, \qquad \alpha + \beta = 1 \tag{1}$$

we get all the points z collinear with x and y, the set of such points being called the *line* determined by x and y; z depends on a real variable α or β. If we impose on (1) the restriction that $\alpha \geq 0$, $\beta \geq 0$, then we get the (closed) *segment* or *interval* $[x, y]$ determined by x and y.

This process can be continued. Let us say that $k + 1$ points $x_0, x_1, \ldots, x_k$ are *linearly dependent* if there exists a relation

$$\alpha_0 x_0 + \alpha_1 x_1 + \cdots + \alpha_k x_k = 0, \quad \alpha_0 + \cdots + \alpha_k = 0,$$

where $\alpha_0, \alpha_1, \ldots, \alpha_k$ are real numbers that do not all vanish simultaneously; otherwise, we say that the points are *linearly independent.* If $x_0, x_1, \ldots, x_k$ are linearly dependent but $x_1, \ldots, x_k$ linearly independent, then α_0 cannot be 0; and, setting $\alpha_0 = -1$, we obtain all the points x_0 linearly dependent on $x_1, \ldots, x_k$ in the form

$$x_0 = \alpha_1 x_1 + \cdots + \alpha_k x_k, \qquad \alpha_1 + \cdots + \alpha_k = 1;$$

these points are determined by $k - 1$ real parameters and, in the usual terminology, they form the $(k - 1)$-dimensional space R_{k-1} determined by $x_1, \ldots, x_k$, and

$$x_0 x_1 x_2 \cdots x_k$$

is called a k-dimensional simplex $(k = 1, 2, 3, \ldots)$.

A set containing, along with any two points x, y, also the entire segment $[x, y]$ is called *convex*; a set containing, along with any $k+1$ linearly independent points, also the simplex determined by them is called S_{k-1}-convex. Obviously, a linear set is S_0-convex, S_1- convex, S_2-convex, $\ldots$, S_{k-1}-convex; a linear space is also obtained by taking all the points whose coordinates satisfy a system of linear equations. Clearly, the intersection of any number of linear or convex sets is itself linear or convex. If E is a linear or convex space and $A \subseteq E$, then the intersection K of all the convex sets containing A is the smallest convex set containing A and is called the *convex hull* of A. Whenever K contains any linearly independent points of A, it contains the whole simplex determined by them and is simply the sum of all these simplexes.

If a is a particular point, then

$$\xi = x + a, \qquad \eta = y + a \tag{2}$$

defines a one-to-one mapping of the points x on each other; this relation is called a *translation*, or a *displacement*. Under such a correspondence, every linear space goes over into another and, in particular, can be made into a (homogeneous, linear) space containing the origin, and we shall now assume this to have been done. Such a space E, if it contains a point x, also contains the points $\alpha x = \alpha x + (1 - \alpha) \cdot 0$, and if it contains the pair of points x, y, also contains all the linear combinations $\alpha x + \beta y$ (and not only those for which $\alpha + \beta = 1$). The translations (2) transform E into itself, provided that a is a point of E. We shall now make E into a metric space in such a way as to make the translations isometric mappings, that is, so that the points x, y have the same distance as $x + a$, $y + a$ and, in particular, the same as $x - y$, 0 or 0, $y - x$. All distances

$$xy = |x - y|$$

are thus reduced to distances of the form $|x| = x0$, and this function $|x|$, the *norm* of the point x, satisfies the *norm axioms*:

(δ_0) $|x| \geq 0$;

(ϵ) $|x| = 0$ only for $x = 0$;

(ζ_0) $|x - y| \leq |x| + |y|$.

The last of these, which we call the *addition axiom*, follows from the triangle inequality by replacement of x, y, z by $x, 0, -y$. Conversely, if $|x|$ satisfies these norm axioms, then

$$xy = |x - y| \tag{3}$$

satisfies the distance axioms; (γ) follows from (ζ_0) because

$$|x - z| = |(x - y) + (y - z)| \leq |x - y| + |y - z|.$$

The norms occurring in the next subsection will satisfy, in addition, the condition

$$|\alpha x| = |\alpha| \cdot |x|$$

(α a real number); that is, they will be positive homogeneous functions of the first degree in the coordinates. In this case, the straight lines of our space will be isometric with E_1 or with the Euclidean straight lines no matter how the norm is defined. This is so because for two points

$$z = \alpha x + \beta y \quad (\alpha + \beta = 1)$$

of the line (1) we have

$$z - z' = (\beta - \beta')(y - x), \quad |z - z'| = |\beta - \beta'| \cdot |y - x|,$$

so that the points z are in isometric relation to the numbers $c\beta$ ($c = |y - x|$ is constant).

3. Examples of linear spaces

The Euclidean space E_n is formed by the assignment to the point

$$x = (x_1, x_2, \ldots, x_n) \tag{4}$$

of the norm

$$|x| = (x_1^2 + x_2^2 + \cdots + x_n^2)^{\frac{1}{2}}. \tag{5}$$

The proof of the addition axiom is obtained most quickly by a well-known argument: since the quadratic form in the real variables u, v

$$\sum_i (x_i u + y_i v)^2 = au^2 + 2buv + cv^2$$

with

$$a = |x|^2, \quad b = xy = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n, \quad c = |y|^2$$

is non-negative, its determinant is ≤ 0,

$$b^2 \le ac,$$

$$\begin{gathered} b \le \sqrt{ac} \le \frac{a+c}{2}, \\ a + 2b + c \le (a+c) \cdot 2, \\ (a + 2b + c)^2 \le (a+c)^2 \cdot 4, \end{gathered}$$

and this is the inequality that was to be proved.

If n is allowed to increase without limit, then the addition inequality states first that when $x_1^2 + x_2^2 + \cdots$ and $y_1^2 + y_2^2 + \cdots$ converge then $(x_1 + y_1)^2 + (x_2 + y_2)^2 + \cdots$ also converges. Therefore, if we now consider sequences of real numbers

$$x = (x_1, x_2, x_3, \ldots), \tag{6}$$

that is, complexes with the index sets $M = \{1, 2, \ldots\}$, and if H is the set of those x for which the sum of the squares converges, then H is a linear space (when x and y belong to it, then so do $x + y$ and $\alpha x + \beta y$), and this becomes a metric space if the norm is defined by

$$|x| = (x_1^2 + x_2^2 + x_3^2 + \cdots)^{\frac{1}{2}}$$

with a corresponding definition (3) of distance. It is called *Hilbert space*, and it is in a certain sense an $\aleph_0$-dimensional Euclidean space. Incidentally, another space can be inserted between the Euclidean spaces E_n and Hilbert space, one formed by the sequences of numbers x in which only finitely many numbers $x_1, x_2, \ldots$ differ from zero; this space can be denoted by $E_1 + E_2 + \cdots$, provided the set of sequences

$$(x_1, \ldots, x_n, 0, 0, \ldots)$$

isometric with E_n is identified with E_n.

This process can be extended to the use of any set M whatsoever as an index set for the complexes, and if, moreover, we do not distinguish between isometric spaces, the result depends only on the cardinality of M. We keep only those $x = (x_m, \ldots)$ in which only a finite number of x_m or an at most countable number — in the latter case, with sum of squares convergent — are different from zero, and then define the norm $|x|$ as above. These two spaces are denoted by $E(M)$ and $H(M)$ and are called the Euclidean and Hilbert spaces of dimension $\overline{\overline{M}} = \aleph_\alpha$, or also of dimension α; Hilbert space is H_0.

For these spaces we have

$$E_1 < E_2 < \cdots < E_n < \cdots < H_0, \tag{7}$$

and every E_n and $H(M)$ contains all the spaces of lower dimension E_m and $H(N)$ ($\overline{\overline{N}} < \overline{\overline{M}}$). All these spaces are distinct.

If we form, for a real number $p \geq 1$, the norm

$$|x| = (|x_1|^p + |x_2|^p + \cdots + |x_n|^p)^{\frac{1}{p}}, \tag{8}$$

then the addition inequality for this norm runs:

$$|x + y| \leq |x| + |y|. \tag{9}$$

This is equivalent to the proposition: if $a_k \geq 0$, $b_k \geq 0$, then

$$\left(\sum (a_k + b_k)^p\right)^{\frac{1}{p}} \leq \left(\sum a_k^p\right)^{\frac{1}{p}} + \left(\sum b_k^p\right)^{\frac{1}{p}}. \tag{10}$$

This follows easily from the theorem:[2] if $a_k > 0$, $b_k > 0$, $p > 1$, then

$$\sum a_k b_k \leq \left(\sum a_k^p\right)^{\frac{1}{p}} \cdot \left(\sum b_k^{\frac{p}{p-1}}\right)^{\frac{p-1}{p}}. \tag{11}$$

In fact, if we set

$$\alpha = \left(\sum a_k^p\right)^{\frac{1}{p}}, \quad \beta = \left(\sum b_k^{\frac{p}{p-1}}\right)^{\frac{p-1}{p}},$$

then the (obvious) inequality

$$\alpha\beta \leq \frac{\alpha^p}{p} + \frac{\beta^{\frac{p}{p-1}}}{\frac{p}{p-1}}$$

gives, for $\alpha = a_k/A$, $\beta = b_k/B$ $(A, B > 0)$,

$$\sum \frac{a_k}{A} \cdot \frac{b_k}{B} \leq \frac{1}{p} \sum \frac{a_k^p}{A^p} + \frac{p-1}{p} \sum \frac{b_k^{\frac{p}{p-1}}}{B^{\frac{p}{p-1}}} = \frac{1}{p} + \frac{p-1}{p} = 1,$$

and from this (11). From (11) we get (10) by setting

$$a_k' = a_k, \quad b_k' = (a_k + b_k)^{p-1},$$

[2] The theorem is valid for any finite or countably infinite number of summands, and with a corresponding generalization for arbitrary (finite or transfinite) sums of non-negative numbers; but the proof is the same.

for then

$$\sum (a_k + b_k)^p = \sum a_k (a_k + b_k)^{p-1} + \sum b_k (a_k + b_k)^{p-1},$$

and by (11)

$$\sum a_k (a_k + b_k)^{p-1} \leq \left(\sum a_k^p\right)^{\frac{1}{p}} \cdot \left(\sum (a_k + b_k)^p\right)^{\frac{p-1}{p}},$$

and similarly for the b_k, so that division by

$$\left(\sum (a_k + b_k)^p\right)^{\frac{p-1}{p}}$$

yields (10).

In the finite-dimensional case, the inequalities just proved give the spaces $E_n^{(p)}$, in which the norm is defined by

$$|x| = \left(\sum_{k=1}^{n} |x_k|^p \right)^{\frac{1}{p}} . \tag{8}$$

For different p, these spaces are distinct; they are all isometric only in the case $n = 1$, in which case they all coincide with E_1. For $n = 2$, the spaces $E_2^{(p)}$ are mapped on one another in an obvious way (the points with coordinates (x_1, x_2) corresponding to the points $(\pm|x_1|, \pm|x_2|)$ with suitable signs), but this mapping is isometric only in the trivial case $p = 2$.

The unit circle $|x| = 1$ of the space $E_2^{(p)}$ is, for $p = 2$, the ordinary Euclidean circle; for $p = 1$, the square with vertices $(1, 0), (0, 1), (-1, 0), (0, -1)$; and for $p = \infty$, the square with vertices $(1, 1), (-1, 1), (-1, -1), (1, -1)$. The curve lies in the intermediate regions for the remaining values of p (Figure 1).

The number p, thus far taken to be > 1, may also be taken $= 1$, thus giving a norm defined by

$$|x| = |x_1| + |x_2| + \cdots + |x_n|. \tag{11}$$

At the other extreme, it can be said that the definition

$$|x| = \max(|x_1|, |x_2|, \ldots, |x_n|) \tag{12}$$

for the norm corresponds to the limiting case $p = \infty$; if this largest of the absolute values $|x_i|$ is denoted by ξ, then it follows from (10) that $\xi \leq |x| \leq n^{1/p} \cdot \xi$, so that if $p \to \infty$ then $|x| \to \xi$. The addition inequality for these last two norms, incidentally, is trivial.

If we interpret x_1, x_2 as rectangular coordinates in the Euclidean plane, so that $E_2^{(p)}$ is mapped on it one-to-one — but isometrically only in the case $p = 2$ — then it is not without interest to visualize the "calibration curve" $|x_1|^p + |x_2|^p = 1$, the image of the "unit circle" $|x| = 1$. For $p = 2$ it is the Euclidean unit circle; for $p = 1$, it is an inscribed square with corners $(1, 0)$ and $(0, 1)$; and for $p = \infty$ it is the circumscribed square with the corners $(\pm 1, \pm 1)$. For the remaining p, the curve stays in the corresponding intermediate regions.

For $n = 3$, we note that the space $E_3^{(1)}$ has as its unit sphere the octahedron with vertices at the unit points on the coordinate axes, while the space $E_3^{(\infty)}$ has the cube with vertices $(\pm 1, \pm 1, \pm 1)$ as its unit sphere.

We mention further examples of metric spaces.

The complex plane or Gaussian plane, in which the distance between two points is defined as the absolute value of their difference (distance in the sense of elementary geometry), is a two-dimensional Euclidean space E_2.

Projective spaces. If we take the rays (half-lines) emanating from a point O of E_n as the elements of a set, then two such rays have a natural distance, namely the angle (between 0 and π) that they form with each other. The first two distance axioms are satisfied; the third holds because, if x, y, z are three rays and if α, β, γ are the angles that they form with one another, then, by a well-known theorem of spherical geometry, the three angles of a spherical triangle satisfy $\gamma \leq \alpha + \beta$. The elements of this metric space are called the *points* of an $(n-1)$-dimensional projective space; for $n = 2$ and $n = 3$, we get the projective line and the projective plane. The metric introduced here makes the projective space into a bounded space in which the distance of any two points is $\leq \pi$.

Riemann surfaces. As is known, a function-theoretic Riemann surface F can be mapped conformally on a (generally branched) covering surface of the sphere. In the small, the mapping is almost isometric; so we can assign a distance to any two points of F (on the same sheet or on different sheets) by defining distance as the lower limit of the lengths of the curves joining the points on the surface. This makes the Riemann surface into a metric space.

Spaces of functions. As elements we take the real functions $x(t)$ continuous in an interval $[a, b]$; distance is defined by

$$|x| = \max_{a \leq t \leq b} |x(t)|. \tag{13}$$

The functions form a linear space; the first two norm axioms are obvious, and the third follows from

$$\max |x(t) + y(t)| \leq \max |x(t)| + \max |y(t)|.$$

If, instead of continuous functions, we take all bounded functions, we get a linear space as well; we get another if distance is defined by

$$|x - y| = \sup_t |x(t) - y(t)|. \tag{14}$$

If we take the integrable functions and define distance by

$$|x - y| = \int_a^b |x(t) - y(t)|\, dt, \tag{15}$$

then the first two distance axioms hold, but not the triangle axiom in its strong form ($xy = 0$ only for $x = y$): two different continuous functions can have the distance zero from each other. In order to have the axioms (δ_0) and (ϵ) hold in this case too, we have to modify the definition of equality: we call two functions "equal" if they coincide almost everywhere (that is, everywhere except on a set of measure zero).

The spaces of infinitely many dimensions $E(M)$, $H(M)$, the function spaces, are no longer subject to the restriction of linear dependence imposed on the finite-dimensional spaces; they are linear spaces in the wider sense, and in them the theorems about linear dependence, dimension, etc., no longer hold. (In the function space, e.g., the powers $1, t, t^2, \dots$ are linearly independent; in H_0, the unit vectors $e_1 = (1, 0, 0, \dots)$, $e_2 = (0, 1, 0, \dots)$, $\dots$)

It can happen that the distance xy is defined not for all pairs of points x, y, but only for the pairs of a subset of (E, E); we then speak of a *partial metric* and a *partially metric space.* For example, the complex numbers of absolute value 1 form, under multiplication, a group in which we can define the distance between two elements ξ, η as the absolute value of their quotient, but only if ξ differs from η; or, the distance of two points of a Riemann surface that belong to different sheets is not defined at all.

We shall continue to develop the theory for metric spaces in the strict sense, but most of what follows carries over to partial metrics with obvious modifications.

§ 21. Convergence

1. Fundamental sequences

Let E be a metric space and $x_1, x_2, x_3, \ldots$ a sequence of its points. We call the sequence a *fundamental sequence* (or a *Cauchy sequence*) if

$$x_m x_n \to 0 \quad \text{as} \quad m, n \to \infty; \tag{1}$$

that is, if for every $\epsilon > 0$ there exists a p such that $x_m x_n < \epsilon$ for all $m > p$, $n > p$. We call the sequence *convergent*, converging to x, and write

$$x_n \to x \quad \text{or} \quad \lim x_n = x,$$

if

$$x_n x \to 0 \quad \text{as} \quad n \to \infty; \tag{2}$$

that is, if for every $\epsilon > 0$ there exists a q such that $x_n x < \epsilon$ for all $n > q$.

A convergent sequence is always a fundamental sequence; for

$$x_m x_n \leq x_m x + x x_n.$$

The converse does not always hold; if it does, that is, if every fundamental sequence is convergent, then the space is called *complete.*

The real line E_1 is complete. For let $x_1, x_2, \ldots$ be a fundamental sequence of real numbers; then the sequence is bounded, that is, there exists an interval containing all the x_n. Since every bounded infinite set of real numbers has at least one point of accumulation, there exists at least one point of accumulation x of the sequence. Since for sufficiently large m, n all the differences $x_m - x_n$ have absolute value $< \epsilon$, all the x_n with sufficiently large n lie in an interval of length $< \epsilon$; since this interval contains x, it follows that $|x_n - x| \leq \epsilon$, that is, $x_n \to x$. Thus E_1 is complete.

Theorem V.1 (I)**.** *If $x_n \to x$ and $x_n \to y$, then $x = y$.*

For $xy \leq x x_n + x_n y \to 0$, so that $xy = 0$, that is, $x = y$.

The *limit* of a convergent sequence is thus uniquely determined. From this it follows that a sequence cannot converge to two different points, and that $x_n \to x$ is equivalent to $|x_n - x| \to 0$.

Theorem V.2 (II)**.** *Every subsequence of a convergent sequence converges to the same limit.*

For if $x_n \to x$ and $n_1 < n_2 < \cdots$, then $x_{n_k}x \leq \epsilon$ for all $n_k > q$, so that $x_{n_k} \to x$.

Theorem V.3 (III)**.** *A fundamental sequence converges if and only if it has at least one point of accumulation.*

For if x is a point of accumulation of the fundamental sequence x_n, then there exist arbitrarily large n with $x_n x < \epsilon$; for $m > p$, $n > p$, we have $x_m x_n < \epsilon$, so that for such n and $m > p$

$$x_m x \leq x_m x_n + x_n x < 2\epsilon,$$

that is, $x_m \to x$. Conversely, a convergent sequence has the limit as a point of accumulation.

2. The space E_ω

The Euclidean spaces E_n are complete. For let $x^{(1)}, x^{(2)}, \ldots$ be a fundamental sequence in E_n, where

$$x^{(m)} = (x_1^{(m)}, x_2^{(m)}, \ldots, x_n^{(m)}).$$

Then for each $k = 1, 2, \ldots, n$, the sequence $x_k^{(1)}, x_k^{(2)}, \ldots$ is a fundamental sequence in E_1, and since E_1 is complete, it converges: $x_k^{(m)} \to x_k$. Then $x^{(m)} \to x = (x_1, \ldots, x_n)$, so that E_n is complete.

Hilbert space H_0 is complete. Let $x^{(1)}, x^{(2)}, \ldots$ be a fundamental sequence in H_0, where

$$x^{(m)} = (x_1^{(m)}, x_2^{(m)}, x_3^{(m)}, \ldots).$$

Then for every k, the sequence $x_k^{(1)}, x_k^{(2)}, \ldots$ is a fundamental sequence in E_1 and therefore converges: $x_k^{(m)} \to x_k$. We assert that $x = (x_1, x_2, x_3, \ldots)$ belongs to H_0 and that $x^{(m)} \to x$.

In fact, given $\epsilon > 0$, we have, for sufficiently large l, m,

$$(x_1^{(l)} - x_1^{(m)})^2 + (x_2^{(l)} - x_2^{(m)})^2 + \cdots < \epsilon^2,$$

and therefore, for every n,

$$(x_1^{(l)} - x_1^{(m)})^2 + \cdots + (x_n^{(l)} - x_n^{(m)})^2 < \epsilon^2,$$

so that for $n > 0$, $l > \lambda$

$$\sum_{k=1}^{n} (x_k^{(l)} - x_k)^2 \leq \epsilon^2,$$

and from this it follows, for $l \to \infty$, that

$$\sum_{k=1}^{\infty} (x_k^{(l)} - x_k)^2 \leq \epsilon^2.$$

From the convergence of the series on the left, we infer that $x - x^{(l)}$, and therefore x as well, belongs to the Hilbert space. Furthermore $|x - x^{(l)}| \leq \epsilon$, which, combined with $|x^{(l)} - x^{(m)}| < \epsilon$ gives $|x - x^{(m)}| < 2\epsilon$ for $n > m$; and this means that $|x - x^{(m)}| \to 0$ or $x^{(m)} \to x$.

The space $E_\omega = E_1 + E_2 + \cdots$ (p. 114) lying between the Euclidean spaces and Hilbert space is incomplete: for if $x = (x_1, x_2, \ldots)$ is a point of Hilbert space, then the points

$$x^{(n)} = (x_1, x_2, \ldots, x_n, 0, 0, \ldots)$$

belonging to E_ω converge to x and constitute a fundamental sequence; but in case infinitely many coordinates x_k are $\neq 0$, this fundamental sequence does not converge in E_ω.

The space of the functions continuous in $[a, b]$ with norm defined by

$$|x| = \int_a^b |x(t)|\,dt$$

is not complete. For every integrable (Riemann integrable and bounded) function ξ a sequence of continuous functions x_n with $|x_n - \xi| \to 0$ can be given; if at the same time there were to be a continuous function x with $|x_n - x| \to 0$, so that $|x - \xi| = 0$, then the null function $x - \xi$ would have to vanish at its points of continuity, and x would have to agree with ξ at the points at which ξ is continuous. The simplest examples show that such a continuous function x need not exist: for example, let $\xi(t)$ have a single jump at c between a and b with $\xi(c-0) \neq \xi(c+0)$, and let it be continuous everywhere else. Even the space of Riemann- integrable functions is not complete, for we can approximate every integrable function by continuous functions, and the fundamental sequence thus obtained converges only if the integrable function is equivalent to a continuous function.

3. Theorems on fundamental sequences

Theorem V.4 (IV). *A set A is everywhere dense in E if and only if every point $x \in E$ is the limit of a sequence of points $x_n \in A$.*

If A is everywhere dense, then for every n there exists in the neighborhood $U(x, 1/n)$ a point $x_n \in A$, and $x_n \to x$. Conversely, if $x_n \to x$ and $x_n \in A$, then in every neighborhood $U(x, \epsilon)$ there are points x_n of A, so that A is everywhere dense.

We prove further theorems:

Theorem V.5 (V). *A fundamental sequence x_n that has infinitely many points of accumulation converges.*

For if x and y are two points of accumulation, then there exist subsequences $x_{m_k} \to x$ and $x_{n_k} \to y$. Since the sequence is fundamental, $x_{m_k}x_{n_k} \to 0$, so that $xy = 0$, that is, $x = y$. The sequence thus has only one point of accumulation and is therefore convergent.

Theorem V.6 (VI). *If x_n and y_n are fundamental sequences, then the sequence of real numbers x_ny_n converges.*

For

$$|x_my_m - x_ny_n| \leq x_my_m + x_my_n + x_ny_n + x_ny_m$$

and the right-hand side tends to 0 as $m, n \to \infty$.

Theorem V.7 (VII). *If $x_n \to x$ and $y_n \to y$, then $x_ny_n \to xy$.*

For

$$|x_ny_n - xy| \leq x_ny_n + x_ny + xy_n + xy \to 0.$$

Theorem V.8 (VIII). *If $x_n \to x$ and $y_n \to y$, then the sequence of real numbers x_ny_n converges to xy.*

In particular, for $y_n = y$, we have: if $x_n \to x$, then $x_ny \to xy$. The distance xy is thus a continuous function of the pair (x, y).

§ 22. Interior Points and Border Points

1. Definitions

Let A be a set in the space E, and let x be a point of E. If there exists a neighborhood U_x that is entirely contained in A, then x is called an *interior point* of A; if there exists a neighborhood U_x that contains no point of A, then x is called an *exterior point* of A; if every neighborhood U_x contains both points of A and points not in A, then x is called a *border point* of A.

The set of interior points of A is called the *interior* or the *open kernel* of A and is denoted by A_i; the set of exterior points is called the *exterior* of A; the set of border points is called the *border* or *boundary* of A and is denoted by A_b. These three sets are mutually exclusive, and their union is the entire space E. The interior A_i is contained in A; the exterior is contained in $E - A$; the border A_b is the intersection of the borders of A and $E - A$:

$$A_b = A_b(E - A).$$

The open kernel A_i is an open set; for if $x \in A_i$, then there exists a neighborhood $U_x \subseteq A$, and every point of U_x is likewise an interior point of A, so that $U_x \subseteq A_i$. The exterior is also open. The border A_b is a closed set; for if x is a point of accumulation of A_b, then every neighborhood U_x contains a border point y of A, and every neighborhood of y contains both points of A and points not in A; thus U_x contains both points of A and points not in A, so that x is a border point of A.

The open kernel of A is the greatest open set contained in A; for if G is an open set contained in A and if $x \in G$, then there exists a neighborhood $U_x \subseteq G \subseteq A$, so that $x \in A_i$, that is, $G \subseteq A_i$. The exterior of A is the open kernel of $E - A$.

Theorem V.9 (I). *The open kernel A_i of A is the union of all open sets contained in A.*

Theorem V.10 (II). $A_i = A - A_b(E - A) = A(E - A_b(E - A))$.

2. Open and closed sets

A set is called *open* if all its points are interior points, that is, if $A = A_i$; a set is called *closed* if it contains all its border points, that is, if $A_b \subseteq A$.

For open and closed sets we have the following theorems:

Theorem V.11 (III). *The complement of an open set is closed, and the complement of a closed set is open.*

For if A is open, then $A = A_i$, so that $A_b \subseteq E - A$; every border point of $E - A$ is thus a point of $E - A$, that is, $E - A$ is closed. If A is closed, then $A_b \subseteq A$, so that no border point of A belongs to $E - A$; every point of $E - A$ is thus either an interior point or an exterior point of $E - A$, that is, $E - A$ is open.

Theorem V.12 (IV). *The union of any number of open sets is open; the intersection of any number of closed sets is closed.*

For if $A = \sum A_\lambda$ and $x \in A$, then $x \in A_\lambda$ for some λ, and since A_λ is open, there exists a neighborhood $U_x \subseteq A_\lambda \subseteq A$, so that $x \in A_i$, that is, $A = A_i$. The second part follows by taking complements.

Theorem V.13 (V). *The intersection of a finite number of open sets is open; the union of a finite number of closed sets is closed.*

For if $A = A_1 A_2 \cdots A_n$ and $x \in A$, then there exist neighborhoods $U_x^{(k)} \subseteq A_k$ $(k = 1, 2, \ldots, n)$; the smallest of these neighborhoods is contained in A, so that $x \in A_i$, that is, $A = A_i$. The second part follows by taking complements.

Theorem V.14 (VI). *For every open set A there exists a function $f(x)$ continuous in E such that $A = [f > 0]$; conversely, every set $[f > 0]$ is open.*

We set, namely, $f(x) = \varrho(x, E - A)$, the distance of the point x from the set $E - A$. If $x \in A$, then there exists a neighborhood $U(x, \delta) \subseteq A$, so that every point y with $xy < \delta$ belongs to A and therefore does not belong to $E - A$; thus $f(x) \geq \delta > 0$. If $x \in E - A$, then $f(x) = 0$. The function $f(x)$ is continuous; for let $x_n \to x$, then

$$|f(x_n) - f(x)| = |\varrho(x_n, E - A) - \varrho(x, E - A)| \leq x_n x \to 0.$$

Thus $A = [f > 0]$.

Conversely, if $f(x)$ is continuous and $x \in [f > 0]$, then there exists a neighborhood $U(x, \delta)$ in which $f(y) > 0$, so that $y \in [f > 0]$; thus $[f > 0]$ is open.

Theorem V.15 (VII)**.** *For every closed set A there exists a function $f(x)$ continuous in E such that $A = [f = 0]$; conversely, every set $[f = 0]$ is closed.*

For if A is closed, then $E - A$ is open, and by VI there exists a continuous function $f(x) \geq 0$ with $E - A = [f > 0]$, so that $A = [f = 0]$. Conversely, if $f(x)$ is continuous and $x_n \to x$ with $x_n \in [f = 0]$, then $f(x) = \lim f(x_n) = 0$, so that $x \in [f = 0]$; thus $[f = 0]$ is closed.

Theorem V.16 (VIII)**.** *A set is open if and only if it is the union of neighborhoods.*

For if A is open, then every point $x \in A$ has a neighborhood $U_x \subseteq A$, and $A = \sum_{x \in A} U_x$. Conversely, every neighborhood is open, and by IV the union of open sets is open.

3. The border theorem

Theorem V.17 (IX). *The border of the union of two sets is contained in the union of the borders:*

$$(A+B)_b \subseteq A_b + B_b. \tag{1}$$

For if $x \in (A+B)_b$, then every neighborhood U_x contains both points of $A+B$ and points not in $A+B$. If x is not a border point of A, then there exists a neighborhood V_x that is entirely contained in A or entirely contained in $E-A$. In the first case, $V_x \subseteq A \subseteq A+B$; in the second case, every neighborhood $U_x \subseteq V_x$ that contains points of $A+B$ must contain points of B, and since it also contains points not in $A+B$, it contains points not in B; thus $x \in B_b$. The same holds if x is not a border point of B.

Theorem V.18 (X). *If A and B are both open or both closed, then*

$$(A+B)_b = A_b + B_b - A_b B_b. \tag{2}$$

For if $x \in A_b + B_b - A_b B_b$, say $x \in A_b$, $x \notin B_b$, then if A and B are open, $x \notin B$ and thus $x \in A \subseteq A+B$ or $x \in E-A-B$; if A and B are closed, $x \in A$ and $x \in E-B$ or $x \in B$. In every case, every neighborhood of x contains both points of $A+B$ and points not in $A+B$, so that $x \in (A+B)_b$.

§ 23. The α, β, and γ Points

1. Points of accumulation

Let A be a set in the metric space E, and let $\mathfrak{m}$ be a cardinal number, finite or transfinite. A point x is called an $\mathfrak{m}$-*point* of A if every neighborhood U_x contains at least $\mathfrak{m}$ points of A; in particular, for $\mathfrak{m} = \aleph_0$, an $\aleph_0$-point is also called a *point of accumulation* of A.

The set of $\mathfrak{m}$-points of A is denoted by $A_{\mathfrak{m}}$. We have

$$A = A_1 \supseteq A_2 \supseteq \cdots \supseteq A_{\aleph_0} \supseteq A_{\aleph_1} \supseteq \cdots .$$

The set $A_{\mathfrak{m}}$ is closed. For if x is a point of accumulation of $A_{\mathfrak{m}}$, then every neighborhood U_x contains a point $y \in A_{\mathfrak{m}}$, and every neighborhood U_y contains at least $\mathfrak{m}$ points of A; thus U_x contains at least $\mathfrak{m}$ points of A, so that $x \in A_{\mathfrak{m}}$.

The set of points of accumulation of A is called the *derived set* of A and is denoted by A_d; the set $A - A_d$ of the points of A that are not points of accumulation is called the *isolated part* of A; its points are called *isolated points* of A.

We now define, following Cantor, the *coherence* of A as the set

$$A_c = A \cdot A_d = A_{\aleph_0};$$

the *adherence* of A as the set

$$A_a = A + A_d;$$

and the set

$$A_\gamma = A_a - A_i = A_b + A_d$$

(union of the border and the derived set) as the set of γ-*points* of A. The points of A that do not belong to the coherence are the isolated points of A; the points of the adherence that do not belong to A are the points of accumulation of A that are not points of A.

Theorem V.19 (I). *The coherence and the adherence are monotonic: if $A \subseteq B$, then $A_c \subseteq B_c$ and $A_a \subseteq B_a$.*

For if $x \in A_c = A \cdot A_d$, then $x \in A \subseteq B$, and every neighborhood U_x contains infinitely many points of A, hence also of B, so that $x \in B_d$ and therefore $x \in B_c$. The proof for the adherence is analogous.

2. The α, β, γ decomposition

The following decomposition of the space E into three disjoint sets is fundamental:

$$E = A_\alpha + A_\beta + A_\gamma, \tag{1}$$

where:

- A_α is the set of points x that have a neighborhood U_x containing only a finite number of points of A (α-*points*, or isolated points of the space relative to A);
- A_β is the set of interior points of A;
- A_γ is the set of γ-points of A, that is, the set of points x every neighborhood of which contains both infinitely many points of A and infinitely many points of $E - A$.

The γ-points are thus the points of accumulation both of A and of $E - A$: $A_\gamma = A_d \cdot (E - A)_d$.

We have:

$$A_\alpha + A_\beta + A_\gamma = E, \quad A_\alpha A_\beta = A_\beta A_\gamma = A_\gamma A_\alpha = 0. \tag{2}$$

The three sets are mutually disjoint and their union is E.

Theorem V.20 (II). *A_α and A_β are open; A_γ is closed.*

For if $x \in A_\alpha$, then there exists a neighborhood U_x containing only finitely many points of A, and every point of U_x has the same property, so that $U_x \subseteq A_\alpha$; thus A_α is open. We already know that $A_\beta = A_i$ is open. The set A_γ is closed as the border of the open set A_α (or of A_β).

The decomposition (1) gives rise, by intersection with A, to a decomposition of A into three disjoint summands:

$$A = A_\alpha A + A_\beta + A_\gamma A. \tag{3}$$

Here $A_\alpha A$ is the set of isolated points of A; A_β is the interior of A; $A_\gamma A$ is the set of points of A that are points of accumulation of $E - A$, that is, the set of border points of A that belong to A.

The sets $A_\alpha A$ and $A_\gamma A$ can also be empty. If A is open, then $A_\gamma A = 0$ and $A = A_\beta$ (but $A_\alpha A$ can still be non-empty). If A is closed, then $A_\gamma \subseteq A$ and $A_\gamma A = A_\gamma$ (but A_β and $A_\alpha A$ can be non-empty). For an open set, the decomposition (3) becomes

$$A = A_\alpha A + A_\beta.$$

For a closed set:

$$A = A_\beta + A_\gamma.$$

Theorem V.21 (III). *If $x \in A_\gamma$ and U_x is a neighborhood, then U_x contains uncountably many points of both A and $E - A$.*

For $x \in A_\gamma = A_d \cdot (E - A)_d$. If U_x were to contain only countably many points of A, then since x is a point of accumulation of A, there would exist in U_x a sequence of distinct points of A converging to x; but then x would be an interior point of the closure of A, which contradicts $x \in A_\gamma$.

Examples in the Euclidean plane E_2:

A_1: a circular disc without circumference: $A_\alpha = A_\beta = A$, $A_\gamma =$ circular disc with its circumference.

A_2: the set of rational points: $A_\alpha = A_\beta = 0$, $A_\gamma = E_2$.

A_3: a single point: $A_\alpha = A$, $A_\beta = A_\gamma = 0$.

A_4: a straight line: $A_\alpha = A_\beta = 0$, $A_\gamma = A$.

A_5: a circular disc without its center: $A_\alpha = 0$, $A_\beta =$ disc without circumference $A_\gamma =$ circumference + center.

3. Dense-in-itself, closed, perfect sets

The condition for a set to be dense-in-itself may be expressed by writing $A \subseteq A_d$. This gives rise to further definitions, as follows: We say that the set A is

- *dense-in-itself* if $A \subseteq A_d$,
- *closed* if $A \supseteq A_d$,
- *perfect* if $A = A_d$,

that is, if every point of A is a point of accumulation, if every point of accumulation of A is a point of A itself, and if both hold at the same time. This can also be expressed, by means of (2), in the form of equations, inasmuch as A becomes equal to one of the summands A_α, A_β, or both: the set A is

- dense-in-itself if $A = A_\gamma A$,
- closed if $A = A_\beta + A_\gamma$,
- perfect if $A_\alpha = A_\beta = 0$, $A_\gamma = A$.

Once more we cite the simplest examples of planar sets.

A_1: a circular disc without circumference: dense-in-itself.

A_2: a circular disc including its circumference and a single external point: closed.

A_3: a circular disc including its circumference: perfect.

Sets without points of accumulation ($A_d = 0$) — in particular, finite sets — count, of course, as closed sets. The null set is everything: isolated, dense-in-itself, closed, perfect (open set, border set).

4. The separated part and the dense-in-itself kernel

It follows from the earlier formula

$$(S)_d = A_d + B_d + \cdots, \quad (D)_d \subseteq A_d B_d \cdots$$

with $A = B = \cdots$, that

Theorem V.22 (IV)**.** *The sum of any number of dense-in-itself sets is dense-in-itself.*

This allows us to form the sum of all the dense-in-itself subsets of A (to which the null set certainly belongs) and this, in turn, is dense-in-itself and is therefore the greatest dense-in-itself subset, or the *dense-in-itself kernel*, of A. We call this kernel A_k; and we call the points not belonging to the dense-in-itself kernel *separated points* and the totality A_s of these points the *separated part* of A. We have thus obtained a new, and final, decomposition of A

$$A = A_k + A_s \tag{4}$$

into disjoint summands. A set consisting entirely of separated points ($A_k = 0$) is called a *separated set*, and we now have still

$$A_s \supseteq A_\alpha A. \tag{5}$$

The coherence $A_c = A \cdot A_d$ is a dense-in-itself set contained in A; for $A_c \subseteq A_d$, and since A_d is the set of points of accumulation of A, every point of A_c is a point of accumulation of A and thus also of A_c, so that $A_c \subseteq (A_c)_d$, that is, A_c is dense-in-itself. Since A_k is the greatest dense-in-itself subset of A, we have $A_c \subseteq A_k$. On the other hand, $A_k \subseteq A$, and since A_k is dense-in-itself, $A_k \subseteq A_d$, so that $A_k \subseteq A \cdot A_d = A_c$. Thus $A_k = A_c = A \cdot A_d$.

The decomposition (4) thus becomes

$$A = A \cdot A_d + A_s, \tag{6}$$

where $A_s = A - A_d = A_\alpha A$ is the isolated part of A.

5. Formulas for the α, β, γ points

The formulas for the behavior of the A-points under formation of sums and intersections are of fundamental importance. Let A, B, ... be any number of sets (not necessarily a finite number), and let

$$S = A + B + \cdots, \quad D = AB \cdots$$

be their sum and intersection respectively, then it follows from the monotonicity that

$$S_\mathfrak{m} \supseteq A_\mathfrak{m} + B_\mathfrak{m} + \cdots, \quad D_\mathfrak{m} \subseteq A_\mathfrak{m} B_\mathfrak{m} \cdots \quad (\mathfrak{m} = \aleph_0, \aleph_1, \ldots).$$

For a finite number of sets we have a stronger result; for two sets, for example:

$$\text{If } S = A + B, \text{ we have } S_\mathfrak{m} = A_\mathfrak{m} + B_\mathfrak{m} \quad (\mathfrak{m} = \aleph_0, \aleph_1, \ldots). \tag{7}$$

It is only necessary to prove that $S_\mathfrak{m} \subseteq A_\mathfrak{m} + B_\mathfrak{m}$ or that a point x that does not belong to $A_\mathfrak{m} + B_\mathfrak{m}$ does not belong to $S_\mathfrak{m}$ either. Now if x is an $\mathfrak{m}$-point neither of A nor of B, then there are neighborhoods U_x and V_x such that AU_x and BV_x are of cardinality $< \mathfrak{m}$, and if U_x, say, is the one having the smaller radius, then $SU_x = AU_x + BU_x$ also has cardinality $< \mathfrak{m}$, whence x is not an $\mathfrak{m}$-point of S.

For the intersection of A with an open set we have:

$$(AG)_\mathfrak{m} \supseteq A_\mathfrak{m} \cdot G \quad (G \text{ open}) \quad (\mathfrak{m} = \aleph_0, \aleph_1, \ldots), \tag{8}$$

while of course the trivial formula $(AG)_\mathfrak{m} \subseteq A_\mathfrak{m} G$ also holds. For if $x \in A_\mathfrak{m} G$ and U_x is a neighborhood, then GU_x is open, and there exists a neighborhood $V_x \subseteq GU_x$; this neighborhood contains at least $\mathfrak{m}$ points of A, which are also points of AG, so that $x \in (AG)_\mathfrak{m}$.

6. Derived sets and closed sets

Theorem V.23 (VII). *A closed set that contains no isolated points of the space is a derived set.*

This condition is thus both necessary and sufficient, for every derived set A_d is closed and $\subseteq E_d$. We base our proof on an auxiliary theorem:

Theorem V.24 (VIII). *If H is the boundary of the open set G, then H can be represented as the derived set B_d of an isolated set $B \subseteq G$.*

We assume H to be non-empty. (If $H = 0$, then B may be taken to be empty.) $H = G_a - G$ is a closed set consisting of those points of G_a which are not points of G. Let A be a maximal well-ordered set of points of G that are pairwise at distances $\geq \delta$ from one another, where $\delta > 0$ is chosen arbitrarily; the existence of such a maximal set follows from the well-ordering theorem. The set A is isolated (its points are isolated from one another).

Now let x be a point of accumulation of A. Then there exists a sequence of distinct points $x_n \in A$ with $x_n \to x$. Since the points of A are at distances $\geq \delta$ from one another, no subsequence of x_n can be a fundamental sequence unless it is eventually constant, which contradicts the distinctness of the x_n. Therefore the sequence x_n cannot be eventually constant, and x is a point of accumulation of A.

We now prove that $A_d = H$. On the one hand, let $x \in A_d$. Then there exists a sequence $x_n \in A$ with $x_n \to x$, $x_n \neq x$. Since $x_n \in G$ and G is open, x is either in G or in $E - G$. If $x \in G$, then since G is open, there exists a neighborhood $U(x, \delta) \subseteq G$, and for sufficiently large n, $x_n \in U(x, \delta)$; but then x_n and x_{n+1} are at distance $< \delta$, which contradicts the construction of A. Thus $x \in E - G$, and since $x \in A_d \subseteq A_a$, we have $x \in G_a - G = H$.

On the other hand, let $h \in H$ and let G be the open set of which H is the boundary. Let $\delta > 0$ be arbitrary. Since $h \in H = G_a - G$, every neighborhood $U(h, \epsilon)$ contains points of G, and for $\epsilon < \delta$ these points have distance $< \delta$ from h. Let A be the maximal set of points of G that are pairwise at distances $\geq \delta$ from one another. If no point of A were in $U(h, \delta)$, then we could adjoin to A a point of $G \cap U(h, \delta)$, contradicting the maximality of A. Thus there are points of A in $U(h, \delta)$, and since δ is arbitrary, $h \in A_d$.

The proof of VII now proceeds as follows. Let $F = F_i + F_b = H + F_i$ be closed, so that H is the boundary of the open set $G = E - F$ and therefore $H = B_d$ with $B \subseteq G$. Since $F \subseteq E_d$, it follows that F_i is dense-in-itself and $F_i \subseteq F_d$. Therefore we have

$$F = F_i + H \subseteq F_d + H \subseteq F + H = F,$$
$$F = F_d + H = F_d + B_d = (F + B)_d;$$

F is the derived set of $F + B$.

Theorem V.25 (IX)**.** *The adherence of the sum is the sum of the adherences:*

$$(A + B)_a = A_a + B_a. \tag{9}$$

For $(A+B)_a = A+B+(A+B)_d = A+B+A_d+B_d = A_a+B_a$.

Theorem V.26 (X)**.** *The coherence of the intersection is contained in the intersection of the coherences:*

$$(AB)_c \subseteq A_c B_c. \tag{10}$$

For $(AB)_c = AB \cdot (AB)_d \subseteq AB \cdot A_d B_d = A_c B_c$.

§ 24. Relative and Absolute Concepts

1. Relative concepts

The concepts of interior, exterior, and border point, of open and closed set, of point of accumulation, and so on, depend not only on the set A under consideration but also on the space E in which A is embedded. If we pass from E to a subset $D \subseteq E$ (a subspace), then the concepts may change. The points of A that are interior points of A relative to D form the *relative open kernel* of A in D, which we denote by $A_i(D)$; similarly, $A_d(D)$ is the *relative derived set* of A in D, and so on.

The relationship between the absolute and relative concepts is given by the following formulas:

$$A_i(D) = D \cdot A_i, \tag{1}$$

$$A_a(D) = D \cdot A_a, \tag{2}$$

$$A_d(D) = D \cdot A_d, \tag{3}$$

$$A_b(D) = D \cdot A_b. \tag{4}$$

Formula (1) states: a point $x \in D$ is a relative interior point of A in D if and only if it is an absolute interior point of A, that is, if there exists a neighborhood U_x with $U_x \subseteq A$; for then $U_x \cdot D \subseteq A \cdot D$, and conversely.

Formula (2): a point $x \in D$ belongs to the relative adherence of A in D if and only if it belongs to D and to the absolute adherence of A.

Formula (3): a point $x \in D$ is a relative point of accumulation of A in D if and only if every neighborhood U_x contains a point of A different from x, and this means exactly that $x \in D \cdot A_d$.

Theorem V.27 (I)**.** *A set that is open (closed) relative to D is the intersection of D with a set that is open (closed) in E.*

If $A \subseteq D$ is open relative to D, then $A = A_i(D) = D \cdot G$, where $G = A_i$ is open in E. If A is closed relative to D, then $D - A$ is open relative to D, so that $D - A = D \cdot G$ with G open in E, and $A = D \cdot (E - G)$ with $E - G$ closed in E.

Theorem V.28 (II)**.** *The sum of any number of sets that are open relative to D is open relative to D; the intersection of any number of sets that are closed relative to D is closed relative to D.*

Theorem V.29 (III). *The intersection of a finite number of sets that are open relative to D is open relative to D; the sum of a finite number of sets that are closed relative to D is closed relative to D.*

2. Connected sets

A set A is called *connected* if it is not the sum of two non-empty disjoint sets that are both closed relative to A (or, equivalently, both open relative to A).

Theorem V.30 (IV)**.** *If A is connected and $A \subseteq B \subseteq A_a$, then B is also connected.*

For if $B = P + Q$ with P, Q non-empty, disjoint, and closed relative to B, then $A = AP + AQ$. Since P is closed relative to B, we have $P = B \cdot F$ with F closed in E, so that $AP = A \cdot F$ is closed relative to A. Since A is connected, either $AP = 0$ or $AQ = 0$; say $AP = 0$. Then $A \subseteq Q$, so that $A_a \subseteq Q_a$, and since Q is closed relative to B, we have $Q = B \cdot G$ with G closed in E, so that $Q_a \subseteq G$ and $B \subseteq A_a \subseteq Q_a \subseteq G$, whence $P = B \cdot (E - G) = 0$, a contradiction.

Theorem V.31 (V)**.** *If A and B are connected and $AB \neq 0$, then $A + B$ is connected.*

For if $A + B = P + Q$ with P, Q disjoint and closed relative to $A + B$, then since A is connected and $A = AP + AQ$, either $A \subseteq P$ or $A \subseteq Q$; say $A \subseteq P$. Since $AB \neq 0$, there exists a point $x \in A \cdot B \subseteq P$, so that $BP \neq 0$. Since B is connected and $B = BP + BQ$, we have either $B \subseteq P$ or $B \subseteq Q$; but $BP \neq 0$, so that $B \subseteq P$. Thus $A + B \subseteq P$ and $Q = 0$, a contradiction.

Theorem V.32 (VI)**.** *The set of all points that can be joined to a point x by connected subsets of A is a connected subset of A; it is called the* component *of x in A.*

For if C is the set of all such points and $C = P + Q$ with P, Q disjoint and closed relative to C, then x belongs to one of them, say $x \in P$. Every point $y \in P$ can be joined to x by a connected set $B_y \subseteq A$; since $x \in P \cap B_y$, $P \cap B_y$ and $Q \cap B_y$ are closed relative to B_y, and since B_y is connected, either $B_y \subseteq P$ or $B_y \subseteq Q$; but $x \in P \cap B_y$, so that $B_y \subseteq P$. Thus every point that can be joined to x by a connected set belongs to P, that is, $C \subseteq P$ and $Q = 0$.

3. Regions and continua

A connected open set is called a *region*; a connected closed set is called a *continuum*. A region that is also a continuum is both open and closed; in a connected space, such a set is either empty or the entire space.

Theorem V.33 (VII). *If G is a region and H is a continuum with $G \cdot H \neq 0$ and $G - H \neq 0$, then $G - H$ is not a region.*

For if $G - H$ were a region, then $G = (G - H) + (G \cdot H)$ would be a decomposition of the connected set G into two non-empty disjoint open sets, which is impossible.

Theorem V.34 (VIII). *The intersection of a decreasing sequence of continua is a continuum.*

Let $H_1 \supseteq H_2 \supseteq H_3 \supseteq \cdots$ be continua and let $H = H_1 H_2 H_3 \cdots$. Then H is closed. If H were not connected, then $H = P + Q$ with P, Q non-empty, disjoint, and closed relative to H. Since P is closed relative to H, $P = H \cdot F$ with F closed in E; similarly $Q = H \cdot G$ with G closed in E. Since P and Q are disjoint and closed, there exist open sets $U \supseteq P$ and $V \supseteq Q$ with $U \cdot V = 0$.

Since $P = H \cdot F = H_1 H_2 \cdots F$ and U is an open neighborhood of P, we have, for sufficiently large n, $H_n \cdot F \subseteq U$; similarly, $H_n \cdot G \subseteq V$ for sufficiently large n. Thus, for sufficiently large n, $H_n = H_n \cdot F + H_n \cdot G + H_n \cdot (E - F - G)$. Since H_n is connected and $H_n \cdot F$ and $H_n \cdot G$ are non-empty and closed relative to H_n, we must have $H_n \cdot (E - F - G) \neq 0$. But then $H_n - U - V \neq 0$, and since H_n is decreasing, $H - U - V \neq 0$, contradicting $H = P + Q \subseteq U + V$.

§ 25. Separable Spaces

1. Definition of separability

A metric space is called *separable* if it contains an everywhere dense countable set (or finite set, since a finite set can be regarded as a countable set in the wider sense). The simplest example of a separable space is E_n: the set of points with rational coordinates is countable and everywhere dense. Hilbert space H_0 is also separable: the set of points with only finitely many rational coordinates, the rest being zero, is countable and everywhere dense.

A separable space has at most the cardinality of the continuum. For let $A = \{a_1, a_2, a_3, \ldots\}$ be an everywhere dense countable set, and let x be an arbitrary point of the space. Then for every n there exists a point a_{k_n} with $xa_{k_n} < 1/n$. The sequence a_{k_n} converges to x, and x is determined by the sequence $k_1, k_2, k_3, \ldots$ of natural numbers. The set of all such sequences has the cardinality of the continuum, so that the space has at most the cardinality of the continuum.

The cardinality of a separable space is thus at most $\mathfrak{c} = 2^{\aleph_0}$.

Theorem V.35 (I)**.** *Every subset of a separable space is separable.*

Let E be separable, with the everywhere dense countable set $A = \{a_1, a_2, a_3, \ldots\}$, and let $B \subseteq E$. For every pair of natural numbers n, m, there exists, if $U(a_n, 1/m) \cdot B \neq 0$, a point $b_{nm} \in U(a_n, 1/m) \cdot B$. The set of all these points b_{nm} is countable (since the pairs (n, m) are countable). We assert that it is everywhere dense in B.

In fact, let $x \in B$ and $\epsilon > 0$. Since A is everywhere dense in E, there exists a point a_n with $xa_n < \epsilon/2$. Choose m so that $1/m < \epsilon/2$. Then $U(a_n, 1/m) \cdot B \neq 0$ (since x belongs to this set), and $b_{nm} \in U(a_n, 1/m) \cdot B$, so that

$$xb_{nm} \leq xa_n + a_n b_{nm} < \epsilon/2 + 1/m < \epsilon.$$

Thus the countable set of the b_{nm} is everywhere dense in B, so that B is separable.

Theorem V.36 (II)**.** *The sum of at most countably many separable sets is separable.*

Let $B_1, B_2, B_3, \ldots$ be separable, with $B_n \subseteq E$, and let $A_n = \{a_{n1}, a_{n2}, a_{n3}, \ldots\}$ be an everywhere dense countable set in B_n. Then the set $A = A_1 + A_2 + A_3 + \cdots$ is countable. We assert that A is everywhere dense in $B = B_1 + B_2 + B_3 + \cdots$.

In fact, let $x \in B$ and $\epsilon > 0$. Then $x \in B_n$ for some n, and there exists a point $a_{nk} \in A_n$ with $xa_{nk} < \epsilon$. Since $a_{nk} \in A$, A is everywhere dense in B.

2. The Lindelöf covering theorem

Theorem V.37 (III). *(Lindelöf covering theorem). If a separable space E is covered by a system of open sets G_λ, then there exists a countable subsystem that already covers E.*

Let $A = \{a_1, a_2, a_3, \ldots\}$ be an everywhere dense countable set. For each point a_n there exists an open set G_{λ_n} of the system that contains a_n. The system of these G_{λ_n} is countable. We assert that it covers E.

In fact, let $x \in E$ and $\epsilon > 0$. Then there exists a point a_n with $xa_n < \epsilon$, and since G_{λ_n} is open and contains a_n, there exists a $\delta > 0$ such that $U(a_n, \delta) \subseteq G_{\lambda_n}$. If we choose $\epsilon < \delta$, then $x \in U(a_n, \delta) \subseteq G_{\lambda_n}$, so that x is covered by G_{λ_n}.

A stronger result holds:

Theorem V.38 (IV). *If a separable space E is covered by open sets G_λ, then there exists a countable subsystem $G_{\lambda_1}, G_{\lambda_2}, G_{\lambda_3}, \ldots$ that covers E, and each point of E has a neighborhood that is contained in one of the G_{λ_n}.*

The countable subsystem constructed in III already has this property; for if $x \in G_{\lambda_n}$, then since G_{λ_n} is open, there exists a neighborhood $U_x \subseteq G_{\lambda_n}$.

3. Borel sets in separable spaces

In separable spaces, the Borel sets have particularly simple properties. The smallest Borel system containing the open sets is called the system of *Borel sets* of the space. Since every open set is the union of a countable number of neighborhoods (the set of all neighborhoods with rational radii and centers belonging to the everywhere dense countable set is countable), the Borel sets are generated by this countable system of neighborhoods.

Theorem V.39 (V). *In a separable space, the system of Borel sets has at most the cardinality of the continuum.*

For the system of Borel sets is generated by a countable system of sets, and the number of Borel sets obtainable from a countable system of sets by the operations of the Borel system is at most the cardinality of the continuum (since each Borel set is determined by a countable transfinite sequence of operations).

Theorem V.40 (VI). *Every closed set in a separable space is the intersection of a countable number of open sets; every open set is the union of a countable number of closed sets.*

For if F is closed, then

$$F = F_1 \cdot F_2 \cdot F_3 \cdots ,$$

where

$$F_n = \sum_{x \in F} U(x, 1/n)$$

is open. Conversely, if G is open, then $E - G$ is closed, and

$$E - G = F_1 \cdot F_2 \cdot F_3 \cdots ,$$

so that

$$G = (E - F_1) + (E - F_2) + (E - F_3) + \cdots ,$$

and $E - F_n$ is closed.

4. Conditionally compact sets

A set A is called *conditionally compact* if every sequence of points of A contains a fundamental subsequence. (In a complete space, this means that every sequence contains a convergent subsequence, and the set is then called *compact.*)

Theorem V.41 (VII). *A conditonally compact set is totally bounded; that is, for every $\epsilon > 0$, it can be represented as the sum of a finite number of sets with diameters $< \epsilon$.*

For if A were not totally bounded, then there would exist an $\epsilon > 0$ such that A could not be represented as the sum of a finite number of sets of diameter $< \epsilon$. Then we could choose a sequence of points $a_1, a_2, a_3, \dots$ of A such that $a_m a_n \geq \epsilon$ for $m \neq n$: having chosen $a_1, \dots, a_n$, the set $A - U(a_1, \epsilon) - \cdots - U(a_n, \epsilon)$ is non-empty, and we choose a_{n+1} from it. This sequence contains no fundamental subsequence, contradicting the conditional compactness of A.

Theorem V.42 (VIII). *In a separable space, every conditionally compact set A has at most the cardinality of the continuum.*

For A is separable (as a subset of a separable space), and by VII it is totally bounded; for every n, it can be covered by finitely many neighborhoods of radius $1/n$. The centers of all these neighborhoods (for all n) form a countable set, and every point of A is the limit of a sequence of these centers, so that A has at most the cardinality of the continuum.

Theorem V.43 (IX). *In a separable space, every set A of cardinality greater than the continuum contains a sequence none of whose subsequences is a fundamental sequence.*

For if every sequence contained a fundamental subsequence, then A would be conditionally compact and thus of cardinality at most the continuum, by VIII.

§ 26. Complete Spaces

1. Completion of a metric space

We have already defined a complete space as one in which every fundamental sequence converges. Every metric space can be *completed*, that is, embedded in a complete space as an everywhere dense subset. The construction is analogous to that of the real numbers from the rational numbers.

Let E be a metric space. We consider all fundamental sequences of points of E. Two fundamental sequences $x = (x_1, x_2, \ldots)$ and $y = (y_1, y_2, \ldots)$ are called *equivalent* if $x_n y_n \to 0$. This is indeed an equivalence relation: reflexivity and symmetry are obvious, and transitivity follows from the triangle inequality:

$$x_n z_n \leq x_n y_n + y_n z_n \to 0.$$

The equivalence classes of fundamental sequences form the elements of a new space $\hat{E}$. We define the distance between two equivalence classes X and Y as follows: if $x = (x_1, x_2, \ldots) \in X$ and $y = (y_1, y_2, \ldots) \in Y$, then

$$XY = \lim_{n \to \infty} x_n y_n. \tag{1}$$

This limit exists (by VI of § 21) and is independent of the choice of representatives; for if $x' \in X$, $y' \in Y$, then $|x'_n y'_n - x_n y_n| \leq x'_n x_n + y_n y'_n \to 0$. The distance axioms are satisfied: the first two are obvious, and the triangle inequality follows from

$$x_n z_n \leq x_n y_n + y_n z_n$$

by passage to the limit.

The space E is embedded in $\hat{E}$ by identifying each point $x \in E$ with the equivalence class of the constant sequence $(x, x, x, \ldots)$. The distance between two points $x, y \in E$ is then the same in E and in $\hat{E}$, so that E is isometric with a subset of $\hat{E}$.

Theorem V.44 (I). *E is everywhere dense in $\hat{E}$.*

For let $X \in \hat{E}$ and $x = (x_1, x_2, \ldots) \in X$. Then the points $x_n \in E$ (considered as points of $\hat{E}$) converge to X; for

$$X x_n = \lim_{m \to \infty} x_m x_n \to 0 \quad \text{as} \quad n \to \infty,$$

since (x_n) is a fundamental sequence.

Theorem V.45 (II). *$\hat{E}$ is complete.*

Let $X^{(1)}, X^{(2)}, \ldots$ be a fundamental sequence in $\hat{E}$. Since E is everywhere dense in $\hat{E}$, for every n there exists a point $x_n \in E$ with $X^{(n)}x_n < 1/n$. Then

$$x_m x_n \leq x_m X^{(m)} + X^{(m)} X^{(n)} + X^{(n)} x_n < 1/m + X^{(m)} X^{(n)} + 1/n \to 0,$$

so that (x_n) is a fundamental sequence in E. Let X be the equivalence class of (x_n) in $\hat{E}$. Then

$$X^{(n)} X \leq X^{(n)} x_n + x_n X < 1/n + x_n X \to 0,$$

since $x_n X = \lim_m x_n x_m \to 0$. Thus $X^{(n)} \to X$, and $\hat{E}$ is complete.

2. Dyadic sets

Suppose that there is given, in a complete space E, a system of closed, bounded, non-empty sets V_{n_1}, $V_{n_1 n_2}$, $V_{n_1 n_2 n_3}$, ..., whose indices take on the possible values 1 and 2; then we have two sets V_1 and V_2 with a single index, four sets V_{11}, V_{12}, V_{21}, V_{22} with two indices and, in general, 2^k sets $V_{n_1 n_2 \cdots n_k}$ with k indices. We suppose further that

$$V_{n_1} \supseteq V_{n_1 n_2} \supseteq V_{n_1 n_2 n_3} \supseteq \cdots \tag{2}$$

and that the diameter of $V_{n_1 n_2 \cdots n_k}$ tends to 0 as $k \to \infty$, uniformly for all indices.

Theorem V.46 (III). *Under these assumptions, the intersection*

$$D = \sum_{n_1 n_2 \cdots} V_{n_1} V_{n_1 n_2} V_{n_1 n_2 n_3} \cdots \tag{3}$$

is non-empty. More precisely: for every sequence $(n_1, n_2, n_3, \ldots)$ *the sets* $V_{n_1}, V_{n_1 n_2}, V_{n_1 n_2 n_3}, \ldots$ *have exactly one point in common, and the set* D *of all these points is closed.*

For each sequence $(n_1, n_2, n_3, \ldots)$ choose a point $x_k \in V_{n_1 n_2 \cdots n_k}$. Since the diameters of these sets tend to 0, the sequence (x_k) is a fundamental sequence; since E is complete, it converges: $x_k \to x$. Since the sets $V_{n_1 n_2 \cdots n_k}$ are closed and decreasing, x belongs to all of them, and since their diameters tend to 0, it is the only point common to all of them.

The set D is closed. For if x is a point of accumulation of D, then there exists a sequence of distinct points $x^{(m)} \in D$ with $x^{(m)} \to x$. Each $x^{(m)}$ corresponds to a sequence of indices $(n_1^{(m)}, n_2^{(m)}, n_3^{(m)}, \ldots)$. Since the first indices $n_1^{(m)}$ can take only the values 1 and 2, there exists an infinite subsequence with the same first index n_1; from this subsequence we extract a further infinite subsequence with the same second index n_2, and so on. The diagonal sequence thus obtained corresponds to a fixed sequence $(n_1, n_2, n_3, \ldots)$, and its limit point x belongs to $V_{n_1 n_2 \cdots n_k}$ for every k, so that $x \in D$.

3. The Baire category theorem

A set is called *nowhere dense* if its closure contains no interior points; a set is called *of the first category* (in the sense of Baire) if it is the sum of countably many nowhere dense sets; otherwise, it is called *of the second category.*

Theorem V.47 (IV)**.** *(Baire category theorem). In a complete space, every set of the first category has an empty interior; that is, its complement is everywhere dense.*

Let $A = A_1 + A_2 + A_3 + \cdots$, where each A_n is nowhere dense. We have to show that in every open set G there exists a point not belonging to A.

Since A_1 is nowhere dense, $G - (A_1)_a$ is non-empty and open; let $x_1 \in G - (A_1)_a$ and let $U_1 = U(x_1, \epsilon_1)$ be a neighborhood with $U_1 \subseteq G - (A_1)_a$ and $\epsilon_1 < 1$. Since A_2 is nowhere dense, $U_1 - (A_2)_a$ is non-empty and open; let $x_2 \in U_1 - (A_2)_a$ and let $U_2 = U(x_2, \epsilon_2)$ be a neighborhood with $U_2 \subseteq U_1 - (A_2)_a$ and $\epsilon_2 < 1/2$. Continuing in this way, we obtain a decreasing sequence of neighborhoods $U_1 \supseteq U_2 \supseteq U_3 \supseteq \cdots$ with diameters tending to 0 such that $U_n \cdot A_n = 0$. Since the space is complete, there exists a point $x \in U_1 \cdot U_2 \cdot U_3 \cdots$, and since $x \in U_n$, we have $x \notin A_n$ for every n, so that $x \notin A$.

Corollary V.48. *A complete space is of the second category in itself.*

For if a complete space E were of the first category in itself, then $E = A_1 + A_2 + A_3 + \cdots$ with A_n nowhere dense, and E would have an empty interior, which is absurd.

Theorem V.49 (V)**.** *The intersection of countably many open everywhere dense sets in a complete space is everywhere dense.*

If $G_1, G_2, G_3, \ldots$ are open and everywhere dense, then $E - G_n$ are nowhere dense, so that $E - (G_1G_2G_3 \cdots) = (E - G_1) + (E - G_2) + \cdots$ is of the first category, and by IV, $G_1G_2G_3 \cdots$ is everywhere dense.

4. Uniform convergence

Let $x_n(t)$ be a sequence of functions defined on a set T with values in a metric space E. The sequence is said to converge *uniformly* to $x(t)$ if for every $\epsilon > 0$ there exists an N such that $x_n(t)x(t) < \epsilon$ for all $n > N$ and all $t \in T$.

Theorem V.50 (VI). *The limit of a uniformly convergent sequence of continuous functions is continuous.*

Let $x_n(t)$ be continuous and converge uniformly to $x(t)$. Let $t_0 \in T$ and $\epsilon > 0$. Choose n so large that $x_n(t)x(t) < \epsilon/3$ for all $t \in T$. Since $x_n(t)$ is continuous at t_0, there exists a neighborhood $U(t_0)$ such that $x_n(t)x_n(t_0) < \epsilon/3$ for all $t \in U(t_0)$. Then

$$x(t)x(t_0) \leq x(t)x_n(t) + x_n(t)x_n(t_0) + x_n(t_0)x(t_0) < \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon,$$

so that $x(t)$ is continuous at t_0.

5. Uniform continuity

A function $f : E \to E'$ between two metric spaces is called *uniformly continuous* if for every $\epsilon > 0$ there exists a $\delta > 0$ such that $f(x)f(y) < \epsilon$ for all $x, y \in E$ with $xy < \delta$.

Theorem V.51 (VII). *A continuous function on a compact set is uniformly continuous.*

Let f be continuous on the compact set $A \subseteq E$. Let $\epsilon > 0$. For each $x \in A$ there exists a $\delta_x > 0$ such that $f(x)f(y) < \epsilon/2$ for all $y \in A$ with $xy < 2\delta_x$. The neighborhoods $U(x, \delta_x)$ cover A, and since A is compact, there exists a finite subcover $U(x_1, \delta_{x_1}), \ldots, U(x_n, \delta_{x_n})$. Let $\delta = \min(\delta_{x_1}, \ldots, \delta_{x_n})$.

If $x, y \in A$ and $xy < \delta$, then $x \in U(x_i, \delta_{x_i})$ for some i, so that $xx_i < \delta_{x_i}$. Furthermore,

$$yx_i \leq yx + xx_i < \delta + \delta_{x_i} \leq 2\delta_{x_i}.$$

Thus

$$f(x)f(y) \leq f(x)f(x_i) + f(x_i)f(y) < \epsilon/2 + \epsilon/2 = \epsilon,$$

so that f is uniformly continuous.

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Chapter VI (continued): Point Sets

Chapter VII: Point Sets and Ordinal Numbers

VI. Point Sets

§26. Complete Spaces *(continued)*

$W_{ik}, W_{ikl}, W_{iklm} \ldots$, where this time each index ranges over a finite set which may possibly depend on the preceding indices but which contains at least two numerals. In W_i, that is, we let $i = 1, 2, \ldots, m$; in W_{ik} we let $k = 1, 2, \ldots, m_i$; in W_{ikl} we let $l = 1, 2, \ldots, m_{ik}$; and so on. $(m, m_i, m_{ik}, \ldots \geq 2.)$ For every sequence $(i, k, l, \ldots)$ of numerals formed in this way, let

$$W_i \supseteq W_{ik} \supseteq W_{ikl} \supseteq \cdots$$

with diameters converging to zero, so that the intersection of the sets consist of a single point; the sum

$$P = \Sigma W_i W_{ik} W_{ikl} \cdots \tag{1}$$

will be called a *polyadic set*. We conclude, as we did above, that

$$P = \Sigma W_i \cdot \Sigma W_{ik} \cdot \Sigma W_{ikl} \cdots$$

and that the convergence of the diameters to zero is uniform. In actual fact, P is really only a dyadic set, as we shall now see.

The sets W_i, W_{ik}, $\ldots$ we denote in general by W; the sets derived from some set W by adjoining another index we denote by W', W'', ... (for example, for $W = W_{ik}$: $W' = W_{ikl}$). Next we form sets V^1, V^2, ... with dyadic indices (1 or 2) which, again, we denote in general by V. The sets derived from the sets V by adjoining a further index, we call V^1, V^2. In this construction we shall require either that V equal some particular W

$$(\alpha) \quad V = W,$$

or that V be the sum of several W with equally many indices

$$(\beta) \quad V = W' + W'' + \cdots$$

where the summands are to be written in an order corresponding to the lexicographic order of the indices. We begin the definition with

$$V^1 = W_1' + W_2' + \cdots , \quad V^2 = W_1'' + W_2'' + \cdots$$

and continue it by induction by defining V^1, V^2 whenever V has already been defined. In the case (α), we let

$$V^1 = W'^1 + W''^1 + \cdots , \quad V^2 = W'^2 + W''^2 + \cdots ;$$

and in the case (β), we let

$$V^1 = W'^1 + W''^1 + \cdots , \quad V^2 = W'^2 + W''^2 + \cdots .$$

As an example, if the first sets W are

$$\begin{array}{cccccc} W_1 & W_2 & W_3 & & & \\ W_{11} & W_{12} & W_{13} & W_{21} & W_{22} & W_{23} \end{array}$$

the first sets V would be derived as follows:

$$\begin{array}{llll} V_1 = W_1 + W_2, & V_2 = W_3 & & \\ V_{11} = W_{11}, & V_{12} = W_{12}, & V_{21} = W_{21} + W_{22}, & V_{22} = W_{23} \\ V_{111} = W_{11}, & V_{112} = W_{12}, & & \\ V_{121} = W_{21}, & V_{122} = W_{22}, & V_{123} = W_{23}. & \end{array}$$

We can now see directly that every V is some W or the sum of a finite number of W, and is therefore closed, bounded, and non-empty. Also $V \supseteq V^1$, $V \supseteq V^2$, so that the inequalities (1) are satisfied for every dyadic sequence of numerals. If V is of type (β) then V^1, V^2 are sums of fewer terms than V, and after the adjunction of sufficiently many indices, some V of type (α) always recurs. Every sequence $V_{p_1}, V_{p_1p_2}, V_{p_1p_2p_3} \ldots$ has a subsequence $W_{i_1}, W_{i_1i_2}, W_{i_1i_2i_3} \ldots$ and therefore has diameters that converge to zero, where the intersections

$$V_{p_1}V_{p_1p_2}V_{p_1p_2p_3}\cdots = W_{i_1}W_{i_1i_2}W_{i_1i_2i_3}\cdots$$

are simultaneously identical. Conversely, every W is equal to a certain V, and every sequence

$$W_{i_1}, W_{i_1i_2}, W_{i_1i_2i_3}, \ldots$$

determines a sequence

$$V_{q_1}, V_{q_1q_2}, V_{q_1q_2q_3}, \ldots$$

of which it is a subsequence (for example, the sequence W_1, W_{11}, … determines the sequence V_1, V_{11}, V_{111}, …). Hence the polyadic set P generated by the W is the same as the dyadic set D generated by the V.

A set P compact in itself can be represented as a polyadic set. For it is the sum of a finite number of sets W_i of arbitrarily small diameters $\leq \delta_1$, where each of these sets may be taken to be closed (in P) and thus to be compact in itself and where the number of these sets can be taken to be ≥ 2. Continuing, we find that

$$P = \Sigma W_i, \quad W_i = \Sigma W_{ik}, \quad W_{ik} = \Sigma W_{ikl}, \quad \ldots$$

Fig. 3

where the W are compact in themselves (and thus complete and bounded), and where the diameters of the sets with l indices are $\leq \delta_l$, with $\delta_l > 0$; P is the polyadic set (4) generated by the sets W.

Dyadic sets and sets compact in themselves are therefore identical.

We next consider in particular a dyadic set where V^1 and V^2 are disjoint and, similarly $V^{p_1 1}$ and $V^{p_1 2}$, $V^{p_1 p_2 1}$ and $V^{p_1 p_2 2}$, ... are disjoint; in general for every n the sets with n indices are to be pairwise disjoint. Such a set

$$D = \Sigma V_{p_1} \cdot \Sigma V_{p_1 p_2} \cdot \Sigma V_{p_1 p_2 p_3} \cdots = \Sigma V_{p_1} + \Sigma V_{p_1 p_2} + \Sigma V_{p_1 p_2 p_3} + \cdots$$

will be called a *dyadic discontinuum*.[1] It is equivalent to the set of dyadic sequences of numerals and is therefore of the cardinality $2^{\aleph_0} = \aleph$ of the continuum; furthermore it is perfect and, even more precisely,

$$D = D' = D'' = D''' = \cdots ;$$

every point of D is a condensation point. For if a finite number of initial numerals are held constant, we obtain the sets DV_{p_1}, $DV_{p_1 p_2}$, $DV_{p_1 p_2 p_3} \ldots$ (for instance $DV_1 =$ the set of points $x \in D$ for which $p_1 = 1$) which are all of cardinality $\aleph$ and have diameters convergent to 0; if x is the point in which these sets intersect, then every neighborhood U_x contains $\aleph$ points of D.

A polyadic discontinuum (where the sets W with n indices are disjoint, for every n) is identical with a dyadic one.

The classical example of a dyadic discontinuum is the following which, because of its relation to the triadic fractions, is also referred to as Cantor's triadic (or ternary) set, or Cantor's discontinuum. Let V be the closed

[1] The name will become clear in §29, XVI. In §36, 1 we shall introduce as a counterpart the dyadic continuum.

Fig. 4

interval $[0,1]$ in the set E of real numbers; furthermore (forming the dyadic sequences with the numerals $0,2$ instead of $0,1$, for a reason that[2] will soon become apparent), let $V_1 = [0, \frac{1}{3}]$ be the left-hand third of V and $V_2 = [\frac{2}{3}, 1]$ the right-hand third and, similarly, let $V_{p_1 1}$ be the left-hand third of V_{p_1} and $V_{p_1 2}$ the right-hand third, and so on. In this case, the intersection $V_{p_1} V_{p_1 p_2} V_{p_1 p_2 p_3} \cdots$ consists, as a moment's reflection will show, of the number

$$\frac{p_1}{3} + \frac{p_2}{3^2} + \frac{p_3}{3^3} + \cdots = 0.p_1 p_2 p_3 \ldots \quad (p_n = 0 \text{ or } 2),$$

and D is the set of those numbers x that admit of at least one such triadic development entirely in terms of zeros and twos without any ones. The qualification *at least* has reference to those numbers between zero and one (triadic rational numbers) that have two triadic developments; if one of them is free of ones, then the number is to belong to D. For example, the numbers

$$\frac{1}{3} = 0.1 = 0.0222\ldots, \quad \frac{2}{3} = 0.2 = 0.1222\ldots, \quad \frac{1}{9} = 0.01 = 0.00222\ldots$$

belong to D, but a number between these two does not, since its first triadic numeral will certainly be $p_1 = 1$.

The open complement $E - D$ consists, apart from the half-lines $(-\infty, 0)$ and $(1, \infty)$, of the middle thirds of V, V_{p_1}, $V_{p_1 p_2}$, ..., and so consists of the open intervals $(\frac{1}{3}, \frac{2}{3})$, $(\frac{1}{9}, \frac{2}{9})$, $(\frac{7}{9}, \frac{8}{9})$, ... As is easily seen, this is dense in E, so that D contains no interval no matter how small (D is nowhere dense in E, §27). The limiting value to which the lengths of

[2] Fig. 3 illustrates the case of closed rectangular areas V only the circumferences of which are shown, the interiors being left unshaded.

the sums of the intervals $2V_{p_1}$, $2V_{p_1p_2}$, ... converge is called the (Lebesgue, one-dimensional) measure of D. Here these lengths are $2^n \cdot 3^{-n}$, and so the measure is zero. It seems paradoxical that a set of measure zero should nonetheless have the cardinality $\aleph$ of the whole interval. We can, however, choose the lengths of the intervals differently: those of the open intervals of $E - D$ smaller and the sums of the lengths λ_1, $\lambda_2 \ldots$ of $2V_{p_1}$, $2V_{p_1p_2} \ldots$ decreasing so slowly that the measure $\lambda = \lim \lambda_n$ of D becomes positive and even as close as we please to (although of course less than) 1. But then we have another paradox: namely, that a set of positive length contains no interval no matter how small.

How to carry over this simple construction to the case of the plane is obvious. For instance, we take a square V (a closed square area defined by, say, $0 \leq x_1 \leq 1$, $0 \leq x_2 \leq 1$; that is, the product of two intervals

[0, 1]) and divide it into nine sub-squares by dividing the sides into three equal pieces, V_1, V_2, V_3, V_4 being the small squares in the corners of V.

Fig. 5

$V - \Sigma V_p$ is a cross formed by the five remaining sub-squares with some of the edges missing. In the same way, we form the sub-squares $V_{p_1 1}$, $V_{p_1 2}$, $V_{p_1 3}$, $V_{p_1 4}$ from V_{p_1}. The polyadic discontinuum generated by the V,

$$D = \Sigma V_{p_1} \cdot \Sigma V_{p_1 p_2} \cdot \Sigma V_{p_1 p_2 p_3} \cdots$$

is the product of two Cantor triadic sets. Its (Lebesgue, two-dimensional) measure — the limit of the areas of $2V_{p_1}$, $2V_{p_1 p_2}$, ... — is zero, but it can also be made positive < 1 by changing the construction slightly.

3. Theorems on Cardinality

VI. (Cantor). *In a complete space, every non-empty perfect set has at least the cardinality* $\aleph$.

VII. (Cantor). *In a complete space, every closed set whose dense-in-itself kernel does not vanish has cardinality at least* $\aleph$.

VIII. (W.H. Young). *In a complete space, every set* $Y = G_\delta$ *(the intersection of a sequence of open sets) whose dense-in-itself kernel does not vanish has cardinality at least* $\aleph$.

We premise the proof with a few explanatory remarks. Each of these theorems is a special case of the one following (a closed set F is the intersection of a sequence of open sets G). On the other hand, VII also follows from VI; and later on (because of the homeomorphism of the G_δ in complete spaces with the complete spaces themselves, §38, III), VIII will also be found to be a consequence of VII. The sets

$$G_\delta = \text{intersection of a sequence of open sets,}$$
$$F_\sigma = \text{sum of a sequence of closed sets}$$

form the next class of Borel sets (§32) following the open sets G and the closed sets F, and in the sequel we shall extend the theorems on cardinality to them. These attributes of sets are of course relative, being dependent on the space E, and actually we ought to write explicitly $F(E)$, and so on. If, however, we once more ignore the argument E, then we have, for $D \subseteq E$,

$$F(D) = DF, \quad G(D) = DG, \quad F_\sigma(D) = DF_\sigma, \quad G_\delta(D) = DG_\delta;$$

the sets with respect to D are the intersections with D of the corresponding sets referred to E. There also exist absolute F_σ and G_δ, sets that have this attribute in every space containing them. This is obvious for the F_σ; an absolute F_σ (considered, for example, as a subset of its complete hull) is the sum of a sequence of absolutely closed (complete) sets belonging to some metric space, and vice versa. At first glance, the non-existence of absolutely open sets appears to militate against the existence of absolute G_δ; But the truth is otherwise: the three statements

A is a G_δ in its complete hull;

A is a G_δ in a complete space;

A is a G_δ in every containing space, that is, an absolute G_δ;

are equivalent. For if A is a G_δ in E, then this is also the case in every smaller space D ($A \subseteq D \subseteq E$); if, on the other hand, A is a G_δ in its closed hull A_0, then $A = A_0 G_\delta$ is a G_δ even in the space E, because A_0 is closed and is thus a G_δ, and the intersection of two G_δ is itself a G_δ. The equivalence of the three statements for a complete space E now follows. In consideration of Young's Theorem VIII, the absolute G_δ are often called *Young sets*.

We can now eliminate the reference to the containing space in the three cardinality theorems and say:

IX. *Every absolutely perfect set, every absolutely closed (complete) set, every absolute G_δ (Young set) has cardinal-*

ity at least $\aleph$, provided that its dense-in-itself kernel is non-vanishing.

Let us now turn to the proof; we need only prove the third theorem but, in the course of doing so, we shall come across the first two. We shall show that the sets in question contain a dyadic discontinuum D; as the generating closed (complete) sets V we shall choose closed spheres. We distinguish here between the

$$\text{open sphere } U_\varrho(x) = \text{set of points } y \text{ for which } xy < \varrho$$
$$\text{closed sphere } V_\varrho(x) = \text{set of points } y \text{ for which } xy \leq \varrho$$

(the open spheres are our old neighborhoods); we speak of them as corresponding, or belonging, to each other whenever they have the same radius and center. Spheres U and V belonging to each other are labelled with the same subscripts. (In Euclidean space, U is then the open kernel of V, and V the closed hull of U.)

Now let $A > 0$ be a dense-in-itself set in the complete space E. We choose two points a_1, a_2 in A and make them the centers of disjoint closed spheres V_1, V_2, to which there belong the open spheres U_1, U_2. Each of the two sets AU_i is non-empty and dense-in-itself (§23, V); we can therefore repeat the construction and circumscribe disjoint closed spheres V_{i1}, $V_{i2} \subseteq U_i$ about two of their points a_{i1}, a_{i2}. Each of the four sets AU_{ij} is non-empty and dense-in-itself; we circumscribe disjoint closed spheres V_{ij1}, $V_{ij2} \subseteq U_{ij}$ about two of their points a_{ij1}, a_{ij2}, and continue in this way.

Nothing in all this prevents us from choosing the radii of these spheres as small as we please — say, choosing the radii of the V_{p_1}, $V_{p_1p_2}$, $V_{p_1p_2p_3} \ldots$ smaller than $1, \frac{1}{2}, \frac{1}{3}, \ldots$. In this way we obtain a dyadic discontinuum D, and this is contained in A; for a point x of D that belongs to $V_{p_1}V_{p_1p_2}V_{p_1p_2p_3}\cdots$ is the limit of the centers a_{p_1}, $a_{p_1p_2}$, $a_{p_1p_2p_3} \ldots$ of the spheres. Therefore:

X. *If A is a dense-in-itself non-empty set in the complete space E, then A contains a dyadic discontinuum D.*

This proves VI (if A is dense-in-itself then A_0 is perfect, and vice versa), and even proves VII: If F is closed and contains A, then $F \supseteq A \supseteq D$. In order to prove VIII, we assume

$$A \subseteq Y = G_1G_2G_3\cdots \quad (G_n \text{ open}),$$

and we are still free to take the radii of the spheres so small that $V_{p_1} \subseteq G_1$ (since, after all, $a_{p_1} \in G_1$), $V_{p_1p_2} \subseteq G_2$, $V_{p_1p_2p_3} \subseteq G_3$, $\ldots$ and thus $D \subseteq Y$.

Our theorems are thus proved; we have even shown somewhat more, namely, that the sets in question contain a perfect subset D. Further — to stay with the most general case (VIII) of a Young set Y — since DV_{p_1} has cardinality $\aleph$ and V_{p_1} may be taken to refer to an arbitrarily small sphere about any point a_i of A, it follows that every point of A is a point of condensation of Y, and since A was an arbitrary dense-in-itself subset of Y, this means: Every point of the dense-in-itself kernel of Y is a point of condensation of Y,

$$Y' \subseteq Y_0. \tag{2}$$

Let us finally combine all this with the facts valid in a separable space (§25): in this case every point set has at most the cardinality $\aleph$; furthermore, we always had $A_0 = A_\varrho$, so that for a Young set $Y_0 = Y_\varrho$, and

$A \setminus A_0$ is at most countable. Thus:

XI. *In a complete separable space every Young set* $Y = G_\delta$ *is either at most denumerable or of cardinal number* $\aleph$, *depending on whether its dense-in-itself kernel* $Y_0 = Y_\varrho$ *vanishes or not.*

For a closed set, $Y_0 = Y_\varrho$; for a perfect set, $Y = Y_\varrho$. (That every set A_0 is perfect is valid, by p. 147, in every separable space; here we also find the converse, that every perfect set is an A_ϱ.)

In a separable space every closed set F can be split by virtue of $F = F_\varrho + F'_\varrho$ into a perfect part F_ϱ and an at most countable part F'_ϱ (the dense-in-itself kernel of a closed set is perfect (p. 141), a fact usually called the *Theorem of Cantor-Bendixson.* The decomposition $A = P + Q$ of a set into a perfect part P and an at most countable part Q, however, is not in general unique; if, for example, E is the space of rational numbers, and P the set of rational numbers $\neq a$ (a arbitrary), then $E = P + (E - P)$ is such a decomposition, for P is perfect in E. But in a complete separable space this splitting, if possible at all, can be made in only one way; for we have $A_\varrho = P_\varrho + Q_\varrho = P + 0 = P$. The splitting, then, can only be performed if $A_\varrho \subseteq A$ (and therefore, in particular, for closed sets), and then only as follows: $A = A_\varrho + (A - A_\varrho) = A_\varrho + A'_\varrho$; for Young sets of this kind, this is identical with $A = A_\varrho + A'_\varrho$.

A countable set of A of the complete space E whose dense-in-itself kernel does not vanish is certainly not a G_δ, for $A_\varrho > 0$ and $A'_\varrho = 0$ contradicts the condition $Y_\varrho = Y'_\varrho$. But as a countable set it is an F_σ. Example: The set of rational points in Euclidean space. Its complement $E - A$ is a G but not an F_σ.

§27. Sets of the First and Second Categories

We consider an arbitrary space E. The closed set F will be said to be *nowhere dense* in E if its open complement G is dense in E. This is equivalent with $G_0 = E$ and thus with $F_i = 0$: closed nowhere dense sets are identical with closed border sets. They are also identical with the *frontiers*, or *boundaries*,[3] of open sets; for if F is nowhere dense, then $F = F_r$ is the frontier of its complement G, and, on the other hand, if G is any open set and F its complement, then the frontier F_r of G is a border set, and furthermore, like every frontier set, it is closed.

An arbitrary set A will be said to be *nowhere dense* in E if its closed hull A_0 is nowhere dense in E. It is both necessary and sufficient for this that the open kernel B_i of its complement $B = E - A$ be dense in E.

[3]We use the terms interchangeably: see the footnote on p. 130.

Ordinary curves in the Euclidean plane E_2, as well as ordinary curves and surfaces in the space E_3, etc. are nowhere-dense sets. More precisely: If $f(x_1, x_2)$ is a real continuous function that can never vanish inside an entire neighborhood of a point — for instance a polynomial whose coefficients do not all vanish — then $f = 0$ represents a nowhere-dense set in E_2, for it is closed and has no interior points.

A dyadic discontinuum D (§26, 2) is nowhere-dense in the complete space E if we require the more stringent inequalities

$$V_{p_1} > V_{p_1 1} + V_{p_1 2}, \quad V_{p_1 p_2} > V_{p_1 p_2 1} + V_{p_1 p_2 2}, \quad \ldots$$

to hold (where equality is excluded). For, arbitrarily close to every point $x \in D$ — which belongs, say, to the intersection $V_{p_1} V_{p_1 p_2} \cdots$ — there are points of

$$(V_{p_1} - V_{p_1 1} - V_{p_1 2}) + (V_{p_1 p_2} - V_{p_1 p_2 1} - V_{p_1 p_2 2}) + \cdots$$

and these points do not belong to D: x is not an interior point of D. In the construction involving spheres on p. 158 this strengthening of the inequalities can clearly be brought about; therefore, in every complete space with a non-vanishing dense-in-itself kernel there are nowhere-dense perfect sets of cardinality $\aleph$. The simplest example is Cantor's triadic set (p. 154), together with the sets derived from it by alteration of the lengths of the intervals.

Let us return to the discussion of the nowhere-dense sets A of E. For brevity, we set

$$E = A_i + A_e,$$

where $A_e = E - A_i$ denotes the set of the exterior points of A (the interior points of the complement $E - A$). Then the statements

$$A \text{ is nowhere dense in } E, \quad A_e \text{ is dense in } E$$

are identical. Let us now replace E by any subset M of the space; $A_i(E) = A_i$ then has to be replaced, of course, by $A_i(M) = MA_i$. It is not required that A be contained in M, but only that A and M be sets in the space E, so that we obtain the more general relation 'A nowhere dense to M' as counterpart to the already familiar relation 'A dense to M'. In short, using the notation

$$P = MA_i, \quad Q = MA_e, \quad M = P + Q, \tag{3}$$

we define the statements

$$A \text{ is nowhere dense to } M, \quad Q \text{ is dense in } M$$

to be identical; by the earlier definition ($Y \subseteq A_0$), the statements

A dense to M, $MA_0 = M$, $Q = 0$, $P = M$ were identical. (The complete counterpart to this would be $Q = M$, $P = 0$, $M \subseteq A_e$, M lies completely outside A; but this is of no particular interest.) In the case $A \subseteq M$, we choose the preposition 'in': A is nowhere dense in M, A is nowhere dense relative to M, or briefly, A is nowhere dense M.

The relation 'A nowhere dense to M' is transitive, that is:

I. *If A is nowhere dense to M, and B is nowhere dense to MA_0, then $A + B$ is nowhere dense to M.*

For the open kernel of the complement of $A + B$ is $(A + B)_i = A_i B_i$, and this, as the intersection of sets dense in M and MA_0 respectively, is dense in M (§23, VI).

The statements 'A nowhere dense to M' and 'A nowhere dense to MA_0' are equivalent. For if Q is dense in M, then $Q \cdot MA_0 = MA_e A_0 = MA_0 \cdot (A_0)_e$ is dense in MA_0; conversely, if Q is dense in MA_0, then $Q + A_e \supseteq A_e$ is dense in M (§23, VII) and, since A_e is open, Q is also dense in M (§23, VIII). From this and from I it follows that:

II. *The sum of finitely many sets that are nowhere dense to M is itself nowhere dense to M.*

III. *If $N \subseteq M$ and A is nowhere dense to M, then A is nowhere dense to N.*

For from (1) we obtain the corresponding decomposition $N = NP + NQ$, and NQ is dense in N because Q is dense in M (§23, V).

A set A that is nowhere dense to itself is called *scattered*. A nowhere-dense set is not necessarily scattered; the set of rational numbers, for example, is nowhere dense (in the space of real numbers), but it is not scattered. For a closed set, the attributes of being nowhere dense and scattered are equivalent: if F is nowhere dense, then $F_i = 0$, and F as a subset of F is nowhere dense to itself; conversely, if F is scattered, then $F_i(F) = FF_i = 0$, so that $F_i = 0$.

IV. *Let G be open. If A is dense to M, then AG is dense to MG. If A is nowhere dense to M, then A is nowhere dense to MG.*

The first part follows from $(AG)_0 \supseteq A_0G \supseteq MG$. The second rests on the first, in that it follows from the second that the complement of A is dense to MG, that is, A_eG is dense to MG (since A_e is open).

If we form the sum of a sequence of sets that are nowhere dense to M:

$$A = A_1 + A_2 + \cdots \quad (A_n \text{ nowhere dense to } M),$$

then A is not in general nowhere dense to M. If, however, the property of being nowhere dense to M does carry over from the summands to the sum, then we say that M is of the *second category*; in the opposite case, M is of the *first category*.

We can also say: M is of the first category if it is the sum of a sequence of sets that are nowhere dense to M, that is, if it can be decomposed into countably many parts that are nowhere dense to the whole; M is of the second category if it cannot be so decomposed.

M is certainly of the first category if it is nowhere dense to itself, that is, if its open kernel vanishes: $M_i = 0$ (a border set). For then $M = M \cdot M$ is itself nowhere dense to M and can be taken as the first term of a sequence of nowhere-dense summands, to which the remaining null sets may be added. If, on the other hand, $M_i > 0$, then M is certainly not nowhere dense to itself, but it may still be decomposable into two nowhere-dense parts, these into four such parts, and so on; if this process can be continued through a sequence of steps, then M is of the first category.

A set that is nowhere dense to M is, by III, also nowhere dense to every subset of M, and so:

V. *Every subset of a set of the first category is itself of the first category (relative to the original set).*

The sum of countably many sets of the first category is of the first category. For if $M = M_1 + M_2 + \cdots$ and $M_n = A_{n1} + A_{n2} + \cdots$, where the A_{nk} are nowhere dense to M_n, then the A_{nk} are also nowhere dense to M (III), and M is the sum of countably many nowhere-dense sets A_{nk}.

VI. *The sum of countably many sets of the first category to M is of the first category to M.*

We are speaking here of course of sets that are nowhere dense not merely to M but to the summands as well, that is, of sets A_n nowhere dense to $M_n \subseteq M$ with $M = \Sigma M_n$. But then A_n is nowhere dense to M (III), and $M = \Sigma A_n$ is the sum of countably many nowhere-dense sets.

If the sum theorem is also valid for an arbitrary (not merely countable) number of summands, that is, if the sum of any number of sets nowhere dense to M is itself nowhere dense to M, then the sets nowhere dense to M form an ideal in the Boolean algebra of the subsets of M (§4, II); and the sets of the first category are precisely the elements of this ideal. In any case, these sets form a *ring*.

A space of the second category cannot be decomposed into countably many nowhere-dense closed sets. In particular, a complete space is of the second category (§26, I).

VII. *A complete space is of the second category.*

This follows from the First Intersection Theorem: If $E = F_1 + F_2 + \cdots$, where the F_n are nowhere-dense closed sets, then $E - F_n = G_n$ are dense open sets, and their intersection would be empty, whereas it is actually dense in E (§26, I).

A closed set F in a complete space is of the second category in itself, that is, relative to F, unless it is nowhere dense (to E). For if $F_i > 0$, then F_i is dense in F, and F as a complete space is of the second category.

A set of the second category cannot be nowhere dense, because a nowhere-dense set is of the first category. Therefore:

VIII. *If M is of the second category, then M_0 is dense in M.*

Since a set of the first category is nowhere dense if and only if it is closed, it follows that:

IX. *If M is of the second category and A is of the first category to M, then $M - A$ is of the second category to M.*

For if $M - A$ were of the first category to M, then $M = A + (M - A)$ would also be of the first category to M (VI).

In particular, in a space E of the second category, the complement of a set of the first category is of the second category.

X. *In a space of the second category, a set A and its* *complement $E - A$ cannot both be of the first category.*

For $E = A + (E - A)$.

A set M of the second category is *everywhere of the second category*, that is, every non-empty subset $N \subseteq M$ that is open in M (every $N = MG$, where $G > 0$ is open) is of the second category (relative to N or to M). For if N were of the first category relative to M, then $M = N + (M - N)$ would also be of the first category (VI), since $M - N$ is nowhere dense in M (it is closed in M and has empty interior there).

Conversely, if every non-empty open subset of M is of the second category (relative to M), then M is everywhere of the second category. In particular, a space is everywhere of the second category if every non-empty open set is of the second category.

In a space of the second category, the sets of the first category form an ideal. The corresponding residue classes are the sets *modulo sets of the first category.* Two sets are called equivalent modulo sets of the first category if their symmetric difference is of the first category. A set that is equivalent to an open set is called a set having the *Baire property.* The sets having the Baire property form a field (§4, III), which contains all the open and all the closed sets, and therefore also the F_σ and G_δ sets, and so on.

An important theorem is:

XI. (Baire). *In a complete space, every non-empty open set is of the second category; in other words, a complete space is everywhere of the second category.*

More generally:

XII. *In a complete space, the intersection of a sequence of dense open sets is dense.*

For the proof, let $G_1, G_2, \ldots$ be dense open sets and H any non-empty open set. We have to show that $H \cdot G_1 G_2 \cdots > 0$. Now HG_1 is open and > 0; we choose a closed sphere $V_1 \subseteq HG_1$ of radius < 1. Since G_2 is dense and open, $V_1 G_2$ contains a closed sphere V_2 of radius $< \frac{1}{2}$; since G_3 is dense and open, $V_2 G_3$ contains a closed sphere V_3 of radius $< \frac{1}{3}$, and so on. The intersection $V_1 V_2 V_3 \cdots$ is a point of $HG_1 G_2 \cdots$ (First Intersection Theorem, §26, I), and so this set is > 0, q.e.d.

From this we obtain anew the theorem that a complete space is of the second category (VII). For if $E = A_1 + A_2 + \cdots$, where the A_n are nowhere dense, and if $G_n = E - (A_n)_0$, then

G_n is dense and open, and the intersection $G_1 G_2 \cdots$ is dense and therefore > 0; but it is the complement of $A_1 + A_2 + \cdots = E$, which is absurd.

But there are also G_δ-sets that are not F_σ-sets. Let E be the Euclidean plane, and let A be the set of irrational points $(x, 0)$ on the x-axis X; then A is a G_δ and $D = E - A$ is also a G_δ-set. However, it is not an F_σ-set, since the set $D \cap X$ which is closed in it, the set of rational points, is not a G_δ in X (because its complement, the set of irrational points, is not an F_σ in X).

§28. Spaces of Sets

1. *The distance of two sets.* In §23, we considered the lower and upper distances between two sets A, B:

$$\begin{aligned} \delta(A, B) &= \inf ab && (a \in A,\ b \in B), \\ d(A, B) &= \sup_{x \in A} \inf_{y \in B} xy && = \inf_{y \in B} \sup_{x \in A} xy, \end{aligned}$$

where $\delta(A, B) = 0$ means that A and B have points of accumulation in common or that one of the sets contains a point of accumulation of the other. $d(A, B) = 0$ means that $A \subseteq B_0$ and $B \subseteq A_0$ (A and B are of the same density class, or, more briefly, A and B are *dense to each other*). Further, we called $\delta(a, B)$ the distance of the point a from the set B, and $d(a, B)$ the distance from a to the farthest point of B.

In general, $\delta(A, B) \neq \delta(B, A)$ and $d(A, B) \neq d(B, A)$; for closed bounded sets > 0, however, we always have $d(A, B) = d(B, A)$.

If $B \subseteq A_0$, then $\delta(A, B) = 0$ means that every neighborhood $U(B, \varrho)$ contains points of A, that is, that B belongs to the closure of the system of neighborhoods $U(A, \varrho)$. More generally, if B has the property that for every $\varrho > 0$ there exists some $a \in A$ for which $ab < \varrho$ or (ii) that b belongs to the neighborhood $U(A, \varrho)$ of A with radius ϱ (p. 135).

We shall now try to define a distance between two sets, one that will satisfy the distance axioms and thus enable us to treat the sets as elements of a new metric space. The upper and lower distances, as we might have suspected and as we can readily confirm by trial, are not suitable for this purpose.

We consider only bounded sets A, $B > 0$. The relation $(\varrho > 0)$

$$B \subseteq U(A, \varrho), \tag{4}$$

for every $b \in B$, $\delta(A, b) < \varrho$,

for every $b \in B$, there is an $a \in A$ for which $ab < \varrho$,

the three forms of which are equivalent, certainly holds for $\varrho > d(A, B)$ (for in this case, all $ab < \varrho$) and does not hold for $\varrho = \delta(A, B)$ (for in this case, all $ab \geq \varrho$). The greatest lower bound of the numbers ϱ for which it holds is a number ≥ 0, which we denote by $\varrho(A, B)$ and which is to be distinguished from $\varrho(B, A)$; from the second form of the relation (ε) we get

$$\varrho(A, B) = \sup_{b \in B} \delta(A, b), \tag{5}$$

and further

$$\delta(A, B) \leq \varrho(A, B) \leq d(A, B). \tag{6}$$

The relation (ε) holds for $\varrho > \varrho(A, B)$; for $\varrho < \varrho(A, B)$, it does not (and for $\varrho = \varrho(A, B)$ it is valid if and only if the supremum in (1) is not attained for any b). $\varrho(A, B) = 0$ means the same as $B \subseteq A_0$, or, A is dense to B; in every other case, $\varrho(A, B) > 0$. In addition, we have

$$\varrho(A, B) + \varrho(B, C) \geq \varrho(A, C). \tag{7}$$

For if $B \subseteq U(A, \varrho)$, $C \subseteq U(B, \sigma)$, then $C \subseteq U(A, \varrho + \sigma)$, from which the conclusion easily follows. If one of the sets consists of a single point, then by virtue of (1),

$$\varrho(a, B) = \sup_{b \in B} ab = d(a, B), \quad \varrho(A, b) = \delta(A, b). \tag{8}$$

Because of its asymmetry, this $\varrho(A, B)$ is still not suitable as a distance; therefore we form

$$AB = \max[\varrho(A, B), \varrho(B, A)] = BA \tag{9}$$

and conclude from (3) that this satisfies the triangle inequality

$$AB + BC \geq AC.$$

We have therefore defined a distance, and thus a metric space of sets, provided that, in analogy with earlier cases, we consider as identical two sets for which $AB = 0$, that is, two sets of the same denseness class; for $AB = 0$, we also have $\varrho(A, C) = \varrho(B, C)$ and $\varrho(C, A) = \varrho(C, B)$ and $AC = BC$ as well.

By virtue of (2), the distance AB lies between the lower and upper distances; when one of the sets contains only a single point, then from (4) we find that

$$Ab = \max[\delta(A, b), d(A, b)] = d(A, b).$$

Hereafter, for the sake of precision of speech, we shall retain only the greatest (closed) set of each density class, thus restricting the domain of definition of $\varrho(A, B)$ and AB to the closed bounded sets > 0 of the space E. In this metric space convergence, $A = \lim A_n$ or $A_n \to A$, must of course be defined by means of $AA_n \to 0$; the sequence is then called *metrically convergent*; A, its *metric limit*; and the limit of a metrically convergent subsequence of A_n, a *metric element of accumulation*.

2. Closed and open limit

Next we must take into account, at first for any sequence of sets A_n of the metric space E, the following limit sets, which are analogous to those for pure sets (upper and lower limit, §3): the upper and lower closed limits

$$F = \overline{\lim} A_n, \quad \underline{F} = \underline{\lim} A_n, \tag{10}$$

and, in the case of equality, the *closed limit*

$$F = \lim A_n; \tag{11}$$

as well as the upper and lower open limits

$$G = \overline{\lim} A_n, \quad \underline{G} = \underline{\lim} A_n, \tag{12}$$

and, in the case of equality, the *open limit*

$$G = \lim A_n, \tag{13}$$

(the abbreviations Fl and Gl may be read, say, as F-limit and G-limit). The points of these various sets are defined by the following properties:

$x \in \overline{F}$: every neighborhood U_x has points in common with infinitely n

$x \in \underline{F}$: every neighborhood U_x has points in common with almost all

$x \in \overline{G}$: there exists a neighborhood U_x that belongs to infinitely many

$x \in \underline{G}$: there exists a neighborhood U_x that belongs to almost all the

We have: $\overline{F} \supseteq \underline{F}$, $\overline{G} \supseteq \underline{G}$. If $B_n = E - A_n$ are the complements of the A_n, then, as a moment's reflection will show,

$$E = \overline{F} + \underline{G} = \underline{F} + \overline{G}; \tag{14}$$

that $\overline{G}$ and $\underline{G}$ are open follows immediately from the definition: $\overline{G}$ and $\underline{G}$ are the sum of all the neighborhoods that belong respectively to infinitely many and almost all of the A_n; nearly as directly, or else from (10), it follows that $\overline{F}$, $\underline{F}$ are closed. The alteration, omission, or introduction of a finite number of sets has no influence on the limit sets; in passing to a subsequence we have, with the Greek letters relating to the latter,

$$\overline{F} \supseteq \Phi \supseteq \underline{\Phi}, \quad \overline{G} \supseteq \Gamma \supseteq \underline{\Gamma}, \tag{15}$$

so that whenever the whole sequence has a closed (open) limit, every subsequence has such a limit also.

Examples. For the sequence A, B, A, B, ... we have

$$\overline{F} = A_0 + B_0, \quad \underline{F} = A_0 B_0,$$

$$\overline{G} = A_i + B_i, \quad \underline{G} = A_i B_i.$$

If the sequences A_n with the limit sets $\overline{F}$, $\underline{F}$, $\overline{G}$, $\underline{G}$, and the sequences B_n with the limit sets Φ, $\underline{\Phi}$, Γ, $\underline{\Gamma}$ are combined into a sequence

$$A_1,\ B_1,\ A_2,\ B_2,\ \ldots$$

then the latter has the limit sets

$$\overline{F} + \Phi, \quad \underline{F} \cdot \underline{\Phi}, \quad \overline{G} + \Gamma, \quad \underline{G} \cdot \underline{\Gamma}.$$

If $A(t)$ is the ellipse in the Euclidean plane defined, for $t > 0$, by

$$x_1^2 + \frac{x_2^2}{t^2} = 1, \quad \text{with vertices } (\pm 1, 0) \text{ and } (0, \pm t),$$

then the closed limit of the sequence $A(n)$ with $n = 1, 2, 3, \ldots$ is the pair of lines $x_2 = \pm 1$, of the sequence $A(\frac{1}{n})$ the interval $-1 \leq x_1 \leq 1$, $x_2 = 0$.

I. (Selection Theorem). *If the space is separable, then every sequence of sets contains a subsequence for which the closed limit exists (and one for which the open limit exists).*

Because of (10), it suffices to prove the theorem in the case of the open limit (besides, a two-fold application of the theorem produces a subsequence for which both the open and the closed limits exist). Instead of using the neighborhoods U_x we make use once more of the countable special neighborhoods V (§25); the V — whose sum is $\overline{G}$ — contained in infinitely many A_n may be called *upper* V; those V whose sum is $\underline{G}$ — contained in almost all the A_n — may be called *lower* V; and those among the upper V that are not lower V, may be called *discrepant* V. In the transition to a subsequence, the set of upper, lower, and discrepant V becomes, respectively, smaller, greater, and smaller (in the wider sense: that is, the possibility of their remaining the same is not excluded); but if the subsequence is chosen suitably, then a particular discrepant V can actually be eliminated. For if V belongs to all of the infinitely many A_p, but not to almost all the A_n, then the transition from the A_n to the subsequence A_p is sufficient to make V into a lower V and thus to eliminate it from the ranks of the discrepant V. Accordingly, in case $\overline{G} > \underline{G}$, that is, in case there are actually discrepant V present, we proceed as follows: We put the (at most countably many) discrepant V in the form of a sequence $V_1, V_2, \ldots$ and eliminate V_1 by passing from A_n to a subsequence $A_p^{(1)}$; in case there are discrepant V for this sequence also, say $V_m, \ldots$, then we eliminate the lowest V_m by passing from $A_p^{(1)}$ to a subsequence $A_p^{(2)}$, and so on. Either a finite number of steps brings us to a subsequence for which there are no further discrepant V, and the open limit exists, or the process can be continued without limit. If that happens to be the case, we form the *diagonal sequence* from the sequence of numbers f, g, h, ... (arranged in ascending order); that is, from

$$\mathfrak{p} = p_1,\ p_2,\ p_3,\ \ldots$$
$$\mathfrak{q} = q_1,\ q_2,\ q_3,\ \ldots$$
$$\mathfrak{r} = r_1,\ r_2,\ r_3,\ \ldots$$

we form the sequence

$$\mathfrak{d} = p_1,\ q_2,\ r_3,\ \ldots$$

Since, apart from a finite number of elements, this latter is a subsequence of every preceding one, it yields a sequence of sets A_n for which there are no longer any discrepant V, so that the open limit exists.

3. The relation between the closed and metric limits

Leaving the open-limit sets aside for the moment, we now ask whether any relations exist between the closed and the metric limits, $\mathrm{Fl}\, A_n$ and $\lim A_n$, or, more generally, between the two closed-limit sets and the metric elements of accumulation. We note that $\overline{F}$ and $\underline{F}$ remain unchanged if the A_n are replaced by their closed hulls, for $A \cdot U_x > 0$ is equivalent with $A_0 \cdot U_x > 0$. Therefore in future we suppose the A_n to be closed and, in addition, bounded and non-empty. If a_n denotes a point of A_n, then we can say, after a certain amount of reflection, that:

$x \in \overline{F}$ means: There exists a sequence a_n with the point of accumulation x (that is, with a subsequence convergent to x).

$x \in \underline{F}$ means: There exists a sequence $a_n \to x$.

We obtain, in addition, the following theorem, where X and Y denote non-empty closed, bounded sets.

II. *If $\varrho(A_n, X) \to 0$, then $X \subseteq \underline{F}$; if $\varrho(Y, A_n) \to 0$, then $Y \subseteq \overline{F}$.*

Proof: Let $\varrho_n = \varrho(A_n, X) \to 0$, $x \in X$; there exists (because the relation (ε) holds for $\varrho > \varrho(A, B)$), a point $a_n \in A_n$, for which $a_n x < \varrho_n + \frac{1}{n}$ so that $a_n \to x$, $x \in \underline{F}$.

Let $\sigma_n = \varrho(Y, A_n) \to 0$, $y \in \overline{F}$; there exists a sequence $a_n \in A_n$ with a_n convergent subsequence $a_{p_n} \to x$ and, for every a_{p_n}, a point $y_n \in Y$ for which $y_n a_{p_n} < \sigma_{p_n} + \frac{1}{n}$. Thus $y_n \to x$; x is an α-point of Y; $x \in Y_\alpha = Y$.

This proves II. From $X \subseteq \underline{F} \subseteq \overline{F} \subseteq Y$ it now follows that, for $X = Y$, this set is also identical with the set $\overline{F} = \underline{F}$, that is:

III. *If the metric limit $X = \lim A_n$ exists, then $X = \mathrm{Fl}\, A_n$ is at the same time the closed limit of the sequence of sets.*

Even a trivial example makes it immediately apparent that the converse of this theorem cannot be obtained without further assumptions: If, in the space E of real numbers, we let A_n be the interval $[-n, n]$, then the closed limit of this sequence exists, namely E itself, but E is not bounded, and EA_n is meaningless. This sequence has no element of metric accumulation at all (since $EE = 0$). Nevertheless, the converse of III can be obtained under two different hypotheses.

IV. *If the space is compact in itself, then the metric limit of every sequence of closed sets exists, provided the sets are non-empty.*

For if A_n is such a sequence, then (§25, VI) the sequence has an element of metric accumulation X; that is, there exists a subsequence A_{p_n} with $A_{p_n} \to X$, so that X is the metric limit of this subsequence. By III, X is also the closed limit of the subsequence, and thus $X = \lim A_{p_n}$ is the closed limit of A_{p_n}. But then, by (11), X is also the closed limit of the whole sequence A_n, and therefore, by III, also the metric limit of the sequence A_n: $A_n \to X$.

V. *If A and all the A_n are contained in a compact set C, then the closed limit $A = \lim A_n$ implies the metric limit $A = \lim A_n$.*

For in this case, metric accumulation and closed accumulation mean the same thing. From $A_n \subseteq C$ it follows that every element of metric accumulation X of the sequence A_n is contained in C; for if $A_{p_n} \to X$, then $X \subseteq C_0 = C$. If the closed limit A exists, then by III it is also the only element of metric accumulation and, since all the A_n are in the compact set C, the sequence has an element of metric accumulation; therefore $A_n \to A$.

From III, IV, V it follows:

VI. *If the space is compact in itself, then the closed and the metric limits of a sequence of sets are identical concepts.*

§29. Connectedness

1. *Connected sets.*

A set $A > 0$ is called *connected* if there exists no decomposition

$$A = A_1 + A_2 \quad (A_1 > 0,\ A_2 > 0) \tag{16}$$

into two disjoint non-empty subsets, each of which is closed in A (or, equivalently, each of which is open in A). The decomposition (1), if it exists, is called a *partition* of A, and a set that is not connected is called *disconnected*. Since the empty set is simultaneously open and closed in every set, a partition of A is a decomposition of A into two disjoint non-empty subsets, each of which is both open and closed in A. In particular, therefore, every non-empty closed set is connected if it cannot be decomposed into two disjoint non-empty closed sets, and a non-empty open set is connected if it cannot be decomposed into two disjoint non-empty open sets. A closed connected set is called a *continuum*; an open connected set is called a *domain*. Whether a given set A is connected or not depends only on itself and not on the containing space E (whereas the concepts of continuum and domain are relative to E).

Examples. An hyperbola is disconnected; its division into two branches is a partition.[4] The set of the positive and negative real numbers exclusive of zero is disconnected. A set consisting of a single point counts as being connected (a continuum). An interval $(a, b]$ in the set of real numbers is connected; for suppose a division $A = A_1 + A_2$ into disjoint sets is given, with $a \in A_1$, and with c the greatest lower bound of the numbers of A_2. If A_2 is closed, then c is the smallest number of A_2, so that $c > a$; and A_1 is not closed, since the half-open interval $[a, c)$ belongs to A_1 while c itself does not.

In order to show that a set $A > 0$ is connected, it must be shown that, in every decomposition (1), one of the sum-

[4] Compare Appendix B.

mands is the null set. It should be noted here that such a decomposition also induces a corresponding decomposition

$$B = B_1 + B_2 = BA_1 + BA_2 = B^1 + B^2$$

for any subset B of A.

I. *If every pair of points of A can be "joined to each other in A," that is, if they belong to a connected subset of A, then A is connected.*

For if a decomposition (1) exists, such that $x_1 \in A_1$, $x_2 \in A_2$, then we join x_1 and x_2 with a connected subset B; but then $B = B_1 + B_2$ is a partition.

II. *The sum of two connected sets whose intersection is non-empty is connected.*

Let $S = A + B$, $D = AB > 0$. To a decomposition $S = S_1 + S_2$ there corresponds $A = A_1 + A_2$, $B = B_1 + B_2$, and $D = D_1 + D_2$. Since A is connected, one of the summands is 0, say $A_2 = 0$, and thus

$D_2 = 0$, $B_2 = 0$; therefore $S_2 = 0$.

III. *The sum of any number of connected sets that have a point in common is connected.*

For the sum S of the connected sets A, B, C, ... containing x is the sum of the connected sets $A + B$ and C whose intersection contains x, etc.

IV. *If $A \supseteq B \supseteq C$ and B and C are connected, then A is connected.*

For if $A = A_1 + A_2$, then $B = B_1 + B_2$, and one of the summands, say $B_1 = 0$; then $C = C_1 + C_2 = C \cdot A_1 + C \cdot A_2$, and $C \cdot A_1 \subseteq B_1 = 0$, so $C_1 = 0$; C is connected.

If C is a connected subset of A and is not contained in any other connected subset of A, then C is called a *component* of A. Every connected subset of A containing a point x belongs to the sum $C(x)$ of all connected subsets of A containing x; this sum is itself connected (III) and is therefore a component. Every point x of A thus belongs to exactly one component $C(x)$; different components are disjoint, and A is the sum of its components.

V. *If A is connected, then every set between A and A_0 ($A \subseteq B \subseteq A_0$) is connected.*

To a decomposition $B = B_1 + B_2$ into relatively closed sets, there corresponds the decomposition $A = A_1 + A_2$. Here we have say, $A_2 = 0$, $A_1 = A$, $A \subseteq B_1 \subseteq A_0$, $B_2 \subseteq A_0$, and since B_1 is closed in B, it follows that $B_2 = 0$.

From IV and V it follows:

VI. *If A is connected, then so are A_0 and A_i.*

For an open set in a linear space or a convex subspace, this condition of connectivity (joining by a polygonal path) is not only sufficient but also necessary. For if G is open and if $G(x)$ is the set of those points y that can be joined to $x \in G$ by a polygonal path $[x, \ldots, y]$, then $G(x)$ is open (if $z \in G(x)$ and $U_z \subseteq G$, then all the points of U_z can be joined to x via

z); $G(x)$ is also closed in G, because if y is a boundary point of $G(x)$ in G, then $y \in G$, and y can be joined to x via a point of $G(x)$ in $U_y \subseteq G$. Therefore $G(x) = G$.

For the components of an open set G we have:

VII. *In every space, the components of an open set are open; or, equivalently: distinct components of an open set have positive lower distance.*

For if C is a component of G and $x \in C$, then there exists a neighborhood $U_x \subseteq G$, and this is the sum of connected sets (each point of U_x can be joined to x); therefore $U_x \subseteq C$, and C is open.

VIII. *If two closed sets F_1, F_2 have a positive lower distance, then there exist disjoint open sets $G_1 \supseteq F_1$, $G_2 \supseteq F_2$.*

In fact, it suffices to choose $G_1 = U(F_1, \varrho)$, $G_2 = U(F_2, \varrho)$ with $2\varrho < \delta(F_1, F_2)$.

IX. *If A is connected and B is a set that is both open and closed in A, then either $B = 0$ or $B = A$.*

For if $0 < B < A$, then $A = B + (A - B)$ is a partition.

X. *The intersection of connected sets that have a point in common is connected.*

For if $x \in \bigcap A_n$, then x belongs to all the A_n, and the intersection is the sum of connected sets containing x.

The following theorem is a counterpart to VIII:

XI. *Two connected sets whose intersection is not connected can be enclosed in two disjoint open sets.*

For let A and B be connected and $AB = C_1 + C_2$ ($C_1 > 0$, $C_2 > 0$, $\delta(C_1, C_2) > 0$). Then $A - C_2$ and $B - C_1$ are open sets containing C_1 and C_2 respectively, and

$$\begin{aligned}(A - C_2)(B - C_1) &= AB - AC_1 - BC_2 + C_1C_2 \\ &= C_1 + C_2 - C_1 - C_2 + 0 = 0.\end{aligned}$$

XII. *If $A \subseteq B \subseteq A_0$ and A is connected, then B is connected.*

For if $B = B_1 + B_2$, then $A = AB_1 + AB_2$; say $AB_2 = 0$; then $B_2 \subseteq A_0 - A \subseteq B - A$; but B_2 is both open and closed in B, so $B_2 = B(B_2)_0$; $(B_2)_0 \subseteq A_0$; if $B_2 > 0$, then $(B_2)_0 > 0$; $(B_2)_0 \subseteq A_0 - A$ is impossible because A is dense in A_0. Therefore $B_2 = 0$.

XIII. *The components of a closed set are closed.*

For the closed hull of a component is, by XII, itself connected and, being the smallest closed connected superset, is identical with the component.

From VII and XIII it follows that in a locally connected space (defined below), the components of open sets are domains and the components of closed sets are continua.

A connected space E is said to be *uniquely arcwise connected* if, for every pair of distinct points x, y, there exists exactly one continuum $K \subseteq E$ such that $x, y \in K$ and no proper subcontinuum of K contains both x and y. A line, a circle, a tree are examples of uniquely arcwise connected spaces. The Euclidean plane is not uniquely arcwise connected.

A set A is called *convex* if, together with any two points x, y, it also contains a segment $[x, y]$ joining them. Convex sets are connected. A space is called *convex* if it is convex as a set.

XIV. *If A and B are connected, then so is $A \times B$ (Cartesian product).*

XV. *A continuous image of a connected set is connected.*

For if A is connected, f continuous on A, and $f(A) = B_1 + B_2$, then $A = f^{-1}(B_1) + f^{-1}(B_2)$, and the $f^{-1}(B_k)$ are disjoint and open in A.

From XV it follows: If A is connected and $f(x)$ is a continuous real function on A, then $f(A)$ is an interval (a connected set of real numbers). If $a, b \in A$ and $f(a) \leq \gamma \leq f(b)$, then there exists $c \in A$ with $f(c) = \gamma$. (Theorem of Bolzano.)

2. *Local connectedness.* A decomposition of A into disjoint connected summands:

$$A = P + Q + R + \cdots \tag{17}$$

a *natural* decomposition if it does not tear apart the connected subsets of A — if its summands, that is, are sums of components or, if connected, components. Natural decompositions thus comprise, among others, the decomposition into components and, further, into a finite number of relatively closed or into any number of relatively open sets (P and $A - P$ are then simultaneously closed or open in A, whence the statement follows easily). If, for example, an open set can be decomposed into open sets then, insofar as they are connected, they are components of G. A domain G with the boundary H is a component of $E - H$, for $G_0 = G + H_0$ is closed, $E - (G + H_0) = G_e$ is open, $E - H = G + G_e$ is a natural decomposition. A domain is uniquely determined by its boundary together with a single one of its points.

The space E will be called *locally connected*[5] if every neighborhood U_x of an arbitrary point x contains a domain G_x containing the point x. The space need not, therefore, be connected (as a whole).[6] A convex space is locally connected, since the neighborhoods themselves are connected. The simplest example of a space that is not locally connected is the set $E = \{1, \frac{1}{2}, \frac{1}{3}, \ldots, 0\}$; the only connected set containing the point 0 is the set $\{0\}$ consisting of this one point alone, and this set is not open in E. An example of a connected space that is not locally connected is given following Theorem XII. Now we have:

IX. *If the space is locally connected and* $A = P + Q + R + \cdots$ *is a natural decomposition of* A*, then*

$$A_i = P_i + Q_i + \cdots\,, \quad A_0 = P_0 + Q_0 + \cdots\,, \tag{18}$$

$$A_r = S_r, \quad S = P_r + Q_r + \cdots\,. \tag{19}$$

The open kernel, the border, and the frontier (boundary) of the whole set depend in this simple way on those of the summands. For the proof of (6) we recall that, in general,

$$A_i \supseteq P_i + Q_i + \cdots\,, \quad A_r \supseteq P_r + Q_r + \cdots\,.$$

On the other hand, if $x \in A_i$, $U_x \subseteq A$, and G_x is a domain for which $x \in G_x \subseteq U_x$, then G_x is contained in a single

[5]Or connected in the small (Hahn).

[6]An isolated space is locally connected; every set $\{x\}$ consisting of only one point is a G_δ.

summand of the decomposition, say $G_x \subseteq P$, $x \in P_i$, and so $A_i \subseteq P_i + Q_i + \cdots$.

In order to prove (7) let us consider the complement $B = E - A$; we have

$$E - P = B + Q + R + \cdots$$
$$(E - P)_i \subseteq B_i + Q_i + R_i + \cdots$$
$$P_r = P_0 + (E - P)_i \subseteq B_0 + P_i + Q_i + R_i + \cdots = B_0 + A_r = A_r,$$

and thus $A_r \supseteq P_r + Q_r + \cdots = S$ and $A_r \subseteq S$, since A_r is closed. On the other hand,

$$A_r \subseteq S, \tag{20}$$

and this half of the assertion is valid even for an arbitrary decomposition (in a locally connected space) $A = P+Q+R+\cdots$ (which need be neither a natural decomposition nor even a decomposition into disjoint summands). For let $x \in A_r$, let U_x be arbitrary, and let $G_x \subseteq U_x$; since x is an α-point of A and of B, the set G_x contains points of both sets; suppose it contains a point of, say, P and one of $B = E - P$; then (by VII) it contains also a point of P_r, or S, so that $x \in S$.

In a natural decomposition into a finite number of summands, (7) simplifies to $A_r = S = P_r + Q_r + \cdots$.

X. *If the space is locally connected, then the components of open sets are always domains, and conversely.*

For if A is open, then, by (6), P, Q, R, … are also open, and this holds in particular for the components. If, conversely, the components of the open sets are domains, then the component of U_x containing x is a domain G_x, and the space is locally connected.

The following theorem is an application of (8):

> *A connected and locally connected space that is also an F_σ-set cannot be decomposed into countably many disjoint non-empty closed sets.*

Assuming that $E = F_1 + F_2 + \cdots$ (F_n closed and non-empty), then since E is connected, F_1 has a boundary $H_1 = F_{1r} > 0$; the sum $H = H_1 + H_2 + \cdots$ (as the complement of ΣF_n) is also closed. For the boundary of

$$E - F_1 = F_2 + F_3 + \cdots$$

it turns out, according to (8), that

$$H_1 \subseteq (H_1 + H_2 + \cdots)_0,$$

that is, $H_1 + H_2 + \cdots$ is dense in H; H_1 and each H_n are nowhere dense in H; and thus H is of the first category in itself, in contradiction to the hypothesis that E be an F_σ-set.

This more stringent kind of connectedness, which excludes a partition not only into a finite number, but even into a countable number of closed summands, is a property of, amongst others, Euclidean spaces; later on we shall prove the result, by another method, for compact continua also (p. 185).

Thus far we have defined local connectedness simply for the whole space. It would seem reasonable to call E locally connected at the single points if every U_x contained a G_x (a domain containing x); however, it is useful to relax the condition somewhat and make the following definition:

The space E is said to be *locally connected at the point* x if every neighborhood U_x contains a connected set C of which x is an interior point ($x \in C_i$); we may take C here as the component of U_x that contains x.

XI. *If the space* E *is locally connected at every point* x, *then it is (simply) locally connected; and conversely.*

For if C_x is the component of U_x that contains x, $y \in C_x$, $U_y \subseteq U_x$ and C_y the component of U_y that contains y, then $C_y \subseteq C_x$, because $C_x + C_y$ is connected, is properly contained in U_x, and thus is identical with C_x. Now y is an interior point of C_y and therefore also of C_x; C_x consists entirely of interior points and is a domain $\subseteq U_x$ containing x. The converse is trivial.

It is possible to formulate local connectedness in x somewhat differently (Mazurkiewicz). Let ϱ be the lower bound of the diameters $d(C)$ of all the bounded connected sets C that contain x and y; when none exist, we leave xy undefined. We have $\varrho \leq xy$; if, furthermore, xy and yz exist, then we also have $xz \leq xy + yz$ as a consequence of the easily verifiable fact that $d(A+B) \leq d(A)+d(B)$ whenever AB is non-empty. This xy then, in so far as it exists, has the properties of a distance, so that we can say:

XII. *The space is locally connected at x if and only if $\varrho \to 0$ whenever $xy \to 0$.*

The condition is intended, of course, to mean: For any $\varepsilon > 0$ and a suitable $\sigma > 0$, ϱ is defined and is $< \varepsilon$ whenever $xy < \sigma$.

If this condition is satisfied, then let U_x and V_x be the neighborhoods of x with the radii ε and σ. Every $y \in V_x$ may be joined to x by means of a connected set C_y for which $d(C_y) < \varepsilon$. The set $C = \bigcup C_y$ is connected, by IV, and is contained in U_x, since each of its points has a distance $< \varepsilon$ from x; it contains V_x, and therefore has x as an interior point; E is locally connected at x.

If, conversely, E is locally connected at x, then let U_x have the arbitrarily chosen radius ε; the component C of U_x that contains x has x as an interior point and therefore contains some neighborhood V_x of radius σ. For $xy < \sigma$, we then have $\varrho \leq d(C) \leq 2\varepsilon$.

Example: In the Euclidean $\xi\eta$-plane, let r_n be the point $\xi = \frac{1}{n}$, $\eta = (-1)^n$ for $n = 1, 2, 3, \ldots$. The sum of the intervals $[r_1, r_2]$, $[r_2, r_3]$, ... is the infinite polygonal path

$$R = [r_1, r_2, r_3, \ldots]$$

which, as it gets closer and closer to the y-axis, travels back and forth infinitely often between the lines $\eta = 1$ and $\eta = -1$, so that every point of the segment

$$S : \xi = 0, \ -1 \leq \eta \leq 1$$

is a point of accumulation of R. The set $E = R + S$, thought of as the space, is not locally connected at the points of S. For if we try to connect a point $s \in S$ to a point $r \in R$ by means of a connected set $C \subseteq E$, then, no matter how close r may be to s, C must cut every line $\xi =$ const. lying between s and r in at least one point; but then C contains a residual portion $[r_n, r_{n+1}, \ldots]$ of the polygonal path and has a diameter > 2, so that $S\varrho$ cannot converge to 0 with sr. R and E, however, are connected.

3. *Separations.* A finite complex of points $(x_1, x_2, \ldots, x_n)$ in which the distances x_1x_2, ..., $x_{n-1}x_n$ between neighboring points are $< \varrho$ will

Fig. 6

be called a ϱ-*chain* ($\varrho > 0$); we say then that x_1 and x_n can be *joined* by a ϱ-*chain*, and in particular, if all the points of the chain belong to A, by a ϱ-chain in the set A. Let us look for all the ϱ-chains joining, in A, two points x and y of A; the greatest lower bound of the numbers ϱ occurring in this will be called the *separation* of the points x and y and will be denoted by xy (although it depends not only on the two points but also on A). For $\varrho > xy$, it is therefore possible to join x to y by a ϱ-chain in A; for $\varrho < xy$, it is not possible; for $\varrho = xy$, it depends on the particular case. Clearly, $xy = yx$. The separations satisfy the triangle inequality in the more stringent form

$$xz \leq \max[xy, yz]; \tag{21}$$

for if x and y can be joined by a ϱ-chain, and y and z by a σ-chain, then x and z can be joined by a τ-chain, where $\tau = \max[\varrho, \sigma]$. If $x_n \to x$, $y_n \to y$, then $x_n y_n \to xy$; the separation xy is a continuous function of x and y.

Of course, we set $xx = 0$; but $xy = 0$ does not mean that $x = y$ but, rather, that x and y can be joined by a ϱ-chain for every $\varrho > 0$. In the set of the real or the rational numbers any two points have the separation 0.[7] Now we have (as regards the lower distances $\delta(P, Q)$, compare pp. 166–7):

XIII. *If $A = P + Q$, where $\delta(P, Q) = \delta > 0$, then every point $p \in P$ has a separation $pq \geq \delta$ from every point $q \in Q$. If two points p, q of A have the separation $pq = \varrho > 0$, then a decomposition $A = P + Q$ for which $p \in P$, $q \in Q$, and $\delta(P, Q) \geq \varrho$ is possible.*

For the first half, note that a ϱ-chain from p to q ($\varrho < \delta$) must contain a link from a point of P to one of Q, and such a link has length $\geq \delta > \varrho$. For the second half, let P be the set of points that can be joined to p by a ϱ-chain in A ($\varrho < pq$); then P and $Q = A - P$ have lower distance $\geq \varrho$, $p \in P$, $q \in Q$.

XIV. *If $p \in A$, then among the subsets of A that contain p and are open and closed in A, there exists a smallest, namely the set P of the points x for which $px = 0$.*

For if $px = 0$, then x can be joined to p by a ϱ-chain for every $\varrho > 0$; if B is open and closed in A and $p \in B$, then $x \in B$; for otherwise $A = B + (A - B)$ would give a partition with $\delta(B, A - B) > 0$, contradicting $px = 0$. Conversely, if $px > 0$, then the decomposition into P and Q given by XIII separates p from x.

The set P of the points with $px = 0$ is called the *component* (quasi-component) of p. For sets compact in themselves, it is identical with the component defined earlier. In general, it may be larger than the component; it is, however, always the intersection of all the sets that are open and closed in A and contain p.

[7] In the set of rational numbers, every two points have separation 0, but the set is not connected.

XIV′. *If A is compact in itself and $p \in A$, then the component of p is the set P of the points x for which $px = 0$.*

For, on the one hand, we always have $P \supseteq C(p)$; on the other hand, $P(0)$ is closed, because the separation px is a continuous function of x; and thus it is compact in itself and therefore connected, since after all any two of its points have the separation 0, whence $P(0) = P$. We have thus shown:

XIV. *A set compact in itself is connected if (and only if) any two of its points have the separation* 0. *The component containing the point p is the set of those points that have a separation* 0 *from p.*

In the proof of XIII, we showed that the set of the points x for which $px < \varrho$, as well as its complement, is closed; the same holds also for the set $P(\varrho)$ of the points $px \leq \varrho$ and its complement $Q(\varrho) = A - P(\varrho)$ for $\varrho > 0$; we see, as we did there, that the lower distance between the two sets is $> \varrho$ (assuming $Q(\varrho) > 0$). For $\varrho = 0$, this no longer holds; it is still true that the component containing p

$$P(0) = \bigcap_{\varrho > 0} P(\varrho) = P(1) \cap P(\tfrac{1}{2}) \cap P(\tfrac{1}{3}) \cap \cdots \tag{22}$$

is closed, but $Q(0)$ need not be, and the lower distance between the two of them may well be 0.

The following three theorems are applications of the separations to compact sets.

XV. *A set compact in itself remains compact in itself if separations replace distances as the metric, and the set becomes a totally disconnected set.*

Split into components, let the set be

$$A = P + Q + R + \cdots\ ;$$

if we make it into a metric space by using the separation in A as distance, then all the points of a component of A coalesce into a single point. By choosing one point from each component it is possible to set

$$A = \{p_1, p_2, \ldots\}.$$

A sequence of points x_n has a point of accumulation x in A, that is, a subsequence for which $xx_n \to 0$, all the more then does $xx_n \to 0$, so that x

has the point of accumulation x in A as well. A is therefore compact in itself, and in the same way there corresponds to every subset B compact in itself of A, a subset B compact in itself of A; if p and q are points of different components of A and if ϱ is positive and $< pq$, then the set $P(\varrho)$ of the points $px \leq \varrho$, as well as its complement $Q(\varrho) = A - P(\varrho)$, is closed in A; the sets corresponding to them, of which one contains the point p and the other the point q, are in turn closed in A, and consequently two different points of A never belong to a connected subset of A; thus, A is totally disconnected (p. 174).

Since the set A is complete and at most separable, it follows that: *The set of components of a set A compact in itself is either at most countable or of the cardinality of the continuum.*

XVI. *The compact, perfect, totally disconnected sets are identical with the dyadic discontinua.*

Suppose, first, only that A is compact in itself. With $\varrho > 0$ we form a greatest set of points whose pairwise separations in A are $> \varrho$ (so that their distances are certainly $> \varrho$). This set is finite and consists of points that we may call $x_1, x_2, \ldots, x_n$. For each point $x \in A$ there exists at least one x_i $(i = 1, 2, \ldots, n)$ for which $xx_i < \varrho$ (otherwise these greatest sets could be extended) and also at most one (otherwise we should have $x_i x_j < \varrho$). If, therefore, A_i is the set of points $xx_i < \varrho$, then

$$A = A_1 + A_2 + \cdots + A_n \tag{23}$$

is decomposed into disjoint, closed, compact sets each of which has a "separation diameter" $< 2\varrho$ — that is, two points of A_i have a separation $< 2\varrho$.

If, second, A is also totally disconnected, then the distance xy $(= xy)$ converges to zero whenever the separation

xy does and even converges uniformly, so that some $\varrho > 0$ corresponds to every $\sigma > 0$ in such a way as to make $xy < \sigma$ whenever $xy < \varrho$. For otherwise there would have to be a sequence of pairs of points for which $x_n y_n \to 0$, $x_n y_n \geq \sigma$, where by restricting consideration to subsequences we may assume $x_n \to x$, $y_n \to y$, so that $xy = 0$, $xy \geq \sigma$; the distinct points x, y would belong to the same component. A can therefore be decomposed for every $\delta > 0$ into a finite number of disjoint sets compact in themselves and of diameter $< \delta$.

If, third, A is perfect as well, then the summands of such a decomposition (11) are also perfect, since they are also open in A and hence dense-in-themselves. The A_i are therefore sets like A, and the process can be repeated;

$$A = \Sigma A_i, \quad A_i = \Sigma A_{ik}, \quad A_{ik} = \Sigma A_{ikl}, \quad \dots$$

in which the diameters of the sets with n subscripts can be taken smaller than $\frac{1}{n}$ or even so small that every sum has at least two summands. Then

$$A = \Sigma A_i \cdot \Sigma A_{ik} \cdot \Sigma A_{ikl} \cdots$$

is a polyadic discontinuum (p. 154) and can be transformed into a dyadic discontinuum.

Conversely, a dyadic discontinuum

$$A = \Sigma V_{p_1} \cdot \Sigma V_{p_1 p_2} \cdot \Sigma V_{p_1 p_2 p_3} \cdots$$

is totally disconnected. For if $C \subseteq A$ is connected then, in order that $C = CV_{p_1} + CV_{p_2}$ not be a partition, C can have points in common with only one V_{p_1}, then with only one $V_{p_1 p_2}$, then with only one $V_{p_1 p_2 p_3}$, and so on; that is, $C = V_{p_1} V_{p_1 p_2} V_{p_1 p_2 p_3} \cdots$ contains only one point. The proof of XVI is thus complete.

XVII. (Janiszewski's Border Theorem). *Let F be closed; let $G = E - F$ be its open complement, $H = F_r$ the boundary of both sets, and C a compact continuum with $CH > 0$.*[8] *Then (i) every component of CF has points in common with H, (ii) for $CG > 0$ every component of CG has points of accumulation in H, (iii) for $CF_i > 0$ every component of CF_i has points of accumulation in H.*

The figure illustrates the case in which F is a closed circular disc in the Euclidean plane E_2, G the exterior of the circle, and H its periphery.

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[8]By virtue of VII, this hypothesis is satisfied, in particular, when $CF > 0$, $CG > 0$.

Proof. (i) If $CF = P + Q$ is decomposed into two closed sets and $P > 0$, then PH is also > 0. For on account of $C = CF + CG$, we have $C = P + (Q + CG)$; in order that the latter not be a partition, the intersection of both summands $P \cdot CG = P \cdot F \cdot G = PH$ must be non-empty. If now $p \in CF$ and $P(\varrho)$ is the set of points x that have a separation $px \leq \varrho$ in the closed compact set CF, then for $\varrho > 0$, $P(\varrho)$ and the

Fig. 7

complement $Q(\varrho) = CF - P(\varrho)$ are, as we know, closed, and so $P(\varrho) \cdot H > 0$. By the First Intersection Theorem (§26, I), the intersection of the sets $P(\varrho) \cdot H$ for $\varrho = 1, \frac{1}{2}, \frac{1}{3}, \ldots$ is also non-empty, and therefore, by (10), $P(0) \cdot H$ is non-empty, where $P(0)$ can denote any component of CF.

(ii) Let F_n be the closed set of points $\delta(x, F) = \frac{1}{n}$ ($n = 1, 2, 3, \ldots$); we have $G = F_1 + F_2 + \cdots$ and, for $p \in CG$, ultimately $p \in CF_n$. By what has been proved in (i), the component of CF_n containing p (C intersects F_n, as well as $E - F_n \supseteq G$ and therefore, by virtue of VII, also the border of F_n) and, a fortiori, that of CG, has a point x_n on the border of F_n, that is, one for which $\delta(x_n, F) = \frac{1}{n}$; and these points have a point of accumulation x in C for which $\delta(x, F) = 0$, $x \in F_0 = H$.

(iii) C intersects F_i as well as $E - F_i \supseteq H$ and thus, by VII, also the boundary of F_i; by (ii), applied to F_i instead of G, every component of CF_i has points of accumulation in this boundary, which is the same as $F_{0i} - F_i \subseteq F - F_i = H$.

Applications: If C is a compact continuum and A a subcontinuum $< C$, then there exists a continuum B for which $A < B < C$. For proof, we choose a point $y \in C - A$ and a positive number $\varrho < \delta(y, A)$, and put for F the set of points $\delta(x, A) \leq \varrho$ and for B the component of CF containing A. C intersects F as well as $E - F$ (at the point y) and thus intersects the border of F; consequently B contains a border point of F for which $\delta(x, A) = \varrho$, and therefore $B > A$; on the other hand, $y \notin B$, and therefore $B < C$.

A compact continuum cannot be decomposed into countably many disjoint, non-empty, closed sets[9] (Sierpiński).

Suppose the compact continuum A to be representable as a sum

$$A = A_1 + A_2 + \cdots$$

of disjoint non-empty closed sets A_n. We show that A contains a continuum B that has points in common with infinitely many A_n, although not with A_1:

$$B = BA_{p_1} + BA_{p_2} + \cdots = B_1 + B_2 + \cdots$$

[9]Compare with the similar theorem on p. 178.

$(p_1 < p_2 < \cdots,\ B_n > 0)$. Once we have this, everything is already proved; for the repetition of the process produces a continuum

$$C = CB_1 + CB_2 + \cdots = C_1 + C_2 + \cdots$$

with non-empty C_n, where the q form a subsequence of the p and $p_1 < q_1 < q_2 < \ldots$, etc. $A \supset B \supset C \supset \cdots$ would therefore be the empty set, in contradiction to the First Intersection Theorem.

In order to prove the statement concerning B, let us understand by $2\varepsilon = \delta(A_1, A_2) > 0$ the lower distance between A_1 and A_2, and (say in the space A itself) by F the closed set of the points for which $\delta(x, A_1) \geq \varepsilon$, so that $A_2 \subseteq F \subseteq A - A_1$. Let then B be the component of F containing any point whatever of A_2, so that $BA_1 = 0$, $BA_2 > 0$. But, according to XVII, B now contains a border point of F, that is, one for which $\delta(x, A_1) = \varepsilon$, and this point does not belong to A_2; B therefore has points in common with at least two A_n and therefore, as a continuum, with infinitely many A_n, as stated.

4. *Sequences of connected sets.*

XVIII. *The intersection of a decreasing sequence $A_1 \supseteq A_2 \supseteq \cdots$ of compact continua is a continuum.*

Let us remark first that two disjoint closed sets F_1 and F_2 can always be enclosed in disjoint open sets G_1 and G_2; all that is necessary is to enclose each point $x_1 \in F_1$ in a neighborhood of radius $\frac{1}{2}\delta(x_1, F_2)$ and similarly, every point $x_2 \in F_2$ in a neighborhood of radius $\frac{1}{2}\delta(x_2, F_1)$. If now $A = A_1 A_2 \cdots$ (a compact set, by the First Intersection Theorem) were disconnected, then $A = F_1 + F_2$ would be a partition of A into two disjoint non-empty closed sets. Let $G_1 \supseteq F_1$, $G_2 \supseteq F_2$ be disjoint open sets. Since the sets $A_n - G_1 - G_2$ are closed and form a decreasing sequence whose intersection is empty, it follows that for sufficiently large n, $A_n \subseteq G_1 + G_2$; but then $A_n = A_n G_1 + A_n G_2$ is a partition of A_n, contradicting the connectedness of A_n.

XIX. *The intersection of a decreasing sequence $A_1 \supseteq A_2 \supseteq \cdots$ of connected sets is connected, provided the A_n are closed and bounded.*

This follows from XVIII if the A_n are compact; for the general case, one must use the fact that the closed limit of connected sets is connected.

XX. *If the sequence A_n of connected sets has the closed limit A, then A is connected.*

For if $A = P + Q$ were a partition with $\delta(P, Q) > 0$, then we could choose disjoint open sets $G_1 \supseteq P$, $G_2 \supseteq Q$ (VIII). Since $P \subseteq \underline{F}$, $Q \subseteq \underline{F}$, almost all the A_n would have to intersect both G_1 and G_2, and therefore be disconnected, which is absurd.

From XX, using III, it follows:

XXI. *If the connected sets A_n have the property that every two of them have a common point, and if the sequence A_n has the closed limit A, then A is connected.*

If the space is compact in itself and the sequence of the connected sets A_n has the closed limit A or — what, by §28, III and V, is the same thing — the metric limit A, then A is a continuum. XVIII is a special case of this. If the space is compact in itself then, by §28, VI, every sequence of connected sets has a metrically convergent subsequence whose limit is itself a continuum. (It is to be recalled that the metric limit was supposed to be closed.) The continua of a space compact in itself thus themselves constitute a space compact in itself.

VII. Point Sets and Ordinal Numbers

§30. Hulls and Kernels

The theory of well-ordering, originally developed by Cantor precisely for the purpose of point-set theory was later somewhat kept back from playing this role — not always for praiseworthy motives — and we have seen for ourselves in the chapter just concluded how much can be done without it. Nevertheless, situations arise in which the ordinal numbers are occasionally indispensable, and others in which they are, at the very least, most welcome as tools in a more elegant exposition of a result that may have been found without them. In particular, they make their appearance, in one or the other of these roles, whenever we are concerned with the formation of sets or systems of sets which are in some way the greatest or smallest of their kind.

1. *Schema for hulls and kernels.* Suppose that in a space E (which may, for the moment, be taken to be a pure set) a set A' is made to correspond uniquely to every set A (both $\subseteq E$; $'$ is a function symbol). Let this set function be *monotone*, that is,

$$\text{for } A \subseteq B \text{ we have } A' \subseteq B'. \tag{24}$$

If then

$$S = A + B + \cdots, \quad D = AB\cdots$$

are the sum and intersection of any number of sets, then

$$S' \supseteq A' + B' + \cdots, \quad D' \subseteq A'B'\cdots.$$

There exist sets A for which $A \subseteq A'$, for example the null set $A = 0$; there also exist sets A for which $A \supseteq A'$, for example the whole space $A = E$. We see at once:

I. *The sum of any number of sets for which $A \subseteq A'$ is itself such a set; the intersection of any number of sets for which $A \supseteq A'$ is itself such a set.*

Accordingly, for an arbitrary set M we can define:

The sum $\underline{M}$ of all the sets $A \subseteq M$ for which $A \subseteq A'$ (to which $A = 0$ certainly belongs) or the greatest set $A \subseteq M$ for which $A \subseteq A'$: the *ϱ-kernel* of M.

The intersection $\overline{M}$ of all the sets $A \supseteq M$ for which $A \supseteq A'$ (to which $A = E$ certainly belongs) or the smallest set $A \supseteq M$ for which $A \supseteq A'$: the *ϱ-hull* of M.

The typical example is given by $A' = A_d$, the set of points of accumulation of A in a (metric) space. Here

$$A \subseteq A_d : \text{ means } A \text{ is dense-in-itself, and}$$
$$A \supseteq A_d : \text{ means } A \text{ is closed,}$$

and we obtain the dense-in-itself kernel $\underline{M} = M_d$, as well as the closed hull $\overline{M} = M_0$.

It can happen that $A \subseteq A'$ always; in that case $\underline{M} = M$, and $\overline{M}$ is the smallest set $A \supseteq M$ for which $A = A'$. Example: $A' = A_\alpha$, the set of α-points of A, and once more $\overline{M}$ turns out to be the closed hull of M.

It can happen that $A \supseteq A'$ always; in that case $\overline{M} = M$, and $\underline{M}$ is the greatest set $A \subseteq M$ with $A = A'$. Example: $A' = A_i$, the set of interior points of A, where $\underline{M}$ turns out to be the open kernel M_i.

It is possible to induce these special cases by considering the functions

$$A_1 = A + A', \quad A_2 = AA', \tag{25}$$

which are monotone simultaneously with A'.[1] Here we always have $A \subseteq A_1$ and $A = A_1$ is equivalent with $A \supseteq A'$; furthermore, we always have $A \supseteq A_2$, and $A = A_2$ is equivalent with $A \subseteq A'$. Thus, in the formation of the kernel, A' can be replaced by A_2; in the formation of the hull, A' can be replaced by A_1; and we have:

The ϱ-kernel $\underline{M}$ is the greatest set $A \subseteq M$ for which $A = A_2$; the ϱ-hull $\overline{M}$ is the smallest set $A \supseteq M$ for which $A = A_1$.

Example: For $A' = A_d$ we find $A_1 = A_0$, and $A_2 = A_d$ (coherence of A); the dense-in-itself kernel $\underline{M_d}$ is the greatest set $A \subseteq M$ for which $A = A_d$; the closed hull $\overline{M}_0$ is the smallest set $A \supseteq M$ for which $A = A_0$.

2. *Utilization of the ordinal numbers.* Let us consider first $\underline{M}$, the greatest set $A \subseteq M$ for which $A = A_2$. This set,

because of the monotonicity of the function A', satisfies the inequality $A = A_2 \subseteq M'$, and is contained not only in M, but also in the (in general) smaller set M_2. It is consequently contained also in M_{22}, M_{222}, ..., where we of course mean the repeated formation of the set function A_2: $M_{222} = (M_{22})_2$, etc.

[1] A_+ does not here denote the separated part of A.

This set is contained also in the intersection $MM_2 \cdot M_{22} \cdots$ of the sets formed thus far and, even further than this, is contained in all the sets arising from the further application of the process. This means that if we assign inductively to each ordinal number ξ a set M_ξ by the rule

$$\begin{aligned} M_0 &= M \\ M_{\xi+1} &= (M_\xi)_2 \\ M_\gamma &= \bigcap_{\xi<\gamma} M_\xi \quad (\gamma \text{ a limit number}), \end{aligned} \tag{26}$$

then our $\underline{M}$ is contained in all the M_ξ.

For every ordinal number $\eta > 0$ the decomposition[1]

$$M = \sum_{\xi<\eta} (M_\xi - M_{\xi+1}) + M_\eta \tag{27}$$

holds.

For if $x \in M$ does not belong to $\underline{M}$, then let M_ξ $(0 < \xi \leq \eta)$ be the set of lowest index to which x does not belong; this ξ, because of the third requirement in (3), may not be a limit ordinal, and therefore we can set $\xi = \zeta + 1$ $(0 \leq \zeta < \eta)$ and $x \in M_\zeta - M_{\zeta+1}$.

The disjoint summands $M_\xi - M_{\xi+1}$ cannot, however, continue to be non-empty indefinitely, since their sum may not exceed the cardinality of M. Let η be the smallest ordinal number for which $M_\eta = M_{\eta+1} = M_{\eta 2}$. Since this set M_η also satisfies the condition $A = A_2$, it must be contained in the greatest set $\underline{M}$ of this sort, and since, on the other hand, $\underline{M}$ was contained in every M_ξ, it follows that $\underline{M} = M_\eta$.

The ϱ-kernel of M is thus obtained in this way: We examine the system of decreasing sets

$$M_0,\ M_1,\ M_2,\ \ldots,\ M_\omega,\ M_{\omega+1},\ \ldots \tag{28}$$

formed according to (3), which begins with M, M_2, M_{22}, ..., and we look for the first set M_η that coincides with $M_{\eta+1}$. It coincides, also, with all the following sets and is the smallest set of the system. It is worth noting that the ϱ-kernel $\underline{M}$, originally defined as the greatest set $A \subseteq M$, or as the sum of all these sets, now appears as the smallest set of the system (5) or as the intersection of all the sets of this system, and is reached not from below, but from above. The "kernel" makes its appearance here as its "shell" is peeled off.

[1]Even for $\eta = \omega$, if we replace $\xi < \eta$ by the null set.

In the example $A' = A_d$, $A_2 = A_d$, the sets (5) are the coherences M, M_d, M_{dd}, ... of M, in which the process $A_2 = A - A_d$ consists in the splitting off of the isolated points; thus the summands $M_\xi - M_{\xi+1}$ of formula (4) are now isolated sets, and we obtain the dense-in-itself kernel or the smallest coherence by continuing the splitting off of the isolated points until the process comes to a standstill.

If, in particular, M is closed, then (5) is the sequence of derived sets M, M', M'', ..., and the dense-in-itself — and, in this case, perfect — kernel is the smallest derived set.

Before exhibiting a further example of the formation of a kernel, we still have to give an analogous discussion of the ϱ-hull $\overline{M}$, the smallest set $A \supseteq M$ for which $A = A_1$. Because of the monotonicity of A', we then have $A = A_1 \supseteq M'$, in containing M, A also contains the (in general) larger sets M_1, and thus M_{11}, M_{111}, etc. as well. If we define the sets M^ξ by the inductive rule

$$\begin{aligned} M^0 &= M \\ M^{\xi+1} &= (M^\xi)_1 \\ M^\gamma &= \bigcup_{\xi<\gamma} M^\xi \quad (\gamma \text{ a limit number}), \end{aligned} \tag{29}$$

where the decomposition

$$M^\eta = M^0 + \sum_{\xi<\eta} (M^{\xi+1} - M^\xi) \tag{30}$$

holds and where the summands $M^{\xi+1} - M^\xi$ must at some point become zero since their sum may not exceed the cardinality of the space, then we find the ϱ-hull of M as the greatest set of the increasing system

$$M^0,\ M^1,\ M^2,\ \ldots,\ M^\omega,\ M^{\omega+1},\ \ldots, \tag{31}$$

that is, as the first set that coincides with its successor (and with all the sets following).

In the example $A' = A_d$, $A_1 = A_0$, the system (8) begins with M, M_0, M_{00}, …, and the second set is already the greatest $M_0 = M_{00} = \cdots$ is the closed hull of M.

If the elements of the space happen to be real functions $x = x(t)$ of a variable t which, in turn, ranges over the real numbers or over any set T whatever, and if A' is then taken to mean the set of the functions $y = y(t)$ that can be represented as the limit functions $y(t) = \lim x_n(t)$ of sequences of functions $x_n \in A$ convergent everywhere (in T), then $A' = A_1$. A set of functions or a *system of functions* $A = A_1$ which is thus not extended by the formation of limits is called a *Baire system of functions.*[10] The smallest Baire system $\overline{M}$ over a given system M appears

[10]More about these in §43.

as greatest member of the sequence (8). Here this greatest member $\overline{M}^\eta$ surely has an index $\eta < \Omega$, where $\Omega = \omega_1$ is the beginning number of $\aleph_1$; for if $x_1, x_2, \ldots$ are functions of $M^2 = \bigcup_{\xi<\eta} M^\xi$ and, say, $x \in \overline{M}^\eta$, then let ξ ($< \Omega$) be the first ordinal exceeding all the ξ_n for which $x_n \in M^{\xi_n}$; since $\xi < \Omega$, the set of the ξ_n is at most countable, and so is ξ; then $x \in M^{\xi+1}$, and $\overline{M}^\eta = M^{\xi+1}$.

3. *The case of a reducible set.* As a further example of the formation of a kernel, we consider a space P that is closed in the complete space E and define A' for the sets $A \subseteq P$ by

$$A' = A \cdot P \cdot A_d = A_d(P) = A_\varrho, \tag{32}$$

that is, as the set of points of accumulation of A in the space P (not in E). The function $A' = A_\varrho$ is monotone; $A_1 = A + A_\varrho = A_0(P)$ is the closed hull of A in P, and $A_2 = A \cdot A_\varrho = A_\delta$ is the dense-in-itself kernel of A in P; $A = A_1$ means that A is closed in P, and $A = A_2$ means that A is dense-in-itself in P (relative to P). If we form the sequence (5) for a set $M \subseteq P$:

$$M_0 = M,\ M_1 = M_\varrho,\ M_2 = M_{\varrho\varrho},\ \ldots \tag{33}$$

then the members of this sequence, beginning with M, are called the *coherences* of M; their formation consists in the successive stripping off of the isolated points of M (relative to P). The greatest coherence M_ϱ, if it vanishes, indicates that M is a *separated* set, that is, contains no subset dense-in-itself relative to P.

If we form the residues

$$\begin{aligned} P_0 &= M_0 - M_1 = F_0 - F_1, \\ P_1 &= M_1 - M_2 = F_1 - F_2, \\ P_\xi &= M_\xi - M_{\xi+1} = F_\xi - F_{\xi+1}, \end{aligned} \tag{34}$$

then, by (4),

$$M = P_0 + P_1 + P_2 + \cdots + P_\xi + \cdots + M_\eta \tag{35}$$

and M_η is the greatest coherence of M.

I. *The coherences of a closed set are closed; the residues of a closed set are differences of two closed sets.*

For if M is closed in P, then $M_1 = M \cdot M_\varrho$ is also closed in P; if M_ξ is closed in P, then so is $M_{\xi+1} = M_\xi \cdot (M_\xi)_\varrho$; if γ is a limit number and all the M_ξ $(\xi < \gamma)$ are closed in P, then so is $M_\gamma = \bigcap_{\xi<\gamma} M_\xi$. The assertion about the residues follows from (10).

The formation of the coherences is only a special case of the general formation of the kernel. We can just as well start from a different monotone function. Let P be a set that is closed in the complete space E, and let

$$A' = A \cdot P \cdot A_i = A_i(P) \tag{36}$$

be the set of interior points of A in the space P. A' is a monotone function (defined for all $A \subseteq P$) of A; let us write

$$B = A \cdot A', \quad A' = P \cdot B_i.$$

Since B is closed in Q, we clearly have $A' = B_i - QB_i = B_i - B = B_e$, and if A is closed in P then, similarly, $B = A_e$, $A'' = A'_e$, A' is closed in A and thus also in P and $A' = A_e = A_\varrho$. If we therefore form the sequence (5) for a set M closed in P, then it follows inductively that all the M_ξ are also closed in P and that $M_{\xi+1} = M_{\xi e}$; these sets M_ξ, the sequence of which begins with M, M_e, M_{ee}, …, we call *residues* of M.

The smallest residue is the ϱ-kernel $\underline{M}$ of M, that is (since it is possible to take $P = M$), the greatest set A closed in M for which $A = A_\varrho$. A set whose smallest residue vanishes will be called *reducible.*

From §17, 3, we find:

II. *The reducible sets are the (finite or infinite) chains of differences formed from closed sets.*

For, first, it follows from (12) that

$$\begin{aligned} P_\xi &= M_\xi - M_{\xi+1} = F_\xi - F_{\xi+1}, \\ F_\xi &= M_\xi, \quad F_{\xi+1} = M_{\xi+1}. \end{aligned} \tag{37}$$

Because of (11), applied to $A = M_\xi$, we have $F_\xi \supseteq F_{\xi+1}$, and from $\xi < \eta$, it follows that $\xi + 1 \leq \eta$, $M_{\xi+1} \supseteq M_\eta$, and $F_{\xi+1} \supseteq F_\eta$. Consequently,

$$F_0 \supseteq F_1 \supseteq F_2 \supseteq \cdots \supseteq F_\xi \supseteq F_{\xi+1} \supseteq \cdots \tag{38}$$

and for a reducible set M, where M_η and F_η ultimately vanish, (4) gives

$$M = \sum_\xi (M_\xi - M_{\xi+1}) = \sum_\xi (F_\xi - F_{\xi+1}), \tag{39}$$

and M is a chain of differences of closed sets.

We preface the converse of this with: If A is closed in $M = F - N$ (F closed), then A_e is closed in N; for since $A_e \subseteq F$, we have $A_e - A = (F - M) \cdot A_e = N \cdot A_e$.

A two-fold application gives: If A is closed in $M = (F - F') + P$ (F and F' closed, $F \supseteq F' \supseteq P$), then A_e is closed in $F' - P$, and $A_{ee} = A_e$ in P. Let us form next from the closed, ultimately vanishing, sets (14) a chain of differences

$$M = F_0 - F_1 + F_2 - F_3 + \cdots$$

and set

$$P_\xi = M_\xi - M_{\xi+1} = (F_\xi - F_{\xi+1});$$

then

$$P_0 = M, \quad P_{\xi+1} = (P_\xi)_e - P_\xi),$$
$$P_\gamma = \bigcap_{\xi<\gamma} P_\xi \quad (\gamma \text{ a limit number}).$$

With a set A closed in M we form the residues

$$A_\xi \quad (A_0 = A,\ A_1 = A_e,\ A_2 = A_{ee},\ \ldots).$$

Then A_0 is closed in P_0; if A_ξ is closed in P_ξ, then it follows from the prefaced remark that $A_{\xi+1}$ is closed in $P_{\xi+1}$; if γ is a limit ordinal and if, for $\xi < \gamma$, A_ξ is closed in P_ξ, then A_γ is closed in P_γ. We have thus shown that every A_ξ is closed in P_ξ and ultimately becomes zero together with P_ξ; A is reducible. Every set closed in the chain of differences M, and in particular M itself, is reducible.

III. *The reducible sets constitute a field.*

This follows from the closing remark of §17, since the intersection of any number of closed sets is itself closed.

The reducible sets include, among many others, the separated sets. For from (9) and (10) it follows that $A_\varrho \subseteq A_\varrho$, and $A_{\varrho\varrho} \subseteq A_{\varrho\varrho} \subseteq A_{\varrho\varrho\varrho}$, and in any case $A_{\varrho\varrho} \subseteq A_\varrho$, from which it follows that the smallest residue of every set is dense-in-itself ($A = A_\varrho = A_{\varrho\varrho}$) and, in the case of a separated set, vanishes. If the M_ξ denote coherences, then from (4), also, we obtain a separated set (whose smallest coherence is 0) directly as a chain of differences of closed sets, since $A_\varrho \supseteq A_{\varrho\varrho} \supseteq A_{\varrho\varrho\varrho}$, and $A - A_\varrho = A_\varrho - A_{\varrho\varrho}$, so that

$$M_\xi - M_{\xi+1} = F_\xi - F_{\xi+1}, \quad F_\xi = M_\xi, \quad F_{\xi+1} = M_{\xi+1}.$$

4. *The case of a separable space.* Here we have the simplification that the smallest coherence or the smallest residue is always attained after at most a countable number of steps. Specifically:

IV. *If the space is separable, then an increasing or a decreasing well-ordered system of open (or closed) sets is at most countable.*

Let

$$\{G_0, G_1, \ldots, G_\xi, G_{\xi+1}, \ldots\} \quad (\xi < \varphi)$$

be an increasing system of open sets, $G_\xi \subset G_\eta$ for $\xi < \eta$; we may suppose its ordinal type φ to be a limit number. Among the special neighborhoods V (p. 146) contained in $G_{\xi+1}$ there surely is one, V_ξ, which is not contained in G_ξ. Then we have $V_\xi \subset V_\eta$ for $\xi < \eta$ (V_ξ is contained in G_η; V_η is not) and the set of the V_ξ, just like that of the G_ξ, is at most countable. The proof for a decreasing system takes a similar course; for closed sets, the theorem is obtained by taking complements.

It is applied, for example, to decreasing sequences of closed sets in the form: Let Ω be the beginning number of $\aleph_1$ and let a closed set F_ξ be assigned to each $\xi < \Omega$; for $\xi < \eta$ let $F_\xi \supseteq F_\eta$. Then an at most countable number of these sets can be distinct, that is, there exists a (first) ξ_0 such that $F_{\xi_0} = F_{\xi_0+1} = \cdots$ is identical with all the sets that follow.

Of course, all this holds also for the sets open and closed in a fixed set $M \subseteq E$, since M is also an (at most) separable space.

In the system (5), therefore, if its sets are all closed in M (for example, if we are dealing with coherences or residues), then the smallest set — that is, the first η, for which $M_\eta = M_{\eta+1}$ — is surely reached for $\eta < \Omega$.

The reducible sets are thus at most countable chains of differences (of type $< \Omega$) or sums of at most countably many sets of the form $F - F'$. Every such set, which can also be written in the form $F(E-F') = FG'$ as the intersection of a closed with an open set, is an F_σ — the sum of a sequence of closed sets (§23, III). Hence a reducible set is an $F_{\sigma\delta} = F_\sigma$; since by virtue of the field property its complement is also reducible, it is at the same time a G_δ — the intersection of a sequence of open sets. Therefore:

V. *The reducible sets of a separable space are simultaneously sets F_σ and G_δ.*

We remark that, as a counterpart:

VI. *If the space is an F_σ-set, then the sets that are simultaneously F_σ and G_δ are reducible.*

Let M be simultaneously an F_σ and a G_δ, and let A be its smallest residue, so that $A = A_{\varrho\varrho}$ or, with $B = A_e$,

$$A_1 = A + B = B_0 = F.$$

A is closed in M and therefore is simultaneously an F_σ and a G_δ; therefore $E - A$ and $B = F(E - A)$ are simultaneously an F_σ and a G_δ. A and B are both dense in F and, as F_σ with dense complement, they are of the first category in F; therefore F is also of the first category in itself; consequently $F = 0$ ($F > 0$ would be an F_σ-set). M is reducible.

In a separable F_σ-space — for example, in a separable complete space — the reducible sets are identical with those that are simultaneously F_σ and G_δ. It is noteworthy that the question whether a set is both F_σ and G_δ

can here be decided by a well-defined (although infinite) procedure, namely, by the formation of residues, while for the sets F_σ, for instance, nothing comparable is known.

Theorem IV allows of the following generalization:

VII. *If the space is separable, then an increasing or a decreasing well-ordered system of reducible sets is at most countable.*

It suffices to consider an increasing system (formation of complements). To every ordinal number $\beta < \Omega$ let us assign a reducible set M_β and, for $\beta < \gamma$, let $M_\beta \subseteq M_\gamma$; we must show that the M_β ultimately become equal ($M_\beta = M_{\beta+1} = \cdots$ for suitable β). Let $M_{\beta\xi}$ be the ξ-th residue of $M_\beta = M_{\beta 0}$ ($\xi < \Omega$, just as all the ordinals yet to appear are also $< \Omega$). Because of (13) and (15), we have

$$M_\beta = \sum_\xi (M_{\beta\xi} - M_{\beta,\xi+1}) = \sum_\xi (F_{\beta\xi} - F_{\beta,\xi+1}),$$

$$F_{\beta\xi} = M_{\beta\xi}, \quad F_{\beta,\xi+1} = M_{\beta,\xi+1}.$$

We now assert: There exists a fixed sequence, independent of β, of closed sets

$$F_0 \supseteq F_1 \supseteq F_2 \supseteq \cdots \supseteq F_\xi \supseteq F_{\xi+1} \supseteq \cdots \tag{40}$$

such that ultimately $F_{\beta\xi} = F_\xi$ and $F_{\beta,\xi+1} = F_{\xi+1}$ (for $\beta \geq \beta_\xi$, where in addition the β_ξ may be assumed to increase with ξ). If this has already been shown for all $\beta < \gamma$ and γ_0 is the first ordinal number following all the β_ξ, then for $\beta = \gamma$, we have

$$M_\gamma = \sum_\xi (F_\xi - F_{\xi+1}) + M_{\gamma 0};$$

the $M_{\gamma 0}$ then increase with increasing γ, and their closed hulls $F_{\gamma 0}$, for which the same is true, ultimately become identical, by IV: $F_{\gamma 0} = F_0$ for $\gamma \geq \gamma_0$. We have, however,

$$M_{\gamma,\xi+1} = F_\gamma - M_{\gamma\xi};$$

these sets and their closed hulls $F_{\gamma,\xi+1}$ decrease with increasing γ and, again because of IV, we have $F_{\gamma,\xi+1} = F_{\xi+1}$ for $\gamma \geq \gamma_0$ $(= \beta_{\xi+1} \geq \gamma_0)$. The assertion is thus proved for γ (the start of the induction, $\gamma = 0$, also is the same).

Again, the sets (16) themselves ultimately become identical: let $F_\xi = F_{\xi+1} = F_{\xi+2} = \cdots = F$. Then for $\beta \geq \beta_\xi$, we find

$$M_{\beta,\xi+1} - M_{\beta,\xi+2} = F_{\xi+1} - F_{\xi+2} = 0,$$

$M_{\beta,\xi+1}$ is the smallest residue of M_β; thus $\underline{M}_\beta = 0$, and

$$M_\beta = \sum_\xi (F_\xi - F_{\xi+1}) = M$$

is independent of ξ, so that VII is proved.

It is of interest to establish that it is possible to form sets having arbitrarily large index ($< \Omega$) for the smallest coherence or the smallest residue. Consider, for example, bounded closed sets of real numbers well-ordered according to size[11]

$$M = \{m_1, m_2, \ldots, m_\xi, \ldots\},$$

where thus the index of m_ξ takes on the values $\xi = 1, 2, \ldots, \eta$ $(1 \leq \xi \leq \eta)$, $m_\xi < m_\eta$ whenever $\xi < \eta$, and m_η is the least upper bound of the numbers m_ξ $(\xi < \eta)$ for a limit ordinal η. Such sets, with prescribed $\eta < \Omega$, can be formed immediately — for instance (η assumed infinite), by ordering a convergent sequence of positive numbers according to the type ω, denoting it by c_α $(0 \leq \alpha < \omega)$, and setting

$$m_\xi = \sum_{\alpha \leq \beta} c_\alpha \quad \text{for } \xi \leq \omega,$$

then continuing with further sequences.

The reader will see without difficulty that the first derived set M' consists of those m_ξ whose indices $\xi = \omega \cdot \alpha$ are limit ordinals, the second derived set M'' of those of the m_ξ just named for which $\alpha_\xi = \omega \cdot \alpha_\xi$ is a limit ordinal also, so that $\xi = \omega^2 \cdot \alpha_\xi$ is a multiple of ω^2, and continuing in this way by induction, that the ξ-th derived set $M^{(\xi)}$ consists of those m_ξ whose index $\xi = \omega^\xi \cdot \alpha_\xi$ is a multiple of ω^ξ. For sufficiently large ξ, such multiples of course no longer occur in the sequence $1, 2, \ldots, \eta$, and $M^{(\xi)}$ vanishes. For example, for $\eta = \omega^\omega$ and $\xi < \omega$, $M^{(\xi)}$ consists of the m_α for which

$$\alpha = \omega^\xi,\ \omega^\xi \cdot 2,\ \ldots,\ \omega^\xi \cdot k;$$

$M^{(\omega)}$ consists of the single m_α for which $\alpha = \omega^\omega$, and $M^{(\omega+1)} = 0$ is the smallest coherence.

[11] Compare Appendix B.

The example of the set

$$M = \{1, \tfrac{1}{2}, \tfrac{1}{3}, \dots, 0\}$$

shows that the smallest coherence may have an arbitrarily large index $< \Omega$. For the first derived set M' consists of the single point 0; if we form the set $A = M - M' + A'$, where A' is a set of type η dense in M', then

$$A_1 = (M_1 - M') + A' = A',$$

and A is dense in M, $A_e = M_e$, and $A_e = M_e - A = M_e - A'$. Here $A_1 \subseteq M_2$, $A_2 = M - M_0$ is dense in M_1, and the repetition of the conclusion gives $A_2 = A_e$. The first residue of A is thus A_e, and the steps of the induction show that A_ξ is the ξ-th residue of A; it ultimately vanishes, but only when $M^{(\xi)}$ is the smallest derived set $(= 0)$ of M; and when this has arbitrarily large index, so does the smallest residue.

§31. Further Applications of Ordinal Numbers

1. *Maximal and Minimal sets.* In the cases considered thus far, ordinal numbers might if necessary have been dispensed with, since the existence of hulls and kernels was already established. Nevertheless they served for the exploration of subtler characteristics, for example, the index of the smallest coherence. In other cases, however, it is not even possible to give an existence proof for certain objects without them.

If in a system of sets $\mathfrak{U}$ there exists no greatest set (the sum of all the sets of the system), there may nonetheless exist (several) maximal sets A, in the sense that no sets $> A$ occur in the system. Such maximal sets will be attained, in general, only in some such way as the following. We choose an arbitrary set A_0 in the system; next, if sets $> A_0$ exist, a set A_1 from among them; if then sets $> A_1$ exist, a set A_2; and so on. If, for some finite ν, we reach a set A_ν for which there exists no set $> A_\nu$, then we have a maximal set. Otherwise, we have a sequence $A_0 < A_1 < A_2 < \cdots$ of sets of type ω. It may now be that no set of the system contains all the A_ν; our enterprise has failed... We therefore make the assumption that an increasing sequence of sets without a last element can always be continued; thus:

$$\begin{array}{l}(\varepsilon)\text{ If }\gamma\text{ is a limit ordinal and if}\\ A_0 < A_1 < \cdots < A_\xi < A_{\xi+1} < \cdots \quad (\xi < \gamma)\end{array} \tag{41}$$

is a sequence of increasing sets of the system $\mathfrak{U}$ of type γ, then there exists a set A_γ in $\mathfrak{U}$ that contains all the A_ξ ($A_\xi < A_\gamma$).

The above process can now be continued, and it must finally lead to a maximal set. Let us therefore define A_γ ($\gamma > 0$) by induction as a set greater than all the A_ξ ($\xi < \gamma$) ($A_\xi < A_\gamma$) in case all the A_ξ have already been defined and there is such a set A_γ available, while otherwise we do not define A_γ; the set of the A_γ so defined then cannot exceed the cardinal number of the system $\mathfrak{U}$, and there exists a smallest ordinal number $\zeta > 0$ for which A_ζ is not defined. Because of (ε), ζ cannot be a limit ordinal, so that $\zeta = \vartheta + 1$; A_ϑ is still defined, and the sets $A_0, \ldots, A_\vartheta$ increase with increasing index. There is no set $> A_\vartheta$ in the system, since otherwise such a set could have been taken as A_ζ; A_ϑ is a maximal set.

The process described here differs from those in §30 mainly in that it lacks direction; in general, we have a choice from among several possible sets for each A_γ, and this may lead, in turn, to different maximal sets. This element of arbitrariness cannot be eliminated but can only be shifted back: we well-order the original system $\mathfrak{U}$ to form a system $\mathfrak{U}^*$ and then for our A_γ we always choose from the sets available the one that is the first in $\mathfrak{U}^*$. A particular well-ordering produces a particular maximal set; different well-orderings may give rise to different maximal sets. If Zermelo's well-ordering process is used in such a way that, whenever possible, a set containing A, B, ... as proper subsets is taken as differential element (§12) of $\{A, B, \ldots\}$, then $\mathfrak{U}^*$ begins with increasing sets $A_0 < A_1 < \cdots < A_\nu$, the last of which will be maximal. But this does not differ essentially from our original method except that there $A_{\nu+1}$ was no longer defined, whereas here the well-ordering $\mathfrak{U}^*$ continues.

It will frequently be possible to form the increasing sets A_γ in such a way that each goes over into its successor $A_{\gamma+1}$ by the addition of a single point x_γ. A simple example is given by the nets $E(\varrho)$ we had occasion to use earlier (p. 145): maximal sets in the system of those sets A in which any two points have a distance $\geq \varrho > 0$. Choose any point x_0 in the space: if for $\gamma > 0$ the points x_ξ (with $\xi < \gamma$) have already been defined, then we call the set of all these points A_γ, and define x_γ as any point whose distance is $\geq \varrho$ from all the x_ξ, if such a point exists; otherwise we leave x_γ undefined. The set of points so defined cannot exceed the cardinality of the space. If γ is the first ordinal number for which x_γ is not defined, then A_γ is a net $E(\varrho)$.

Similarly, we can construct, following Hamel, a basis for the set E of the real numbers.[12] If A is a set of real numbers we let $[A]$ be the set of those real numbers

$$y = \sum_{a \in A} r_a \cdot a \tag{42}$$

that can be represented as linear combinations of finitely many numbers $a \in A$ with rational coefficients r_a (so that only finitely many $r_a \neq 0$ are

[12] Or, more generally, for a linear space E.

allowed in the sum). If at least one equation

$$0 = \sum_{a \in A} r_a \cdot a \tag{43}$$

holds in which not all the r_a vanish, then we say that A is *rationally dependent*, otherwise *rationally independent*, or a *basis* for $[A]$. If A is rationally independent, then every number y of $[A]$ can be represented in the form (1) in only one way. We obtain a basis for E, which is to be conceived as a maximal set in the system of all bases (and incidentally also as a minimal set in the system of sets A for which $[A] = E$) as follows: We choose a real number $x_0 \neq 0$, then an x_1 that is not a rational multiple $r_0 x_0$ of x_0, then an x_2 that is not a rational combination $r_0 x_0 + r_1 x_1$ of x_0 and x_1; in general: if $\gamma > 0$, if the numbers x_ξ, the set of which is called A_γ, are already defined for $\xi < \gamma$, and if, finally, $[A_\gamma] < E$, then let x_γ be a number of $E - [A_\gamma]$, and otherwise leave x_γ undefined. If γ is the smallest ordinal number for which x_γ is not defined, then A_γ is a basis for E.

If A is a basis for E, and if the real function $f(x)$ is defined for $x \in A$ in any way whatever and is then extended to every real y in accordance with the unique representation (1) by the formula

$$f(y) = \sum_{a \in A} r_a \cdot f(a),$$

then we always have $f(y + z) = f(y) + f(z)$. This functional equation thus has $2^{\aleph}$ solutions (since the basis A is obviously of the cardinality of the continuum). If we require, in addition, that $f(y)$ be continuous (at a single point, and consequently everywhere) then, as is well known, it has only the solutions $f(y) = cy$, with constant c.

For minimal sets (sets A for which there are in the system $\mathfrak{U}$ no sets properly contained in A), the existence proof proceeds analogously, with an assumption (δ) corresponding to (ε) concerning the possibility of the continuation of decreasing sequences whose type is a limit number.

As an example, we consider all the continua A (connected, closed sets) in the space E that contain the two fixed distinct points x and y. A minimal set — one, therefore, that contains as a subset no smaller continuum joining the points x and y — is called, following Zoretti, a *continuum irreducible between* x and y.[13] The periphery of a circle on which x and y lie is reducible, while each of the two circular arcs that have x and y as end points is irreducible.

[13]This concept has no relation to the reducibility defined in §30, 3. Compare §39.

The existence of such minimal sets is of course not assured in all cases. Nevertheless it is true (Janiszewski) that:

> *A compact continuum E containing the points x and y has (at least) one subcontinuum irreducible between x and y.*

For $\xi < \Omega$ (Ω a beginning number of $\aleph_1$, as usual) we define the sets A_ξ as follows:

$$A_0 = E;$$
$$A_{\xi+1}, \text{ a continuum } < A_\xi \text{ containing } x, y, \text{ if there are any such continu}$$
$$\text{otherwise } A_{\xi+1} = A_\xi;$$
$$A_\gamma = \bigcap_{\xi<\gamma} A_\xi \quad (\xi < \gamma,\ \gamma \text{ a limit ordinal}).$$

We see that the A_ξ decrease with increasing index (equality not excluded) and are continua containing x and y; for the A_ξ with limit index this follows from Theorem XVIII of §29, by representing γ as limit of a sequence $\xi_1 < \xi_2 < \cdots$ and taking $A_\gamma = A_{\xi_1} A_{\xi_2} \cdots$ into account. Since E is separable, Theorem IV of §30 applies, and at some time we must have $A_\xi = A_{\xi+1}$, and A_ξ irreducible between x and y.

It is possible, incidentally, to proceed in such a way that, at the very latest, A_ω is irreducible. We consider the open complements G (open in the space E) of the sets A and look for a maximal set in the collection of the G. Let $V_1, V_2, \ldots$ be the special neighborhoods (§25) in the space E. If E is reducible, so that some non-vanishing G_0 exists, then we choose the latter so that the index of the first $V_p \subset G_0$ is as small as possible. If $A_1 = E - G_0$ is reducible, so that some $G_1 > G_0$ exists, then we choose G_1 so that the index of the first V_q $(q > p)$ contained in it but not in G_0 is as small as possible, and we continue in this way. Either some

G_n with finite index or, at the latest, $G_\omega = G_1 + G_2 + \cdots$ is a maximal G, and thus $A_\omega = E - G_\omega$ or $A_n = E - G_n$ is irreducible.

We give still another example of the utilization of the ordinal numbers, one which no longer concerns maximal or minimal sets.

2. *Totally imperfect sets.* Let E be a perfect, separable, complete space ($E = E_0$). A set A containing no non-empty perfect subset is called *totally imperfect*; its perfect kernel A_0 (§23) vanishes. Conversely, if $A_0 > 0$, then A contains a perfect subset (for example, A_0 itself). The question arises whether there exist totally imperfect sets of the second category in E (that is, sets that are not the sum of countably many nowhere-dense sets). Bernstein showed that:

> *There exist totally imperfect sets whose complements are also totally imperfect; such a set, as well as its complement, is of the second category in E.*

For the proof, we well-order the non-empty perfect sets of the separable space (of which there are $\aleph$ many) into a transfinite sequence

$$P_0, P_1, \ldots, P_\xi, \ldots \quad (\xi < \Omega)$$

of type Ω. By induction, we choose two distinct points a_ξ, b_ξ in each P_ξ as follows: If, for all $\eta < \xi$, the points a_η, b_η have already been chosen, then these points form a set of cardinality $< \aleph$ ($\xi < \Omega$); since P_ξ has cardinality $\aleph$, there exist points in P_ξ different from all the a_η, b_η; let a_ξ, b_ξ be two such points. Now form

$$A = \{a_0, a_1, \ldots, a_\xi, \ldots\}, \quad B = \{b_0, b_1, \ldots, b_\xi, \ldots\}.$$

Then A and B are disjoint, and every non-empty perfect set P contains a point of A (namely a_ξ, if $P = P_\xi$) and a point of B (namely b_ξ). Therefore A contains no non-empty perfect subset (otherwise such a subset would have to contain a point of B, contradicting $AB = 0$); similarly, $B = E - A$ contains no non-empty perfect subset. If A were of the first category, then $E - A = B$ would be of the second category (§27, IX); but B is totally imperfect, and a set of the first category has a perfect kernel 0; therefore B would be of the first category, contradicting §27, X. Thus both A and B are of the second category.

§ 6.31. Further Applications of Ordinal Numbers

The existence of such minimal sets is of course not assured in all cases. Nevertheless it is true (Janiszewski) that:

A compact continuum E containing the points x and y has (at least) one subcontinuum irreducible between x and y.

For $\xi < \Omega$ (Ω a beginning number of $\mathrm{Z}(\aleph_0)$, as usual) we define the sets A_ξ as follows:

$$\begin{aligned} A_0 &= E; \\ A_{\xi+1} &= \text{a continuum } \subset A_\xi \text{ containing } x, y, \text{ if there are any such continua;} \\ &\qquad \text{otherwise } A_{\xi+1} = A_\xi; \\ A_\eta &= \bigcap A_\xi \quad (\xi < \eta, \eta \text{ a limit ordinal}). \end{aligned}$$

We see that the A_ξ decrease with increasing index (equality not excluded) and are continua containing x and y; for the A_η with limit index this follows from Theorem XVIII of §29, by representing η as limit of a sequence $\xi_1 < \xi_2 < \cdots$ and taking $A_\eta = A_{\xi_1} A_{\xi_2} \cdots$ into account. Since E is separable, Theorem IV of §30 applies, and at some time we must have $A_\xi = A_{\xi+1}$, and A_ξ irreducible between x and y.

It is possible, incidentally, to proceed in such a way that, at the very latest, A_{Ω_1} is irreducible. We consider the open complements G (open in the space E) of the sets A and look for a maximal set in the collection of the G. Let $V_1, V_2, \ldots$ be the special neighborhoods (§25) in the space E. If E is reducible, so that some non-vanishing G_0 exists, then we choose the latter so that the index of the first $V_i \subset G_0$ is as small as possible. If $A_1 = E - G_0$ is reducible, so that some $G_1 \supset G_0$ exists, then we choose G_1 so that the index of the first V_i $(i > 0)$ contained in it but not in G_0 is as small as possible, and we continue in this way. Either some G_ξ with finite index or, at the latest, $G_{\Omega_1} = G_1 + G_2 + \cdots$ is a maximal G, and thus $A_{\Omega_1} = E - G_{\Omega_1}$ or $A_{\Omega_1} = E - G_\xi$ is irreducible.

We give still another example of the utilization of the ordinal numbers, one which no longer concerns maximal or minimal sets.

2. Totally imperfect sets. Let E be a perfect, separable, complete space (for example, a Euclidean space). The system of (non-empty) perfect sets P has the cardinality $\aleph$ (§25, V) and

$$\aleph = \aleph_1^{\aleph_0} \tag{6.1}$$

is the beginning number of the class $Z(\aleph_0)$. We can arrange all the (non-empty) perfect sets P in a sequence of type

$$P_0, P_1, \dots, P_\xi, \dots \quad (\xi < \aleph). \tag{6.2}$$

We choose two distinct points $x_0, y_0 \in P_0$. If, for $0 < \xi < \aleph$, all the points x_ζ and y_ζ $(\zeta < \xi)$ have already been defined and if we call the sets formed by them A_ξ and B_ξ, then $A_\xi + B_\xi$ is of some cardinality $< \aleph$, and thus there remain infinitely many $(\aleph)$ points of P_ξ that do not belong to the set $A_\xi + B_\xi$. We define x_ξ and y_ξ as two distinct such points. It follows from this definition that for $\zeta < \xi$ the four points $x_\zeta, y_\zeta, x_\xi, y_\xi$ are all different. Let A be the set of all points x_ξ; its complement $B = E - A$ contains all the points y_ξ (where it is possible, incidentally, to arrange matters so as to make B into just the set of the points y_ξ). Both the sets A and B are of cardinality $\aleph$ and are totally imperfect, for since AP_ξ contains the point x_ξ and BP_ξ the point y_ξ, we never have $P_\xi \subseteq B$ or $P_\xi \subseteq A$.

We have thus decomposed E into two totally imperfect sets A and B. This is of interest for the theory of connectedness as well; for A and B are discontinuous (p. 174), and yet, if E is a Euclidean space of dimension at least two, they are connected. For we have: *In a Euclidean space of dimension at least two the complement of a totally imperfect set is connected* (Sierpiński).

If we take E to be, say, the Euclidean plane, then we must show the impossibility of a decomposition $E = A + B$

in which A is totally imperfect and B disconnected. Let $B = P + Q$ be a partition into two relatively closed sets $P = BP$, and $Q = BQ$; since $PQ = BP \cdot BQ = 0$, the closed set $F = P_a Q_a$ is contained in A. We join two points $p \in P$ and $q \in Q$ by a polygonal path, say $C = [p, m, q]$, where m lies on the perpendicular bisector of p and q. Every (closed) sub-segment of C is perfect, is therefore not contained in A, and consequently contains points of B. CB is dense in C, $C \subseteq B_a$, $C = CB_a = CP_a + CQ_a$, where now, in order that this not be a partition of C, the closed set $F = P_a Q_a$ must have points on C. But then F, and therefore A, contains a perfect set (a sub-segment of C), contrary to hypothesis.

§ 6.32. Borel and Suslin Sets

The Borel and Suslin sets generated by the closed sets F (§§18 and 19) are called the *Borel and Suslin sets of the space E.*

Since the intersection F_δ of a sequence of closed sets (and even of arbitrarily many closed sets) is also closed, we shall begin the generation of the Borel sets with the F_σ, as was done in §18, (3) or according to the rule (α) there given. We therefore have the following definition for the ordinal numbers $\xi < \Omega$:

The sets F^0 are the sets F.

(α) $\boxed{\text{The sets } F^\eta \text{ are, for odd } \eta\text{, the sums and for even } \eta > 0\text{, the intersections, of sequences of sets } F^\xi \ (\xi < \eta).}$

The sets $F^0, F^1, F^2, F^3, \ldots$ are the $F, F_\sigma, F_{\sigma\delta}, F_{\sigma\delta\sigma}, \ldots$; next come the F^ω as the intersections of sequences of the earlier sets, etc.

Let us consider, in addition, the Borel sets generated by the open sets G, beginning with the G_δ as in §18, (4) or in accordance with the rule (β); then we have the following definition:

The sets G^0 are the sets G.

generated by closed sets F, and so forth, in the notation familiar from §19. We recall that all the Borel sets are also Suslin sets and that the iteration of the Suslin process yields nothing new: every F_{SS} is an F_S. The Suslin sets generated by the open sets

$$G_S = \mathfrak{S}\, G_{n_1} G_{n_1,n_2} G_{n_1,n_2,n_3} \cdots \tag{6.4}$$

are in their totality identical with the F_S; for every G is an F_σ and thus an F_S, and G_S is an $F_S = F'_S$, while similarly F_S is a G_S. The G_S are not, however, the complements of the F_S, which rather look as follows:

$$E - F_S = \mathfrak{D}\,(G_{n_1} + G_{n_1,n_2} + G_{n_1,n_2,n_3} + \cdots) \tag{6.5}$$

and as we shall shortly see, the Suslin sets do not in general form a field.

The property of being a Borel or a Suslin set is *relative* and depends on the space E; we ought therefore to write, more explicitly, $F^\xi(E)$, and so forth. We have already spoken of this in the case of the F_σ, G_δ, $F_{\sigma\delta}$, and $G_{\delta\sigma}$ (§26, 3); if $D \subseteq E$ and the argument E is again omitted, then

$$F(D) = D \cdot F, \qquad G(D) = D \cdot G, \tag{6.6}$$

that is, the Borel sets of the space D are the intersections of D with the corresponding Borel sets of the space E. The proof, by induction, is immediate. The same holds for the Suslin sets:

$$F_S(D) = D \cdot F_S, \qquad G_S(D) = D \cdot G_S. \tag{6.7}$$

There also exist *absolute* Borel and absolute Suslin sets — sets, that is, that have this character in every containing space E. A set has this character absolutely if it possesses it in any complete space and, specifically, in its complete hull $\bar{A}$. We have already satisfied ourselves, for the case $\xi = 1$, that the non-existence of absolute G_δ does not contradict the existence of absolute G_δ^ξ ($\xi \geqq 1$); even in G_δ only summands of the form $G_\delta = G^0$ appear.

We prove now the **Theorem on Cardinality**:

Theorem 6.32.1. *In a complete separable space every Suslin set — and thus, in particular, every Borel set — is either at most countable or of cardinality* $\aleph$.

This is a generalization of §26, XI; but it is no longer the case that, as happened there, the vanishing or non-vanishing of the dense-in-itself kernel is decisive for the cardinality. The set of rational numbers (an F_σ) is dense-in-itself and yet only countable. We shall show, as was done then, that an uncountable Suslin set contains a dyadic discontinuum D as a subset, but in the construction of the spheres we must now use points of condensation instead of points of accumulation. Let

$$A = \mathfrak{S}\, F(i) \cdot F(i,k) \cdot F(i,k,l) \cdots \tag{6.8}$$

be a Suslin set generated by closed sets F, where the sum is taken over all the sequences of natural numbers $(i, k, l, \ldots)$ while, as we wish to point out immediately, $(j, g, h, \ldots)$ is to denote a dyadic sequence of numbers formed from the numerals 1 and 2. For simplicity, we assume all the $F(i, k, \ldots)$ to be non-empty (the vanishing ones contribute nothing to A). Furthermore, as we know from §19, the F may be supposed decreasing with an increasing number of indices and their diameters converging to 0.

If A is uncountable, then there exists a sphere U such that $A \cdot U$ is uncountable (the point of condensation of A used here may even be a point of A itself, and this is, in fact, necessary if A has no point of condensation other than points of A itself). Among the summands

$$A(i) = F(i) \cdot F(i,k) \cdot F(i,k,l) \cdots \tag{6.9}$$

there must then be at least one, say $A(i_1)$, for which $A(i_1) \cdot U$ is uncountable. We choose two condensation points α_1, α_2 of, and in, this set and enclose them in disjoint closed spheres $V_1, V_2 \subseteq U$; for the open spheres U_1, U_2 belonging to them, each of the four sets $A(i_1, k) \cdot U_p$ is uncountable. Then at least one summand of $A(i_1) \cdot U_p = \sum_k A(i_1, k) \cdot U_p$, is uncountable — say, $A(i_1, k_p) \cdot U_p$. We continue in this way, and thus there corresponds to each dyadic sequence of numbers $(j, g, h, \ldots)$ a sequence of natural numbers $(i_1, k_{j_1}, l_{j_1 g_1}, \ldots)$ such that the sets

$$A(i_1) \cdot U_j, \quad A(i_1, k_{j_1}) \cdot U_{j_1 g}, \quad A(i_1, k_{j_1}, l_{j_1 g_1}) \cdot U_{j_1 g_1 h}, \ldots$$

are uncountable and therefore all the more so are the closed sets

$$F(i_1) \cdot V_j, \quad F(i_1, k_{j_1}) \cdot V_{j_1 g}, \quad F(i_1, k_{j_1}, l_{j_1 g_1}) \cdot V_{j_1 g_1 h}, \ldots$$

since $A(i) = F(i)$, and so forth. If, now, we had taken the radii of the spheres $V_j, V_{j_1 g}, V_{j_1 g_1 h}, \ldots$ to be $< 1, 1/2, 1/3, \ldots$ — and there was nothing to prevent our doing so — then the closed sets just referred to constitute a decreasing sequence whose diameters $\to 0$. Therefore they have a single point of intersection x which belongs to the dyadic discontinuum

$$D = \mathfrak{D}\, V_j \cdot \mathfrak{D}\, V_{j_1 g} \cdot \mathfrak{D}\, V_{j_1 g_1 h} \cdots \tag{6.10}$$

as well as to the set A. This holds for every dyadic sequence, and thus for every point $x \in D$, and D is a subset of A.

Theorem I is the most comprehensive theorem on cardinality that we know. But nevertheless it applies only to a vanishingly small subsystem of the system of all point sets. In a separable space there are $\aleph$ closed sets (§25, IV) and therefore $\aleph^{\aleph_0} = \aleph$ sequences of closed sets and hence $\aleph$ Suslin sets, while there are $2^{\aleph}$ point sets all together.

§ 6.33. Existence Proofs

We now turn to the question whether, in the process by which the Borel sets F^{ξ}, G^{ξ} are generated, genuinely new sets appear at every step, or whether at some time the process comes to a standstill. For trivial spaces E, this is certainly not the case. If E consists entirely of isolated points, then every point set is closed and open at the same time, and even the very first step of the process is superfluous. If the space is countable, then every point set is an F_σ and a G_δ at the same time, and the $F_{\sigma\delta}$, and so forth, give us nothing new. If, therefore, we say that a set G^η is *exactly* a G^η or belongs exactly to the Borel class η if it is not already a G^ξ $(\xi < \eta)$, then there remains the question, under what circumstances does there exist for each index $\eta < \Omega$ sets that are exactly sets G^η. As regards this, it is to be noted that in

$$G^0 \subset G^1 \subset G^2 \subset \cdots \subset G^\xi \subset G^{\xi+1} \subset \cdots$$

there are either only strict inequalities or, from some index on, nothing but equalities, for if $G^\eta = G^{\eta+1}$, then G^η is already the whole Borel system (p. 101).

Furthermore, every Borel set B has already been shown to be a Suslin set S, and we may ask whether sets S exist that are not sets B. In the above trivial spaces all the point sets were indeed Borel sets, and so every S was also a B. In order that an S be a B it is in any event necessary that its complement $E - S$ also be (a B, and so) an S, inasmuch as the B form a field; and therefore an S whose complement is not an S is certainly not a B.

Both these questions can be discussed together, and their answer is given in the existence theorem:

Theorem 6.33.1. *In a complete space whose dense-in-itself kernel does not vanish, there exist Suslin sets that are not Borel sets, and for every ordinal number $\eta < \Omega$ there exist sets that are exactly G^η.*

Proof. The proof is based on the construction of a Suslin set C, generated by Borel (indeed, by open) sets, and of its complement $D = C - C$, which is not a Suslin set. We choose the basic space to be the Cantor discontinuum, whose points we represent as usual by dyadic sequences $x = (j_1, j_2, j_3, \ldots)$ with $j_n = 1$ or 2. Let D be the set of those x having only finitely many $j_n = 1$; this is the set of left endpoints of the removed intervals and is therefore countable. Its complement $C = D - D$ remains the set of those x that have infinitely many $j_n = 1$, and this set is a G_δ, since it is obtained from the perfect set D by the removal of a set F_σ (which is countable).

We should like to represent the points $x \in C$ somewhat differently — by means of sequences of natural numbers $(i_1, i_2, \ldots)$ rather than by dyadic sequences. For a natural number i let $[i]$ denote a dyadic complex consisting of a sequence of $i-1$ twos followed by the numeral 1, for example

$$[1] = (1), \quad [2] = (2,1), \quad [3] = (2,2,1), \ldots,$$

and further, let $[i_1, i_2, \ldots, i_k]$ denote the dyadic complex resulting from adjoining $[i_1], [i_2], \ldots, [i_k]$ in this order — for example, $[2,3,1] = (2,1,2,2,1,1)$ — and similarly, denote by $[i_1, i_2, \ldots]$ the sequence of numerals, formed in a similar way, that has infinitely many ones (namely, $p_1 = i_1$, $p_2 = i_1 + i_2$, $p_3 = i_1 + i_2 + i_3$, …, with the remaining $p_n = 2$). If we then set

$$F_{i_1 i_2 \cdots i_k} = V_{[i_1, i_2, \ldots, i_k]}, \tag{6.11}$$

then for every sequence of natural numbers

$$F_{i_1} \supseteq F_{i_1 i_2} \supseteq F_{i_1 i_2 i_3} \supseteq \cdots$$

with diameters $\to 0$, the sets F with k indices are disjoint (e.g., $F_1 = V_1$ and $F_2 = V_{2,1}$), and we have

$$C = \mathfrak{S}\, F_{i_1} \cdot F_{i_1 i_2} \cdot F_{i_1 i_2 i_3} \cdots, \tag{6.12}$$

where every point x corresponds bi-uniquely to a sequence $(i_1, i_2, \ldots)$ of natural numbers (such that x is the only point of the intersection $F_{i_1} \cdot F_{i_1 i_2} \cdots$).

It will now suffice to prove Theorem I for the (not complete) space C. For, Suslin sets in C and Borel sets G^ξ in C ($\xi \geqq 1$) are Suslin and Borel sets in E as well, because C is a $G_\delta = G^0$ in E.

Let $\mathfrak{M}$ be the system of sets $M \subseteq C$ that are of the form

$$M = A_{n_1 n_2 \cdots n_k} = A(n_1, n_2, \ldots, n_k), \tag{6.13}$$

where the $A(n_1, \ldots, n_k)$ are open in C (and their number is therefore at most countable). The Suslin sets S generated by $\mathfrak{M}$ are identical with those of the space C, since even the sets

M_n (and thus surely the M^*) include all the open sets G of this space. Every B^ξ is a G^ξ and every G^ξ at least a $B^{\xi+1}$ (for $\xi = 0$, a B^0).

Now let $\nu = (m_1, m_2, \ldots)$ be an increasing sequence of natural numbers, N a collection of such sequences, and

$$X = \mathfrak{S}\, M_{\nu_1} M_{\nu_2} \cdots = \Phi(M_{\nu_1}, M_{\nu_2}, \ldots) \quad (M_{\nu_p} \in \mathfrak{M}), \tag{6.14}$$

where the $\mathfrak{S}$-function Φ of its arguments (which are arbitrary sets of $\mathfrak{M}$), is determined by the set N. From Theorem II of §18 and Theorem II of §19, we know that for suitable N or Φ the set X represents exactly the sets S or even the sets B^ξ $(1 \leqq \xi < \Omega)$. We choose and hold fixed any particular function Φ whatever; then all the sets representable in the form (3) by means of this function we call the sets Φ.

By means of the sequence of natural numbers $n_1, n_2, \ldots$ determined by x, we can now assign uniquely to each point $x \in C$ the set

$$\Phi(x) = \Phi(U_1, U_2, \ldots) \tag{6.15}$$

determined by x. This is therefore a set Φ, and every set Φ can be so represented, for $M_n \in \mathfrak{M}$ means precisely that $M_n = U_{n_p}$ is to be one of the U_n. (The natural numbers $n_1, n_2, \ldots$ are quite arbitrary and not, say, increasing or pairwise distinct.) Every Φ is therefore some $\Phi(x)$ where, incidentally, it may very well happen that $\Phi(x) = \Phi(y)$ even though $x \neq y$.

The point x can now either belong to the set $\Phi(x)$ assigned to it, or not. Let A be the set of $x \in \Phi(x)$, B the set of $x \notin \Phi(x)$; $A + B = C$. Then B is different from all the $\Phi(x)$. For if we had $B = \Phi(x_0)$, then, according as $x_0 \in B$ or $x_0 \notin B$, we would have $x_0 \in A$ or $x_0 \in B$; the first case contradicts $x_0 \in B$, the second contradicts $B = \Phi(x_0)$ (since $x_0 \in B$ would mean $x_0 \notin \Phi(x_0) = B$).

Now B can certainly not be a set Φ, and in particular, if Φ represents the Suslin sets, B is not a Suslin set.

We now show: if Φ represents the sets B^ξ $(1 \leqq \xi < \eta)$, then A is a set B^η and B a set $B^{\eta+1}$. Since

$$x \in U_n \text{ means that } n_k = n \text{ for at least one } k,$$

it follows that U_n is the sum of the sets $C - F_{n_1 \cdots n_k n}$ (for $n = 1$, to the set $C - F_n$), with arbitrary natural numbers $n_1, \ldots, n_{k-1}$; $A_{n_1 \cdots n_k}$ is the sum of these sets over the $n_1, \ldots, n_{k-1}$, and is thus an F_σ (in C). Finally,

$$A = \mathfrak{S}\, A_{n_1 n_2 \cdots} \cdot U_n$$

is thus also a Borel set; for $x \in U_n$ means, after all, that there exists a natural number k for which $x \in F_{n_1 \cdots n_k}$, $n_k = n$, that is, $x \in A_{n_1 \cdots n_k} \cdot U_n$.

With this we have reached our goal, which is the following:

If Φ represents exactly the Suslin sets S, then by (5), A_η is a Suslin set generated by Suslin sets (and even Borel sets) A_ξ, and so is itself a set S. But its complement B is not an S, and therefore A is not a Borel set.

If Φ represents exactly the Borel set B^ξ, then, by (5), A is a Borel set generated by the Borel sets A_ξ, and so is itself a Borel set of the space C. Its complement B is then also a Borel set, although not a B^ξ, rather it is exactly some B^η with arbitrarily high index, and the same, then, holds for G^η. But then for every index η there are sets that are exactly G^η. This holds, first, for indices $\xi + 1$: if every $G^{\xi+1}$ were a G^ξ, then $G^\xi = G^{\xi+1}$ would be the whole Borel system, and all the G^η would already be sets G^ξ. But this holds also for the limit number η: if every G^η were a G^ξ ($\xi < \eta$), and thus an $F^{\xi+1}$, then every $G^{\eta+1}$ would have to be an F^η, every $F^{\eta+1}$ a G^η and every $G^{\eta+1}$ a G^η. This concludes the proof of the existence theorem.

§ 6.34. Criteria for Borel Sets

1. Necessary conditions. The existence theorem has taught us that in a suitable space there exist Suslin sets that are not Borel sets. We therefore ask for criteria by which to recognize whether a given Suslin set is a Borel set. We already know one:

I. *If a Suslin set S is a Borel set, then so is its complement $E - S$.*

For $E - S$ is even a Borel set. We have already used this obvious fact in the existence proof.

II. *If the Suslin set S is a Borel set, then it can be represented by means of disjoint summands.*

The proof rests on the following conclusions, in which, for the sake of brevity, a Suslin set representable by means of disjoint summands will be called an *L-set*, and an L-set A whose complement $E - A$ is also an L-set will be called a *B-set.*

(A). *The sum of a sequence of disjoint L-sets is an L-set.*

These L-sets can be represented in the form

$$A^{(n)} = \mathfrak{S}\, F^{(n)}_{m_1} F^{(n)}_{m_1 m_2} F^{(n)}_{m_1 m_2 m_3} \cdots ;$$

if, in addition, we set $F_{m_1} = E$ (the whole space), then

$$A = \mathfrak{S}\, F_{m_1} \cdot F_{m_1 m_2} \cdot F_{m_1 m_2 m_3} \cdots = \mathfrak{S}\, A^{(m_1)}_{m_2 m_3 \cdots}$$

is an L-set.

(B). *The intersection of a sequence of L-sets is an L-set.*

Let

$$\begin{aligned} A &= \mathfrak{S}\, A_{n_1} A_{n_1 n_2} A_{n_1 n_2 n_3} \cdots , \\ B &= \mathfrak{S}\, B_{n_1} B_{n_1 n_2} B_{n_1 n_2 n_3} \cdots , \\ C &= \mathfrak{S}\, C_{n_1} C_{n_1 n_2} C_{n_1 n_2 n_3} \cdots \end{aligned}$$

be countably many L-sets (A_{n_1} etc. closed). We combine all the sequences of indices into a single sequence, say by means of the diagonal process:

$$n_1, \quad n_1, n_2, \quad n_1, n_2, n_3, \quad \ldots$$

$= m_1, m_2, m_3, m_4, \ldots$ Then we find that

$$\begin{aligned} ABC \cdots &= A_{m_1} \cdot A_{m_1 m_2} \cdot B_{m_1} \cdot A_{m_1 m_2 m_3} \cdot B_{m_1 m_2} \cdot C_{m_1} \cdots \\ &= \mathfrak{S}\, F_{m_1} F_{m_1 m_2} F_{m_1 m_2 m_3} \cdots , \end{aligned}$$

where $F_{m_1 \cdots m_k}$ denotes that one among the sets written above it whose highest index is m_k (so that $F_{m_1 \cdots m_k}$ depends only on $m_1, \ldots, m_k$, although not, perhaps, on all of them; for instance, $F_{m_1 m_2 m_3} = B_{m_1}$). $ABC \cdots$ has thus been represented as an L-set. Of course, the intersection of a finite number of L-sets is also an L-set.

(C). *The sum and the intersection of a sequence of B-sets are B-sets.*

Let the sets $A_1, A_2, \ldots$ and their complements $B_1, B_2, \ldots$ be L-sets. As a consequence of (B), the intersection $A_1 A_2 \cdots$ and — bringing in (A) — the sum

$$A_1 + A_2 + A_3 + \cdots = A_1 + B_1 A_2 + B_1 B_2 A_3 + \cdots$$

as well, are L-sets — and even B-sets, since the same is true of the B_n.

(D). *The sets F_σ are L-sets.*

A set F_σ

$$A = F_1 + F_2 + F_3 + \cdots$$

can be supposed to be made up of increasing closed summands, so that

$$A = F_1 + (F_2 - F_1) + (F_3 - F_2) + \cdots$$

with disjoint summands which, as differences of closed sets, are (special) sets F_σ. In this way we find

$$A = \mathfrak{S}\, F_{n_1} = \mathfrak{S}\, A_{n_1}, \quad A_{n_1} = F_{n_1} - F_{n_1 - 1},$$
$$A_{n_1} = \mathfrak{S}\, F_{n_1 n_2} = \mathfrak{S}\, A_{n_1 n_2}, \quad A_{n_1 n_2} = F_{n_1 n_2} - F_{n_1 n_2 - 1},$$

and so forth ($F_0 = F_{00} = \cdots = 0$). Because of

$$F_{n_1} \supseteq A_{n_1} \supseteq F_{n_1 n_2} \supseteq A_{n_1 n_2} \supseteq \cdots$$

we have a representation of A

$$A = \mathfrak{S}\, A_{n_1} A_{n_1 n_2} \cdots = \mathfrak{S}\, F_{n_1} F_{n_1 n_2} \cdots$$

as an L-set.

To this we add a comment which, although superfluous at the moment, will be used later: If the space is separable, then the representation of A by disjoint summands can be so arranged that the diameters of the sets $F_{n_1 \cdots n_k}$ converge to 0 as $k \to \infty$. For E can be represented as sum of a sequence of closed sets with arbitrarily small diameters, so that we can suppose the closed sets F_n to have diameters < 1, the closed sets $F_{n_1 n_2}$ to have diameters $< 1/2$, and so forth.

As a result of (D), the closed sets and the open sets, in particular, are L-sets, that is, B-sets; and since (C) shows that the B-sets form a Borel system, the Borel sets of the space are surely B-sets, thus proving II.

2. The indices. Let

$$A = \mathfrak{S}\, F(n_1) \cdot F(n_1, n_2) \cdot F(n_1, n_2, n_3) \cdots \qquad (6.16)$$

be a Suslin set of the space E, and $B = E - A$ its complement. We have thus assigned here to every finite complex of natural numbers, which we denote for brevity by

$$r = (n_1, n_2, \ldots, n_p) \qquad (6.17)$$

a closed set

$$F(r) = F(n_1, n_2, \ldots, n_p), \qquad (6.18)$$

and the sum in (1) extends over all sequences $(n_1, n_2, \ldots)$ of natural numbers. We may suppose here, as we did earlier, that

$$F(n_1) \supseteq F(n_1, n_2) \supseteq F(n_1, n_2, n_3) \supseteq \cdots . \qquad (6.19)$$

If we adjoin one more natural number to the complex (2), we obtain a complex

$$(r, n) = (n_1, \ldots, n_p, n),$$

which we call a *successor* of r; r is called the *predecessor* of (r, n).

Let R_0 be any set of complexes r. If r, but no successor of r, belongs to R_0, we call r a *final element* of R_0. By means of transfinite induction, we now define a sequence of sets of complexes:

$$\begin{aligned} R_{\xi+1} &= R_\xi - \{\text{final elements of } R_\xi\}, \\ R_\eta &= \textstyle\bigcap_{\xi<\eta} R_\xi \quad (\eta \text{ a limit number}). \end{aligned} \tag{6.20}$$

Thus $R_{\xi+1}$ is obtained from R_ξ by removing the final elements; R_η consists of those complexes that belong to all R_ξ with $\xi < \eta$. Obviously

$$R_0 \supseteq R_1 \supseteq R_2 \supseteq \cdots \supseteq R_\xi \supseteq R_{\xi+1} \supseteq \cdots ,$$

and this decreasing sequence comes to a standstill at the latest at $\xi = \Omega$; for the set of all complexes r is countable, and if at some time the final elements are exhausted, then no further decrease takes place. That is, from some index γ on, $R_\gamma = R_{\gamma+1} = \cdots$; this set R_γ can thus no longer contain final elements, although it may, of course, be empty. Let us call this γ the *index* of the set R_0; then, for $\xi < \gamma$, we still have $R_\xi > R_{\xi+1}$. The set $R = R_\gamma = R_{\gamma+1} = \cdots$ will be called, say, the *kernel* of R_0 (it is the greatest set $\subseteq R_0$ that has no final elements). Processes of this kind are familiar to us from §30; the present one greatly resembles the splitting off of the isolated points and the formation of the smallest coherence.

Every point x of the space now determines the set $R_0(x)$ of those r for which $x \in F(r)$, and if we perform on this set the process just described, then we reach the sets $R_\xi(x)$, ending with the kernel $R(x) = R_\gamma(x) = R_{\gamma+1}(x) = \cdots$; the index $\gamma = \gamma(x)$ of the set $R_0(x)$, which depends on x, is also called the *index of the point* x. We therefore have for the sets A and B a splitting according to the indices of their points:

$$A = \textstyle\sum_\gamma A_\gamma, \qquad B = \sum_\gamma B_\gamma \qquad (\gamma < \Omega), \tag{6.21}$$

where A_γ and B_γ are the sets of those points of A and B whose index is $\gamma(x) = \gamma$. We shall now see that these sets are Borel sets.

Every set $R_\xi(x)$ contains the predecessors of its elements, or

$$\text{for } (r, n) \in R_\xi(x), \text{ we have } r \in R_\xi(x); \tag{6.22}$$

for this holds for $\xi = 0$, and if it holds for all $\xi < \eta$, then it holds for η as well: For $\eta = \zeta + 1$, if $(r, n) \in R_{\zeta+1}(x)$, then $(r, n) \in R_\zeta(x)$ and has a successor $(r, n, m) \in R_\zeta(x)$; by the induction hypothesis, $(r, n) \in R_\zeta(x)$, and so $r \in R_\zeta(x)$. Since r has a successor $(r, n) \in R_\zeta(x)$, r is not a final element of $R_\zeta(x)$, and so $r \in R_{\zeta+1}(x) = R_\eta(x)$. For limit η, $R_\eta(x)$ is the intersection of all $R_\xi(x)$ with $\xi < \eta$, and the property is obvious.

Furthermore,

$$r \in R_{\xi+1}(x) \text{ means: } (r, n) \in R_\xi(x) \text{ for at least one } n. \tag{6.23}$$

In fact, $r \in R_{\xi+1}(x)$ means that r is not a final element of $R_\xi(x)$, i.e., r has a successor $(r, n) \in R_\xi(x)$.

We now define, for every ξ and every complex r, a set $F_\xi(r)$ by means of the following recursion:

$$\begin{aligned} F_0(r) &= F(r), \\ F_{\xi+1}(r) &= \textstyle\sum_n F_\xi(r, n), \\ F_\eta(r) &= \textstyle\bigcap_{\xi<\eta} F_\xi(r) \quad (\eta \text{ a limit number}). \end{aligned} \tag{6.24}$$

The first of these equations arises from the definition of $R_0(x)$ and the third from that of $R_\eta(x)$, so that (7) holds for η if and only if it holds for all $\xi < \eta$. The middle equation in (9) follows thus: $x \in F_{\xi+1}(r)$ or $r \in R_{\xi+1}(x)$ means that r is an element, though not a final element, of $R_\xi(x)$ and therefore has a successor $(r, n) \in R_\xi(x)$; or, that some (r, n) occurs in $R_\xi(x)$ (so that, by (6), the assertion $r \in R_\xi(x)$ becomes superfluous); or, that x belongs to one of the sets $F_\xi(r, n)$.

Furthermore,

$$S_\xi = \mathfrak{S}\, F_\xi(r) \tag{6.25}$$

is the set of points x for which $R_\xi(x) > 0$; for this means just that there exists some r for which (7) holds. And

$$T_\xi = \mathfrak{S}\, [F_\xi(r) - F_{\xi+1}(r)] \tag{6.26}$$

is the set of the points x for which $R_\xi(x) > R_{\xi+1}(x)$; for this means that there is some r which is a final element of $R_\xi(x)$, for which, therefore,

$$r \in R_\xi(x), \quad r \notin R_{\xi+1}(x),$$

i.e., $x \in F_\xi(r)$, $x \notin F_{\xi+1}(r)$ or $x \in F_\xi(r) - F_{\xi+1}(r)$.

Now for these sets, we have

$$\begin{aligned} S_\xi &= A_{\xi+1} + A_{\xi+2} + \cdots + B_{\xi+1} + B_{\xi+2} + \cdots\ , \\ T_\xi &= A_{\xi+1} + A_{\xi+2} + \cdots + B_\xi. \end{aligned} \tag{6.27}$$

For, first of all, the kernel $R(x)$ of $R_0(x)$ is non-empty if and only if $x \in A$. If $x \in A$, then $R(x) > 0$, and conversely. Now $x \in S_\xi$ means $R_\xi(x) > 0$, i.e., there exists $r \in R_\xi(x)$. This is certainly the case when $\gamma(x) > \xi$, since then $R_\xi(x) \supseteq R_\gamma(x) = R(x) \neq 0$. If $\gamma(x) \leqq \xi$, then $R_\xi(x) = R_{\xi+1}(x) = \cdots = R(x) = 0$ (since $x \in B$), so that $x \notin S_\xi$.

For $x \in B$, and in this case only, $R(x) = 0$.

Now, because of the definition of the indices $\gamma(x)$, we have the equivalences of

$$\begin{aligned} &R_\xi(x) > R_{\xi+1}(x) \quad \text{with } \xi < \gamma(x), \\ &R_\xi(x) = R_{\xi+1}(x) = R(x) \quad \text{with } \xi \geqq \gamma(x). \end{aligned}$$

Hence T_ξ is the set of points whose index is $> \xi$, while S_ξ (defined by $R_\xi(x) > 0$) contains all the points of A as well as those of B whose index is $> \xi$, and no others. (12) has thus been proved. From it we obtain further

$$S_\xi = S_{\xi+1} + T_\xi, \qquad E - S_\xi = B_0 + B_1 + \cdots + B_\xi. \tag{6.28}$$

Starting from the closed sets $F_0(r)$, it now follows from (9), using induction, that all the sets $F_\xi(r)$ are Borel sets of the space and from (10) and (11) that the S_ξ and T_ξ are Borel sets; the same follows for the A_ξ and B_ξ, say from (13) by induction. By (5), therefore, the Suslin set A and its complement B are represented as sums of $\aleph_0$ Borel sets;

both sets are then also intersections of $\aleph_0$ Borel sets, and in particular, from (12) we have simply

$$A = \mathfrak{D}\, S_\xi. \tag{6.29}$$

The sets S_ξ and T_ξ decrease with increasing index; the sets (13) increase. The intersection of all the T_ξ is

$$\bigcap_\xi T_\xi = B_0 + B_1 + B_2 + \cdots = B, \tag{6.30}$$

that is, the points of B are those and only those whose index is $> \xi$ for every ξ, i.e., $\gamma(x) = \infty$ (which is to mean that $R_\xi(x) > R_{\xi+1}(x)$ for every ξ).

The intersection of the S_ξ is A. Since $S_\xi = A + (B_{\xi+1} + B_{\xi+2} + \cdots)$, the points of the intersection are those of A and, in addition, those points of B that belong to every S_ξ, i.e., for which $R_\xi(x) > 0$ for every ξ. But for points of B, $R(x) = 0$, and so $R_\xi(x) = 0$ from some index on; hence no point of B belongs to all S_ξ, and the intersection of the S_ξ is exactly A.

3. Sufficient conditions. Let us begin with two auxiliary considerations.

(a) If to every $\xi < \Omega$ there are assigned countably many sets D_n^ξ (say, for $n = 1, 2, \ldots$) which decrease as ξ increases ($D_n^\xi \supseteq D_n^\eta$ for $\xi < \eta$), and if for each ξ

$$\textstyle\sum_n D_n^\xi > 0,$$

then there exists an n such that for every ξ we also have that $D_n^\xi > 0$.

For, there exists for each ξ first some $n(\xi)$ for which $D_{n(\xi)}^\xi > 0$. The function $n(\xi)$, which can only take on a countable number of values, must take on at least one value uncountably often. But then $D_n^\xi > 0$ for uncountably many ξ, and thus (because of the property of monotonicity) for all ξ.

(b) If A is a Suslin set with disjoint summands, then it can also be represented as a Suslin set

$$A = \mathfrak{S}\, F_{m_1} \cdot F_{m_1 m_2} \cdot F_{m_1 m_2 m_3} \cdots, \tag{6.31}$$

where the $F_{m_1 \cdots m_k}$ are closed and have diameters converging to 0 as $k \to \infty$; if we then form the intersection of this with the set (1), assign the natural numbers n bi-uniquely to the number pairs (m_1, m_2), and set

$$F_{n_1 n_2 \cdots n_k} = E(m_1, \ldots, m_k) \cdot F_{n_1} \cdot F_{n_1 n_2} \cdots F_{n_1 \cdots n_k},$$

then

$$A = \mathfrak{S}\, F_{n_1} \cdot F_{n_1 n_2} \cdot F_{n_1 n_2 n_3} \cdots$$

becomes a representation of the form desired; furthermore, as we should like to point out, it has disjoint summands whenever this is true for (1). The subsidiary condition $F_{n_1} \supseteq F_{n_1 n_2} \supseteq \cdots$ too can be maintained, if we assume $E(m_1) \supseteq E(m_1, m_2) \supseteq \cdots$.

These considerations out of the way, we shall next invert Theorems I and II.

Theorem 6.34.1. *If the space is complete and separable, then a Suslin set whose complement is a Suslin set is a Borel set. Two disjoint Suslin sets are Borel sets considered as subsets of their sum.*

For the proof, we consider two Suslin sets A, $\bar{A}$ with the corresponding Borel sets $F_\xi(r)$, $\bar{F}_\xi(r)$ and S_ξ, $\bar{S}_\xi$. Let the representations (1) of A and $\bar{A}$ satisfy condition (8) on diameters. Then we show that if $S_\xi \bar{S}_\xi$ is non-empty for every ξ, then $A\bar{A}$ is non-empty.

In fact, we have by (15), for every ξ (and in what follows, as well)

$$\textstyle\sum_r F_\xi(r) \cdot \bar{F}_\xi(r) > 0,$$

where the sum extends over all complexes $r = (n_1, \ldots, n_k)$ of length $k = k(\xi)$; and, more than that, there exists at least one complex r_ξ for which

$$F_\xi(r_\xi) \cdot \bar{F}_\xi(r_\xi) > 0. \tag{6.32}$$

For $\xi = 0$, this is simply $F(r_0) \cdot \bar{F}(r_0) > 0$. For $\xi = 1$, there exists, by (a), at least one n for which $F_1(n) \cdot \bar{F}_1(n) > 0$, since

$S_1\bar{S}_1 > 0$ means that $\sum_n F_1(n) \cdot \bar{F}_1(n) > 0$. But $F_1(n) = \sum_m F(n,m)$, and so on; in general, from $F_\xi(r) \cdot \bar{F}_\xi(r) > 0$ it follows that at least one summand of the sum

$$\textstyle\sum_m F_\xi(r,m) \cdot \bar{F}_\xi(r,m) = F_{\xi+1}(r) \cdot \bar{F}_{\xi+1}(r)$$

is positive, say for $m = m_\xi$, so that $F_{\xi+1}(r, m_\xi) \cdot \bar{F}_{\xi+1}(r, m_\xi) > 0$. In this way, from the complex r_ξ we obtain a complex $r_{\xi+1} = (r_\xi, m_\xi)$ for which (17) holds. For a limit number η, r_η is to be the complex consisting of all the complexes r_ξ with $\xi < \eta$, or the common initial part of all the r_ξ with $\xi < \eta$.

Continuing in this way, we obtain two (because $r > 0$) different sequences $(m_1, m_2, \ldots)$ and $(\bar{m}_1, \bar{m}_2, \ldots)$ of numbers such that $(\xi = 0, 1, 2, \ldots)$

$$F(m_1, \ldots, m_\xi) \cdot \bar{F}(\bar{m}_1, \ldots, \bar{m}_\xi) > 0.$$

These closed sets, decreasing for increasing ξ, and with diameters that approach zero, have by the Second Intersection Theorem, a (unique) common point of intersection x which belongs to A as well as to $\bar{A}$; therefore $A\bar{A}$ is non-empty.

If, for every ξ, $S_\xi\bar{S}_\xi$ is non-empty then, by (12) or (14), $S_\xi \supseteq S_\xi$ is a fortiori non-empty; and consequently so is $A\bar{A}$. Therefore, conversely:

If $A\bar{A} = 0$, then at some time for some ξ (and for all the ones following) we must have $S_\xi\bar{S}_\xi = 0$, so that

$$A = A\bar{S}_\xi = (A + \bar{A})\bar{S}_\xi,$$

and A is the intersection of $A + \bar{A}$ with a Borel set. Two disjoint Suslin sets are thus Borel sets in their sum. *If*, in particular, the complement $B = E - A$ of a Suslin set is also a Suslin set, then both are Borel sets (in $A + B = E$), and from a certain index on, we even have $A = S_\xi$.

III has now been proved; the Suslin condition is sufficient provided the space is complete and separable. The second half of III shows, incidentally, that the theorem also holds if the space is a Suslin set M in a complete separable space E.

Similarly, the Lusin criterion can be shown to be sufficient:

Theorem 6.34.2. *If the space is complete and separable, then every Suslin set which can be represented by means of disjoint summands is a Borel set.*

This is, in any event, a consequence of III; for if S is representable by means of disjoint summands, then so is $E - S$, and by II of the present section, $E - S$ is then an L-set, hence a Suslin set, and so, by III, S is a Borel set.

The theorem can also be proved directly, without reference to the complement, by modifying the proof of III. We consider, for every ξ, the sets $F_\xi(r)$ in (9), which are Borel sets, and we wish to show that S_ξ becomes constant from some index on, so that $A = S_\xi$ is a Borel set. Now S_ξ decreases as ξ increases; if the S_ξ were all different, then their complements would form an increasing sequence of $\aleph_1$ different Borel sets, which in a separable space is impossible (by the theorem on cardinality, I of §32); therefore $S_\xi = S_{\xi+1} = \cdots$ from some ξ on. If $S_\xi = S_{\xi+1}$, then, because of (13), $T_\xi = 0$; hence no point has index $> \xi$, i.e., $A_{\xi+1} = A_{\xi+2} = \cdots = 0$ and $B_{\xi+1} = B_{\xi+2} = \cdots = 0$, while $A = S_\xi$ and $B = E - S_\xi$. This concludes the proof of IV.

Chapter 7

Mappings of Two Spaces

§ 7.1. Continuous Mappings

1. Foundations. As in §2, we begin with an already given set of ordered pairs (x, y); let the sets consisting of the elements x and y that enter into these pairs be called respectively A and B. Thus there correspond to each $x \in A$ one or more *images* (or image points, transforms, or maps) $y = \varphi(x)$, namely those y that in combination with x form a pair (x, y) of the set of pairs; similarly, there correspond to each y one or more *inverse images* (or pre-images) $x = \psi(y)$. What we have here are two functions φ and ψ inverse to each other — in general, many-valued — and a *mapping* between the spaces A and B, where the asymmetry induced by the ordered pairs and fixed in the terminology (image, inverse image) is, at this point, immaterial. From the point functions φ and ψ it is a natural step to set functions Φ and Ψ. A set $P \subseteq A$ determines its *image* (or image set) $\Phi(P)$, the set of all the images of the points of P, where $\Phi(0) = 0$ always; a set $Q \subseteq B$ determines its *inverse image* $\Psi(Q)$, the set of all the inverse images of the points of Q, where $\Psi(0) = 0$ always. The following formulas hold:

$$\Phi(P) \subseteq Q \text{ and } P \subseteq \Psi(Q) \text{ are equivalent,} \tag{7.1}$$

that is, all the points of P have images in Q if and only if all the points of Q have inverse images in P; further,

$$\begin{aligned}\Psi(B-Q) &= A-\Psi(Q),\\ \Phi(P_1+P_2+\cdots) &= \Phi(P_1)+\Phi(P_2)+\cdots,\\ \Psi(Q_1\cdot Q_2\cdots) &= \Psi(Q_1)\cdot\Psi(Q_2)\cdots;\end{aligned}\tag{7.2}$$

this is most easily understood by noting that the set A now decomposes into disjoint subsets $\Psi(\{y\})$, the inverse images of the individual points y of B. (It would be more accurate to write $\Psi(\{y\})$, the inverse image of the set $\{y\}$ consisting of only one point.) In future we consider only the case in which $\varphi(x)$ is single-valued without, however, assuming this to be the case for the inverse function; a genuine asymmetry between A and B has thus entered.

Now let A and B be metric spaces. The function $\varphi(x)$ is said to be *continuous at the point* x if, for every sequence $x_n \to x$, we also have $\varphi(x_n) \to \varphi(x)$.

At the isolated points of A, every function is continuous. A continuous function of a continuous function is itself continuous. In more detail: Let $y = \varphi(x)$ and $z = \psi(y)$ be single-valued functions, where x, y, and z range over the sets A, B, and C, and suppose that $\varphi(x)$ is continuous at $x_0 = \psi(y_0)$ and $\psi(y)$ at $y_0 = \varphi(x_0)$; then the composite function $z = \chi(x) = \psi(\varphi(x))$ is continuous at x_0. The proof is familiar.

I. *B is a continuous image of A if and only if the inverse image of every set open (closed) in B is open (closed) in A.*

The corollary expressed by the words written in parentheses is proved by taking complements, since $\Psi(B-Q) = A-\Psi(Q)$.

We already know a special case of the theorem: Theorem I of §22. If $\varphi(x)$ is a real continuous function, so that B is a set of real numbers, then the set denoted in §22 by $[\varphi > 0]$ is nothing but the inverse image of the set Q consisting of the intersection of B with the half-line $y > 0$ and which is thus a set open in B.

Because of the formulas (2), even more follows from I:

II. *If B is a continuous image of A, then the inverse images of the Borel and Suslin sets of the space B are Borel and Suslin sets of the space A, and in fact, the sets $F^{\xi}(B)$ and $G^{\xi}(B)$ have sets $F^{\xi}(A)$ and $G^{\xi}(A)$ as their inverse images.*

This actually follows from the very fact that the inverse images of sums and intersections are, respectively, sums and intersections of the inverse images. We recall, incidentally, that the converse of I is not true, even for open and closed sets; there exist single-valued functions whose inverse images of open sets are open, and which nevertheless are not continuous ("discontinuous solutions of the functional equation $f(x+y) = f(x) + f(y)$").

If, however, both functions $y = \varphi(x)$ and $x = \psi(y)$ are single-valued and continuous, then each of them is called *bicontinuous* or *continuous both ways* (*fonction bicontinue*). B is also called a *homeomorphic image* of A (and A a homeomorphic image of B), and the one-to-one mapping between the two sets is called a *homeomorphism* (or *topological mapping*); furthermore, we say that A and B are *homeomorphic* to one another, which we shall express by the notation

$$B \simeq A \quad \text{or} \quad A \simeq B.$$

Once more — to summarize these concepts — we have the following cases depending on the nature of the function $x = \psi(y)$ ($y = \varphi(x)$ is assumed continuous):

Image of A	*Inverse function*
B a continuous image of A	$\psi(y)$ may be many-valued
B a one-to-one continuous image of A	$\psi(y)$ single-valued
B a homeomorphic image of A	$\psi(y)$ single-valued and continuous

2. The conservation of compactness and connectedness. In addition to the topological characteristics, the metric ones too are, in part, conserved under continuous mappings.

III. *The continuous image of a set compact in itself is itself compact in itself; the continuous image of a bounded closed set of a Euclidean space is itself bounded and closed.*

Let B be a continuous image of A and let $Q \subseteq B$ be infinite; we are to prove that Q has an accumulation point in B. The inverse image $P = \Psi(Q) \subseteq A$ is infinite and therefore has an accumulation point $x \in A$; every neighborhood of x contains infinitely many points of P, and therefore every neighborhood of $y = \varphi(x)$ contains infinitely many points of Q, so that y is an accumulation point of Q.

It follows further from this that a real function continuous in a set A compact in itself ranges over a closed and bounded set of numbers and therefore has a greatest and smallest value (that is, the least upper bound is an actually attained maximum, the greatest lower bound an actually attained minimum).

If we denote by K a set compact in itself, and by K_σ the sum of a sequence of such sets, then the continuous image of a K is itself a K, that of a K_σ a K_σ. (Compact sets in Euclidean spaces have the character K, F_σ sets in such spaces the character K_σ.)

IV. *A continuous function in a set A compact in itself is uniformly continuous.*

V. *Every set compact in itself is the continuous image of a compact, perfect, totally disconnected set. Two compact, perfect, totally disconnected sets are always homeomorphic.*

A compact, perfect, totally disconnected set can be represented, by §29, XVI, as a dyadic discontinuum

$$A = \mathfrak{D}\, V_{n_1} \cdot V_{n_1 n_2} \cdot V_{n_1 n_2 n_3} \cdots$$

(§26, 2), where the sets V may be assumed closed in A and therefore compact in themselves (by replacing V by $\bar{V}$). For fixed n, any pair of disjoint sets V with n indices have a positive lower distance between them, the smallest of which let us denote by ϵ_n. If x and ξ are distinct points of A, and if the corresponding dyadic sequences of numerals first differ at the nth place, then every connected set containing x and ξ contains at least one point of each of the 2^{n-1} sets V with n indices, and so has diameter $\geqq \epsilon_n$. Thus A is totally disconnected.

Now let A be any set compact in itself. Since A is separable, it contains a countable everywhere dense subset $\{a_1, a_2, \ldots\}$, and we define a function $\varphi(x)$ on the dyadic discontinuum just described by associating to every set $V_{n_1 \cdots n_k}$ one of the points a as its image, in such a way that the images of two intersecting sets V are arbitrarily close to each other. The function $\varphi(x)$ so defined is continuous on A, and $B = \Phi(A)$ is the continuous image of A; since B contains all the points a_n and is closed, $B = A$.

VI. *The continuous image of a connected set is itself connected. The continuous image of an arbitrary set has no greater number of components than the set has.*

If B is a continuous image of A and $B = B_1 + B_2$ a partition of B into two closed sets (closed in B), then there corresponds to this partition a partition of $A = \Psi(B_1) + \Psi(B_2) = A_1 + A_2$ into two closed sets (closed in A). The disconnectedness of B implies that of A; the connectedness of A implies that of B. Since every partial function $\varphi(x \mid P)$ is continuous whenever the function $\varphi(x) = \varphi(x \mid A)$ is continuous, there corresponds to every connected subset of A a connected subset of B; the image of a component of A falls completely inside a component of B, and the system of components of B has at most the same cardinality as that of the components of A. If A and B are homeomorphic, then the components of the two sets correspond to each other; if one of the sets is connected, then so is the other.

VII. *If the set A is compact in itself and is locally connected at every inverse image point $x = \psi(y)$ of y, then its continuous image B is locally connected at y.*

B, like A, is compact in itself and so at any rate is bounded. Let $y_n \to y$; in accordance with §29, XII we have to prove that the numbers $\delta_y^{y_n}$ there defined converge to 0. Suppose that $x_n = \psi(y_n)$ are any of the inverse images of the

y_n; the x_n have a convergent subsequence $x_p \to x$, whence $y_p = \varphi(x_p) \to \varphi(x)$, so that $y = \varphi(x)$, and therefore $x = \psi(y)$ is an inverse image of y. A is locally connected at x; $\xi_p \to x$; this means that, except for a finite number of p, x and x_p are joined by a connected set $U_p \subseteq A$ whose diameter converges to 0 as $p \to \infty$. Then y and y_p are joined by the connected image $B_p = \Phi(U_p)$, the diameter of which also converges to 0 because of the uniform continuity (Theorem IV), so that $\delta_y^{y_p} \to 0$. Therefore the $\delta_y^{y_n}$ (although some of them may not be defined) have a subsequence convergent to 0, and the same is true of every subsequence of the $\delta_y^{y_n}$, so that the whole sequence converges to 0.

§ 7.2. Interval-Images

As we know (p. 151), the convergence is uniform, that is, the greatest diameter δ_n of the sets V with n indices converges to 0 for $n \to \infty$. When we assumed the sets with n indices to be disjoint we obtained a dyadic discontinuum. Now we shall assume the other extreme: the sets to be discussed are to constitute a *chain* when arranged in lexicographic order; that is, in

$$\mathfrak{S}\, V_n = V_1 + V_2, \quad \mathfrak{S}\, V_{n_1 n_2} = V_{11} + V_{12} + V_{21} + V_{22}, \ldots \tag{7.3}$$

neighboring sets are to have points in common. The dyadic set

$$C = \mathfrak{D}\, V_{n_1} \cdot V_{n_1 n_2} \cdot V_{n_1 n_2 n_3} \cdots \tag{7.4}$$

generated by the V, for which, as we know, we might also write

$$C = \mathfrak{D}\, \mathfrak{S}\, V_{n_1} \cdot \mathfrak{S}\, V_{n_1 n_2} \cdot \mathfrak{S}\, V_{n_1 n_2 n_3} \cdots \tag{7.5}$$

will then be called a *dyadic continuum*; justification for the name will appear at once.

I. *Interval-images and dyadic continua are identical.*

Let $T_1 = [0, 1/2]$ and $T_2 = [1/2, 1]$ be the left-hand and right-hand halves of the interval T; similarly, T_{11} and T_{12} the left-hand and right-hand halves of T_1; T_{21} and T_{22} the left-hand and right-hand halves of T_2; and so forth. Every system of intervals $T_{n_1 \cdots n_k}$ with k indices is a chain in the order of the real numbers. The interval T itself is

$$T = \mathfrak{D}\, T_{n_1} \cdot T_{n_1 n_2} \cdot T_{n_1 n_2 n_3} \cdots ,$$

and the points $t \in T$ correspond bi-uniquely to the dyadic sequences $(n_1, n_2, \ldots)$.

Now let B be a continuous image of T, so that every $t \in T$ has precisely one image $y = \varphi(t)$, and let $V_{n_1 \cdots n_k}$ be the image of $T_{n_1 \cdots n_k}$. These are closed sets, and two sets with the same number of indices and neighboring subscripts have a point in common (at least the image of the common endpoint of the corresponding intervals). If t belongs to the intervals $T_{n_1}, T_{n_1 n_2}, T_{n_1 n_2 n_3}, \ldots$, then $y = \varphi(t)$ belongs to their images $V_{n_1}, V_{n_1 n_2}, V_{n_1 n_2 n_3}, \ldots$; therefore $y \in C$, and $B \subseteq C$. Conversely, let $y \in C$, and thus $y \in V_{n_1}, y \in V_{n_1 n_2}, y \in V_{n_1 n_2 n_3}, \ldots$; let t_k be the midpoint (or an arbitrary point) of $T_{n_1 \cdots n_k}$. Then t_k converges to a point $t \in T$, and because of the continuity of φ, the images $y_k = \varphi(t_k) \in V_{n_1 \cdots n_k}$ converge to $\varphi(t)$. Since the diameters of the $V_{n_1 \cdots n_k}$ converge to 0, y is the only point belonging to all the $V_{n_1 \cdots n_k}$, and so $y = \varphi(t) \in B$. Therefore $C \subseteq B$, and so $B = C$.

An interval-image is therefore a dyadic continuum.

Conversely, let C be a dyadic continuum, formed by means of the sets V in accordance with (2), where, as we may assume, the diameters of $V_{n_1 \cdots n_k}$ are $< \epsilon_k$ and $\epsilon_k \to 0$. To every point $x \in C$ there corresponds a dyadic sequence $(n_1, n_2, \ldots)$, and to this the number

$$t = \frac{n_1 - 1}{2} + \frac{n_2 - 1}{2^2} + \frac{n_3 - 1}{2^3} + \cdots$$

of the interval $T = [0, 1]$. This correspondence between x and t is single-valued, for if two different sequences correspond to x, then they are lexicographical neighbors, and since neighboring sets V have points in common, x belongs to both; the corresponding t are then equal (dyadic fractions). If t is given and t_k is a sequence of numbers converging to t, then the corresponding points x_k form a sequence whose distance from x converges to 0 whenever $t_k \to t$ does. For as soon as $|t - t'| < 1/2^n$, t and t' belong either to one T^* (subintervals with n indices) or to two neighboring T^*; so that x and x' belong to one V^* (sets V with n indices) or to two lexicographically neighboring V^*, and so $xx' \leqq 2\epsilon_n$. A dyadic continuum is therefore an interval-image.

Let us take note of the following corollary to I. If in all the chains (1) the intersections of lexicographically neighboring V and these only are non-empty, then C is a one-to-one and thus homeomorphic image of T. A point x can then belong to two dyadic sequences of numerals only if they are lexicographical neighbors, and so in that case as well it has only a single inverse image $t = \psi(x)$. If, for example, x belongs to the sequences of numerals $(1, g_1, g_2, \ldots)$ and $(2, h_1, h_2, \ldots)$, then all the dyadic sequences beginning with $1, g_1, \ldots, g_{k-1}, 2$ correspond to the same point of C as the sequence beginning with $2, h_1, \ldots, h_{k-1}, 1$; the corresponding t are equal, and x has only one inverse image.

A point of an interval-image can therefore have one or two (but never more than two) inverse images; it has precisely two when the sequences corresponding to it are not lexicographical neighbors, i.e., when there are points of C that are separated by this point. These points are characterized by the fact that $C - x$ is disconnected (for a partition of T into two subintervals at the point $t = \psi(x)$ corresponds to a partition of C into two relatively closed sets at the point x).

II. *Every connected, locally connected, compact set A is an interval-image.*

The proof uses the fact that such a set is *arcwise connected* (§29, XIV). Let a and b be two points of A, $K(a, b)$ the continuum (in fact, the irreducible continuum) between a and b. We divide $K(a, b)$ into a finite number of subcontinua with arbitrarily small diameters; this is possible because of the local connectivity and the compactness of A. Let

$$A = A_1 + A_2 + \cdots + A_p = K(a_0, a_p)$$

$(a_0, a_p$ any points of A; $a_0 \in A_1$, $a_p \in A_p)$; next,

$$A_p = A_{p1} + \cdots + A_{pq} = K(a_{p-1}, a_{pq})$$

$(a_{p-1}, a_{pq}$ any points of A_p; $a_{p-1} \in A_{p1}$, $a_{pq} \in A_{pq})$; and so forth. If we choose here

$$\begin{gathered} a_0 = x \in A_1, \quad a_p = y \in A_p, \\ a_p = a_{p-1} \in A_{p1}, \quad a_{pq} = a_{pq-1} \in A_{pq}, \ldots, \\ a_{p\cdots q} = x_{p\cdots q} \in A_{p\cdots q}, \quad a_{p\cdots qr} = y_{p\cdots q} \in A_{p\cdots qr}, \end{gathered}$$

then the sets $A_{p\cdots qr}$, in lexicographical order, form a chain, since, to take one example, $A_{p_1}A_{p_2}$ contains the point $y_1 =$

x_2, and since it is evident that the procedure can be continued without limit. A is therefore a polyadic continuum (p. 230) and thus an interval-image, and II has been proved.

III. (*Hahn-Mazurkiewicz Theorem*). *The set A is an interval-image if and only if it is a compact, locally connected continuum.*

To prove the condition sufficient, we show that a compact, locally connected continuum satisfies the hypothesis of II; we need to show that it is connected and locally connected. Let A be compact, locally connected, and connected. Then A is a continuum, and, by §29, XIV, it is arcwise connected. Let a and b be any two points of A. The irreducible continuum $K(a,b) \subseteq A$ between them is, by the local connectivity of A, all of A; for if $K(a,b) < A$, then the local connectivity at a point $x \in A - K(a,b)$ would give a neighborhood U of x whose component in A is a neighborhood of x, and this component would have to meet $K(a,b)$, contradicting the irreducibility. So $A = K(a,b)$, and A is an interval-image by II.

§ 7.3. Images of Suslin Sets

1. In §§35–36 we discussed the conservation of the simplest properties of sets under continuous mappings — compactness, connectedness, and local connectivity. For the moment, there seems to be even no prospect of improving on this one result, since we have already found that even an absolutely closed set F can have any set whatever (though of course one of the same cardinality as F) as its continuous image. If, however, we consider sets in separable, complete spaces, then we can still prove a relatively strong result: *The continuous images of Suslin sets are Suslin sets, and the one-to-one continuous images of Borel sets are Borel sets.* The Suslin and Borel sets of a complete space are of this character in every space — they are absolute Suslin and Borel sets — so that we may here speak of separable absolute S and B.

Before giving a precise statement of the theorem and proving it, we should like to make the following preliminary remarks.

Let A be a set in a space X, and let B be a subset of a space Y. The points $x \in A$ are to be mapped into points $y = \varphi(x) \in B$. In general, this is not a mapping of the space X into the space Y; for the points $x \in X - A$ have no images at all, and for points $y \in Y - B$ no inverse images need exist. We therefore say: a *partial* mapping between X and Y is given; the space X is mapped only partially (namely, on the subset A) into the space Y. In order nevertheless to have mappings between whole spaces, we can proceed as follows: Let A be closed in X (which we can always arrange by replacing X with the complete hull of A if necessary); we define a new space $Z = X \times Y$ (§21) whose points are the pairs $z = (x, y)$, and in this space we consider the set A^* of pairs $z = (x, \varphi(x))$, i.e., the "graph" of the function φ. The set A^* is in one-to-one correspondence with A, and the projection of Z onto Y maps A^* onto B. If φ is continuous, then A^* is closed relative to $A \times Y$; if A is closed in X, then A^* is closed in Z.

After these preliminary remarks, let us turn to the theorem. Let A be a Suslin set (in the space X) and let B be its continuous image (in the space Y). We say that B is *generated* by A, and write

$$B = \Phi(A) \tag{7.6}$$

if this relation is established by the continuous mapping $y = \varphi(x)$. Since B is not necessarily closed in Y, we form the closed hull $\bar{B}$ of B in Y, and for every set $P \subseteq A$ we denote by $\Phi(P)$ the set of the images of the points of P, considered as a subset of $\bar{B}$ (and not of Y); the closed hull of $\Phi(P)$ in $\bar{B}$ will be denoted by $\bar{\Phi}(P)$. Thus $\bar{\Phi}(P)$ is closed in $\bar{B}$ (and hence in Y, since $\bar{B}$ is closed in Y); (1) means that we look for the image of $A \cap P$ and extend it in $\bar{B}$ to its closed hull. Let us express this relation by $Q = \bar{\Phi}(P)$. Then we have:

I. *If the sets $P_1 \supseteq P_2 \supseteq \cdots$ form a decreasing sequence with diameters converging to zero and have a (unique) point of* *intersection x belonging to A, then the sets $Q_n = \bar{\Phi}(P_n)$ have a unique point of intersection $y = \varphi(x)$, the image of x.*

Since $x \in A \cap P_n$, $y \in \Phi(A \cap P_n) \subseteq Q_n$ for every n, so that $y \in Q = Q_1 Q_2 \cdots$. It only remains to show that Q has no point other than y. Let $z \in Q_n$, $z \in Q_n$, so that z is an ϵ-point of $\Phi(A \cap P_n)$. Then there exists a point $y_n \in \Phi(A \cap P_n)$ for which $zy_n < 1/n$. This in turn indicates that there is a point $x_n \in A \cap P_n$ for which $y_n = \varphi(x_n)$. Since z and x belong to P_n, we have $zx_n \leqq \delta(P_n)$, and thus z

II. *The continuous image of a Suslin set is a Suslin set; the one-to-one continuous image of a Borel set is a Borel set.*

For the proof, let A be represented as a Suslin set

$$A = \mathfrak{S}\, P_{n_1} \cdot P_{n_1 n_2} \cdot P_{n_1 n_2 n_3} \cdots ,$$

where the P are closed sets in X, and let $Q_{n_1 \cdots n_k} = \bar{\Phi}(P_{n_1 \cdots n_k})$. Then, by I,

$$B = \mathfrak{S}\, Q_{n_1} \cdot Q_{n_1 n_2} \cdot Q_{n_1 n_2 n_3} \cdots$$

is a Suslin set.

If A is a Borel set, so that (§34, II) it can be represented with disjoint summands

$$A = \mathfrak{S}\, P_{n_1} \cdot P_{n_1 n_2} \cdot P_{n_1 n_2 n_3} \cdots ,$$

and if B is a one-to-one continuous image of A, then

$$B = \mathfrak{S}\, Q_{n_1} \cdot Q_{n_1 n_2} \cdot Q_{n_1 n_2 n_3} \cdots$$

can also be represented with disjoint summands. Since B as well as A is separable, and the space Y can therefore be taken as complete and separable, it follows that B is a Borel set.

The differences between Theorem II above and Theorem II of §35 should be considered carefully. There it was a question of relative Borel and Suslin sets (in A and in B), inferences were drawn about the inverse images of sets of the space B; here we are concerned with absolute Borel and Suslin sets. There it was a question of arbitrary single-valued mappings of A on B, while here we require that the mapping be continuous.

III. *Every separable, absolute Suslin set is the continuous image of the Baire null space A (or the set I of the irrationals). Every separable, absolute Borel set is the one-to-one continuous image of a set closed in A (or in I).*

As above, we let A be the Baire space of the sequence

$$n_1, \quad n_1, n_2, \quad n_1, n_2, n_3, \quad \ldots$$

and we consider, in a complete separable space E, an absolute Suslin set S. Since S is a Suslin set in E, it has a representation

$$S = \mathfrak{S}\, F_{n_1} \cdot F_{n_1 n_2} \cdot F_{n_1 n_2 n_3} \cdots ,$$

where the F are closed in E and can be taken to be decreasing with diameters converging to 0. To every point $x \in A$ there corresponds a sequence $(n_1, n_2, \ldots)$ and thus a sequence of closed sets $F_{n_1} \supseteq F_{n_1 n_2} \supseteq \cdots$ with diameters converging to 0; their (non-empty) intersection is a point $y = f(x) \in S$. The function $y = f(x)$ maps A continuously onto S: for if $x_p \to x$, and if the corresponding sequences of numerals first differ at the kth place from some p on, then the points y_p and y all belong to the same set $F_{n_1 \cdots n_k}$, whose diameter is arbitrarily small for large k; therefore $y_p \to y$. Thus S is a continuous image of A.

If S is an absolute Borel set, then (§34) it can be represented as a Suslin set with disjoint summands; the function $y = f(x)$ is then one-to-one, since the summands are disjoint. But the whole space A is not necessarily mapped; rather, only the set of those x for which the corresponding intersection $F(x) = F_{n_1} \cdot F_{n_1 n_2} \cdots$ is non-empty. This set B of points x is closed in A. For the condition that $F(x)$ be non-empty is equivalent to the condition that for every k

$$F_{n_1 \cdots n_k} > 0,$$

and since the F are closed, this condition is preserved under passage to the limit: if $x_p \to x$ and $F(x_p) > 0$ for every p, then $F(x) > 0$. Therefore B is closed in A, and S is the one-to-one continuous image of the set B closed in A.

IV. The theorems just proved admit of an important application. Let E be the Euclidean plane and let B be the projection of a planar Borel set A onto a line. Then B is a Suslin set (by II), and if A is a G_δ, then B is the one-to-one continuous image of a Borel set; B is indeed a Borel set, but the class is nevertheless not necessarily conserved; the planar G_δ under one-to-one projection give rise to one-dimensional Borel sets of arbitrarily high class. Thus even the simple problem of the nature of the projections of Borel sets leads as a matter of course past the Borel sets to the Suslin sets.

§ 7.4. Homeomorphism

A homeomorphism between two sets A and B ($A \simeq B$) was defined as a one-to-one mapping $y = \varphi(x)$, $x = \psi(y)$, continuous in both directions, between the two sets. Despite the symmetry of this relation, we shall continue to use the names image and inverse image. Starting from the considerations of the preceding section, which will be considerably recast here, we first prove a theorem approximately analogous to II:

I. *If A is a set in a complete space X and B a set in a complete space Y, and if A and B are homeomorphic, then there exist absolute G_δ-sets $U \subseteq X$ and $V \subseteq Y$ such that $A \subseteq U$, $B \subseteq V$, and such that the homeomorphism between A and B can be extended to a homeomorphism between U and V.*

Proof. Let $y = \varphi(x)$ and $x = \psi(y)$ be the two continuous functions that establish the homeomorphism between A and B. For every natural number n and every point $x \in A$ we choose a neighborhood $U_n(x)$ of radius $< 1/n$ in X, and for the image point $y = \varphi(x)$ a neighborhood $V_n(y)$ of radius $< 1/n$ in Y, in such a way that

$$\varphi(A \cap U_n(x)) \subseteq V_n(y), \quad \psi(B \cap V_n(y)) \subseteq U_n(x).$$

This is possible because of the continuity of φ and ψ. We now define, for every n,

$$U_n = \sum_{x \in A} U_n(x), \qquad V_n = \sum_{y \in B} V_n(y),$$

so that U_n is open in X and V_n is open in Y, and

$$U = U_1 \cdot U_2 \cdot U_3 \cdots, \qquad V = V_1 \cdot V_2 \cdot V_3 \cdots$$

are absolute G_δ-sets containing A and B respectively.

Now as a matter of fact $Q_n \subseteq B$ for $P_n = A$. Let $m < n$ be numbers of the sequence ν. If $z \in Q_n$, then, for every n, $z \in Q_{n_\nu}$; thus $z \in V_n(y_\nu)$ for suitable $y_\nu \in B_\nu$, $zy_\nu < 1/n$, and $y_\nu \to z$. If m is held fixed, y_ν as well as z lies ultimately in $V_m(y_m)$. Then the inverse image $x_\nu = \psi(y_\nu) \in A$, ultimately lies in $U_m(x_m)$, so that $x_\nu x_\mu < 1/m$ for sufficiently large ν, and thus the x_ν form a fundamental sequence whose limit $x \in X$ lies in the closed hull of $U_m(x_m)$ and therefore in P_m. Since this holds for every m, $x \in P_\nu \subseteq A$; x therefore has an image $y = \varphi(x) \in B$, and it follows from $x_\nu \to x$ that $y_\nu \to y$, so that $z = y \in B$, which is what we wished to prove.

Now we consider, as we have done several times before, a set N of increasing sequences $\nu = (m_1, m_2, \ldots)$ of natural numbers and form from them the Suslin set

$$A = \mathfrak{S}\, A_\nu = \mathfrak{S}\, P_{m_1} \cdot P_{m_1 m_2} \cdots,$$

and for each $\nu \in N$ we define the set

$$B_\nu = \prod_{n=1}^{\infty} V_n(y_n),$$

where $y_n = \varphi(x_n)$ and x_n is the point of A determined by the first n terms of ν. The sets B_ν are closed in Y, and

$$B = \mathfrak{S}\, B_\nu$$

is a Suslin set containing the image of A.

II. (*Lavrentiev's Theorem*). *If $A \subseteq X$ and $B \subseteq Y$ are homeomorphic and X, Y are complete spaces, then the homeomorphism can be extended to sets $U \supseteq A$ and $V \supseteq B$ which are G_δ in X and Y respectively.*

III. *A Young set A, that is, a G_δ in a complete space E, is homeomorphic to a complete space.*

To begin with, let E be merely a metric space, let F be closed $(0 < F < E)$, and let $\delta(x, F)$ be the lower distance of the point x from the set F. We consider points x, y, and z of the complement $G = E - F$ and define

$$\varrho(x, y) = \frac{xy}{xy + \delta(x, F) + \delta(y, F)}. \tag{7.7}$$

This expression, symmetric in x and y, has the attributes of a distance: it vanishes for $x = y$, is otherwise positive (< 1), and satisfies the triangle inequality. To prove the latter, we abbreviate $\delta(x, F)$ by δ_x and, taking into account that $\delta_z \leqq \delta_x + xz$ and $\delta_z \leqq \delta_y + yz$, we have

$$\frac{xy}{xy + \delta_x + \delta_y} \leqq \frac{xz + zy}{xz + zy + \delta_x + \delta_y} \leqq \frac{xz}{xz + \delta_x + \delta_z} + \frac{zy}{zy + \delta_z + \delta_y}.$$

Applying $\delta_z \leqq \delta_x + zy$ also in (1), we obtain

$$\frac{xy}{xy + \delta_x + \delta_y} \leqq \frac{xz}{xz + \delta_x + \delta_z} + \frac{zy}{zy + \delta_z + \delta_y},$$

i.e., $\varrho(x, y) \leqq \varrho(x, z) + \varrho(z, y)$.

We now transform all the distances xy of the space G into distances $\varrho(x, y)$; this generates a new space $\hat{G}$ homeomorphic to G. The points of $\hat{G}$ are the same as those of G; the neighborhoods, however, are different. A sequence of points x_n of G that is fundamental in the distance ϱ is also fundamental in the original distance; for from

$$\frac{x_\mu x_\nu}{x_\mu x_\nu + \delta_{x_\mu} + \delta_{x_\nu}} < \epsilon$$

it follows that $x_\mu x_\nu < \epsilon(x_\mu x_\nu + \delta_{x_\mu} + \delta_{x_\nu}) < \epsilon(M + \delta_{x_\mu} + \delta_{x_\nu})$, where M is a bound for the diameters of the set of points under consideration. Conversely, a sequence fundamental in the original distance is fundamental in the distance ϱ as well, provided the distances δ_{x_n} are bounded away from 0. If x_n is a fundamental sequence in $\hat{G}$, then it is also fundamental in G, and so it has a limit x in E (if E is complete). This limit cannot be a point of F; for if $x \in F$, then $\delta_{x_n} \to 0$, and from the fundamentality of x_n in $\hat{G}$ it would follow that for $\mu, \nu \geqq n(\epsilon)$

$$\varrho(x_\mu, x_\nu) = \frac{x_\mu x_\nu}{x_\mu x_\nu + \delta_{x_\mu} + \delta_{x_\nu}} < \epsilon,$$

so that, for fixed μ, letting $\nu \to \infty$,

$$\frac{x_\mu x}{x_\mu x + \delta_{x_\mu}} \leqq \epsilon.$$

But then x_n would not be a fundamental sequence in $\hat{G}$. Therefore $x \in G$, and $\hat{G}$ is complete.

This argument is now applied to the case of a G_δ:

$$A = G_1 \cdot G_2 \cdot G_3 \cdots = E - (F_1 + F_2 + F_3 + \cdots).$$

We set $G^{(0)} = E$, $\varrho_0(x, y) = xy$, and define successively

$$\varrho_n(x, y) = \frac{\varrho_{n-1}(x, y)}{\varrho_{n-1}(x, y) + \delta_{n-1}(x, F_n) + \delta_{n-1}(y, F_n)},$$

where δ_{n-1} denotes the lower distance in the space $G^{(n-1)}$. This too is a distance and generates a third space $\hat{A}$ homeomorphic to A and $\hat{A}$. This space $\hat{A}$ is complete, provided E is complete — which concludes the proof of III.

For let x_n be a fundamental sequence in $\hat{A}$, so that (since $\varrho_n \leqq \varrho_{n-1}$) it is also a fundamental sequence in A and in $\hat{A}$. Then it has a limit, in the sense of the original distance xy, in E. This limit cannot be a point of B; for if, say, $t \in F_m$, then for $m < n$ it would follow from (3) that

$$\varrho_m(x_m, x_n) \geqq \frac{x_m x_n}{x_m x_n + \varrho_{m-1}(x_m, F_m)},$$

so that for $n > n_0$ ($x_m x_n < \epsilon$, $x_m t > 0$, $\varrho_{m-1}(x_n, F_m) \to 0$)

$$\liminf \varrho_m(x_m, x_n) > 0.$$

But then x_n would not be a fundamental sequence in $\hat{A}$ nor, all the more, would it be a fundamental sequence in $\hat{A}$, since otherwise even $\limsup \varrho_m(x_m, x_n)$ would have to converge to 0 with increasing m. Therefore there exists a limit $x \in A$, and $\hat{A}$ is complete.

IV. *If A is homeomorphic to B, and if A is a Suslin set in a complete space X, then B is a Suslin set in every complete space Y in which it is embedded; and if A is a Borel set F^ξ or G^ξ, then B is also a Borel set of the same kind.*

Proof. Let $y = \varphi(x)$ and $x = \psi(y)$ be the functions establishing the homeomorphism. For every $y \in B$ and every n we consider the set of points $x \in A$ for which $\varphi(x)$ lies in the sphere $V_n(y)$ of radius $1/n$ about y. Let $U_n(x)$ be a neighborhood of x of radius $< 1/n$ such that $\varphi(U_n(x) \cap A) \subseteq V_n(y)$. Let A_0 be the set of those $x \in A$ for which there exists a sequence of points $a_p \in A$ converging to x such that the images $b_p = \varphi(a_p)$ form a fundamental sequence. The set A_0 contains A; for if $x \in A$, then for every sequence $a_p \to x$ we have $b_p = \varphi(a_p) \to \varphi(x)$, so that the b_p form a fundamental sequence. But A_0 may also contain points not in A, namely those points x that are limits of sequences from A for which the image sequence is fundamental without converging to a point of B. If, however, we assume that every fundamental sequence of points of B has a limit in Y (which is the case when Y is complete), then A_0 is exactly the set of those $x \in \bar{A}$ for which the image of every convergent sequence is fundamental.

We now show that A_0 is a G_δ in $\bar{A}$. Let A_n be the set of those $x \in \bar{A}$ for which there exists a sequence $a_p \to x$ such that the images $b_p = \varphi(a_p)$ satisfy $b_\mu b_\nu < 1/n$ for all sufficiently large μ, ν. Then A_n is open in $\bar{A}$, and $A_0 = A_1 \cap A_2 \cap A_3 \cdots$ is a G_δ in $\bar{A}$. Since $\bar{A}$ is absolutely closed in X (it is the complete hull of A), A_0 is an absolute G_δ.

In the same way, the set B_0 of those $y \in B$ for which $\psi(y) \in A_0$ is an absolute G_δ in $\bar{B}$, and A_0 and B_0 are homeomorphic. If now A is a Suslin set in X, then A_0 is an absolute Suslin set, and so is B_0; therefore B is a Suslin set in Y. If

A is a Borel set, then A_0 is an absolute Borel set of the same kind, and the same holds for B_0 and hence for B.

V. (*Kuratowski's Theorem*). *If $A \subseteq X$ and $B \subseteq Y$ are Borel sets in complete separable spaces X, Y, and if A and B are homeomorphic, then there exist Borel sets $P \subseteq X$ and $Q \subseteq Y$ such that $A \subseteq P$, $B \subseteq Q$, and $P \simeq Q$.*

VI. *The homeomorphic image of an absolute Borel set F^ξ ($\xi \geqq 2$) is itself such a set (with the same index ξ).*

For after the homeomorphism has been extended to the absolute G_δ-sets X and Y, we can apply Theorem II of §35. If A is an absolute G_δ ($\delta \geqq 1$), then it is a G_δ in X. Then, by virtue of the continuity of $x = \psi(y)$, B is a G_δ in Y, and is therefore an absolute G_δ, since Y is an absolute G_δ.

In the same way, if A is an absolute F^ξ ($\xi \geqq 2$), then B is an F^ξ in Y and is therefore an absolute F^ξ, since Y is an absolute G_δ and F^ξ. The proof for the absolute Suslin sets proceeds along similar lines.

Some further set characteristics invariant under homeomorphism can be obtained from IV by considering the complements $X - A$ and $Y - B$.

VII. *If A is homeomorphic to B, then:*

(a) *If A is a G_δ in X, then B is a G_δ in Y.*

(b) *If A is an F_σ in X, then B is an F_σ in Y.*

(c) *If A is dense-in-itself, then so is B.*

(d) *If A is nowhere dense, then so is B.*

§ 7.5. Simple Curves

1. Conditions for simple curves. Our interval-image, that is, the continuous map of a closed interval or of the interval $T = [0, 1]$ of the real numbers did not necessarily have much resemblance to our intuitive notion of a curve.

Somewhat closer to this conception comes the *simple curve*,[1] namely the homeomorphic image of T. In order that C be such a curve, the following properties are certainly necessary, and the list could easily be extended; the concepts closed, continuum, and so forth, refer to C itself as the space.

(a) C is compact in itself,

(b) C has two points a and b between which it is an irreducible continuum; that is, C itself is a continuum, but no continuum $< C$ contains the points a and b.

(c) C has two points a and b between which it is irreducibly connected; that is, C itself is connected, but no connected set $< C$ contains the points a and b.

(d) C is connected and has two points a and b of the following kind: For every point $x \in C$ there exists a decomposition $C = A + B$ for which A and B are closed and have only the point x in common, and $a \in A$, $b \in B$.

(e) C is locally connected.

As a matter of fact, the necessity of (a) and (e) follows from III of §36. The remaining conditions result if we take a and b to mean the images of the end-points of 0 and 1; no connected sets $< T$ contain the points 0 and 1, whence (c) and, *a fortiori*, (b) follows; and for every $t \in T$, $T = [0, t] + [t, 1]$ yields a decomposition which, carried over to C, has the property (d) ($[0, 0]$ and $[1, 1]$ are to mean the sets $\{0\}$ and $\{1\}$ consisting of a single point).

We shall now show that conditions (a), (b), and (e), or (a) and (c), or (a) and (d) are also sufficient.

I. *Conditions (c) and (d) are equivalent.*

That (c) follows from (d) is very easy to see. If $D \subset C$ contains the points a and b, let $x \in C - D$, and let $C = A + B$

[1] Also called a simple arc, or a Jordan arc. The homeomorphic image of the circumference of a circle is called a simple closed curve, or a Jordan curve.

be, in accordance with (d), a decomposition belonging to x; then $D \subseteq A + B = C$, $D \cap A \cap B = D \cap \{x\} = 0$, and $D = D \cap A + D \cap B$ is a partition, and so D is not connected.

Conversely, let (c) — or even merely the following partial assumption — be satisfied: Let C be a continuum irreducible between a and b in the sense of (b), and let it become disconnected by elimination of a point other than a or b. In order to prove (d) from this, we may suppose x different from a and b, since for $x = a$ and $x = b$ the decompositions $C = a + C = C + b$ satisfy the given conditions (we now denote a set $\{x\}$ consisting of one point simply by x). Then $C - x$ is disconnected and can therefore be partitioned into $C - x = P + Q$, where P and Q are closed in $C - x$. Then the sets $A = P + x$ and $B = Q + x$ are closed (in C) and in addition, by III of §29, they are connected, since their sum C and their intersection x are connected. Neither of the continua $A < C$ and $B < C$ can contain both of the points a and b; therefore we have, say,

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open, provided it has no first element. Both at once would mean a partition of C, so that either C_1 has a last element or C_2 a first element.

If, furthermore, C is separable, so that a countable set R is dense in C (in the metric sense), then R is also dense in C in the ordinal sense (p. 63). For (x_1, x_2), where $x_1 < x_2$, is open and therefore contains a point r of R; that is, $x_1 < r < x_2$. By Theorem V of §11, $C - \{a, b\}$ is therefore of the type η of the open interval $(0, 1)$ and C itself of the type of the closed interval $T = [0, 1]$. There exists, in consequence, a similarity transformation

$$c = \varphi(t), \quad t = \psi(c)$$

of C on T ($0 \leq t \leq 1$), with $a = \varphi(0)$ and $b = \varphi(1)$. Here the function $c = \varphi(x)$ is continuous. For, to every interval open in T (the intersection of T with an open interval), namely $[0, \tau)$, $(\tau, 1]$, and (τ_1, τ_2), there corresponds a set open in C, namely $[a, x)$, $(x, b]$, and (x_1, x_2); and thus to every set open in T there corresponds a set open in C — which is precisely the condition for the continuity of $\varphi(x)$ of Theorem I of §35. This proves II.

In the nature of a converse to Theorem II, moreover, we have the following:

If C is connected and T is a one-to-one continuous image of C, then C satisfies condition (c) or (d). For the decomposition $T = [0, \tau] + [\tau, 1]$ gives a decomposition $C = A + B$ such as is required by (d). From the mere fact that T is a one-to-one continuous image of C, neither connectedness nor separability of C follows. C can, after all, be (p. 223) any set of cardinality $\aleph$ in which the distance between every two points is 1.

If C now is compact in itself and T is a one-to-one continuous image of C, then the mapping is a homeomorphism (§35, III). Thus:

III. In order that C be a simple curve, the conditions (a) and (c) or the conditions (a) and (d) are necessary and sufficient.

Conditions (a) and (c) are due to Lennes; (a) and (d) to Sierpiński.

If condition (c) is weakened to (b), then strengthening in some other respect is necessary; this is accomplished by the condition of local connectedness; and we have:

IV. In order that C be a simple curve, conditions (a), (b), and (e) are both necessary and sufficient.

It is only necessary to show that for every point x a decomposition $C = A + B$ in accordance with (d) is possible, where x is again assumed to be distinct from a and b. Let U_n $(n = 1, 2, 3, \ldots)$ be the neighborhood of x of radius $1/n$, and let $F_n = C - U_n$ be closed. For n sufficiently large, a and b belong to F_n, but, by virtue of (b), not to the same component of F_n; let P_n be the component of F_n containing a, and Q_n the one containing b (as long as $a \in U_n$, we set $P_n = 0$, and analogously for Q_n). The sets

$$P = \mathfrak{S}P_n, \quad Q = \mathfrak{S}Q_n$$

(where the summands increase with n) are disjoint and connected. Because of the Border Theorem (Theorem XVIII) of §29, P_n contains a point that is on the border of F_n, and so is at distance $1/n$ from x; x is a point of accumulation of P and Q; and $P + Q + x$, as the sum of the connected sets $A = P + x$ and $B = Q + x$, is connected. If we now show, in addition, that A and B are closed, then the proof is complete; for it then follows from the irreducibility (b) that $C = P+Q+x = A+B$, which is the desired decomposition. Therefore what we must show is that, if $c \neq x$ is an x-point of P, then $c \in P$. Let R_n be the component of F_n containing c, and let $R = \mathfrak{S}R_n$ (ultimately $c \in F_n$; until this occurs we again let $R_n = 0$). Now c ultimately becomes an interior point of F_n, and, by virtue of the local connectedness, an interior point of R_n (the component of F_n containing c already has c as an interior point); R_n thus contains points of P. Therefore R_nP is non-empty, RP is non-empty and, for sufficiently large n, R_nP_n is non-empty as well; that is, $R_n \subseteq P_n$ and $R \subseteq P$, so that $c \in P$.

This proves IV. A compact, locally connected continuum was always the continuous image of an interval; if, in addition, it is irreducible between a certain two of its points, then it is the homeomorphic image of an interval.

2. Prime parts of a continuum.[1]

Let us take C to have more than one point, and to be connected, and let us again take it as the space to which the relative concepts refer. The points y at which C is locally connected may be called *regular* points and the remaining, *singular* points; if R is the set of regular points and S that of singular points, then

$$C \cdot H = R + S = R + S_i.$$

The points (sets of one point) of R_i and the components of S_i are called the *prime parts* of C (Hahn). In case $R_i = 0$ and S is dense in C, C has itself as its only prime part and will be called a *prime continuum.* If $R_i > 0$, then this dense-in-itself set is locally connected, since any one of its points may be joined to one sufficiently near it by a connected set of arbitrarily small diameter $\subseteq C$ and therefore $\subseteq R_i$; its components are open in R_i and have more than one point, so that R_i is thus (by §29, VII) of cardinality at least $\aleph$. In this case C therefore has at least $\aleph$ prime parts and will be called a *composite continuum*; the prime parts are themselves continua.

Let us introduce for two points x and y a concept analogous to the (always vanishing) separation in C, the *singular separation*, which we denote once more by xy. A g-chain (p. 180)

$$(x, s_1, s_2, \ldots, s_n, y), \qquad xs_1 < \varrho,\ s_1s_2 < \varrho, \ldots,\ s_ny < \varrho, \tag{3}$$

the "inside" points $s_1, s_2, \ldots, s_{n-1}$ of which are singular points — we also include, for $xy < \varrho$, the chain (x, y) without any inside points — will be called a singular g-chain. Let us join x and y by all possible singular ϱ-chains; the greatest lower bound of the numbers ϱ so defined will be called the *singular separation* xy. We have $xy \leq xy$, and the triangle inequality $xy + yz \geq xz$ holds; for if x and y can be joined by a singular ϱ-chain, and y and z by a singular σ-chain, then

[1] Compare Appendix C.

x and z can be joined by a singular $(\varrho + \sigma)$-chain. If, instead of the singular ϱ-chains, we use chains

$$(x, t_1, t_2, \ldots, t_n, y), \qquad xt_1 < \varrho,\ t_1t_2 < \varrho, \ldots,\ t_ny < \varrho, \tag{4}$$

whose inside points are taken from S_i, then the greatest lower bound of the ϱ is unchanged; for a g-chain (3) is a special g-chain (4) and, on the other hand, to a given g-chain (4) there belongs a $(\varrho + \delta)$-chain (3) obtained by choosing, for each t_ν, a s_ν for which $s_\nu t_\nu < \delta/2$ ($\delta > 0$ arbitrarily small).

The singular separation xy of course vanishes for $x = y$; for $x \neq y$ it does so if and only if both points belong to S_i and have separation 0 in this set. If from now on we take C to be a compact continuum, then $xy = 0$ means the same as: x and y belong to the same prime part.

By virtue of these separations, C becomes a new metric space Γ in which the points belonging to the same prime part are to be considered identical. This can also be expressed as follows: To every point $x \in C$ there is made to correspond a uniquely defined point $\xi = \varphi(x)$ with the stipulation that $\varphi(x) = \varphi(y)$ if and only if x and y belong to the same prime part; the set Γ of the points ξ becomes the single-valued image of C while, conversely, the inverse image of ξ is an entire prime part. With the distance $\xi\eta = xy$, Γ becomes a metric space and — since $\xi = \varphi(x)$ — the continuous image of C and so, once more, a compact continuum. Let us call it the *continuum of the prime parts* of C.

The inverse image of a continuum $\Delta \subseteq \Gamma$ is a continuum $D \subseteq C$. For D is closed; if $D = D_1 + D_2$ can be partitioned, then no prime part can have points in common with both summands (because it is a continuum). The images Δ_1 and Δ_2 of D_1 and D_2 are then disjoint and are compact closed sets, and $\Delta = \Delta_1 + \Delta_2$ can be partitioned. From this it follows that: If C is irreducible between x and y, then Γ is irreducible between the image points ξ and η (irreducible in the sense of (b), and thus an irreducible continuum). We assume here that $x \neq y$, with x and y belonging to different prime parts; in the excluded case, C would be a prime continuum and Γ would consist of a single point.

The image $\varphi(G)$ of a set G open in C need not be open in Γ. But if G contains the entire prime part A, the inverse image of the point ξ, then $\varphi(G)$ has ξ as an interior point. For if $F = C - G$, then the image $\varphi(F)$ is closed (since it is the continuous image of a compact closed set), does not contain ξ, and therefore has a positive lower distance ϱ from ξ; therefore all the points η for which $\xi\eta < \varrho$ belong to $\varphi(G)$.

We should now like to get to the true goal of this investigation, which is, to prove that the continuum of the prime parts is locally connected.

V. If C is a compact continuum and if F is closed (in C) and P is a component of F, then every point x of $F \cdot P(F-P)_q$ is singular, and the prime part belonging to x intersects the border F_b of F.

Let us first note the following. In order that this theorem have content, that is, that a point x exist, we must have $F_b > 0$ and $F > P$, so that F is not connected, and in particular $F < C$. Since the space C is connected, F cannot be open at the same time, and therefore F_b is non-empty. The Border Theorem (Theorem XVII of §29) on which the proof of the present theorem mainly rests, states that every component of $C \cdot F = F$ meets the border F_b, and further, that if Q is a continuum intersecting F_b and F_i, then every component of $Q \cdot F_i$ has points of accumulation in F_b.

Now x cannot be regular; for if $U_x \subseteq F_i$, then x must be an interior point of the component of U_x containing it, and all the more of P, and so could not be a point of accumulation of $F - P$.

Further, let $F = P + \mathfrak{S}Q_n$ be the decomposition of F into components. Since x is supposed to be an accumulation point of $F - P$ and $\delta(x, Q_n) > 0$, there must be a sequence of components Q_n for which $d(x, Q_n) \to 0$; by the Axiom of Choice, we can assume that the closed limit $Q = \mathrm{Fl}\, Q_n$ exists. It is, again, a continuum $\subseteq F$ (Zoretti's Theorem) and contains x and is therefore contained in P and, furthermore, of course in $(F-P)_q$; from what has already been proved it is clear that all the points of $Q \cdot F_i$ (which includes x) are thus singular, $Q \cdot F_i \subseteq S$. As we have already remarked above, all the Q_n intersect the border F_b, so that $Q \cdot F_b$ is non-empty also, and X_n, the component containing x of $Q_n \cdot F_i$, has points of accumulation in F_b; $X_n \cdot F_b$ is non-empty.

Now since $X_n \subseteq S$, $X_n \subseteq S_i$, and X_n is connected, X_n is contained in the prime part A belonging to x, and $A \cdot F_b$ is non-empty.

VI. If A is a prime part of the compact continuum C, F the set of the points for which $d(x, A) \leq \varrho$ ($\varrho > 0$), and P the component of F containing A, then $A = P_i$.

For if $x \in A = F \cdot P$, then x must not belong to $(F-P)_q$, since otherwise, by V, A would have to intersect the border of F; therefore, a neighborhood $U_x \subseteq F_i$ exists which is disjoint from $F - P$, so that $U_x \subseteq P$, and x is an interior point of P.

VII. (Moore's Theorem). The continuum of the prime parts of a compact continuum is locally connected.

Let $\xi \in \Gamma$, and let the inverse image of ξ be the prime part A of C; let Γ_0 be a neighborhood of ξ and its inverse image a set $G_0 \supseteq A$ open in C. If ϱ is positive and smaller than the lower distance between A and the boundary[2] of G_0 (both are compact continua), then the set F of Theorem VI is contained in G_0; as in that theorem, let P be the component of F containing A, and let $A \subseteq G = P_i \subseteq G_0$, where $G = P_i$ is open. For the images it follows that $\varphi(G) \subseteq \varphi(P) \subseteq \varphi(G_0) = \Gamma_0$. $\varphi(G)$, as we saw above, contains the point ξ as an interior point, $\varphi(P)$ is a continuum containing ξ as an interior point, and we have thus proved the local connectedness of Γ at ξ.

VIII. (Hahn's Theorem). If C is a compact, composite continuum irreducible between certain two of its points, then the continuum of its prime parts is homeomorphic with the interval T $(0 \leq t \leq 1)$. T is the continuous image of C, and the mapping is such that the inverse images of the points t are the prime parts of C.

For: Γ, the continuum of the prime parts, is compact, has more than one point, is irreducible between two points, is locally connected, and thus by IV, is homeomorphic to T. T is a continuous image of Γ, Γ a continuous image of C, and therefore T a continuous image of C, where there corresponds to each point t exactly one point of Γ and one prime part of C.

A simple example in the (x_1, x_2)-plane is the set $C = R + S$ formed from the two sets

$$R: \; 0 < x_1 < 1, \; x_2 = \sin \pi / x_1, \qquad S: \; x_1 = 0, \; -1 \leq x_2 \leq 1,$$

where at the same time, R is the set of regular points and S the set of singular points. The prime parts are S and the points of R; the projection on the x_1-axis gives $[0, 1]$ as the continuous image of C. C is an irreducible continuum between every point $a \in S$ and the point b $(x_1 = 1,\; x_2 = 0)$; and between all other pairs of points, it is reducible. But it is not irreducibly connected even between a and b, in the sense of (c), since $a + R$ is connected; the set $a + R$, however, is

[2]When this vanishes $(G_0 = C)$, then $\varrho > 0$ may be arbitrary.

irreducibly connected between a and b and, by II, T is its one-to-one continuous image (again by projection).[3]

[3]The same holds for the set in Fig. 6 (p. 180).

8.40 Topological Spaces

Homeomorphisms are also called *topological mappings* and homeomorphic sets are called *topologically equivalent*; properties common to homeomorphic sets are called *topological invariants*. Thus, the property of being an absolute G_δ ($\varrho = 1$) is topologically invariant, but that of absolute closure (completeness) is not; relative closure and relative openness, on the contrary, are topologically invariant; that is, if A is closed (open) in E and E is homeomorphic to $\bar{E}$, while A is homeomorphic to a corresponding subset $\bar{A}$, then $\bar{A}$ is closed (open) in $\bar{E}$.

The mathematical discipline concerned with these matters is called *Topology*, or *Analysis Situs*. (The latter term, due to Leibniz, was reintroduced by Riemann.) We have not been able to go very deeply into this subject nor to prove more than a very few general theorems about it. But this seems like a suitable occasion to touch, in all brevity, on those point-set theories that emphasize the topological point of view from the very beginning and work only with topologically invariant concepts; a topological space defined in this way is to be distinguished from its homeomorphs in the same way in which, in our own theory, we distinguished between a metric space and the spaces isometric with it. It is not a mere matter of a formal transformation of the metric theory but rather of a new and wider conception of a space; the metrizable spaces — that is, those spaces that are homeomorphic to metric spaces — constitute merely a special class among the topological spaces.

I. Sum and Intersection Axioms. The closed sets must, regardless of anything else, satisfy the following conditions:

(1) The space E and the null set 0 are closed.

(2) The sum of two closed sets is closed.

(3) The intersection of any number of closed sets is closed.

As a consequence, the closed hull A_a can be defined as the intersection of all the closed sets $\supseteq A$ (which in any case includes E); it has the following properties:

$$
\begin{aligned}
&\text{(a)} \quad 0_a = 0 \\
&\text{(b)} \quad A_a \supseteq A \\
&\text{(c)} \quad A_{aa} = A_a \\
&\text{(d)} \quad (A+B)_a = A_a + B_a
\end{aligned}
$$

It would also be possible to make these properties our starting point (Kuratowski) by taking them as our axioms for the set function A_a, defining closed sets by means of $A_a = A$, and proving in this way the theorems about closed sets; for example, (3) follows from the fact that because of (d), A_a is a monotone set function, that is, if $A \subseteq B$ then $A_a \subseteq B_a$.

For the open sets, the corresponding theorems hold:

(1) The space E and the null set 0 are open.

(2) The intersection of two open sets is open.

(3) The sum of any number of open sets is open.

The sum of all the open sets $\subseteq A$ is called the *open kernel* A_i; and has properties (a), (b), (c), and (d) analogous to those of A_a. Of course, these properties also, or the theorems about open sets, could be taken as our axiomatic starting point. One modification of this is formed by the axioms about neighborhoods. If we denote by G_x the open sets containing x (including, of course, E) and select from them a subsystem U_x, in such a way that every G_x contains a U_x, then these U_x are called *neighborhoods* of x, and the neighborhoods of all the points constitute a *complete system of neighborhoods* for the space E. In general, there will be many such systems possible; the largest of them consists of all the non-empty open sets; in metric spaces, the spherical neighborhoods $U_\varrho(x)$ with the positive radii ϱ, introduced in §22, form such a system, and they do so even if we restrict ϱ to rational values or to the values 1, 1/2, 1/3, ...; in separable spaces, the "special" neighborhoods V of §25 form a complete system of neighborhoods; in locally connected spaces, there exists a complete system of connected neighborhoods. Two such complete systems, with the neighborhoods U_x and V_x, are related as follows: Every U_x contains some V_x, and every V_x contains some U_x. If the space E with the neighborhoods U_x is homeomorphic to $\bar{E}$, then the images of the U_x give rise to a complete system of neighborhoods for $\bar{E}$. From the theorems about open sets we derive the following properties of the neighborhoods:

(A) Every point x has at least one neighborhood U_x; and U_x always contains x.

(B) For any two neighborhoods U_x and V_x of the same point, there exists a third, $W_x \subseteq U_x V_x$.

(C) Every point $y \in U_x$ has a neighborhood $U_y \subseteq U_x$.

It is now again possible to treat neighborhoods as unexplained concepts and to use them as our starting point, postulating Theorems

(A), (B), and (C) as *neighborhood axioms.* Open sets G are then defined as sums of neighborhoods or as sets in which every point $x \in G$ has a neighborhood $U_x \subseteq G$ (the null set included). Theorems (1), (2), and (3) about open sets are then provable.

Whether one form or another is preferred of the axioms discussed so far, the space defined by them is of course still very deficient in properties. We have not even seen to it thus far that sets consisting of one point (and consequently, by (2), sets consisting of any finite number of points)[4] are closed. This can be achieved either by an axiom requiring just this or, better still, by imposing the following condition:

For two distinct points x and y there exists a closed set containing x but not y.

For, the closed hull of the set $\{x\}$ containing only one point is then this set itself.

In this condition, the word closed can also be replaced by open; it then goes over into the first of the following separation axioms.

II. Separation Axioms.

(4) If $x_1 \neq x_2$, then there exists an open set G_1 containing x_1 but not x_2.

(5) If $x_1 \neq x_2$, then there exist two disjoint open sets G_1 and G_2 for which $x_1 \in G_1$ and $x_2 \in G_2$.

(6) If F_2 is a closed set not containing the point x_1, then there exist two disjoint open sets G_1 and G_2 for which $x_1 \in G_1$ and $F_2 \subseteq G_2$.

(7) If F_1 and F_2 are two disjoint closed sets, then there exist two disjoint open sets G_1 and G_2 for which $F_1 \subseteq G_1$ and $F_2 \subseteq G_2$.

(8) If A_1 and A_2 are two disjoint sets closed in their sum, then there exist two disjoint open sets G_1 and G_2 for which $A_1 \subseteq G_1$ and $A_2 \subseteq G_2$.

Each of these axioms, retaining (4), is a strengthening of the preceding one, and in this way topological spaces of increasing specialization are obtained. In metric spaces the last separation axiom holds, and therefore so do all of them (compare p. 186; the proof also suffices for (8)).

III. Axioms on Cardinality. Here we are concerned primarily (if the trivial finite spaces are excluded) with the relations to the

[4]Such a program was carried through in the first edition of this book, [*Grundzüge der Mengenlehre.* (Leipzig, 1914; repr. New York, 1949)].

countable; we express them by means of neighborhoods in the two *Axioms of Countability.*

(9) There exists a complete system of neighborhoods for which every point has at most a countable number of neighborhoods.

(10) There exists a countable complete system of neighborhoods.[5]

The first is satisfied in every metric space, the second in every separable space.

The list of axioms we have given can of course be extended; but the axioms of Group I and at least one of the separation axioms are perhaps the least that can be required of a topological space if it is not to become altogether pathological. It is possible to base on the second separation axiom and on the "First Axiom of Countability," that is, on (1), (2), (3), (5), and (9) a theory close to the theory of metric spaces. But in order to metrize the space, the strongest separation axiom, and so at least (1), (2), (3), (8), and (9) are necessary; we mention here without proof that for homeomorphism to a separable metric space the validity of Axioms (1), (2), (3), (6), and (10) is both necessary and sufficient. (Urysohn's Theorem, somewhat sharpened by Tychonoff).

An interesting category of topological spaces has been defined by Fréchet; it is based on the concept of a convergent sequence or of a limit (limit point). To certain sequences $(x_1, x_2, \ldots)$ of points of the space E, there are to correspond uniquely defined points x of the space E; such a sequence is called *convergent* (to x), and the corresponding point $x = \lim x_n$ is called its *limit.* The following two limit axioms are required to hold:

(α) Every constant sequence $(x, x, x, \ldots)$ converges to x.

(β) Every subsequence of a sequence converging to x converges to x.

A space having such a definition of limit will be called an *L-space.*

If $A \subseteq E$ and a sequence of points of A converges to x, then x will be called a *limit point*[6] of A; let the set of the limit points of A be called A_l. Because of Axiom (α), we have $A_l \supseteq A$; the sets for which $A_l = A$ are defined to be *closed.* Then Theorems (1), (2), and (3) concerning closed sets hold; this follows easily from the fact that A_l is a monotone set function and that, because of (β), we obviously

[5]First Axiom of Countability and Second Axiom of Countability are the commonly accepted terminology for (9) and (10), respectively.

[6]The limit points here, as we shall see at once, are not necessarily identical with the α-points (points of the closed hull) but constitute only a part of them.

have $(A+B)_l = A_l + B_l$. The closed hull A_a of A is then found, by §30, 2, as the greatest set of the sequence $A^{(0)}, A^{(1)}, \ldots, A^{(\alpha)}, \ldots$ defined inductively by

$$A^{(0)} = A, \quad A^{(\alpha+1)} = A_l^{(\alpha)}, \quad A^{(\lambda)} = \mathfrak{S}A^{(\alpha)} \ (\lambda \text{ a limit number})$$

and which begins with $A, A_l, A_{ll}, \ldots$; moreover, because of the same considerations as in the case of the Baire system of functions (p. 192) we have in any case $A_a = A^{(\Omega)}$; that is, the greatest set $A_a = A^{(\alpha)}$ is already reached for some index $\alpha < \Omega$.

Incidentally, the systems of functions just referred to constitute the simplest example of L-space — an example that shows immediately that in general A_a is not closed, as it is in metric spaces, and that the limit points constitute only some of the α-points that are obtained when the formation of limits is repeated without bound. Let the elements of E ("points") be the real functions $x = x(t)$ defined in a space T, and define convergence $x = \lim x_n$ by means of $x(t) = \lim x_n(t)$; that is, $x_n(t)$ is to converge — in the ordinary sense — to $x(t)$ everywhere, for $t \in T$. A set A, contained in E, is thus a system of functions, a closed set a Baire system of functions (p. 191 and §43), and the closed hull the smallest Baire system over A. If now T is, for example, the set of real numbers, A the system of continuous functions x, A_l that of the functions $y = \lim x_n$, A_{ll} that of the functions $z = \lim y_n$, then $A_l < A_{ll}$; for, as we shall see in §42, the functions y still have points of continuity, whereas the functions z can already be everywhere discontinuous. The closed hull $A_a = A^{(\Omega)}$ is here reached only for $\alpha = \Omega$.

The metric counterpart should be kept in mind; if we make the system E of bounded functions into a metric space by means of the distance $xy = \sup\limits_t |x(t) - y(t)|$, then $\lim x_n = x$ means the uniform convergence of $x_n(t)$ to $x(t)$ for all t; A_l is then always closed, and

$$A_l = A = A_a.$$

Another departure of the L-spaces from the metric spaces is that even the second separation axiom may fail to hold (the first, (4), does hold, since the sets containing just one point are closed, because of Limit Axiom (α)). It may even be violated in a completely flagrant way.

Let us call the space E *decomposable* if it can be represented as a sum $F_1 + F_2$ of two closed sets (properly) contained in E (if, in addition F_1 and F_2 are disjoint, then we have a partition, so that unconnected spaces are certainly decomposable and indecomposable ones certainly connected, although not conversely). In our theory we had no occasion to mention this concept because, apart from one-point spaces, which are, of course, indecomposable, every metric space is decomposable. As a matter of fact, every space in which the second separation axiom (5) holds is decomposable between any two of its points $x_1 \neq x_2$; that is, decomposable in such a way that x_1 belongs only to F_1 and not to F_2, and x_2 only to F_2 and not to F_1: one need only take F_1 and F_2 to be the complements ($F_1 = E - G_2$ and $F_2 = E - G_1$) of the open sets named in (5).

Since clearly the continuous image of an indecomposable space is itself indecomposable (just as that of a connected space was itself connected), then this image, in case it is a metric space, must consist of a single point: in particular, in an indecomposable space every real continuous function is constant. With the indecomposable spaces having this paradoxical nature, it is especially remarkable that an L-space (consisting of more than one point) can be indecomposable, as the following example shows.

Let E be the periphery of a circle, with a positive sense chosen for its rotations about the center; let the symbol φ_θ denote a rotation through the fixed angle $2\pi\theta$, where θ is irrational; let the point x go into the point $x\varphi_\theta$ under this rotation and the set A, contained in E, into $A\varphi_\theta$. We define as convergent, first, the constant sequences $(x, x, x, \ldots)$ having the limit x, second, the sequences consisting entirely of distinct points that converge in the ordinary sense — on the basis of the distances of elementary geometry — to a point x, and to these sequences we assign the limit $x\varphi_\theta$. The two limit axioms are satisfied.

Then $A_l = A + (A')\varphi_\theta$, where A' denotes the set of points of accumulation of A in the ordinary sense. We now claim that for $E = A + B$ at least one of the equations $A_a = E$ and $B_a = E$ holds, which will show the indecomposability of E. If on each circular arc there are points of A, then $A' = E$ and $A_l = E$ and, *a fortiori*, $A_a \supseteq E$. If the opposite is the case, then B contains a circular arc C (including its endpoints). Then $C \subseteq C_a \subseteq C_{l,l}$, $C_{l,l,l} \subseteq C_{l,l,l,l}$, etc., and this set is the whole periphery of the circle, for if x is the point half way along the arc C, then the set $\{x, x_l, x_{ll}, \ldots\}$ is dense in E (in the ordinary sense). Thus $C_a = E$ and $B_a = E$.

Chapter 9

Real Functions

9.1 Functions and Inverse Image Sets

1. Inverse image sets.

Let a single-valued, real function $f(x)$ be defined in the space A (which, for the time being, may be a pure set and not a metric space), so that a real number $f(x)$ is made to correspond to every point $x \in A$. The set of points x at which $f(x) > y$ (y a given real number) will be denoted for brevity, as in §22, by $[f > y]$; we define similarly sets such as $[f \geq y]$, $[y < f < z]$, and so on. If B is the set of values, or range, of $f(x)$, that is, the set of numbers taken on by $f(x)$, then these sets are nothing but the inverse images of certain subsets of B, in the terminology of §35. Since real functions can exhibit[1] different kinds of behavior "from above" and "from below" — e.g., they may be bounded from above and unbounded from below — it seems advisable to consider simultaneously the sets $[f > y]$ and $[f < y]$, say; in place of the latter, we prefer their complements and call the sets

$$[f > y] \quad \text{and} \quad [f \geq y]$$

the *inverse image sets* (or *Lebesgue sets*) belonging to the function f.

But the two kinds of sets are related, for we have

$$[f > y] = \mathfrak{S}_n \left[f \geq y + \tfrac{1}{n}\right], \quad [f \geq y] = \mathfrak{D}_n \left[f > y - \tfrac{1}{n}\right] \qquad (n = 1, 2, 3, \ldots). \tag{1}$$

[1] If, in addition, x is a real variable, then a further distinction has to be made as between left and right.

A function determines its inverse image sets, but the converse is also true, and in fact we can see from (1) that the sets $[f > y]$ are themselves sufficient to determine f; for, having them, we then also know the sets $[f = y]$ and, as differences, the sets $[f = y]$. Neither need all the sets $[f > y]$ be known, since further relations hold among them: for example, it suffices to know the sets $[f > r]$ for rational r, since

$$[f > y] = \mathfrak{S}_{r>y}\,[f > r].$$

Among the sets $[f > r]$ there subsist, further, the relations

$$[f > r] = \mathfrak{S}_{\varrho > r}\,[f > \varrho], \quad A = \mathfrak{S}_r\,[f > r], \quad 0 = \mathfrak{D}_r\,[f > r],$$

and the reader can easily convince himself that a system of sets $M(r)$ satisfying these conditions really defines one (and only one) function f for which $[f > r] = M(r)$.

The interrelations between the properties of a function and those of its inverse image sets will be the principal subject of this chapter. We are already familiar with one result of this kind: If f is a continuous function, then every set $[f > y]$ is open (in A) and every set $[f \geq y]$ is closed. The converse of this also holds: If all the sets $[f > y]$ and $[f < z]$ are open, then so are their intersections $[y < f < z]$ and, since every one-dimensional open set G is a sum of open intervals, the inverse image of G is open and, by I of §35, $f(x)$ is thus continuous (B is again the range (set of values) of f).

We now consider a system of functions f all of which are defined in the same space A.

Given two functions f_1 and f_2 we obtain, by considering at each point x the maximum and minimum of the two numbers $f_1(x)$ and $f_2(x)$, the two additional functions

$$f = \max\,[f_1, f_2] \quad \text{and} \quad f = \min\,[f_1, f_2].$$

Then we obviously have

$$\begin{aligned} &[f > y] = [f_1 > y] + [f_2 > y], \qquad &[f \geq y] = [f_1 \geq y] + [f_2 \geq y], \\ &[f > y] = [f_1 > y] \cdot [f_2 > y], \qquad &[f \geq y] = [f_1 \geq y] \cdot [f_2 \geq y]. \end{aligned} \tag{2}$$

For the sum

$$f = f_1 + f_2$$

$f > y$ or $f_1 + f_2 > y$ asserts the existence of a rational number r (which depends on the point x under consideration) satisfying $f_1 >$

$r > y - f_2$, or $f_1 > r$ and $f_2 > y - r$, from which the first of the two formulas

$$[f > y] = \mathfrak{S}_r\,[f_1 > r] \cdot [f_2 > y - r] \tag{3}$$

is obtained; the second results from the corresponding treatment of $[f < y]$ and formation of complements.

In addition, let $f_1, f_2, \ldots$ be a sequence of functions, and let its least upper bound and greatest lower bound be

$$f_g = \sup f_n, \quad f_s = \inf\,[f_1, f_2, \ldots] = \inf f_n, \tag{4}$$

where, for every x, the sequence of the numbers $f_n(x)$ is, in the first case, bounded from above, and, in the second case, bounded from below. (We consider only finite functions and so exclude the improper values $\pm\infty$, which we can discuss separately whenever necessary.) Here we have

$$[f_g > y] = \mathfrak{S}\,[f_n > y], \qquad [f_s \geq y] = \mathfrak{D}\,[f_n \geq y], \tag{5}$$

whereas the two middle formulas in (2) cannot be carried over to an infinite number of functions. (5) would seem to suggest that it is not unsuitable to denote the functions $\sup f_n$ by f_g and the functions $\inf f_n$ by f_s. The least upper bound f_g of a sequence of functions f_n is itself an f_g, for in place of

$$g = \sup_n \sup_m f_{n,m} = \sup_{n,m} f_{n,m}$$

we can write

$$g = \sup_k \; f_{n_k,m_k} = \sup_k\,[f_{n_k}, f_{m_k}, f_{n_k,k}, \ldots]$$

with the double sequence transformed into a simple one; similarly, f_{sg} is an f_s.

The upper limit and the lower limit of a sequence, as is well known, are defined by:

$$\begin{aligned} \overline{\lim}\, f_n &= \lim g_n, \quad g_n = \sup\,[f_n, f_{n+1}, f_{n+2}, \ldots], \\ \underline{\lim}\, f_n &= \lim h_n, \quad h_n = \inf\,[f_n, f_{n+1}, f_{n+2}, \ldots]. \end{aligned} \tag{6}$$

In the first case, f_n is to be assumed bounded from above, and g_n from below; in the second, f_n from below and h_n from above (these assumptions about boundedness will not always be repeated in future). We have $g_1 \geq g_2 \geq \ldots$ and $h_1 \leq h_2 \leq \ldots$, so that it is also possible to write

$$\overline{\lim}\, f_n = \inf g_n, \quad \underline{\lim}\, f_n = \sup h_n;$$

the g_n are functions f_g, $\overline{\lim}\, f_n$ is thus an f_s and, similarly, $\underline{\lim}\, f_n$ is an f_g. The limit function $\lim f_n$ of a convergent sequence is both at the same time.

In the case of uniform convergence, an essential simplification appears: The limit function $\lim f_n$ of a uniformly convergent sequence is both an f_g and an f_s, assuming that the functions f are taken from a system which contains $f + \text{constant}$ whenever it contains f. For if $|f - f_n| \leq \varepsilon_n$ and $\varepsilon_n \to 0$, then $f = \inf (f_n + \varepsilon_n) = \sup (f_n - \varepsilon_n)$.

In order to put our statements about inverse image sets in convenient form, let us agree as follows: The sets M and N are to range over given systems of sets $\mathfrak{M}$ and $\mathfrak{N}$. If then $[f > y]$ is an M for every y, we say that the function f is of the class $(\mathfrak{M}, *)$. If $[f \geq y]$ is always an N, then we say that f is of the class $(*, \mathfrak{N})$. When both hold, we say f is of the class $(\mathfrak{M}, \mathfrak{N})$. For example, the continuous functions of the metric space A are of the class (G, F), where G ranges over the open sets of the space A and F over the closed sets, and vice versa. When, in particular, the N are the complements $A - M$ of the M, then the statements "f is of the class $(\mathfrak{M}, *)$" and "$-f$ is of the class $(*, \mathfrak{N})$" have the same meaning.

Let us now assume that the M as well as the N form a ring (the sum and the intersection of two M is an M; the sum and the intersection of two N is an N). Then we can state the following theorems:

I. If the functions f are of the class $(\mathfrak{M}, *)$, then

$$\begin{aligned}
&\max [f_1, f_2] \text{ and } \min [f_1, f_2] \text{ are also of the class } (\mathfrak{M}, *);\\
&f_g = \inf f_n \text{ are of the class } (*, \mathfrak{M}_\delta);\\
&f_s = \sup f_n \text{ and } f_1 + f_2 \text{ are of the class } (\mathfrak{M}_\sigma, *).
\end{aligned}$$

II. If the functions f are of the class $(*, \mathfrak{N})$, then

$$\begin{aligned}
&\max [f_1, f_2] \text{ and } \min [f_1, f_2] \text{ are also of the class } (*, \mathfrak{N});\\
&f_s = \sup f_n \text{ are of the class } (\mathfrak{N}_\sigma, *);\\
&f_g = \inf f_n \text{ and } f_1 + f_2 \text{ are of the class } (*, \mathfrak{N}_\delta).
\end{aligned}$$

All these assertions follow immediately from (2), (3), and (5), although for the two middle ones we must also make use of (1). If the f are of the class $(\mathfrak{M}, *)$, then they are at the same time of the class $(*, \mathfrak{M}_\delta)$, and f_g is of the class $(*, \mathfrak{M}_{\delta\delta}) = (*, \mathfrak{M}_\delta)$. If the f are of the class $(*, \mathfrak{N})$, then they are also of the class $(\mathfrak{N}_\delta, *)$, and f_s is of the class $(\mathfrak{N}_{\delta\sigma}, *) = (\mathfrak{N}_\sigma, *)$.

We shall call a system of functions f an *ordinary function system* if it satisfies the following axioms:

(α) Every constant function is an f.

(β) The maximum and the minimum of two f is an f.

(γ) The sum, difference, product, and quotient (with nowhere-vanishing denominator) of two f is an f.

Because of the identities

$$|f| = \max[f, -f], \qquad \min[f_1, f_2] = \tfrac{1}{2}(f_1 + f_2) - \tfrac{1}{2}|f_1 - f_2|,$$

condition (β) can be replaced by: The absolute value of every f is an f.

An ordinary function system is called *complete* if it also satisfies the following postulate:

(δ) The limit of a uniformly convergent sequence of functions f is an f.

We then have the following theorem:

III. Let the sets M, to which there belong the whole space and the null set, form a σ-ring,[2] their complements $N = A - M$ therefore forming a δ-ring. Then all the functions f of the class $(\mathfrak{M}, \mathfrak{N})$ form a complete system.

For (α) is satisfied, since A and 0 are sets M and N; (β) holds because, by I and II, $\max[f_1, f_2]$ and $\min[f_1, f_2]$ are of the class $(\mathfrak{M}, \mathfrak{N})$. Next, f_g and $f_1 + f_2$ are of the class $(\mathfrak{M}, *)$ and f_s and $f_1 + f_2$ are of the class $(*, \mathfrak{N})$, so that $f_1 + f_2$ and the uniform limit of functions f are of the class $(\mathfrak{M}, \mathfrak{N})$; ($\delta$) is satisfied. That the difference of two f is also an f follows from the corresponding statement for the sum, since $-f$ is an f. The square of an f is an f; for, $[f^2 > y]$ is the whole space for $y < 0$, is the sum of the two sets $[f > \sqrt{y}]$ and $[f < -\sqrt{y}]$ for $y \geq 0$, and is thus an M; in the same way, $[f^2 \geq y]$ is an N. Therefore the product of two f,

$$f_1 f_2 = \tfrac{1}{4}(f_1 + f_2)^2 - \tfrac{1}{4}(f_1 - f_2)^2,$$

---

[2] That is, a ring and a σ-system (every M_σ is an M).

is an f. Provided $f \not\equiv 0$, $1/f$ is an f; for $[1/f > y]$ is identical with $[0 < f < 1/y]$ for $y > 0$, the latter set being the intersection of two M; for $y = 0$, it is identical with $[f > 0]$; and for $y < 0$, with the sum of the sets $[f > 0]$ and $[f < 1/y]$. Thus $1/f$ and $-1/f$ are of the class $(\mathfrak{M}, *)$ and $1/f$ is of the class $(\mathfrak{M}, \mathfrak{N})$. The validity of (γ) is thus proved.

2. Extension of ordinary systems.

Let f range over an ordinary function system. Next let f^* denote the limit of an (everywhere) convergent sequence of functions f, let g denote the particular case of an increasing sequence $(f_1 \leq f_2 \leq \ldots)$, let h denote that of a decreasing sequence $(f_1 \geq f_2 \geq \ldots)$, and let k denote a function which is both a g and an h. In the table

$$f, \quad g, \quad h, \quad k, \quad f^*$$

every function is a special case of the one that follows to the right. Incidentally, this scheme is identical with the following one:

$$\begin{array}{ccccc} f_s, & g, & k, & h, & f_g \\ f_g, & h, & k, & g, & f_s \end{array}$$

From now on, of the two halves of the results concerning g and h, we shall most of the time prove only one.

Every g is an $f_s = \sup f_n$, and also conversely, inasmuch as

$$\sup f_n = \lim \max [f_1, f_2, \ldots, f_n].$$

Included among the k (which are simultaneously f_s and f_g) are, in particular, the limits of uniformly convergent sequences f_n. The limit of uniformly convergent g_n, h_n, and k_n is a g, h, and k ($\lim g_n$ is a $g_s = g$; $\lim h_n$ is an $h_g = h$).

The maximum, the minimum, and the sum of two g is a g. For if f_n and f'_n are increasing sequences that converge to g and g', then $\max [f_n, f'_n]$, $\min [f_n, f'_n]$, and $f_n + f'_n$ are increasing sequences that converge to $\max [g, g']$, $\min [g, g']$, and $g + g'$.

The maximum, the minimum, the sum, the difference, and the product of two f^* is each, of course, an f^*, and so is their quotient (with nowhere-vanishing denominator). It suffices to show that $1/g$ is an f^* whenever $g \neq 0$ is an f^*. If $g > 0$ everywhere and $g = \lim f_n$, then it is possible also to assume that $f_n > 0$ by replacing it by

$\max[f_n, 1/n]$, which converges to $\max[g, 0] = g$, and then we have $1/g = \lim 1/f_n$. If $g \geq 0$, then $1/g = 1/g \cdot g$ is the product of two f^* and is thus an f^*.

The f^* therefore form an ordinary system, but over and above this we have:

IV. The limit of a uniformly convergent sequence of functions f^* is an f^*; the f^* form a complete system.

First, it should be noted that if $f^* = \lim f_n$ has absolute value $\leq c$, then we may assume that $|f_n| \leq c$; for $\bar{f}_1 = \max[f_1, -c]$ and $\bar{\bar{f}}_2 = \min[\bar{f}_1, c]$ as well converge to f^*.

Now let F be the uniform limit of the functions $F_1, F_2, \ldots$ that are functions f^*; by restriction to a subsequence, it is possible to assume that F_n deviates from all the following F by at most ε_{n+1} $(n = 0, 1, 2, \ldots)$ where $\varepsilon_1 + \varepsilon_2 + \cdots$ is a convergent series of positive numbers and, in consequence, to write

$$F - F_1 = (F_2 - F_1) + (F_3 - F_2) + \cdots$$

or $F = F_1 + g_2 + g_3 + \cdots$, where the g_m $(m = 1, 2, \ldots)$ are functions f^* of norm $\leq \varepsilon_m$; it is to be shown that F is an f^*. By what was said above, we can suppose that

$$g_m = \lim_n f_{mn}, \quad |f_{mn}| \leq \varepsilon_m.$$

But then

$$f_n = f_{1n} + f_{2n} + \cdots + f_{nn}$$

converges to F, which proves our result. In fact, for $l > m$, the absolute value of

$$f_{ml} + f_{m+1,l} + \cdots + f_{ll} - (f_{mn} + f_{m+1,n} + \cdots + f_{ln})$$

is $\leq \varepsilon_m + \varepsilon_{m+1} + \cdots + \varepsilon_l < \sigma_m$, if we set $\delta_m = \varepsilon_{m+1} + \varepsilon_{m+2} + \cdots$; from

$$f_{1n} + \cdots + f_{mn} - \delta_m < f_n < f_{1n} + \cdots + f_{nn} + \delta_m$$

it follows, for $l \to \infty$, that

$$g_1 + \cdots + g_m - \delta_m \leq \underline{\lim}\, f_n \leq \overline{\lim}\, f_n \leq g_1 + \cdots + g_m + \delta_m$$

and from this, for $m \to \infty$, that $\lim f_n = g_1 + g_2 + \cdots = F$.

Now we denote the inverse image sets

$$\begin{array}{ccccc} [f > y], & [g > y], & [h > y], & [k > y], & [f^* > y], \\ M & P & Q & R & M^* \end{array}$$

by M, N, P, Q, R, M^*, N^*, so that M, for example, denotes all the sets appearing among the sets $[f > y]$ when f ranges over the system of functions under consideration and y over all the real numbers and where, moreover, y can be assumed to have a fixed value, say $y = 0$, since $f - y$ is also an f.

By definition, the functions f, g, h, k, and f^* are of the classes $(\mathfrak{M}, \mathfrak{N})$, $(\mathfrak{P}, *)$, $(*, \mathfrak{Q})$, $(\mathfrak{P}, \mathfrak{Q})$, and $(\mathfrak{M}^*, \mathfrak{N}^*)$. It will be our task to invert these statements insofar as possible, first expressing the $\mathfrak{P}$, $\mathfrak{Q}$, $\mathfrak{M}^*$, and $\mathfrak{N}^*$ in terms of the $\mathfrak{M}$ and the $\mathfrak{N}$.

The $\mathfrak{M}$ and the $\mathfrak{N}$ are complements of each other, as are the $\mathfrak{P}$ and the $\mathfrak{Q}$, and also the $\mathfrak{M}^*$ and the $\mathfrak{N}^*$. All six systems of sets are rings; the sum and the intersection of two P, for example, are themselves sets P, by (2), because the maximum and the minimum of two g is a g.

In addition, there follows from I and II:

V. The sets $\mathfrak{P}$, $\mathfrak{Q}$, $\mathfrak{M}^*$, and $\mathfrak{N}^*$ are sets $\mathfrak{M}_\sigma$, $\mathfrak{N}_\delta$, $\mathfrak{Q}_\sigma$, and $\mathfrak{P}_\delta$.

For, $g = f_s$ is of class $(\mathfrak{M}_\sigma, *)$, so that every P is an $\mathfrak{M}_\sigma$; f_g is of the class $(\mathfrak{P}, *)$ so that f_{gg} is of the class $(*, \mathfrak{P}_\delta)$; the latter holds in particular for $\lim f_n$ and *a fortiori* for f^*, so that every N^* is a $\mathfrak{P}_\delta$. The other statements are proved in the same way.

We give four simple lemmas that will lead to the converse of V.

(A) For every M, there exists an f that is positive in M and otherwise (that is, in $A - M$) vanishes.

For there exists a function f for which $M = [f > 0]$; then $f^+ = \max[f, 0]$ satisfies the condition. Moreover $f^{+\varepsilon} = \min[f^+, \varepsilon]$ with $\varepsilon > 0$ serves the same purpose; that is, the function called for in (A) may be assumed in addition to be as small as we please $(0 \leq f \leq \varepsilon)$.

(B) For every M, there exists a function F which is the limit of a uniformly convergent sequence of functions f and which, in addition, is positive in M_σ and otherwise vanishes.

Let $M_\sigma = M_1 + M_2 + \cdots$, and let $\varepsilon_1 + \varepsilon_2 + \cdots$ be a convergent series of positive numbers; in accordance with (A) we determine a function f_n for which $0 \leq f_n \leq \varepsilon_n$ and which is positive in M_n and otherwise vanishes. Then

$$F = f_1 + f_2 + \cdots$$

is a function of the required kind.

(C) For every M there exists a g which equals 1 in M and is otherwise 0.

We choose f as in (A); then

$$g = \lim_n \frac{nf}{1+nf} = \lim_n \min[nf, 1],$$

say, are functions of the kind required.

(D) For every M_σ, there exists a g which is 1 in M_σ and is otherwise 0.

If $M_\sigma = M_1 + M_2 + \cdots$, then let g_n be a function which, in accordance with (C), is 1 in M_n and is otherwise 0; then

$$g = \sup g_n$$

(which is also a g) is a function of the kind required.

Now we can form the converse of V.

VI. The sets $\mathfrak{M}_\sigma$, $\mathfrak{N}_\delta$, $\mathfrak{Q}_\sigma$, and $\mathfrak{P}_\delta$ are sets $\mathfrak{P}$, $\mathfrak{Q}$, $\mathfrak{M}^*$, and $\mathfrak{N}^*$.

By (D), every $M_\sigma = [g > 0]$ is a $\mathfrak{P}$ and also, of course, every N_δ is a $\mathfrak{Q}$. But at the same time $M_\sigma = [g \geq 1]$ is an N^* (since g is an f^*), and thus every $\mathfrak{P}$ is an $\mathfrak{N}^*$, every $\mathfrak{Q}$ an $\mathfrak{M}^*$ and every $\mathfrak{Q}_\sigma$ an $\mathfrak{M}^*_\sigma$. If we apply (B), not to the f but rather to the f^*, where the function F named in (B) is now, by IV, an f^*, then it follows that every $M^*_\sigma = [F > 0]$ is an $\mathfrak{M}^*$. Thus every $\mathfrak{Q}_\sigma$ is an $\mathfrak{M}^*$ and similarly every $\mathfrak{P}_\delta$ an $\mathfrak{N}^*$.

Thus the $\mathfrak{P}$, $\mathfrak{Q}$, $\mathfrak{M}^*$, and $\mathfrak{N}^*$ are identical with the $\mathfrak{M}_\sigma$, $\mathfrak{N}_\delta$, $\mathfrak{Q}_\sigma$, and $\mathfrak{P}_\delta$ or (eliminating the $\mathfrak{P}$ and $\mathfrak{Q}$) the $\mathfrak{M}^*$ and $\mathfrak{N}^*$ are identical with the $\mathfrak{N}^*_\delta$ and the $\mathfrak{M}_{\sigma\delta}$; the transition from the f to the f^* induces the σ and δ process in the inverse image sets.

3. Inversion of the theorems on class.

We now come to the principal result of the whole theory, namely:

VII. The functions of the class $(\mathfrak{P}, \mathfrak{Q})$ are identical with the functions v that form the smallest complete system containing the system of the f.

One half of the statement follows from III. The $\mathfrak{P}$ form a ring which — because of postulate (β) for ordinary systems — contains the space A and the null set. In addition, because of their identity with the $\mathfrak{M}_\sigma$, they form a σ-system ($\mathfrak{P}_\sigma$ is a $\mathfrak{P}$). The corresponding result holds for their complements $\mathfrak{Q} = A - \mathfrak{P}$. The functions of the class $(\mathfrak{P}, \mathfrak{Q})$ therefore constitute a complete system containing the system of the f; the smallest complete system of this kind must be contained in this system; that is, every function v is of the class $(\mathfrak{P}, \mathfrak{Q})$.

In order to show that, conversely, every function φ of the class $(\mathfrak{P}, \mathfrak{Q})$ is a v, we first prove the following: If $y_1 < y_2$, then there always exists a function v such that for

$$P_1Q_2, \quad P_1P_2, \quad Q_1P_2$$

we have $v = 0$, $0 < v < 1$, $v = 1$.

For let $P_1 = [\varphi > y_1]$ and $P_2 = [\varphi < y_2]$. Corresponding to these sets $P = M_\sigma$ there exist, by Lemma (B), functions v_1 and v_2 (limits of uniformly convergent sequences f_n) that are positive in these sets and vanish in their complements $Q_1 = [\varphi \le y_1]$ and $Q_2 = [\varphi \ge y_2]$. Since $Q_1Q_2 = 0$, the v_1 and v_2 do not vanish simultaneously, we have $v_1 + v_2 > 0$, and the function $v = v_1/(v_1 + v_2)$ satisfies the required conditions.[3] In fact, in the three sets above we have

$$\begin{array}{ccc} Q_1P_2 & P_1P_2 & P_1Q_2 \\ v_1 = 0 & v_1 > 0 & v_1 > 0 \\ v_2 > 0 & v_2 > 0 & v_2 = 0 \\ v = 0 & 0 < v < 1 & v = 1. \end{array}$$

If, for the time being, φ is now bounded — say, $0 \le \varphi \le 1$ — then we choose a natural number n, as large as we wish, determine a function v_m for $m = 1, 2, \dots, n$ such that for

$$P_1^{(m)} = \left[\varphi > \tfrac{m-1}{n}\right], \quad P_2^{(m)} = \left[\varphi < \tfrac{m}{n}\right],$$

$v_m = 0$, $0 < v_m < 1$, $v_m = 1$, and set $v = (v_1 + \cdots + v_n)/n$. (The meaning of f, v, and v differs from that above.) If, at some point x, we find $(m-1)/n \le \varphi \le m/n$, then we have

$$v_1 = \cdots = v_{m-1} = 1, \quad 0 < v_m < 1, \quad v_{m+1} = \cdots = v_n = 0$$

at that point, so that $(m-1)/n \le v \le m/n$, and therefore $|\varphi - v| \le 1/n$ everywhere. Therefore it is possible to approximate φ uniformly by some v, so that φ must itself be a v.

[3] v need no longer be the uniform limit of functions f but belongs to the smallest complete system generated by the f.

If φ is of the class $(\mathfrak{P}, \mathfrak{Q})$ but is not bounded, then

$$\psi = \frac{\varphi}{1 + |\varphi|} \tag{7}$$

is bounded $(-1 < \psi < 1)$ and is of the same class. For

$$y = \frac{Y}{1 + |Y|}, \qquad Y = \frac{y}{1 - |y|}$$

is a similarity mapping of the open interval $-1 < y < 1$ onto the set of all the real numbers Y (in the ordinal sense, so that it is a continuous and monotone mapping). Every set $[\psi > y]$ is, for $-1 < y < 1$, a set $[\varphi > Y]$, and conversely; the former also contains, for $y = -1$ and $y \geq 1$, the whole space and the null set, which — as already remarked — are also sets $\mathfrak{P}$. Thus ψ is of the class $(\mathfrak{P}, *)$ whenever φ is of this class, and conversely; the same holds for $(*, \mathfrak{Q})$. Therefore, once again, ψ is a v, and consequently

$$\varphi = \frac{\psi}{1 - |\psi|} \tag{8}$$

is also a v, and Theorem VII is proved. This "bounding" transformation, the replacing of the unbounded function φ by the bounded function ψ, will be made use of later on also.

VIII. If the functions f form a complete system, then the $\mathfrak{M}$ form a σ-ring and the $\mathfrak{N}$ a δ-ring, and the functions of the class $(\mathfrak{M}, \mathfrak{N})$ are identical with the f.

For, Lemma (B), where F is now an f, shows that every $\mathfrak{M}_\sigma$ is an $\mathfrak{M}$. The $\mathfrak{P}$ and $\mathfrak{Q}$ then become identical with the $\mathfrak{M}$ and $\mathfrak{N}$, the functions v with the f, and the balance of the result follows from VII.

Theorem VIII is, at the same time, the converse of III.

By IV, the hypothesis of completeness certainly holds for the f^*, and therefore:

IX. The functions of the class $(\mathfrak{M}^*, \mathfrak{N}^*)$ are identical with the f^*.

Let us now look for the converse to the fact that every g is of the class $(\mathfrak{P}, *)$. By definition, for every g there exists a $f \leq g$ — a minorant f. Conversely, we have:

X. Every function of the class $(\mathfrak{P}, *)$ that has a minorant f is a g.

Let φ be of the class $(\mathfrak{P}, *)$ and $\varphi \geq f$ or, if we write $1 - f$ in place of f, $\varphi \geq t + f > 0$. $\varphi + f$ is also of the class $(\mathfrak{P}, *)$, as follows from (3) with $\mathfrak{P} = \mathfrak{M}_\sigma$, and so — altering our notation slightly — we can assume that $\varphi \geq 0$. For some $\delta > 0$ and $n = 1, 2, \ldots$, noting that the set $[\varphi > n\delta]$ is a $P = \mathfrak{M}_\sigma$, we look for a g_n that, by Lemma (D), is 1 in this set and is 0 otherwise. The everywhere-convergent series

$$g = \delta(g_1 + g_2 + g_3 + \cdots),$$

whose elements in fact ultimately vanish for every x, represents a g, since it is the limit of increasing g_n. If at some point $(n-1)\delta < \varphi \leq n\delta$, then at that point

$$g_1 = \cdots = g_{n-1} = 1, \quad g_n = g_{n+1} = \cdots = 0, \quad g = (n-1)\delta$$

and therefore the relation $0 \leq \varphi - g < \delta$ holds everywhere. Thus φ can be uniformly approximated by functions g and is itself a g.

XI. Every function of the class $(\mathfrak{P}, *)$ is the limit of an increasing sequence of functions v of the class $(\mathfrak{P}, \mathfrak{Q})$.

For a bounded function, this is contained in X. If φ is unbounded and of the class $(\mathfrak{P}, *)$, then once more we make the "bounding" transformation (7); ψ is of the class $(\mathfrak{P}, *)$ and is bounded $(-1 < \psi < 1)$, and it is therefore a g; $\psi = \lim v_n$ with $f_1 \leq f_2 \leq \ldots$, where we may assume $-1 \leq f_n \leq 1$ by replacing f_n by $\max[f_n, -1]$. To return to φ however, we must have, instead of the functions f_n, functions that do not attain the lower bound -1. If we set

$$v_n = f_n + \tfrac{1}{2}|f_1 - f_2| + \tfrac{1}{2}|f_2 - f_3| + \cdots + \tfrac{1}{2}|f_{n-1} - f_n| + \tfrac{1}{2n},$$

then this function, as the sum of a uniformly convergent series of functions f, is a v (and even a k). Obviously, $\psi \leq v_1 \leq v_2 \leq \ldots$ and v_n is an increasing sequence converging to φ. In $v_n = f_n$ the equality can hold only if $f_n = f_{n+1} = f_{n+2} = \cdots$, so that $v_n = \varphi > -1$; otherwise $v_n > f_n = -1$. Therefore since $-1 < v_n < 1$ always holds, we can form the functions

$$V_n = \frac{v_n}{1 - |v_n|}$$

which, once more, are functions v and converge from below to

$$\frac{\varphi}{1 - |\varphi|} = \varphi.$$

The corresponding theorems hold for the functions of the class $(*, \mathfrak{Q})$; if they have a majorant f, they are functions h, and, in any case, they are limits of decreasing sequences of functions v. Every function f that lies entirely between two functions v of the class $(\mathfrak{P}, \mathfrak{Q})$ is both a function g and a function h, and is thus a function k. Since, in particular, every bounded function v is a k, it follows from the bounding transformation that every function v can be represented in the form k/k' with $k < k'$. In this representation, moreover, $|k| = \max[k, -k]$, as the maximum of two k, is itself a k, and the same is true of $1 - |k|$; that is, the functions v can be represented as quotients of two k. Conversely, every quotient of two k, as the quotient of two k, is itself a v.

If the system of the f is complete, then it follows from XI that every function of the class $(\mathfrak{M}, *)$ is a g. We summarize here all the simplifications that occur in this case:

XII. If the f form a complete system, then the f, g, h, and f^* are identical with the functions of the classes $(\mathfrak{M}, \mathfrak{N})$, $(\mathfrak{M}, *)$, $(*, \mathfrak{N})$, and $(\mathfrak{N}_\delta, \mathfrak{M}_\sigma)$, respectively.

Here the k and v are identical with the f and the sets $\mathfrak{P}$, $\mathfrak{Q}$, $\mathfrak{M}^*$, and $\mathfrak{N}^*$ with the sets $\mathfrak{M}$, $\mathfrak{N}$, $\mathfrak{N}_\delta$, and $\mathfrak{M}_\sigma$.

If the f form merely an ordinary system, then XII is applicable to the v; the v, v_g, v_s, and v^* are identical with the functions of the classes $(\mathfrak{P}, \mathfrak{Q})$, $(\mathfrak{P}, *)$, $(*, \mathfrak{Q})$, and $(\mathfrak{Q}_\sigma, \mathfrak{P}_\delta)$. The f^* here are, by IX, identical with the v^*, while the f, g, and k (or the f, f_s, and f_g) constitute only some of the v, v_g, and v_s; the g are identical with those v_g that have a minorant f; the h, with those v_s that have a majorant f; and the k (of which, again, the f constitute only a subclass), with those v that lie between two f.

We suggest that the reader now review the discussion up to this point, keeping in mind some simple example like the following: Let the f be the functions that take on only a finite number of values; then the g are the functions that are bounded from below, the h those bounded from above, the k those bounded from both above and below, and the v and f^* the completely arbitrary functions. That every function bounded from below, say $g \geq 0$, is in fact a g — the converse is trivial — is seen as follows: In the set

$$R_n = \left[\tfrac{m}{n} \leq g < \tfrac{m+1}{n}\right]$$

let f_n be the constant rational number m/n; f_n is an f and converges to g since, for $n > g$, we have $f_n \leq g < f_n + 1/n$. Since $R_0 < R_{2n}$ for f_n, f_{2n}; the functions $f_1, f_2, f_3, f_4, \ldots$ converge to g from below.

The rest is easy to see. The sets $\mathfrak{M}$ and $\mathfrak{N}$ are already the arbitrary subsets of A, and all the more is this true of $\mathfrak{P}$, $\mathfrak{Q}$, $\mathfrak{M}^*$, and $\mathfrak{N}^*$.

4. Interposition and Extension Theorem.

XIII. (Interposition Theorem). If g is a function g, h a function h, and $g \leq h$ everywhere, then there exists a function k for which $g \leq k \leq h$.

For brevity, we set

$$\{t\} = \max[t, 0] = \tfrac{1}{2}(|t| + t) \tag{9}$$

for real t; this is a continuous function of t which is non-negative and which increases (more exactly: does not decrease) with increasing t.

We now make the following analysis, which we shall have occasion to revert to later (in the proof of XVI). Let

$$g = \lim g_n, \quad h = \lim h_n, \quad g_n \leq g, \quad h_n \geq h$$

and $g \leq h$. Then

$$g_n - h_n \leq g - h \leq 0, \quad h_n - g_{n+1} \geq h - g \geq 0, \tag{10}$$

so that, in addition,

$$\{g_n - h_n\},\ \{h_n - g_{n+1}\},\ \{g_{n+1} - h_{n+1}\},\ \ldots \to 0.$$

The alternating series

$$\omega = g_1 + \{h_1 - g_1\} - \{h_1 - g_2\} + \{h_2 - g_2\} - \{h_2 - g_3\} + \cdots \tag{11}$$

therefore converges everywhere. We claim that $g \leq \omega \leq h$. Let us distinguish between the points for which $g = h$ and $g > h$. If $g = h$, then all the members of the sequence (10) are ≤ 0, the braces may be removed, and we have

$$\begin{aligned}\omega &= g_1 + (h_1 - g_1) - (h_1 - g_2) + (h_2 - g_2) - (h_2 - g_3) + \cdots \\ &= \lim g_n = \lim h_n = g = h.\end{aligned}$$

If $g > h$, then the terms in (10) ultimately become negative. If the first negative term is $g_n - h_n$, so that $g = h_{n-1} \geq h_n = h$, then

$$\omega = g_1 + \{h_1 - g_1\} - \cdots - \{h_{n-1} - g_n\} = h_n = h \leq \omega \leq g.$$

If the first negative term is $h_n - g_{n+1}$, so that $g > g_{n+1} \geq h_n = h$, then

$$\omega = g_1 + \{h_1 - g_1\} - \cdots + \{h_n - g_{n+1}\} = h_n = h \leq \omega \leq g.$$

Now, in order to prove XIII, let $g = \bar{g}$ and $h = \bar{h}$; let the functions g_n and h_n be functions f. The individual terms and the partial sums of series ω are also functions f; since the partial sums with an odd number of terms form an increasing sequence while those with an even number of terms form a decreasing sequence, ω is both a g and an h and is thus a k.

In the following theorem we are concerned with the question of "extending" a function g defined in the set $B < A$ to a function v defined in A (which of course means that $v = g$ in B). We say that g is of the class $(\mathfrak{M}, *)$ if every set $[g > y]$ is the intersection of B with an M; and correspondingly for the classes $(*, \mathfrak{N})$ and $(\mathfrak{M}, \mathfrak{N})$.

XIV. (Extension Theorem). If Q_0 is a set Q, then a function of the class $(\mathfrak{P}, \mathfrak{Q})$ defined in Q_0 can be extended to a function of the class $(\mathfrak{P}, \mathfrak{Q})$ defined in the whole space A; that is, it can be extended to a function v.

Let $P_0 = A - Q_0$; let φ be defined in Q_0, let it be of the class $(\mathfrak{P}, \mathfrak{Q})$, and let it, for the time being, be bounded — say, $-1 \leq \varphi \leq 1$. We define the function h by:

$$h = \varphi \text{ in } Q_0, \quad h = -1 \text{ in } P_0,$$

and we claim that this is an h. For it is of the class $(*, \mathfrak{Q})$ since, for $y > -1$, $[h \geq y]$ coincides with $[\varphi \geq y]$ (which is a $Q \cdot Q_0$ and thus a Q) and, for $y \leq -1$, is the whole space; furthermore, h is bounded and therefore, by X, is an h. In the same way, we conclude that the function g,

$$g = \varphi \text{ in } Q_0, \quad g = 1 \text{ in } P_0,$$

is a g ($-g$ is an h). Since $g \leq h$, a function k can be interposed between them, $g \leq k \leq h$; in Q_0, $k = \varphi$; $-\varphi$ can, in particular, be extended here to a k.

We now wish to show that for $|\varphi| < 1$ we can also make $|v| < 1$. If we determine k as we did just now, then $|k| \leq 1$; the equality can hold only at points of P_0. Now, using Lemma (B), we determine for P_0, which is an $\mathfrak{M}_\sigma$, a function p_0 (also a k) which is positive in P_0 and vanishes in Q_0. The function $v = k/(1 + p_0)$ is certainly a v

(by virtue of its boundedness it is still a k, however); in Q_0 we have $v = k = \varphi$, in P_0 $|v| < |k|$ or $v = 0$, and thus, everywhere, $|v| < 1$.

If, finally, φ is defined in Q_0, is of the class $(\mathfrak{P}, \mathfrak{Q})$, and is not bounded, then we make the bounding transformation (7). The previous assumptions apply to ψ; it can be extended to a v for which $|v| < 1$; and φ can be extended to $V = v/(1 - |v|)$, that is, to a function v.

5. Absolute convergence.

The functions f^* can be represented as sums of convergent series of functions f:

$$f^* = \lim f_n = f_1 + (f_2 - f_1) + (f_3 - f_2) + \cdots = g_1 + g_2 + g_3 + \cdots .$$

If we require absolute convergence here rather than simple convergence (that is, $|g_1| + |g_2| + |g_3| + \cdots$ is to converge everywhere), then we obtain only some of the functions f^*; we denote these functions by d. They, also, constitute an ordinary system. For, to begin with, the sum, difference, and product of two d is itself a d. Next, the absolute value $|d|$ of a d is a d, for by virtue of $||\beta| - |\alpha|| \leq |\beta - \alpha|$ the series obtained from the absolutely convergent series $f_1 + (f_2 - f_1) + \cdots$ by interchanging f_n with $|f_n|$ is absolutely convergent when $f_1 + (f_2 - f_1) + \cdots$ is absolutely convergent. It remains only to prove that $1/d$, with $d \neq 0$, is also a d. First, let $d = \lim f_n > 0$; if f_n is set equal to $\max[f_n, 1/n]$, which is a function that also converges to d, then we obtain the estimate $|f_{n+1} - f_n| \leq |f_{n+1} - f_n| + (1/n - 1/(n+1))$ most easily from the obvious inequality

$$\max[\alpha_1, \alpha_2] - \max[\beta_1, \beta_2] \leq \max[\alpha_1 - \beta_1, \alpha_2 - \beta_2] \leq |\alpha_1 - \beta_1| + |\alpha_2 - \beta_2|;$$

f_n can therefore be replaced by f_n^+, or f_n may be taken > 0 to begin with. For $1/d = \lim 1/f_n$, and

$$\frac{1}{f_{n+1}} - \frac{1}{f_n} = \frac{f_n - f_{n+1}}{f_n \cdot f_{n+1}}$$

is the general term of an absolutely convergent series whenever this is true of $f_{n+1} - f_n$; $1/d$ is a d. If, finally $d \equiv 0$, then $1/d = d \cdot 1/d^2$, as the product of two d, is itself a d.

From the decomposition

$$d = d^+ - (-d)^+ = \{d\} - \{-d\}$$

it follows that, in the case of absolute convergence,

$$d = g - g' = \{d\}^+ - \{-d\}^+$$

is the difference of two convergent series with terms ≥ 0; that is, every function d can be represented as the difference $g - g'$ of two functions g, in place of which it is also possible to set $k - k'$ or $g + h$ (with $g = -h'$ and $g' = -h$) — the difference of two functions h or the sum of a g and an h. (Whereas $g - h = g + g'$ is again a g and $h - g = h + h'$ again an h.) Conversely, it is clear that the functions g and h and their sums and differences are functions d. With the restriction to absolute convergence, the f thus give rise to the functions

$$d = g - g' = h - h' = g + h. \tag{12}$$

Let us call any function whose range is isolated a *step function.* Then we have:

XV. Every function f^* which is also a step function is a function d.

We preface the proof by the following remark. If we denote by $\mathfrak{R}$ those sets that are both $\mathfrak{M}^*$ and $\mathfrak{N}^*$, then every function φ which is a step function f^* is of the class $(\mathfrak{R}, \mathfrak{R})$.

For the set $[\varphi > y]$ is identical with $[\varphi \geq y]$ if φ does not take on the value y; and otherwise, for sufficiently small $\delta > 0$, it is identical with $[\varphi = y + \delta]$; similarly, the set $[\varphi \geq y]$ is identical with $[\varphi > y]$ or with $[\varphi > y - \delta]$; both sets are $\mathfrak{M}^*$ and $\mathfrak{N}^*$ at one and the same time. The sets $\mathfrak{R}$ obviously form a field. For every function of the class $(\mathfrak{R}, \mathfrak{R})$, the sets $[\varphi = c_m]$ are sets $\mathfrak{R}$ as well.

Now let φ be a step function f^* or, more generally, a function of the class $(\mathfrak{R}, \mathfrak{R})$ with a range consisting of at most a countable number of values; let $c_1, c_2, \dots$ be the various values taken on by φ. The sets $[\varphi = c_m]$ are sets $\mathfrak{R}$ and, in particular, sets $\mathfrak{M}^* = \mathfrak{Q}_\sigma$, and they can therefore be represented with increasing summands Q as follows:

$$\begin{aligned}[\varphi = c_m] &= Q_{m1} + Q_{m2} + Q_{m3} + \cdots \\ &= Q_{m1} + (Q_{m2} - Q_{m1}) + (Q_{m3} - Q_{m2}) + \cdots \\ &= D_{m1} + D_{m2} + D_{m3} + \cdots .\end{aligned}$$

We now set $c_m = a_m - b_m$, where the positive numbers a_m and b_m increase without bound together with m, for example,

$$a_m = |c_m| + b_m + m, \quad b_m = |c_m| - c_m + m$$

and define the functions g and g' by means of

$$g(x) = a_m + n, \quad g'(x) = b_m + n \quad \text{for } x \in D_{mn} \tag{13}$$

so that $\varphi = g - g'$ everywhere. But these functions are functions g. For if y is given, then the inequality $g \leq y$ can hold for only a finite number of m ($a_m \leq y - 1$) and then for only a finite number of $n < y - a_m$; therefore, if n_m denotes the greatest integer $< y - a_m$,

$$[g \leq y] = \mathfrak{S}_{m \leq y-1} \mathfrak{S}_{n \leq n_m} D_{mn} = \mathfrak{S}_{m \leq y-1} Q_{m,n_m}$$

is the sum of a finite number of Q and is thus itself a Q (or 0, if no $a_m \leq y - 1$ exists). g is thus of the class $(\mathfrak{P}, *)$ and is > 0 and therefore, by X, is a function g; the same is true for g', so that XV is proved.

The functions g, h, and d can be approximated as close as we please by means of step functions of the same kind. Let g be a function g, and let $\delta > 0$ and $m = 0, \pm 1, \pm 2, \ldots$; the set $[g > m\delta] = P_m$ is a $\mathfrak{P}$. We define the function g_δ to be equal to $m\delta$ in the set $P_{m-1} - P_m = [(m-1)\delta < g \leq m\delta]$. We see at once that every set $[g_\delta > y]$ is a $\mathfrak{P}$, so that g_δ is of the class $(\mathfrak{P}, *)$; since g has a minorant f, so does $g_\delta = g$, and g_δ is a g (Theorem X). Furthermore, $0 \leq g - g_\delta < \delta$, and g_δ is a special step function taking on only the values $m\delta$. By means of step functions h_δ and $d_\delta = g_\delta + h_\delta$ of the same kind, we can approximate the functions h and $d = g + h$.

On the other hand, the general functions f^* can be approximated by functions d. For another interposition theorem holds:

XVI. If φ and ψ are two functions f^*, and $\varphi > \psi$ everywhere, then there exists a function $\omega = d$ for which $\psi \leq \omega \leq \varphi$.

We can refer back to the proof of XIII. By (6), every f^* can be represented as a limit of decreasing g or increasing h; let us therefore take the functions ψ_n of the above proof to be functions g and the φ_n to be functions h. The functions (10) are again functions g and retain this property upon being enclosed in braces, since $\max[g, 0]$ is a g. Because this time we made the stronger assumption $\varphi > \psi$ (rather than $\psi \leq \varphi$), the terms of the sequence (10) ultimately become negative at every point and those of the series (11) ultimately become 0; therefore both of the series

$$\bar{g} = g_1 + \{h_1 - g_1\} - \{h_1 - g_2\} + \{h_2 - g_2\} - \cdots ,$$
$$\bar{h} = h_1 - \{h_1 - g_2\} + \{h_2 - g_2\} - \{h_2 - g_3\} + \cdots$$

converge and, as limit functions of increasing g, represent functions g; then we have $\psi \leq \bar{g} - \bar{h} \leq \varphi$ and $\bar{g} = \omega = \bar{h}$, proving the theorem.

If we apply it, in particular, to the case $\psi = \varphi - \delta$ ($\delta > 0$ constant), then it follows that every function f^* can be approximated by functions d, and therefore also by step functions d, as closely as we please. It can therefore also be represented by an absolutely convergent series with terms d; that is, restricting ourselves to absolute convergence, we reach the f^* from the f in two steps, with the d intermediate between the two.

9.2 Functions of the First Class

1. Introduction.

We assume the space A to be metric and identify the functions f of the preceding paragraph with the continuous functions; since they constitute a complete system — the limit of a uniformly convergent sequence of continuous functions is itself continuous — Theorem XII above is applicable. The $\mathfrak{M}$ are identical with the open sets G of the space and the $\mathfrak{N}$ with the closed sets F (§22, 1 and II). Therefore:

I. The continuous functions f and the limit functions g, h, and f^* of increasing, decreasing, and convergent sequences of continuous functions are identical with the functions of the classes (G, F), $(G, *)$, $(*, F)$, and (F_σ, G_δ), respectively.

The f^* are called *functions of the first (Baire) class*, and in §43 this will be extended to higher classes. The functions of the classes $(G, *)$, $(*, F)$ are called *semi-continuous* and, specifically, those of the class $(G, *)$ *lower semi-continuous* and those of the class $(*, F)$ *upper semi-continuous*. The significance of these names will be made clear shortly. In view of the fact that the sets $\mathfrak{P}$ and $\mathfrak{Q}$ coincide here with the $\mathfrak{M}$ and $\mathfrak{N}$ and the functions k with the f, the Interposition and the Extension Theorems take on the following forms:

II. If g is lower semi-continuous, h upper semi-continuous, and $g \leq h$ everywhere, then there exists a continuous function f for which $g \leq f \leq h$.

III. A continuous function defined in the closed set F can be extended to a function continuous in the whole space A.

The names "upper semi-continuous" and "lower semi-continuous" derive from the fact that the conditions for continuity are here split into two parts. A function $f(x)$ is continuous at the point a if, for every $\varepsilon > 0$, there exists a neighborhood U_a for whose points $x \in U_a$

$$|f(x) - f(a)| < \varepsilon.$$

If, for every $\varepsilon > 0$, a neighborhood U_a exists in which

$$f(x) - f(a) < \varepsilon,$$

then $f(x)$ is said to be *upper semi-continuous* at the point a; if there is always one in which

$$f(x) - f(a) > -\varepsilon,$$

then $f(x)$ is said to be *lower semi-continuous* at a. To make this more intuitive, we remark that, e.g. an upper (lower) semi-continuous function can be obtained from a continuous function by increasing (decreasing) the functional value $f(a)$ alone, without alteration of the neighboring values $f(x)$. If $f(a) = +1$ and if $f(x) = 0$ everywhere else, then $f(x)$ is upper semi-continuous at the point a. If $f(x)$ is upper semi-continuous, then $-f(x)$ is lower semi-continuous.

If $f(x)$ is lower semi-continuous at the point a, then a is an interior point of every set $[f > y]$ to which it belongs, and vice versa. For if $f(a) > y$ and if we choose $0 < \varepsilon < f(a) - y$, then in a certain neighborhood U_a, $f(x)$ is still $> f(a) - \varepsilon > y$, so that a is an interior point of $[f > y]$. If, conversely, this condition is met, then for every $\varepsilon > 0$ a is an interior point of $[f > f(a) - \varepsilon]$, and there exists a neighborhood U_a in which $f(x) > f(a) - \varepsilon$; and $f(x)$ is lower semi-continuous at a. From this it follows that, for $f(x)$ to be lower semi-continuous at every point, it is necessary and sufficient that every set $[f > y]$ be open. And for an upper semi-continuous function f, that is, for a lower semi-continuous function $-f$, it is necessary and sufficient that $[f < y]$ be open and $[f \geq y]$ closed. The reason for the nomenclature "lower semi-continuous" and "upper semi-continuous" for functions of the classes $(G, *)$, $(*, F)$ is thus made clear. These functions were introduced by Baire, who was also the first to prove that every lower semi-continuous function is the limit of an increasing sequence of continuous functions or, in our notation, that every function of the class $(G, *)$ is a g. The converse, that every g is of the class $(G, *)$, moreover, holds here in a much wider sense than is ordinarily the case; the least upper bound not only of a countable number but of an arbitrary number of continuous or lower semi-continuous functions

$$g = \sup g_\alpha$$

remains lower semi-continuous, since

$$[g > y] = \mathfrak{S}\,[g_\alpha > y],$$

as a sum of open sets, is itself open (of course, the $g_\alpha(x)$ are supposed to be bounded from above at every point x). Consider, for example, the lower semi-continuous minorants $g \leq \varphi$ belonging to an arbitrary function φ; if such functions exist at all, then there is one among them that is the greatest, namely the least upper bound of all of them. Corresponding results hold for upper semi-continuous functions.

One might perhaps wish to see Theorem I supplemented by a method of construction in accordance with which the functions of the classes $(G, *)$, $(*, F)$, and (F_σ, G_δ) can actually be represented as the limit functions of increasing, decreasing, and convergent sequences of continuous functions. Such methods of construction, valid for the general case of an arbitrary initial function f, are of course, already implicit in the proofs of the converses of the theorems on classes (§41, 3); there remains, however, the possibility that they here can be simplified. For the Baire Theorem, that a lower semi-continuous function φ is a limit of an increasing sequence of continuous functions, it is in fact possible to give an exceedingly simple proof, at least when φ is bounded from below — say $\varphi \geq 0$. Let ε be a positive number and

$$f_\varepsilon(x) = \inf_z [\varphi(z) + \varepsilon \cdot xz],$$

where the greatest lower bound is taken for all points z of the space; obviously $0 \leq f_\varepsilon$ ($f_\varepsilon(x) \leq \varphi(x)$ is obtained by choosing $z = x$). For two points x and y of the space,

$$\varphi(z) + \varepsilon \cdot zy \leq \varphi(z) + \varepsilon \cdot zx + \varepsilon \cdot xy,$$

and thus, taking the greatest lower bound with respect to z,

$$f_\varepsilon(y) \leq f_\varepsilon(x) + \varepsilon \cdot xy$$

or, interchanging the two points, $|f_\varepsilon(x) - f_\varepsilon(y)| \leq \varepsilon \cdot xy$; the function f_ε is therefore continuous. If we now let ε range over all the natural numbers, the functions

$$f_n(x) = \inf_z [\varphi(z) + n \cdot xz]$$

form an increasing sequence for which $f_n \leq \varphi$, and thus $g = \lim f_n \leq \varphi$; we show that also $g \geq \varphi$, so that $g = \varphi$. For preassigned $\varepsilon > 0$, let us choose the point z_n so that

$$f_n(x) > \varphi(z_n) + n \cdot xz_n - \varepsilon$$

holds; then (since $\varphi \geq f_n$, $\varphi \geq 0$), $\varphi(x) > n \cdot xz_n - \varepsilon$, so that $xz_n \to 0$; but since φ is lower semi-continuous, we finally have

$$\varphi(z_n) > \varphi(x) - \varepsilon, \quad f_n(x) > \varphi(x) - 2\varepsilon,$$

so that $g(x) \geq \varphi(x)$.

2. Examples of functions of the first class.

Immediately after the lower semi-continuous functions g and the upper semi-continuous functions h, the functions

$$d = g + h = g - h' = h - h''$$

— which we discussed in some generality in §41, 5 — are worthy of our attention; they arise from the summation of absolutely convergent series of continuous functions. Every function of the first class can be approximated with arbitrary closeness by functions d and, in particular, by step functions d.

Every function with at most a countable number of points of discontinuity is of the first class.[4] Let C be the set of the points of continuity of f and D the set of the points of discontinuity of f; let $M = [f > y]$. If now $x \in MC$, then the inequality $f > y$ still holds in a certain neighborhood of x, and thus $MC \subseteq M$; and $MD \supseteq M_i$. If D is at most countable, then so is M_i, and $M = MC + M_i$ is composed of an open set and an at most countable set and is thus an F_σ. Since the same is true for $[f < y]$, f is of the class (F_σ, G_δ).

Worthy of special mention among the functions of the first class of a real variable x are, on the one hand the derivatives $g'(x) = \lim n\,[g(x+1/n) - g(x)]$ of differentiable functions and, on the other hand, the functions $f(x)$ for which the left-hand and right-hand limiting values $f(x+0)$ and $f(x-0)$ exist at all points (and so, in particular, the monotone functions and their sums and differences: the functions of bounded variation). As a little reflection will show, these functions with one-sided limiting values have at most a countable number of points of discontinuity and are thus indeed functions of the first class; moreover, they are even functions d. For since $f(x+0)$, as a function of x, has the same one-sided limiting values as $f(x)$, and $\max\,[f(x), g(x)]$ has the right-hand limiting value $\max\,[f(x+0), g(x+0)]$, we have for the function

$$\varphi(x) = \max\,[f(x), f(x+0), f(x-0)]$$

that $\varphi(x+0) = f(x+0) \leq \varphi(x)$, and $\varphi(x)$ is upper semi-continuous; similarly,

$$\psi(x) = \min\,[f(x), f(x+0), f(x-0)]$$

[4] We now drop the requirement that the letter f shall always stand for a continuous function.

is lower semi-continuous; $\varphi + \psi$ is a function d. If in this $f(x)$ is replaced by

$$f(x+0), \quad f(x+0) - f(x), \quad f(x-0) - f(x),$$

then we obtain

$$f(x+0) + f(x-0), \quad f(x+0) - f(x), \quad f(x-0) - f(x)$$

as functions d and then, by means of linear combinations, $f(x)$ itself as well.

3. Point of continuity.

If f is, for the time being, any real function defined in the space A, let C be the set of its points of continuity and $D = A - C$ the sets of its points of discontinuity. C is a G_δ; D is an F_σ (with respect to the space A). For, continuity can also be characterized in this way: $f(x)$ is continuous at the point a if and only if for every $\varepsilon > 0$ there exists a neighborhood U_a for whose points x and y

$$|f(x) - f(y)| < \varepsilon$$

always holds. For fixed ε let $C(\varepsilon)$ be the set of the points a having such a neighborhood U_a; every point $b \in U_a$ then also has such a neighborhood U_b ($\subseteq U_a$), so that $U_a \subseteq C(\varepsilon)$, and $C(\varepsilon)$ is open. Then C is the intersection of all the $C(\varepsilon)$ for $\varepsilon > 0$, or

$$C = C(1) \cdot C(1/2) \cdot C(1/3) \cdots,$$

and C is a G_δ (W. H. Young).

One extreme is represented by the (everywhere) continuous functions, with $C = A$ and $D = 0$; the other extreme, by the everywhere discontinuous functions, with $C = 0$ and $D = A$. An example of the latter is given, for example, by the so-called Dirichlet function of the real variable x, which is 1 for rational x and 0 for irrational x. Closest to the continuous functions are the functions for which C is dense in A; following Hankel, we call them *pointwise discontinuous* (or better, at most pointwise discontinuous, thus including the continuous functions as well).

As an F_σ with a dense complement, D is — by the result on p. 166 — of the first category in A (an A_1); and if thus A itself is of the second category in itself (an A_2) — say, a Young set — then C is an A_2. It may also happen here that D is dense in A, but that the two sets are of different categories and cannot possibly exchange roles. Thus it can happen — for example, if A is the set of real numbers — that C is the set of irrationals and D the set of rationals (if $D = \{r_1, r_2, \ldots\}$ and $\varepsilon_1 + \varepsilon_2 + \cdots$ is a convergent series of positive numbers, then

$$f(x) = \sum_{r_n < x} \varepsilon_n$$

is a monotone function of this kind), but not conversely.

Next, let $f_n \to f$ be a sequence of functions convergent everywhere (in A); we ask, under what circumstances can we infer the continuity of the limit function from that of the f_n. We say that a sequence converges *uniformly at the point a* if for every $\varepsilon > 0$ there exists a natural number m and a neighborhood U_a such that

$$|f_n(x) - f(x)| \leq \varepsilon \quad (x \in U_a). \tag{1}$$

Then we have:

IV. If $f_n \to f$ and every f_n is continuous at the point a, then f is continuous at the point a if and only if a is a point of uniform convergence.

Proof. If a is a point of uniform convergence, we satisfy (1) by a suitable choice of m and U_a, where because of the continuity of f_m at a, it is possible to choose U_a so small that

$$|f_m(x) - f_m(a)| \leq \varepsilon \quad (x \in U_a).$$

Since, by (1), we have in particular that $|f_m(a) - f(a)| \leq \varepsilon$, it follows from these three inequalities that

$$|f(x) - f(a)| \leq 3\varepsilon \quad (x \in U_a),$$

that is, that f is continuous at a.

If, conversely, f is continuous at a, we first determine an m for which

$$|f_m(a) - f(a)| \leq \varepsilon,$$

next, using the continuity of both functions, a U_a for which

$$|f_m(x) - f_m(a)| \leq \varepsilon, \quad |f(x) - f(a)| \leq \varepsilon \quad (x \in U_a);$$

from these three inequalities it now follows that

$$|f_n(x) - f(x)| \leq 3\varepsilon \quad (x \in U_a),$$

which is precisely uniform convergence at a.

The set K of the points of uniform convergence may be formed in this way: Let $G(\varepsilon)$ be the set of the points a for which (1) may be satisfied by a suitable choice of m and U_a. We see once again that $G(\varepsilon)$ is open ($U_a \subseteq G(\varepsilon)$) and K is the intersection of all the $G(\varepsilon)$ for $\varepsilon > 0$, or

$$K = G(1) \cdot G(1/2) \cdot G(1/3) \cdots ;$$

the set of the points of uniform convergence, again, is a G_δ.

4. The Theorem of Baire.

If the functions f_n are everywhere continuous and $f = \lim f_n$ is a function of the first class, then the points of continuity of f coincide with the points of uniform convergence: $C = K$.

Let us retain this as our assumption. Then for some fixed $\varepsilon > 0$, let $F(\varepsilon)$ or, more concisely, F_m, be the set of the points x that satisfy all the inequalities

$$|f_n(x) - f_m(x)| \leq \varepsilon \quad \text{for } n > m. \tag{2}$$

For each particular n, this inequality defines a closed set; F_m is the intersection of these closed sets for $n = m+1, m+2, \ldots$, and therefore is itself closed. Because of the convergence of the sequence, every point x satisfies the inequalities (2) for sufficiently large m and thus

$$A = F_1 + F_2 + \cdots . \tag{3}$$

On the other hand, if a is an interior point of F_m, so that the inequalities (2) and the inequality $|f(x) - f_m(x)| \leq \varepsilon$ that follows from it for $n \to \infty$ hold not only for a but also for a certain neighborhood ($x \in U_a$), then a belongs to the set $G(\varepsilon)$ defined earlier, and so

$$G(\varepsilon) \supseteq F_{1i} + F_{2i} + \cdots . \tag{4}$$

From this, it follows that: If $G(\varepsilon) = 0$, then every $F_m = 0$, and F_m, as a closed set without interior points, is nowhere dense in A and, by (3), A is of the first category in itself. Therefore, conversely:

If A is an A_2, then every $G(\varepsilon)$ is non-empty.

Furthermore: If A is a G_δ-set (p. 165), so that every non-empty set G open in A is of the second category in itself, then we consider the partial functions $f_n(x|G)$ and $f(x|G)$ (the restrictions to G); the $G(\varepsilon)$ mentioned above then has to be replaced by $G \cdot G(\varepsilon)$. We thus find that for non-empty G, $G \cdot G(\varepsilon)$ is also non-empty, that is, $G(\varepsilon)$ is dense in A. The closed complement $F(\varepsilon)$ of this set is thus nowhere dense in A, the set

$$D = F(1) + F(1/2) + F(1/3) + \cdots$$

of the points of discontinuity of f is an A_1, and the set C of the points of continuity is dense in A and is an A_2. We have thus discovered the theorem:[5]

V. If the space A is a G_δ Set (that is, every non-empty open set G is of the second category in itself or in A), then every function of the first class is at most pointwise discontinuous, and the set of its points of continuity is dense in A.

This thus applies in particular if A is a Young set and, specifically, a complete space. Thus, if A is the set of real numbers, a function of the first class can only be pointwise discontinuous; the Dirichlet function $f(x)$ ($= 1$ for x rational, $= 0$ for x irrational) is not of the first class, although it is of the second class (§43); that is, it is a limit of functions of the first class, as the formula

$$f(x) = \lim_m \lim_n (\cos m!\, \pi x)^{2n}$$

shows.

Theorem V fails when the assumption about A does not hold. In the countable set $A = \{a_1, a_2, \ldots\}$ of real numbers, every function is of the first class since, for any arbitrarily prescribed $b_n = f(a_n)$ it is always possible to determine a continuous function $f_n(x)$ — a polynomial, for example — such that $f_n(a_1) = b_1, \ldots, f_n(a_n) = b_n$ and $f_n \to f$. If A is the set of rational numbers and, say, $f(x) = 1$ or 0 depending on whether x is a dyadic rational number ($x = m/2^n$, m an integer) or not, then f is everywhere discontinuous.

A flat converse to Theorem V, in the sense that pointwise discontinuous functions are of the first class as well, is out of the question, if only because of considerations of cardinality. If, for instance, A is the set of real numbers, then there are only $\aleph$ continuous functions

[5] Compare §45, 3.

and $\aleph^{\aleph_0} = \aleph$ functions of the first class; but there are $\aleph^{\aleph} = 2^{\aleph}$ pointwise discontinuous functions. For if D is a perfect, nowhere-dense set (of cardinality $\aleph$) and C its open complement dense in A, then every function which vanishes in C and does not vanish in D (and which is thus continuous at $x \in C$ and discontinuous at $x \in D$) is pointwise discontinuous.

But let us apply V to the partial functions $f(x|B)$ contained in $f(x) = f(x|A)$ which are, after all, themselves functions of the first class in their space B; if B is a G_δ-set, then the points of continuity of the partial function (which need not be points of continuity of the total function) are dense in B. This holds, in particular, if A is an F_σ-set (p. 165) and B is closed in A and is thus also an F_σ-set. Therefore we can deduce from V:

VI. If the space A is an F_σ-set (that is, if every non-empty closed set F is of the second category in itself) and if f is a function of the first class, then every non-empty closed set F (closed in A) contains at least one point of continuity of the partial function $f(x|F)$.

We next have the following counterpart to VI:

VII. If the space A is separable, if f is a function defined in A, and if every non-empty set F closed in A contains at least one point of continuity of the partial function $f(x|F)$, then f is of the first class.

We have to prove that f is of the class (F_σ, G_δ), that is, that all the sets $B = [f > y]$ and $C = [f < z]$ are sets F_σ. Let us consider two such sets for which $y < z$, so that $A = B + C$. Now suppose F non-empty and closed and a a point of continuity of $f(x|F)$. If $a \in B$ and $f(a) > y$, then there exists a neighborhood U_a for which $f > y$ remains true in $F \cdot U_a$, that is, $F \cdot U_a \subseteq B$; in the same way, there exists for $a \in C$ a neighborhood for which $F \cdot U_a \subseteq C$, and at least one of these cases holds (and both when $a \in BC$). If we set $F - F \cdot U_a = F_1$, then it follows that:

Every non-empty closed set F contains a smaller closed set $F_1 < F$ such that the difference $F - F_1$ is contained either in B or in C. If, in addition, F_1 is non-empty, then we find in the same way a set F_2 properly contained in F_1, and, continuing the procedure transfinitely, we define the closed sets F_ξ for all the ordinal numbers $\xi < \Omega$ in the now familiar manner, namely (except for $F_0 = F$): If F_ξ is non-empty, we let $F_{\xi+1}$ be a set properly contained in F_ξ such that the difference $D_\xi = F_\xi - F_{\xi+1}$ is contained either in B or in C (and possibly in both); for $F_\xi = 0$, we let $F_{\xi+1} = 0$); for a limit ordinal

λ, we let $F_\lambda = \mathfrak{D}F_\alpha$. Since the space is supposed to be separable, there must exist (§30, IV) a smallest set $F_\alpha = F_{\alpha+1} = F_{\alpha+2} = \cdots = 0$; because of the method of construction, this must be the null set, and with this, $F = \mathfrak{S}D_\xi$ is split into at most countably many summands, which are contained in B or in C and which, furthermore, as differences of closed sets, are special sets F_σ. If we recombine the summands belonging to B and to C, we obtain a decomposition $F = Y + Z$ into two disjoint sets F_σ, for which $Y \subseteq B$ and $Z \subseteq C$; in particular, the whole space admits such a decomposition $A = Y + Z$.

Finally, we hold y fixed while allowing z to describe a sequence $z_1 > z_2 > \cdots$ with $z_n \to y$. Corresponding to $B = [f > y]$ and $C_n = [f < z_n]$, we determine a decomposition of the space $A = Y_n + Z_n$ into two disjoint sets F_σ for which $Y_n \subseteq B$, $Z_n \subseteq C_n$. Let

$$Y = \mathfrak{S}Y_n, \quad Z = \mathfrak{D}Z_n, \quad C = \mathfrak{D}C_n;$$

then we find $C = [f \leq y] = A - B$. From $A \supseteq B + C \supseteq Y + Z$ and $Y \subseteq B$, $Z \subseteq C$ it follows that $Y = B$ and $Z = C$; that is, the set $B = [f > y]$ is an F_σ. The same, of course, holds for the sets $[f < z]$, and we have thus proved VII.

From VI and VII, we obtain:

VIII. (Baire's Theorem). If the space A is a separable F_σ-set, then the function f defined in A is of the first class if and only if every non-empty set F closed in A contains at least one point of continuity of the partial function $f(x|F)$.

The word closed can be replaced here without further ado by perfect, since an isolated point of F is *eo ipso* a point of continuity of $f(x|F)$.

5. Application of Baire's Theorem.

Let A be the set of real numbers; $f(x)$ is thus a real function of the real variable x; in the plane E with the rectangular coordinates x, y we consider the set ("curve") C defined by $y = f(x)$. We denote the points of the plane by $z = (x, y)$ and those of C by $\zeta = (x, f(x)) = \varphi(x)$. We then ask when C is connected. For this to be the case, the following condition is surely necessary:

(α) Every point of C is a point of accumulation, both from the left and from the right; that is, for every x there exists a sequence $x_n < x$ for which $\varphi(x_n) \to \varphi(x)$ and another such sequence $x_n > x$.

For if we split the set C by means of $x < x_0$ and $x \geq x_0$ into C_1 and C_2, where C_2 is closed in C, then in order that no partition occur, C_1 must have a point of accumulation in C_2, which obviously can be none other than $\varphi(x_0)$. That condition (α) does not suffice by itself is shown by a function such as the Dirichlet function, which takes on only the values 0 and 1, each in a set dense in A.

If, however, $f(x)$ is of the first class, then (α) is also sufficient.

Let us suppose that $C = C_1 + C_2$ can be partitioned into two relatively closed sets; projected onto the x-axis, this gives

$$A = A_1 + A_2 = G_1 + G_2 + F,$$

where we have set $G_1 = A_{1i}$, $G_2 = A_{2i}$, $F = A_{1b} + A_{2b}$.[6] According to VI, the partial function $f(x|F)$ has a point of continuity x, say $x \in A_1$. Then x cannot be a limit of points $x_n \in A_2F$, since otherwise $\varphi(x) \in C_1$ would be a limit of points $\varphi(x_n) \in C_2$; thus there must exist a neighborhood U of x disjoint from A_2F. But this U cannot then be also disjoint from G_2, since x is, after all, an x-point of $A_1 = A_1F + G_1$, and so $U \cdot G_2$ is non-empty. But condition (α) prohibits this. For if the open interval (a, b) is a component of the open set $U \cdot G_2 \subseteq U$, then at least one of the end points, say a, lies in U; now since the point $\varphi(a)$, because of (α), is a point of accumulation of C from the right and thus of C_2, we would have $a \in A_2$ and $a \in A_1 - G_1 = A_1F$, and U would actually not be disjoint from A_2F. Therefore C cannot be partitioned.

We can form here still another example (other than that of §31, 2) of a connected discontinuous set. Let us ask, independent of the foregoing, When does C contain a continuum consisting of more than one point (relative to the plane E and thus a connected set perfect in E)? The answer is: If and only if the points of continuity fill up a whole interval. One part of this statement is trivial: If $f(x)$ is continuous for $a \leq x \leq b$, then C contains a simple curve. The other we formulate thus: If K is a continuum of more than one point contained in C, and if $\varphi(a)$ and $\varphi(b)$ are two points of K $(a < b)$, then $f(x)$ is continuous in $[a, b]$ (continuous from the right at a, from the left at b, from both sides in between).

For suppose we let ${}_aK_a$, ${}_aK_b$ and ${}_bK_b$ be the subsets of K defined by $x \leq a$, $a \leq x \leq b$, and $x = b$. They are closed (in K, and therefore in E) and, by §29, III, they are continua because ${}_aK_a$ and ${}_aK_b$ have

[6] If the relative terms are not qualified, they refer to the space A.

the sum ${}_aK_b$ and a one-point intersection and similarly, ${}_aK_b$ and ${}_bK_b$ have the sum K_b and a one-point intersection. The projection of ${}_aK_b$ on the y-axis is connected, and $f(x)$ therefore takes on in (a, b) every value between $f(a)$ and $f(b)$, if these are distinct; the same is true for every subinterval. But then $f(x)$ must be bounded in $[a, b]$. For if, as we may assume, we had $f(x_n) \to +\infty$, $x_1 < x_2 < \cdots$ and $f(x_1) > y > f(a)$, then at some point ξ_n between x_1 and x_n, f would have to take on the value y; this would give $f(\xi_n) = y$ and $\xi_n \to x_1$, and the point (x_1, y) distinct from $\varphi(x_1)$ would have to belong to the closed set K, which is not the case. Therefore ${}_aK_b$ is a bounded (compact) continuum. We now consider its projection on the x-axis; this, again, is connected, and therefore identical with the entire interval $[a, b]$; it is the one-to-one continuous image of ${}_aK_b$ and so, conversely (§35, III), ${}_aK_b$ is also the one-to-one continuous image of $[a, b]$. This means precisely that $f(x)$ is continuous in $[a, b]$, in the sense defined above.

From all this we can see that C is discontinuous (in E) if and only if the points of discontinuity of $f(x)$ are dense in A. And C is simultaneously connected and discontinuous if $f(x)$ is a function of the first class, satisfies the condition (α), and has its points of discontinuity dense in A. Such a function is easy to construct. The function defined by

$$s(x) = \sin \pi/x \quad (x \neq 0), \quad s(0) = 0$$

is discontinuous only at the point $x = 0$, but even there it satisfies the condition (α), since $s(1/n) = s(-1/n) = 0$ for $n = 1, 2, 3, \ldots$. If $\{r_1, r_2, r_3, \ldots\}$ is the set of rational numbers and $c_1 + c_2 + c_3 + \cdots$ is a convergent series of positive numbers, then, as follows easily from the uniform convergence, the function

$$f(x) = \sum c_n s(x - r_n)$$

is discontinuous only at the rational points, although even there it satisfies the condition (α) and is a function of the first class (since it has only countably many points of discontinuity).

9.3 Baire Functions

1. Baire systems.

We consider real functions all of which are defined in the same space A (which, for the time being, can be taken as a pure set). A system of such functions f is called a *Baire system* if the limit of every convergent sequence of functions f belonging to the system itself belongs to the system. For a given system $\mathfrak{S}$ of functions f there exists a Baire system — for instance, that of all the functions defined in A — and a smallest such system, the intersection of all the Baire systems containing $\mathfrak{S}$; the functions of this system are called the *Baire functions generated by* f.

Let us, for the time being, represent them in the following form, corresponding to the representation of Borel sets of pp. 97–98. Letting a, b, c, ... range over the natural numbers, we consider all the functions of the form

$$g = \lim_a g_a, \quad g_a = \lim_b g_{ab}, \quad g_{ab} = \lim_c g_{abc}, \ldots$$

with the requirement that for every sequence of natural numbers a, b, c, ... in the sequence of the functions g_a, g_{ab}, g_{abc}, ... there are ultimately to occur only functions f. These functions g are identical with the Baire functions b generated by the f. For, on the one hand, the g obviously form a Baire system over $\mathfrak{S}$, so that every b is a g. On the other hand, every g is also a b; if some g were not a b, there would also have to be at least one g_a which is not a b, and then at least one g_{ab}, and then a g_{abc}, etc., each of which is not a b, in contradiction with the fact that the sequence of these functions is ultimately to contain only terms f.

A step-by-step construction, again, is preferable to this representation. Among the Baire functions are the functions themselves, next the functions $f^1 = \lim f_n$, then the functions $f^2 = \lim f_n^1$, and so forth, with finite or infinite repetition, which need not, however, extend beyond the ordinal numbers $\xi < \Omega$. In the inductive definition of the f^ξ it is advisable, in order to arrive at as polished a final result as possible, to proceed as follows (a complete system of functions has been defined on pp. 267–8):

The f^0 are the functions of a complete initial system $\mathfrak{S}^0$;

The $f^{\xi+1}$ are the limits of convergent sequences of functions f^ξ;

The f^λ (λ a limit ordinal) are the functions of the smallest complete system to which all the functions f^ξ ($\xi < \lambda$) belong.

Accordingly, the functions f^0, f^1, f^2, ..., with finite indices, for example, are not followed immediately, as functions f^ω by the limits of convergent sequences of functions f^n, but rather by the functions of the smallest complete system containing the f^n. This departure from the otherwise usual notation — which we assumed provisionally even in §30, 2 — is justified, as we shall see, by better agreement between the function indices and the set indices. For the f^ξ now constitute a complete system for every index (and not only for the $\xi + 1$, §41, IV) and this, as we know, entails simplification.

Every f^ξ is a special f^η for $\eta > \xi$. The f^ξ of all indices constitute the smallest Baire system over $\mathfrak{S}^0$; for fixed ξ they form a Baire class of functions which is also called, for brevity, the *class* ξ. The classes increase with increasing indices, equality not excluded; but if $\xi = \zeta + 1$, then $\mathfrak{S}^\xi$ is already the whole Baire system, and $\mathfrak{S}^\eta = \mathfrak{S}^\xi$ for $\eta > \xi$. A function belongs exactly to the class ξ if it belongs to this class but to no earlier one.

We make the further definitions:

$$g^\xi = \text{limit of an increasing sequence of functions } f^\xi,$$
$$h^\xi = \text{limit of a decreasing sequence of functions } f^\xi;$$

in addition, for a limit ordinal λ,

$$g_\lambda = \text{limit of an increasing sequence of functions } f^\xi \ (\xi < \lambda),$$
$$h_\lambda = \text{limit of a decreasing sequence of functions } f^\xi \ (\xi < \lambda).$$

The inverse image sets are to be denoted by

$$\mathfrak{M}^\xi = [f^\xi > y], \quad \mathfrak{N}^\xi = [f^\xi \geq y]; \tag{1}$$

we shall see in a moment that no others are needed.

If we identify the f of §41 with the f^0 which, after all, always form a complete system, then Theorem XII of that section yields:

I. The functions f^ξ, g^ξ, and h^ξ, are identical with the functions of the classes $(\mathfrak{M}^\xi, \mathfrak{N}^\xi)$, $(\mathfrak{M}^\xi, *)$ and $(*, \mathfrak{N}^\xi)$.

In addition, the functions $f^{\xi+1}$ (the former functions f^*) are identical with those of the class $(\mathfrak{N}^\xi_\delta, \mathfrak{M}^\xi_\sigma)$; hence this gives

$$\mathfrak{M}^{\xi+1} = \mathfrak{N}^\xi_\delta, \quad \mathfrak{N}^{\xi+1} = \mathfrak{M}^\xi_\sigma. \tag{2}$$

Furthermore, let λ be a limit ordinal; we identify the earlier f with the functions f^ξ of all the classes $\xi < \lambda$. The $\mathfrak{M}$ are now the $\mathfrak{M}^\xi$, the $\mathfrak{N}$ are the $\mathfrak{N}^\xi$; the $\mathfrak{P}$ and $\mathfrak{Q}$, however, are the $\mathfrak{M}^\lambda$ and $\mathfrak{N}^\lambda$, since the functions f^ξ have now taken on the role of the earlier v. We conclude thus from $\mathfrak{P} = \mathfrak{M}_\sigma$ and $\mathfrak{Q} = \mathfrak{N}_\delta$ that:

$$\begin{aligned} \mathfrak{M}^\lambda &= \text{sum of a sequence of sets } \mathfrak{M}^\xi \ (\xi < \lambda) \\ \mathfrak{N}^\lambda &= \text{intersection of a sequence of sets } \mathfrak{N}^\xi \ (\lambda \text{ a limit ordinal}). \end{aligned} \tag{3}$$

In this, as follows from the remarks made in connection with Theorem XII of §41, 3, the limits of convergent sequences of function f^ξ are identical with the $f^{\xi+1}$; the g_λ are those g^λ that have a minorant f^ξ; the h_λ are those h^λ that have a majorant f^ξ; the k_λ (which are both g_λ and h_λ) are those f^ξ that remain between two functions f^ξ. The f^λ are the quotients of two k_λ. On the other hand, the k^ξ (which are both g^ξ and h^ξ) are always identical with the f^ξ.

Formulas (2) and (3) determine the $\mathfrak{M}^\xi$ and $\mathfrak{N}^\xi$ inductively, starting from the sets $\mathfrak{M}^0 = [f^0 > y]$ and $\mathfrak{N}^0 = [f^0 \geq y]$ of the initial system. If we leave off the superscript 0, we thus have

$$\begin{aligned} \mathfrak{M}_0 &= \mathfrak{M}, \quad \mathfrak{N}_0 = \mathfrak{N}, \\ \mathfrak{M}_1 &= \mathfrak{N}_\delta = \mathfrak{N}, \quad \mathfrak{N}_1 = \mathfrak{M}_\sigma = \mathfrak{M}_1, \\ \mathfrak{M}_2 &= \mathfrak{M}_{1\sigma} = \mathfrak{M}_2, \quad \mathfrak{N}_2 = \mathfrak{N}_{1\delta} = \mathfrak{N}_2, \\ \mathfrak{M}_3 &= \mathfrak{N}_{2\delta} = \mathfrak{N}_3, \quad \mathfrak{N}_3 = \mathfrak{M}_{2\sigma} = \mathfrak{M}_3, \\ \mathfrak{M}_\lambda &= \mathfrak{S}\mathfrak{M}_{\xi_n} = \mathfrak{S}\mathfrak{N}_{\xi_n} \quad (\xi_n < \lambda) \\ \mathfrak{M}_{\xi+1} &= \mathfrak{N}_{\xi\delta}, \quad \mathfrak{N}_{\xi+1} = \mathfrak{M}_{\xi\sigma}. \end{aligned} \tag{4}$$

In addition, we make note of the Interposition and Extension Theorems:

II. If $g^\xi \leq h^\xi$, then there exists a function f^ξ for which $g^\xi \leq f^\xi \leq h^\xi$.

For limit ordinals λ we have, in addition: If $g_\lambda \leq h_\lambda$, then there exists a function k_λ for which $g_\lambda \leq k_\lambda \leq h_\lambda$.

III. If N is a set $\mathfrak{N}^\xi$, then a function of the class $(\mathfrak{M}^\xi, \mathfrak{N}^\xi)$ defined in N can be extended to a function f^ξ defined in the whole space.

2. The Baire functions of the space.

If, in particular, the functions continuous in the space A are taken as the initial functions $f^0 = f$, then the Baire functions generated by them are called the *Baire functions* (or the *analytically representable functions*) of the space A. The functions $\mathfrak{M}$ and $\mathfrak{N}$ are here the G and F, and the $\mathfrak{M}^\xi$ and $\mathfrak{N}^\xi$ become the *Borel sets* of the space A; by the definition of the G_δ and F_σ (§32, (α) and (β)), we find from (4) that

$$\begin{aligned} &\mathfrak{M}_0 = G, \quad \mathfrak{N}_0 = F, \\ &\mathfrak{M}_1 = F_\delta = F^1, \quad \mathfrak{N}_1 = G_\sigma = G^1, \\ &\mathfrak{M}_2 = G_\sigma^1 = G^2, \quad \mathfrak{N}_2 = F_\delta^1 = F^2, \\ &\mathfrak{M}_3 = F_\delta^2 = F^3, \quad \mathfrak{N}_3 = G_\sigma^2 = G^3, \\ &\mathfrak{M}^* = G_\sigma, \quad \mathfrak{N}^* = F_\delta, \\ &\mathfrak{M}^{\xi+1} = \mathfrak{N}^\xi_\delta, \quad \mathfrak{N}^{\xi+1} = \mathfrak{M}^\xi_\sigma \end{aligned} \tag{5}$$

and in general,

$$\begin{aligned} &\mathfrak{M}^\xi = G_\xi, \quad \mathfrak{N}^\xi = F^\xi \quad (\xi \text{ even}) \\ &\mathfrak{M}^\xi = F^\xi, \quad \mathfrak{N}^\xi = G_\xi \quad (\xi \text{ odd}). \end{aligned} \tag{6}$$

It should be remarked here that for a limit ordinal λ it is quite indifferent whether $\mathfrak{M}^\lambda$ is considered as a sum of sets $\mathfrak{M}^\xi$ or $\mathfrak{N}^\xi$ or G_ξ or F_ξ ($\xi < \lambda$) since, indeed, every $\mathfrak{M}^\xi$ is a special $\mathfrak{N}^{\xi+1}$ etc.

IV. The Baire functions of the space A are identical with the functions of the class $(\mathfrak{B}, \mathfrak{B})$, where $\mathfrak{B}$ ranges over the Borel sets of the space A.

For every Baire function is of the class $(\mathfrak{B}, \mathfrak{B})$. If, conversely, f is of the class $(\mathfrak{B}, \mathfrak{B})$, then we consider for rational r the countably many Borel sets $[f > r]$ and $[f \geq r]$ and choose the ordinal number ξ so large that all these sets are simultaneously sets G_ξ and sets F_ξ and hence are simultaneously sets $\mathfrak{M}^\xi$ and sets $\mathfrak{N}^\xi$. Then for every y

$$[f > y] = \mathfrak{S}_{r>y}\,[f > r]$$

is an $\mathfrak{M}^\xi_\sigma = \mathfrak{M}^\xi$ and

$$[f \geq y] = \mathfrak{D}_{r<y}\,[f \geq r]$$

is an $\mathfrak{N}^\xi_\delta = \mathfrak{N}^\xi$, so that f is of the class $(\mathfrak{M}^\xi, \mathfrak{N}^\xi)$, or is an f^ξ.

From Theorem I of §33, there follows directly the *Existence Theorem* for Baire functions:

V. In a complete space whose dense-in-itself kernel does not vanish, there exist for every $\xi < \Omega$ Baire functions f^ξ that belong exactly to the class ξ (and to no earlier class).

In a separable space there exists $\aleph$ continuous functions (at least $\aleph$, because the constant functions — which are of this cardinality — are included among them, and at most $\aleph$, because a continuous function is already determined by its values at a countable dense subset). There then exist also only $\aleph$ functions f^ξ for every ξ, as follows by induction, and $\aleph$ (more specifically, at most $\aleph \cdot \aleph = \aleph$) Baire functions altogether. Thus, in case the space has cardinality $\aleph$ (for example, if it is complete and has a non-empty dense-in-itself kernel), the Baire functions constitute a vanishingly small subsystem of the system of all ($\aleph^\aleph = 2^\aleph$) functions.

That the Baire functions, even apart from considerations of cardinality, are still very close to the continuous functions and, in fact, if sets of the first category are neglected, are themselves continuous, is shown by the following theorem:[7]

VI. If the space A is a G_δ-set, then for every Baire function f there exists a set C — G_δ dense in A — such that the partial function $f(x|C)$ is continuous.

This is true for the continuous functions themselves ($C = A$) and can be extended to all the Baire functions by going from the functions f_n of a convergent sequence to their limit f. Let $f_n(x|C_n)$ be continuous and let C_n be a G_δ dense in A, so that $D_n = A - C_n$ is an F_σ of the first category in A; then $D_0 = D_1 + D_2 + \cdots$ is an F_σ and an A_1; $C_0 = A - D_0 = C_1 \cdot C_2 \cdots$ is a G_δ dense in A and, again, is a G_δ set (p. 165). Now all the functions $f_n(x|C_0)$ are continuous; their limit $f(x|C_0)$ is thus, by V of §42, at most pointwise discontinuous — that is, the set C of its points of continuity is dense in C_0 and is a G_δ in C_0, so that it is dense in A and is a G_δ in A; $f(x|C)$ is continuous.

For example, the Dirichlet function f (p. 287) is constant and equal to 0 in the set C of the irrational numbers and is thus continuous; that is, the irrational numbers are points of continuity of the partial function $f(x|C)$ — but not of the total function.

VI can be expressed concisely as follows: Every Baire function is continuous "up to" sets of the first category; we mean by this that the space allows of a decomposition $A = C + D$ in which D is an A_1 and $f(x|C)$ is continuous. It is then possible *eo ipso* to assume C a G_δ and D an F_σ, by replacing the nowhere-dense sets whose sum is D by their closed hulls.

[7] Compare this with §45, 3.

Theorem VI has a certain resemblance to Theorem V of §42 about functions of the first class, and also admits of an analogue to Theorem VI of §42:

If the space A is an F_σ-set and f is a Baire function, then for every non-empty set F closed in A the partial function $f(x|F)$ is also continuous up to sets of the first category.

That is, there exists a decomposition $F = C + D$ for which D is an F_σ and $f(x|C)$ is continuous. This follows from the fact that F is itself an F_σ-set and therefore (p. 165) a G_δ-set and $f(x|F)$ a Baire function of the space F.

One might now conjecture that, corresponding to Theorem VII of §42, in a separable space the above continuity property might perhaps also suffice; but this conjecture is false. To begin with, let us note that the continuity property is not weakened if, instead of for all closed sets, it is required to hold merely for all perfect sets — for it then holds for all closed sets. For let

$$F = F_1 + F_2 = F_1 + (F_2 - F_1) + F_3$$

be closed, and let $F_1 > 0$ be the isolated part, F_2 the separated part, F_3 the dense-in-itself (perfect) kernel, and F_2 the derived set. The derived set $F_{1d} \subseteq F_2$ of the isolated part is nowhere dense in F (its complement $F - F_{1d} \supseteq F_1$ is dense in F, since its closed hull contains $F_{1d} + (F - F_{1d}) = F$) and it is therefore an F_σ; it consists of $F_1 - F_3$ and of those points of F_1 that are points of accumulation of F_2. Now since $f(x|F_3)$ is by hypothesis continuous up to sets of the first category, there exists a set D_3 of the first category in F_3, and *a fortiori* in F, such that $f(x|F_3 - D_3)$ is continuous. If we now remove from F the set $F_{1d} + D_3$ of the first category, then there remains $F_1 + C$, where $f(x|C)$ is continuous and C contains no point of accumulation of F_1; as a result, $f(x|F_1 + C)$ is also continuous at the points of C and also, of course, at the (isolated) points of F_1; that is, $f(x|F)$ is continuous up to sets of the first category. (In the case $F_1 = 0$, $f(x|F_1)$ is continuous, and F_2 is an F_σ.)

We now show, following Lusin, that it is possible (in the space A of the real numbers) for $f(x|P)$ to be continuous up to sets of the first category for all non-empty perfect sets P, while f is nevertheless not a Baire function. To do this, we construct an uncountable set L such that LP is always a P_1. Assuming this to have been done, we see that L is not a Borel set (nor even a Suslin set), for otherwise it would have to contain a non-empty perfect set P, and $LP = P$ would be a P_2. The characteristic function f of L ($f = 1$ in L; $f = 0$ in $A - L$) is thus not a Baire function, whereas $f(x|P)$ is nevertheless always continuous up to sets of the first category; specifically, $f(x|P-LP) \equiv 0$ is continuous.

In order to form such a set L, we consider sequences of natural numbers

$$X = (x_1, x_2, \ldots, x_n, \ldots)$$

and define $X < Y$ (X "finally" smaller than Y) if ultimately, for $n = n_0$, we have $x_n < y_n$; the smallest number n_0 for which this is the case will be denoted by $n(X, Y)$. This relation is transitive: If $X < Y$ and $Y < Z$, then $X < Z$ and, in fact, it is obvious that

$$n(X, Z) \leq \max\,[n(X, Y), n(Y, Z)].$$

For every finite or countable set of number sequences $X, Y, Z, \ldots$ there exists a "finally greater" U; it is necessary only to choose

$$U_1 > x_1, \quad U_2 > \max\,[x_2, y_2], \quad U_3 > \max\,[x_3, y_3, z_3], \ldots.$$

Accordingly, it is possible to construct a set of number sequences $X_0 < X_1 < \cdots < X_\xi < \cdots$ of type Ω.

On the other hand, we assign to each number sequence X the continued fraction

$$t_X = \cfrac{1}{x_1 + \cfrac{1}{x_2 + \cfrac{1}{x_3 + \ddots}}}$$

(t_X an irrational number between 0 and 1), and to the above sequences X_ξ the numbers t_ξ ($\xi < \Omega$). Then the set $L = \{t_0, t_1, \ldots, t_\xi, \ldots\}$ has the required properties.

Writing $n_{\xi\eta}$ as an abbreviation for $n(X_\xi, X_\eta)$ ($\xi < \eta$), let us remark the following: If a sequence of numbers t_η converges to t_ξ, then the number of initial numerals common to the two continued fractions, and consequently $n_{\xi\eta}$ as well, tends to 0. We now choose some $\gamma > 0$ and set

$$M = \{t_\xi : \xi < \gamma\}, \quad N = \{t_\eta : \eta \geq \gamma\},$$

and $L = M + N$; furthermore, we split up the set N of the numbers t_η $(\eta > \gamma)$ into $N = N_1 + N_2 + \cdots$, where N_n is the set of the t_η for which $n_{\gamma\eta} = n$. Then no point of M is a point of accumulation of N_n. For if $\xi < \gamma$, then

$$n_{\xi\eta} \leq \max\,[n_{\xi\gamma}, n_{\gamma\eta}],$$

so that for $t_\eta \in N_n$, $n_{\xi\eta} \leq \max\,[n_{\xi\gamma}, n]$,

$$n_{\xi\eta} \text{ is bounded, so that } t_\eta \text{ cannot converge to } t_\xi.$$

The same holds for $\xi = \gamma$.

We can now (L separable) choose γ so great that M is dense in L; then N_n is nowhere dense in L. For $M \cdot N_n = 0$, and $L - L \cdot N_n \supseteq M$ is dense in L. Therefore N is of the first category in L; that is, there exists a decomposition $L = M + N$, where M is countable and N is an A_1.

Finally, suppose $P > 0$ perfect. If LP is uncountable, then this is a set of the same kind as L itself (corresponding to a set of number sequences $X_\alpha < X_\beta < \cdots < X_\xi < \cdots$) and admits of a decomposition $LP = M' + N'$, where N' is of the first category in LP and thus in P, and M' is countable and therefore also of the first category in P (every countable set contained in G_δ is a G_δ (p. 164): LP is a P_1. The same holds when LP is at most countable. Our investigation has now reached its goal. A function f need not be a Baire function even if $f(x|F)$ is always continuous up to sets of the first category.

It should also be mentioned that the complement $K = A - L$ of the Lusin set is an F_σ-set; that is (p. 165), every non-empty set perfect in K is of the second category in itself. Such a set, which is thus dense-in-itself and closed in K, is of the form KP, where P is perfect (in the set A of the real numbers); because of $P = KP + LP$, KP is of the second category in P and thus, all the more so, in itself. Thus there exists F_σ-sets that are not absolute Borel sets, and thus a statement made earlier (p. 165) to the effect that an F_σ-set need not be a Young set, is proved in a far wider context.

If a function f (whether a Baire function or not) is continuous up to sets of the first category in the space A, thought of as a G_δ-set — so that at least one decomposition $A = C + D$ exists, where D is an A_1 and $f(x|C)$ is continuous — then among all such decompositions $A = C_1 + D_1 = C_2 + D_2 = \cdots$ there exists (Sierpiński) one with greatest C and smallest D, namely

$$C = C_1 \cdot C_2 \cdots, \quad D = D_1 + D_2 + \cdots.$$

It need only be shown that $f(x|C)$ is continuous (D is of course an A_1).

Let $x \in C$ and, say, $x \in C_1$; since $f(x|C_1)$ is continuous, there exists for any $\varepsilon > 0$ a neighborhood U_x for which

$$|f(z) - f(x)| < \varepsilon \quad \text{for } z \in C_1 \cdot U_x. \tag{7}$$

Next, let $y \in C \cdot U_x$; if, say, $y \in C_2$ (where C_2 can equal C_1), then there exists a neighborhood U_y for which

$$|f(z) - f(y)| < \varepsilon \quad \text{for } z \in C_2 \cdot U_y. \tag{8}$$

The non-empty (it contains y) open set $U_x \cdot U_y$ is an A_2 and is therefore not contained in $D_1 + D_2$; consequently $U_x \cdot U_y \cdot C_1 \cdot C_2$ is non-empty, and there exists at least one z satisfying both (7) and (8), so that

$$|f(x) - f(y)| < 2\varepsilon \quad \text{for } y \in C \cdot U_x,$$

and since x was any point of C, it follows that $f(x|C)$ is continuous.

3. Extensions of the space.

It is possible to characterize the Baire functions of a real variable by means of only two plane sets rather than by means of the system of its one-dimensional inverse image sets and, in general, to characterize the Baire functions of the space A by means of two sets of the extended space $A^* = (A, Y)$ — the product of A with the set Y of the real numbers in which distances may be defined by $\sqrt{x\bar{x} + (y - \bar{y})^2}$. Let the Borel sets (6) of the extended space be denoted by $\bar{\mathfrak{M}}^\xi$ and $\bar{\mathfrak{N}}^\xi$. The inverse image sets

$$\mathfrak{M}(\delta) = [f > \delta), \quad \mathfrak{N}(\delta) = [f \geq \delta) \tag{9}$$

considered thus far were the sets of the points x satisfying the inequalities in brackets; by

$$M = [f > y], \quad N = [f \geq y] \tag{10}$$

we now mean the sets (no longer depending on a parameter) of the points (x, y) satisfying the inequalities in question. In the same sense we let

$$C = N - M = [f = y]. \tag{11}$$

Geometrically interpreted — in the case in which x is a real variable also — C is the "curve" represented by $y = f(x)$, M is that part of the xy-plane lying below it, and $\mathfrak{M}(\delta)$ is the projection on the x-axis of that part of C lying in the half-plane $y > \delta$ (or of that part of M lying on the line $y = \delta$); corresponding statements can be made concerning N and $\mathfrak{N}(\delta)$.

We now have the theorem:

VII. $f(x)$ is a Baire function $f^\xi(x)$ if and only if the set M is an $\bar{\mathfrak{M}}^\xi$ and the set N is an $\bar{\mathfrak{N}}^\xi$. If $f(x)$ is an $f^\xi(x)$, then the set C is an $\bar{\mathfrak{N}}^\xi$.

For the proof, we consider the Baire functions $f^\xi(x)$ of the space A and the Baire functions $f^*(x, y)$ of the extended space A^*. We have:

Every $f^*(x, 0)$ is an $f^\xi(x)$. This holds true for 0, since from a function continuous in both the variables x and y the substitution $y = 0$ produces a function continuous in x, and we then proceed by induction. The induction from ξ to $\xi + 1$ is trivial; furthermore, if λ is a limit ordinal and the statement is proved for all $\xi < \lambda$, then the conclusion follows for λ via the representation of f^λ as quotient of two k_λ (p. 293).

Similarly, there follows:

Every $f^\xi(x)$ is an $f^\xi(x, y)$, and $f^\xi(x) - y$ is an $f^\xi(x, y)$ — the latter because y is a continuous function of x and y and is thus an $f^\xi(x, y)$ of every class.

If, therefore, $f(x)$ is an $f^\xi(x)$ then $f(x) - y$ is an $f^\xi(x, y)$ and is therefore of the class $(\bar{\mathfrak{M}}^\xi, \bar{\mathfrak{N}}^\xi)$; the set M is an $\bar{\mathfrak{M}}^\xi$, N is an $\bar{\mathfrak{N}}^\xi$ and C is an $\bar{\mathfrak{N}}^\xi$. If, conversely, M is an $\bar{\mathfrak{M}}^\xi$ and N is an $\bar{\mathfrak{N}}^\xi$, then $f(x, y) = f(x) - y$ is of the class $(\bar{\mathfrak{M}}^\xi, \bar{\mathfrak{N}}^\xi)$, for this function has the peculiarity that its inverse image sets (in A^*) arise from M and N by translations $\eta = y + \delta$ which transform the whole space A^* isometrically into itself. Therefore $f(x, y)$ is an $f^\xi(x, y)$, and $f(x, 0) = f(x)$ is an $f^\xi(x)$.

The first part of Theorem VII does not, in general, have a converse. Nevertheless, the following theorem provides a limited converse, which is remarkable because the sets $C(\delta) = [f = \delta]$ of the space A allow of no inference as to the character of the function $f(x)$:

VIII. If the space A is a separable absolute Suslin set and if C is a Suslin set in the extended space (A, Y), then $f(x)$ is a Baire function.

Let A be a Suslin set in the separable complete space X; then

$A^* = (A, Y)$ is a Suslin set in the separable complete space (X, Y) and is thus a separable S ($S =$ absolute Suslin set), C and the half-space $y > \delta$ open in A^* are Suslin sets in A^*, and their intersection is therefore an at most separable S; $\mathfrak{M}(\delta)$ is the projection of this on the space A, is therefore a continuous image, and by Theorem II of §37, thus is also an S. A consideration of the half-space $y \leq \delta$ also shows the complement $A - \mathfrak{M}(\delta)$ to be an S; both sets are Borel sets in their sum A (§34, III). $\mathfrak{N}(\delta)$ turns out, just like $\mathfrak{M}(\delta)$, to be a Borel set in A; $f(x)$ is a Baire function. In fact, as we know from VII, C was therefore actually a Borel set in A^* and not merely a Suslin set. Incidentally, we cannot infer the Baire class of $f(x)$ from a knowledge of the Borel class of C; if C is an $\bar{\mathfrak{N}}^{\xi}$, $f(x)$ can be an $f^{\eta}(x)$ with $\eta \geq \xi$.

The projection of C on Y is the range B of the values of $f(x)$, the (real, single-valued) image of A under $y = f(x)$; we call this set, when $f(x)$ is a Baire function, the *Baire image* of A.

We have the following:

IX. The Baire image of a separable absolute Suslin set is a Suslin set. The one-to-one Baire image of a separable absolute Borel set is a Borel set. If B is the one-to-one Baire image of a real Suslin set A, then A is also the one-to-one Baire image of B.

The proof is already contained for the most part in the proof of VIII. If A is a separable S, then we saw that A^* is a separable S, as are C and B (§37, II). If A is separable and is an absolute Borel set or a Borel set in X, then so is (A, Y) in (X, Y), so that A^* is a separable absolute Borel set and so are C (as a Borel set in A^*) and B, provided the projection is one-to-one. For the third statement, let $x = g(y)$ be the function, defined in B, inverse to $y = f(x)$. Now B was a separable S, and C was an S and therefore a Suslin set in the space $B^* = (X, B)$, where X is now the set of real numbers. Once more, therefore, with an interchange of the variables x and y, we have the condition of Theorem VIII, and $g(y)$ is a Baire function (whose class we cannot deduce from that of $f(x)$).

Theorem IX is a generalization of Theorem II of §37, although, for the time being, only for real functions (compare XIII). Moreover, the (real) continuous functions defined in the Baire null space or in the space $\mathfrak{J}$ of the irrational numbers already (§37, III) produce all the (real) Suslin sets as their ranges. But if x is to have as its domain the set A of all the real numbers, then the continuous functions $\{f(x)\}$ do not suffice to yield as their images all the Suslin sets (the continuous image of A is an F_σ); but the functions of the first class do suffice, and in fact:

X. Every real Suslin set is the range of a function $f(x)$ of the first class of the real variable x (and even of a function with an at most countable number of points of discontinuity).

For we were able to represent the real Suslin set B as the continuous image of $\mathfrak{J}$ or, what amounts to the same, by means of a continuous function $f(i)$, where i ranges over the irrational numbers. We extend this function to all the real numbers (without extending its range) by arranging the rational numbers in a sequence $\{r_1, r_2, \ldots\}$, choosing for each r_n an irrational number i_n with $|i_n - r_n| < 1/n$, and setting $f(r_n) = f(i_n)$. Then the total function $f(x)$ is still continuous at every irrational point, since for $r_n \to i$ it follows that $i_n \to i$ and $f(r_n) = f(i_n) \to f(i)$. Therefore $f(x)$ is discontinuous at most at the rational points and so, by p. 283, is of the first class.

4. Non-real Baire functions. Let A and Y be metric spaces; we consider all the single-valued functions $y = \varphi(x)$ defined in A for which $y \in Y$, where thus the image $B = \varphi(A)$ is a subset of Y. Let us call A the *space*, and Y the *image space* of this system of functions. (When Y is the set of real numbers we have the real functions $\varphi(x)$, to which this chapter has been devoted.) A sequence of functions $y_n = \varphi_n(x)$ which converges for all x generates a new function $y = \varphi(x)$. Therefore we can again define the Baire functions φ^ξ for $\xi < \Omega$, with the exception that limit ordinals must be excluded wherever the concepts of an ordinary and a complete system of functions (§ 41,1) — formulated specifically for the real functions — cannot be carried over to an arbitrary image space. Let the φ^0 be the continuous functions; if the φ^ξ have already been defined, we let the $\varphi^{\xi+1}$ be the limit functions of convergent sequences of functions φ^ξ; if η is a limit ordinal, then we define (not the φ^η but) the $\varphi^{\eta+1}$ as the limit functions of convergent sequences of functions φ^ξ ($\xi < \eta$). For real functions, the indices of these classes coincide with those defined earlier, except that the φ^ω still allow of being interposed.

Since no distinction between above and below is necessary here, the Lebesgue inverse image sets are most expeditiously defined as follows. First, for Q contained in Y, we define $[y \in Q]$ to be the set of those x for which $y = \varphi(x) \in Q$, that is, the inverse image of Q (B the image of A). The sets

$$[y \in G] \quad \text{and} \quad [y \in F],$$

where G ranges over all the sets open in Y and F over all the sets closed in Y, will then be called the inverse image sets of the function $y = \varphi(x)$; they are complements of each other, and it would suffice to consider only one kind. As M ranges over a given system of sets contained in A and $N = A - M$ over their complements, we say that the function φ is of the class (M, N) if every set $[y \in G]$ is an M and every set $[y \in F]$ is an N. The following theorem then holds, where among several possible statements as to class, we have chosen the one that takes on the simplest form in the case of Borel sets:

XI. *Let the functions $\varphi_n \to \varphi$ be of the class (M, N). Then φ is of the class $(N_{\sigma\delta}, M_{\delta\sigma})$ in the case of uniform convergence, of the class (M_σ, N_δ). If the M form a σ-system and the N a δ-system, then φ is of the class (N_δ, M_σ) and, in the case of uniform convergence, of the class (M, N).*

Let the non-empty set F be closed in Y; for $\nu = 1, 2, 3, \ldots$, let F_ν be the closed set of the points y for which $\delta(y, F) \leq 1/\nu$, G_ν the open set of the points $d(y, F) < 1/\nu$, and H_ν an arbitrary set between the two ($G_\nu \subseteq H_\nu \subseteq F_\nu$). Thus

$$F = \mathfrak{D}G_\nu = \mathfrak{D}H_\nu = \mathfrak{D}F_\nu.$$

Further, let $y_n = \varphi_n(x) \to y = \varphi(x)$. Then the following conditions on x are equivalent:

(a) $y \in F$;

(b) For every ν, $y_n \in H_\nu$ for almost all n;

(c) For every ν, $y_n \in H_\nu$ for infinitely many n.

(b) follows from (a). If $y \in F$ then, for every ν, $y \in G_\nu$, and thus (since G_ν is open) $y_n \in G_\nu \subseteq H_\nu$ for almost all n.

(b) implies (c).

(c) implies (a). For every ν, $y_n \in H_\nu \subseteq F_\nu$ holds for infinitely many n, so that (since F_ν is closed) $y \in F_\nu$, and consequently $y \in F$.

As a result, using (a) and (c), we obtain

$$[y \in F] = \mathfrak{D}_\nu \varlimsup_n [y_n \in H_\nu].$$

If we choose $H_\nu = F_\nu$, then the set $[y_n \in H_\nu]$ is an N, its upper limit for $n \to \infty$ is an $M_{\sigma\delta}$, and $[y \in F]$ is an $M_{\sigma\delta}$. The function φ is of the class $(M_{\sigma\delta}, N_{\delta\sigma}) = (N_{\sigma\delta}, M_{\delta\sigma})$.

If the convergence is uniform, then (a) is equivalent with the stronger statement:

(b*) for every ν, $y_n \in H_\nu$ for $n \geq n_\nu$

(where n_ν depends only on ν and not on x). For if n_ν is so chosen that $\delta(y_n, F) < 1/\nu$ for $n \geq n_\nu$, then it follows from (a), for every ν and $n \geq n_\nu$, that $d(y_n, F) \leq \delta(y_n, F) < 1/\nu$, $y_n \in G_\nu \subseteq H_\nu$, and therefore (b*); conversely, (a) already follows from (b*). Hence

$$[y \in F] = \mathfrak{D}_\nu \mathfrak{S}_{n_\nu} \mathfrak{D}_{n \geq n_\nu} [y_n \in H_\nu];$$

if we now choose $H_\nu = F_\nu$, $[y \in F]$ is then an N_δ, and φ is of the class (M_σ, N_δ). Thus XI is proved.

If once again we define the Borel sets $\mathfrak{G}^\xi_\delta$ of the space A, then we have:

XII. *The Baire functions φ^ξ are of the class $(M^\xi_\sigma, N^\xi_\delta)$.*

This follows from XI by induction, because the M^ξ form a σ-system and their complements N^ξ form a δ-system. The induction from ξ to $\xi + 1$ is trivial; that from $\xi < \eta$ to $\eta + 1$ (η a limit ordinal) rests on the fact that all the functions φ^ξ are of the class (M^ξ, N^ξ).

XIII. *The Baire image, real or not, of a separable absolute Suslin set is an at most separable, absolute Suslin set. The one-to-one Baire image, real or not, of a separable absolute Borel set is a separable absolute Borel set.*

First we have to say: If $\varphi_n \to \varphi$ and if every φ_n gives rise to an image B_n of A which is at most separable, then φ also yields an at most separable image B, since B is obviously a subset of $(\mathfrak{S} B_n)_\alpha$. (B is already contained, by the way, in the lower closed limit $\underline{\mathfrak{F}} B_n$.) If, therefore, A is separable, then its continuous images (footnote, p. 238) and its Baire images are all at most separable. As a consequence, the image space Y of a Baire function $y = \varphi(x)$ can be replaced by an at most separable Y_0, and this can be done in such a way that not only does the image B of A lie in Y_0, but φ also remains a Baire function in the image space Y_0. To this end, we make use, say, of the representation

$$\varphi = \lim_p \varphi_p, \quad \varphi_p = \lim_q \varphi_{pq}, \quad \varphi_{pq} = \lim_r \varphi_{pqr}, \ldots$$

[2] mentioned at the beginning of the present section, in which for every natural sequence of numbers, the functions $\varphi_n, \varphi_{nq}, \varphi_{pqr}, \ldots$

[2]Compare this with Appendix D.

are ultimately continuous (this form of φ is, as we saw, necessary and sufficient for Baire functions). If $B, B_n, B_{nq}, B_{pqr}, \ldots$ are the Baire images (ultimately continuous) traced out by these functions, then Y can obviously be replaced by

$$Y_0 = B_1 + B_2 + B_{12} + B_{123} + \cdots .$$

Accordingly, we may assume the image space separable, and even complete.

Furthermore: If A and Y are arbitrary and $\varphi(x)$ is a Baire function, then the set C of the points (x, y) for which $y = \varphi(x)$ is a Borel set (cf. VII and VIII) in the extended space $A^* = (A, Y)$. For let us denote the distances $d(y_1, y_2)$ in the space Y, as in linear spaces, by $|y_1 - y_2|$, without thereby assuming Y to be a linear space and, for any function $\varphi(x)$ of the image space Y defined in A, let us consider the distance

$$f(x, y) = |y - \varphi(x)|;$$

then this is a real function in the extended space A^*. If $\varphi(x)$ is continuous, then so is $f(x, y)$; and since, in addition, $\varphi_n \to \varphi$ implies

$$|y - \varphi_n(x)| \to |y - \varphi(x)|,$$

we see at once that when φ is a Baire function φ^ξ, f is a real Baire function f^ξ. The set $C = [f = 0]$ is therefore an N_δ^ξ of the extended space A^*.

The rest of the proof now continues as it did in IX. Let A be a Suslin set in the separable complete space X, and let Y be separable and complete; (A, Y) is a Suslin set in (X, Y) and is therefore a separable S; the same for C; and the same for B, as a projection of C. If A is, in particular, a Borel set, then so is (A, Y), so is C, and — provided the projection is one-to-one — so is B as well. In this connection, compare § 46.

§ 8.44. Sets of Convergence

Let a sequence of continuous and — once again — real functions $f_n = f_n(x)$ be defined in the space A; we seek the set of convergence C of this sequence, that is, the set of the points x (points of convergence) at which $\lim f_n(x)$ exists. For this it is necessary and sufficient that for every $\epsilon > 0$ there exist a term $f_m(x)$ of the sequence that differs by at most ϵ from all those that follow, that is, one for which

$$|f_n(x) - f_m(x)| \leq \epsilon \quad \text{for } n > m. \tag{8.1}$$

From this it follows that: Let $F_m(\epsilon)$ (as in § 42,4) be the set of the points x at which (1) holds for fixed ϵ and m, and let

$$C(\epsilon) = F_1(\epsilon) + F_2(\epsilon) + F_3(\epsilon) + \cdots$$

be the set of those x at which, with ϵ fixed, (1) holds for any m whatever; then C is the intersection of all the $C(\epsilon)$ or — since the sets $F_m(\epsilon)$ and $C(\epsilon)$ decrease with decreasing ϵ —

$$C = C(1) \cdot C(\tfrac{1}{2}) \cdot C(\tfrac{1}{3}) \cdots .$$

$F_m(\epsilon)$ is closed, $C(\epsilon)$ is an F_σ, and the set of convergence C is an $F_{\sigma\delta}$.

This can also be understood this way: If we assume that

$$\overline{f} = \overline{\lim}\, f_n \quad \text{and} \quad \underline{f} = \underline{\lim}\, f_n$$

exist everywhere (that is, are finite), then these are functions of the second class, and in fact, by § 41, (6), $\overline{f}$ is an h^2 of the class $(*, G_\delta)$, and $\underline{f}$ is a g^2 of the class $(F_\sigma, *)$. Their difference

$$\omega(f) = \overline{f} - \underline{f}$$

which is called the *oscillation* of f_n for $n \to \infty$ is, as a function of the second class,[3] of the class $(G_{\delta\sigma}, F_{\sigma\delta})$; the set of the points

[3]For the upper limit of a sequence of h^2 is itself an h^2 and for the lower limit, a g^2.

of divergence $\omega(f) > 0$ is thus a $G_{\delta\sigma}$; the set of the points of convergence $\omega(f) = 0$ is an $F_{\sigma\delta}$. The set of points at which $\alpha < \underline{\lim}\, f_n < \overline{\lim}\, f_n < \beta$ is also an $F_{\sigma\delta}$, specifically, it is the intersection $[\overline{f} = \underline{f}][\underline{f} > \alpha][\overline{f} < \beta] = F_{\sigma\delta} \cdot F_\sigma \cdot F_\sigma$. We can now free ourselves from the hypothesis of finite $\overline{f}$ and $\underline{f}$ by means of the bounding transformation $g_n = f_n/(1 + |f_n|)$; the set of the points at which f_n converges is identical with the set of points at which $-1 < \lim g_n < 1$ (whereas $g_n \to 1$ is equivalent with $f_n \to \infty$ and $g_n \to -1$ with $f_n \to -\infty$).

Moreover, the set $[\overline{f} = 0] = [\underline{f} = 0] = [\omega = 0]$ is a G_δ, and so is the set $[\overline{f} = -\infty]$, and also the sum of the two; that is, the set D_∞ of the points at which $f_n(x)$ is not bounded is a $G_{\delta\sigma}$. If — as, for example, in the theory of Fourier series — we form a sequence of continuous functions of a real variable that diverge in this way at all the rational points (so that $f_n(x)$ is not bounded), then the same behavior occurs automatically also at $\aleph$ irrational points; for, by XI of § 26, D_∞ — as a dense G_δ — is of the cardinality of the continuum.

The set of points at which f_n converges to a pre-assigned value c is also an $F_{\sigma\delta}$, for it is the intersection $[\omega = 0][\lim f_n \geq c][\lim f_n \leq c]$. The set of points at which $\lim f_n = +\infty$ is a $G_{\delta\sigma}$, since

$$[\lim f_n = +\infty] = [\underline{f} = +\infty] = [\overline{f} > 0] \cdot [\overline{f} > 1] \cdot [\overline{f} > 2] \cdots .$$

But the set at which f_n *diverges* to $+\infty$ or $-\infty$ is also an $F_{\sigma\delta}$, since

$$[f_n \to \infty] = [\underline{f} = \infty] = [\overline{f} > 0] \cdot [\underline{f} > 1] \cdot [\underline{f} > 2] \cdots .$$

We have already proved the first half of the theorem:

I. (*Hahn*). *The set of convergence of a sequence of real continuous functions is always an $F_{\sigma\delta}$, and for every set $C = F_{\sigma\delta}$ it is possible to give a sequence of real continuous functions for which C is the set of convergence.*

For the proof of the second half, we first take C to be an F_σ,

$$C = F_1 + F_2 + F_3 + \cdots , \quad F_1 \subseteq F_2 \subseteq F_3 \subseteq \cdots$$

(F_n closed); we let $D = A - C$ be the complement. The function h, which is defined in

$$F_1, \quad F_2 - F_1, \quad F_3 - F_2, \ldots, \quad D$$

to be 1, $\frac{1}{2}$, $\frac{1}{3}$, ..., 0 is upper semi-continuous; for $[h > \alpha]$ is one of the sets F_n, when it is neither the null set nor the whole space. It is the limit of a decreasing sequence of continuous functions $y_1 \geq y_2 \geq \cdots$; since the functions

$$y'_n = \min[y_n, 1] \quad \text{and} \quad y''_n = \max[y_n, \tfrac{1}{n}]$$

also converge as a decreasing sequence to h, we may as well assume that $1/n \leq y_n \leq 1$. The continuous functions $g_n = 1/y_n$ for which $1 \leq g_n \leq n$ hold form an increasing sequence and converge in C to $1/h$ — that is, to integer limiting values — while they diverge to $+\infty$ in D. Thus this would already constitute a solution of the problem for the case $C = F_\sigma$; to proceed further, we transform it in the following way. We have $0 \leq g_{n+1} - g_n \leq n$; if we interpose between g_n and g_{n+1} further functions, with fractional index,

$$g_{n+\frac{k}{2^n}} = g_n + \tfrac{k}{2^n}(g_{n+1} - g_n) \quad (k = 1, 2, \ldots, 2n - 1)$$

and change the notation by numbering the original and the interposed fractions consecutively with natural numbers, then we obtain a sequence of continuous functions $0 = g_1 \leq g_2 \leq \cdots$, $g_{n+1} - g_n \leq 1/2$. In C, $\lim g_n$ is a natural number; in D, $\lim g_n = +\infty$. The functions

$$f_n = \sin(a \cdot g_n)$$

(a a positive constant) converge to 0 in C and diverge in D; for (if $x \in D$) at least one g_n falls into each of the intervals $[k + 1/4, k + 3/4]$ of length $1/2$ ($k = 1, 2, 3, \ldots$), and then $(-1)^k \sin(a \cdot g_n) = \sin(a/4)$, so that $f_n \geq a/\sqrt{2}$ infinitely often and, also infinitely often, $f_n \leq -a/\sqrt{2}$.

Therefore there exists, for every $C = F_\sigma$, a sequence of continuous, uniformly bounded, functions f_n (of absolute value $\leq a$) that converge to 0 in C and diverge in $D = A - C$. (*Sierpiński.*)

Now let $C = C_1 C_2 \cdots$ be the intersection of a sequence of sets $C_m = F_\sigma$; let D_m be the complement of C_m and $D = D_1 + D_2 + \cdots$ that of C. For every $m = 1, 2, \ldots$ we determine a sequence of continuous functions f_{mn} of absolute value $\leq 1/m$, so that the limiting value $\lim_n f_{mn}$ exists and is 0 in C_m, and does not exist in D_m. Let the double sequence of these functions f_{mn} be written as a simple sequence f_n (say $f_1, f_2, f_3, \ldots = f_{11}, f_{12}, f_{21}, f_{22}, f_{31}, \ldots$). Then the limiting value $\lim f_n$ exists and $= 0$ in C, and does not exist in D. For if $x \in C$, the inequality $|f_{mn}| \geq \epsilon > 0$ can hold for only a finite number of m, because $|f_{mn}| \leq 1/m$, and then each time for only a finite number of n, because $\lim_n f_{mn} = 0$; for $x \in D$ and, say, $x \in D_m$, the functions $f_{m1}, f_{m2}, \ldots$ already form a divergent subsequence of the f_n. I is thus proved.

If, instead of being continuous, the functions f_n are taken from the system of all the functions of the class (M, N), where the M form a σ-ring and their complements N form a δ-ring, then their set of convergence is an $N_{\delta\sigma}$; the proof is as at the beginning of this section. This also applies to non-real functions (§ 43, 4), provided the image space is complete.

Let us return to real continuous functions and consider, instead of a sequence of functions $f_n(x)$, a family of functions $f(x, y)$ depending on a real positive parameter y (and which are therefore, for fixed y, continuous functions of $x \in A$) and whose behavior for $y \to 0$ is to be examined. The set of convergence C of the points x at which $\lim_{y \to 0} f(x, y)$ exists is itself an $F_{\sigma\delta}$. For in order that this be the case, it is necessary and sufficient that for every $\epsilon > 0$ there exist some $\eta > 0$ such that

$$|f(x, y_1) - f(x, y_2)| \leq \epsilon \quad \text{for } y < y_1, y_2 < \eta. \tag{8.2}$$

For fixed ϵ and η the set $F(\epsilon, \eta)$ of the points x at which (2) holds is closed (as the intersection of the sets in which the inequality (2) holds for a particular pair y_1 and y_2). The set of points x at which (2) holds for fixed ϵ and any η is $C(\epsilon) = \mathfrak{S}_\eta F(\epsilon, \eta)$, for which we can write

$$C(\epsilon) = F(\epsilon, 1) + F(\epsilon, \tfrac{1}{2}) + F(\epsilon, \tfrac{1}{3}) + \cdots$$

since the $F(\epsilon, \eta)$ increase as η decreases; finally, C is again the intersection of the $C(\epsilon)$, or

$$C = C(1) \cdot C(\tfrac{1}{2}) \cdot C(\tfrac{1}{3}) \cdots$$

from which we see that C is an $F_{\sigma\delta}$.

The upper limit

$$\overline{f}(x) = \overline{\lim_{y \to 0}} f(x, y)$$

is defined by means of

$$\overline{f}(x) = \lim_{\eta \to 0} g(x, \eta), \qquad g(x, \eta) = \sup_{0<y<\eta} f(x, y).$$

If it is to exist (for fixed x) — that is, if it is to be finite — then, for sufficiently small y, $f(x, y)$ must be bounded from above and $g(x, y)$ from below; $g(x, y)$ decreases with decreasing y. If $\overline{f}(x)$ exists for every x, then $\overline{f}(x)$ is again a function h^2 of the class $(*, G_\delta)$. In order to see this, we can assume $f(x, y)$, with x fixed, to be bounded from above for all $y > 0$ by considering in its stead $\min[f(x, y), 1/y]$ — a function that coincides for sufficiently small y, say $y < \eta$, with $f(x, y)$ and is otherwise, for $y > \eta$, in any case, $\leq 1/\eta$. Under this assumption, $g(x, y)$ exists for every y and, as the least upper bound of continuous functions, is lower semi-continuous (§ 42, 1); furthermore

$$\overline{f}(x) = \lim_{n} g(x, 1/n),$$

as the limit of decreasing functions of the first class, is then an h^2. The function

$$\underline{f}(x) = \underline{\lim}_{y \to 0} f(x, y)$$

to be defined analogously is — if it is everywhere finite — a g^2 of the class $(F_\sigma, *)$.

For example, for

$$f(x, y) = \frac{\psi(x + y) - \psi(x)}{y}$$

where $\psi(x)$ is a continuous function of the real variable x, we have the following: The set of points at which $\psi(x)$ is differentiable at the right is an $F_{\sigma\delta}$; the upper right-hand derivate (which is assumed everywhere finite) is an h^2; the lower right-hand derivate is a g^2. The same, of course, holds at the left and, in addition, for both sides (if we do not distinguish between left and right).

In the case of a family of functions, these conclusions (as to the classes of $\overline{f}$ and $\underline{f}$ and also as to the convergence set) can not be carried over from the continuous functions to those of the class (M, N); after all, they are based essentially on the fact that the intersection of an arbitrary collection (and not only a countable collection) of F is an F and that the sum of an arbitrary number of G is a G. Thus the fact that, when $f(x, y)$ is continuous in x for fixed y (where $f(x, y)$ is bounded from above for fixed x),

$$g(x) = \sup_y f(x, y)$$

is lower semi-continuous, has no analogue for the Baire functions of higher class. If $f(x, y)$ is a Baire function of both variables in the extended space (A, Y), and A is a separable S ($S =$ absolute Suslin set), then it results that $g(x)$ is a function of the class (S, S); for the set $[g > c]$ is obviously the projection of the set $[f > c]$ on the space A, and is thus an S; and in the same way, $[g \geq c] = \mathfrak{D}_n[f > c - 1/n]$. The sets $[g < c]$ and $[g = c]$ are complements $A - S$. (When, in particular, the space A is Euclidean and $f(x, y)$ is of the first class, then $g(x)$ is of the class $(F_\sigma, *)$ and so is of the second class. For here the projection of an F_σ is itself an F_σ.)

It is instructive to consider some other sets of convergence as well. We suppose x real and y positive; let the real function $f(x, y)$ defined in the upper half-plane $y > 0$ be continuous in x for fixed y. We let (x, y) converge from the upper half-plane to a point $(\xi, 0)$ of the x-axis; depending on the way in which we allow (x, y) to approach $(\xi, 0)$,

$$(x, y) \to (\xi, 0), \tag{8.3}$$

the set C of the numbers ξ, or of the points $(\xi, 0)$, for which $\lim f(x, y)$ exists can have different character.

Linear approach. In the case of approach parallel to the y-axis, where it is thus a matter of $\lim_{y\to 0} f(\xi, y)$, we know that C is an $F_{\sigma\delta}$. The same is the case when the approach takes place along the fixed line

$$x - \xi = ty, \tag{8.4}$$

which forms the angle α with the y-axis ($-\pi/2 < \alpha < \pi/2$, $t = \tan\alpha$); here we are concerned with the limiting value

$$\lim_{y\to 0} f(\xi + ty, y) = c(\xi, t) \tag{8.5}$$

of a function which, if y and t are fixed, is continuous in ξ; and the set $C(t)$ of the ξ where this limit exists is a $F_{\sigma\delta}$. This, then, is the result obtained in an approach along a fixed direction.

The sum $C = \mathfrak{S}C(t)$ is the set of points at which there is convergence when they are approached in at least one direction. C is a Suslin set. For $f(\xi + ty, y)$ is, for y fixed, a continuous function of the variables ξ and t; the set of the points (ξ, t) at which the limit (5) exists is an $F_{\sigma\delta}$ in the ξt-plane, and C is the projection of Γ on the ξ-axis.

The intersection $C = \mathfrak{D}C(t)$ is the set of points at which there is convergence when they are approached in any direction (where the limit (5) may, however, depend on the direction). C is the complement of a Suslin set — namely, of the projection of $E - \Gamma$ on the ξ-axis, where Γ is the set just mentioned and E is the ξ-plane.

Under approach in a fixed sector

$$|x - \xi| \leq ty \quad (t > 0),$$

the set of convergence is an $F_{\sigma\delta}$, in the case of approach in at least one (sufficiently small) sector it is an $F_{\sigma\delta\sigma}$, and in the case of approach in every sector (no matter how wide), an $F_{\sigma\delta\sigma\delta}$; the proofs of these statements are left to the reader.

In the case of a completely arbitrary approach (3), the set of convergence C is a $G_{\delta\sigma}$, and this without any assumptions about $f(x, y)$, which therefore need no longer be continuous in x. This is so because it is necessary and sufficient for convergence that for every $\epsilon > 0$ the point $(\xi, 0)$ have a plane neighborhood U in the upper half $(y > 0)$ of which

$$|f(x_1, y_1) - f(x_2, y_2)| \leq \epsilon$$

holds for every pair of points. The set $G(\epsilon)$ of the points $(\xi, 0)$ which, for fixed ϵ, have such a neighborhood, is open in the ξ-axis; C is the intersection of the $G(\epsilon)$ and is therefore a $G_{\delta\sigma}$.

Chapter 9

Supplement

§ 9.1. The Baire Condition

9.1.1 Modules and congruences

All the sets to be considered will be subsets of a fixed set E, which we call the "space" and elements of which we call "points." The *symmetric difference* of two sets A and B is the set

$$[A, B] = (A + B) - AB = (A - AB) + (B - AB) = [B, A]$$

of those points that belong to one of these sets but not to the other. If $a(x)$ and $b(x)$ are the characteristic functions (p. 22) of A and B, then $|a(x) - b(x)|$ is the characteristic function of $[A, B]$.

For the three sets A, B, and C, we have

$$[A, C] \subseteq [A, B] + [B, C]. \tag{9.1}$$

For let $x \in [A, C]$, say, $x \in A$, $x \notin C$; depending on whether $x \in B$ or $x \notin B$, x belongs to $[A, B]$ or to $[B, C]$.

Furthermore, it is obvious that

$$[E - A, E - B] = [A, B]. \tag{9.2}$$

If to every element (index) ν of an arbitrary set[1] N we assign a pair of sets A_ν and B_ν, then

$$[\mathfrak{S}A_\nu, \mathfrak{S}B_\nu] \subseteq \mathfrak{S}[A_\nu, B_\nu], \quad [\mathfrak{D}A_\nu, \mathfrak{D}B_\nu] \subseteq \mathfrak{S}[A_\nu, B_\nu]. \tag{9.3}$$

[1]This need not, of course, be a subset of E.

To prove the first formula, let x be a point of the set on the left, and say $x \in \mathfrak{S}A_\nu$, $x \notin \mathfrak{S}B_\nu$; thus there exists an n for which $x \in A_n$, and, at the same time, $x \notin B_n$, $x \in [A_n, B_n]$. The second formula in (3) is proved in the same way, or with the use of (2).

In the Suslin construction (p. 105), let

$$A = \mathfrak{S}A_n, \quad A_n = A_{n_1} \cdot A_{n_1 n_2} \cdots ;$$

$$B = \mathfrak{S}B_n, \quad B_n = B_{n_1} \cdot B_{n_1 n_2} \cdots .$$

Then

$$[A, B] \subseteq \mathfrak{S}[A_{n_1}, B_{n_1}] + \mathfrak{S}[A_{n_1 n_2}, B_{n_1 n_2}] + \cdots . \tag{9.4}$$

This follows from (3) because

$$[A, B] \subseteq \mathfrak{S}[A_n, B_n],$$

$$[A_n, B_n] \subseteq [A_{n_1}, B_{n_1}] + [A_{n_1 n_2}, B_{n_1 n_2}] + \cdots .$$

A non-empty system $\mathfrak{M}$ of subsets M of the space E is called a *module* when the following conditions are satisfied:

(α) Every subset of a set of $\mathfrak{M}$ itself belongs to $\mathfrak{M}$;

(β) The sum of a finite number of sets of $\mathfrak{M}$ itself belongs to $\mathfrak{M}$.

If, instead of (β), we impose the stricter requirement

(β_σ) The sum of a (finite or) countable number of sets $\mathfrak{M}$ itself belongs to $\mathfrak{M}$,

then the system $\mathfrak{M}$ that satisfies (α) and (β_σ) is called a *σ-module.*

Every module contains the null set 0.

The finite sets or (if E is a metric space) the sets nowhere dense in E form a module; the at most countable sets or the sets of the first category in E (the E_{I}) form a σ-module.

On the model of Number Theory and Algebra, we shall call two sets A and B *congruent modulo* $\mathfrak{M}$

$$A \equiv B \pmod{\mathfrak{M}}$$

if they "differ only by sets $M \in \mathfrak{M}$" or, more precisely, if their symmetric difference $[A, B]$ belongs to $\mathfrak{M}$. This can be expressed as follows: The two summands

$$M_1 = A - AB = (A + B) - B,$$

$$M_2 = B - AB = (A + B) - A,$$

of which the symmetric difference is composed are sets M of the module $\mathfrak{M}$, and since

$$B = AB + M_2 = (A - M_1) + M_2 = (A + B) - M_1 = (A + M_2) - M_1,$$

one of the congruent sets is obtained from the other by subtracting one set M and adding another. We can also say that A and B agree "up to sets M" or "to within sets M" or are identical "except for sets M" or "neglecting sets M." $A \equiv 0$ $(\mathfrak{M})$ means that A belongs to $\mathfrak{M}$. In the following we shall drop the explicit mention of the fixed module in congruences, and instead of $A \equiv B$ $(\mathfrak{M})$ we shall simply write $A \equiv B$.

That $A \equiv A$ and that $A \equiv B$ implies $B \equiv A$ — that is, that the congruence relation is reflexive and symmetric — is trivial; because of (1), it is also transitive; that is, $A \equiv B$ and $B \equiv C$ imply $A \equiv C$. It therefore allows of the division of the subsets of E into *classes of congruent sets* in which two classes are either disjoint or identical. If $\mathfrak{M} = \mathfrak{P}(E)$, then all the sets are $\equiv 0$, and there is only one class; the other extreme would be for $\mathfrak{M}$ to consist only of the null set 0 and every individual set to form a class by itself.

From (2) it follows that $A \equiv B$ implies $E - A \equiv E - B$.

From (3) it follows that $A_\nu \equiv B_\nu$ implies

$$\mathfrak{S}A_\nu \equiv \mathfrak{S}B_\nu, \quad \mathfrak{D}A_\nu \equiv \mathfrak{D}B_\nu$$

for a finite set of indices and, in the case of a σ-module, for a countable set also.

In the case of a σ-module, (4) yields: if $A_{n_1 n_2 \cdots n_k} \equiv B_{n_1 n_2 \cdots n_k}$, then $A \equiv B$; the Suslin process leaves congruences unaltered.

Now let E be a metric space; we call every open set containing the point x a neighborhood U_x of x. Let $\mathfrak{M}$ again be a fixed module, and let A be an arbitrary set of the space. For the local behavior of A to $\mathfrak{M}$ at a point x of the space we make the following distinction:

x will be called a *null point* of A (for $\mathfrak{M}$) if it has a neighborhood U_x for which

$$AU_x \equiv 0; \tag{9.5}$$

otherwise — and therefore when we have

$$AU_x \not\equiv 0$$

for every neighborhood U_x — we call x a *positive point.* The set of positive points will be called A_+ and that of the null points A_0; therefore $E = A_+ + A_0$. A_0 is open; A_+ is closed. For if $x \in A_0$, so that x has a neighborhood U_x for which (5) holds, then U_x is contained in A_0, because every point of U_x also has such a neighborhood (namely U_x). We thus have

$$A_0 = \mathfrak{S} U_x,$$

if a U_x is assigned to each null point x in accordance with (5), and

$$AA_0 = \mathfrak{S} AU_x,$$

in which every summand is $\equiv 0$. When $\mathfrak{M}$ is a σ-module and E — or, at any rate, A — is separable, then it follows that

$$AA_0 \equiv 0, \tag{9.6}$$

since, by virtue of VI of § 25, the sum with respect to the U_x (or the AU_x) can be restricted to countably many of these summands. But equation (6) can also hold when E is not separable (compare Theorem I below).

A point $x \in E - A_+$ is a null point of A, since it has a neighborhood $U_x = E - A_+$ for which $AU_x = 0$; from this it follows that

$$A_+ \subseteq A_0.^2 \tag{9.7}$$

[2] If all the finite subsets of A belong to $\mathfrak{M}$, then $A_+ \subseteq A_0$; if all the countable subsets do, then $A_+ = A_0$ (compare § 23).

If $A \equiv B$, then $AU_x \equiv BU_x$ also, and every null point of one of the sets is also a null point of the other, so that $A_0 = B_0$ and $A_+ = B_+$. (The converse does not hold.)

9.1.2 The Baire condition for sets

Let $\mathfrak{M}$ now be the σ-module of the sets of first category (in E). The null points and the positive points of A are here called *points of the first* and *second categories* of A. We show first that formula (6) holds in every case, even for non-separable spaces E.

I. (Theorem of Banach). *The points of the first category of A lying in A form a set of the first category.*

First we consider the special case that A is contained in A_0 so that we are to prove: *If all the points of A are of the first category, then A is of the first category.* We form a maximal system — that is, one that cannot be extended — of disjoint open sets U_ν for which $AU_\nu \equiv 0$; the existence of such a system follows as in § 31 by means of well ordering and because of considerations of cardinality. ν runs through a suitable subset of A_0; $V = \mathfrak{S}U_\nu$ is open; $V' = E - V$ is closed. Then V is nowhere dense, and the open kernel $V_i = 0$; for were $V_i > 0$, then — depending on whether $AV_i = 0$ or a is a point of AV_i — there could be adjoined to the sets U_ν either V_i itself or — since A is of the first category at a — some suitable $U_a \subseteq V_i$ for which $AU_a = 0$. We now have $V' = 0$,

$$A = AU + AV' = AU,$$

and it remains to show that $AU \equiv 0$. As a set of the first category,

$$AU_\nu = A_{\nu 1} + A_{\nu 2} + A_{\nu 3} + \cdots = \mathfrak{S}A_{\nu m}$$

is the sum of a countable number of nowhere-dense sets $A_{\nu m}$, which (with ν fixed) can be taken as disjoint. Then all the sets $A_{\nu m}$ are pairwise disjoint, and if we can show that every

$$A_m = \mathfrak{S}_\nu A_{\nu m}$$

is nowhere dense, then from

$$AU = \mathfrak{S}_m A_m = \mathfrak{S}_{\nu,m} A_{\nu m}$$

the result $AU \equiv 0$ follows. Now we obviously have

$$A_m \subseteq A_m U_\nu;$$

from this it follows for the closed hulls (p. 137, (13)) that

$$\overline{A}_m \supseteq \overline{A}_{\nu m} \cdot \overline{U}_\nu,$$

and if G denotes the open kernel $\overline{A}_m$ of $\overline{A}_m$: $\overline{A}_m \supseteq G \cdot \overline{U}_\nu$, and thus $G \cdot \overline{U}_\nu = 0$, since $A_{\nu m}$ is nowhere dense. Consequently, $G \cdot \mathfrak{S}\overline{U}_\nu = 0$, $G \cdot V = 0$, $G \cdot V_i = 0$, $G = 0$, and A_m is nowhere dense.

The special case $A = AA_0$ of Theorem I is thus proved; in the general case, we observe that for every subset B of A we have $BA_+ \subseteq B_+$ clearly holds and, in particular, for $B = AA_0$ we have $B = B_0$, and thus $B \equiv 0$.

The boundary of every open set G or of every closed set F is nowhere dense, so that $G_\alpha - G = F - F_i = 0$; every open set is congruent with a closed set, and conversely. ($G \equiv G_\alpha$, $F \equiv F_i$.)

We say that a set A is a *B-set* or *satisfies the Baire condition* — in the wider sense (Kuratowski) — if it is congruent to an open set or to a closed set.

The complement of a B-set is a B-set. For $A \equiv G$ yields $E - A \equiv E - G = F$.

The B-sets form a Borel system. For if $A_n \equiv G_n = F_n$ holds for $n = 1, 2, 3, \ldots$, then it follows that

$$\mathfrak{S}A_n \equiv \mathfrak{S}G_n = G, \quad \mathfrak{D}A_n \equiv \mathfrak{D}F_n = F.$$

All the Borel sets of the space are thus B-sets.

Every B-set is obtained from an open set by leaving off one E_{I} and adding another. If only one of these two operations is to be admitted, then we have: The B-sets are of the form

$$G_\delta + E_{\mathrm{I}} \quad \text{or} \quad F_\sigma - E_{\mathrm{I}}. \tag{9.8}$$

For let $A \equiv G$ be a B-set; the symmetric difference $[A, G]$, as a set of the first category, is contained in a set $D = F_\sigma$ of the first category; if we set $C = E-D = G_\delta$, then $C[A,G] = [CA, CG] = 0$, $CA = CG$, and

$$A = EA = CA + DA = CG + DA,$$

where CG is a G_δ and DA is an E_{I}. Taking complements yields the second form in (8).[3]

A is a B-set if and only if there exists a set $C = E$ ($E - C$ an E_{I}) such that CA is open, or closed, or simultaneously open and closed in C.

It is evident that this suffices; if, for example, $CA = CG$ and $C = E$, then $A = EA = CA = CG \equiv EG = G$. The necessity follows thus: If $A \equiv H$ and $C \supseteq E - [A, H] = E$, then as above, $C[A, H] = [CA, CH] = 0$ and $CA = CH$. If H is replaced by an open or a closed set, then there exists a $C_1 = E$ for which $C_1A = C_1G$, a $C_2 = E$ for which $C_2A = C_2F$ and a $C = C_1C_2 = E$ for which $CA = CG = CF$.

Further necessary and sufficient conditions for B-sets are found with the help of the two sets A_0 and A_+ of the points of the first and second categories of A. Let us set $B = E - A$. For every set A we have, by (6),

$$A = A(A_+ + A_0) = AA_+ = A_+ - BA_+. \tag{6*}$$

As a consequence, the congruences

$$BA_+ \equiv 0 \tag{9.9}$$

and

$$A \equiv A_+ \tag{9.10}$$

mean the same; they are sufficient for a B-set A (A is congruent to a closed set A_+), but they are also necessary, since (7) and (6*) imply, for a closed set F that F_+ is contained in F, so that $F = FF_+ = F_+$, and if $A \equiv F$ is a B-set, then $A_+ = F_+$ and

[3] And even a Suslin system (Nikodym).

thus $A \equiv A_+$. (9) states that the points not in A at which A is of the second category form only a set of the first category.

From (10) there follow conditions analogous to (8): A is of the form $A_+ + E_{\mathrm{I}}$ and $A_+ - E_{\mathrm{I}}$, where A_+ is a closed set determined by A itself.

9.1.3 The Baire condition for functions

Let E now be an arbitrary metric space and H an arbitrary metric image space (not necessarily the space of real numbers); let $y = \varphi(x)$ be an arbitrary function defined in E whose values lie in H.

In order to define Baire functions here as well, we consider all the functions that are continuous at the points of a set C which is the complement of a set of first category (in E). We call $\varphi(x)$ a *Baire function* or *B-function* if there exists a decomposition $E = C + D$ for which D is of the first category and for which $\varphi(x|C)$ is continuous. As we showed in § 43, 2, it can be assumed that $C = G_\delta$ and that D is an F_σ of the first category.

For every subset M of E, $\varphi(x|M)$ is called a *partial function.* A function $\varphi(x)$ is called an $\mathfrak{A}$-*function* if there exists a set C which is dense in E and for which $\varphi(x|C)$ is continuous. The points of continuity of $\varphi(x)$ form a set C whose complement $D = E - C$ is of the first category; therefore every B-function is an $\mathfrak{A}$-function, while the converse does not hold.

For the characteristic function of a set A, the condition of being an $\mathfrak{A}$-function means that A is an N-set (p. 295), that is, a set with nowhere-dense boundary; the condition of being a B-function means that A satisfies the Baire condition.

In place of N-sets, let us now consider sets A whose boundary is a set of the first category; we call them, in the terminology just introduced, *sets of the first category at their boundary points.* For a set A, this condition means: A_0 (the set of the null points of A) differs from $E - A$ by a set of the first category, that is, the boundary points of A that do not belong to A are of the first category. For the complement $B = E - A$, this means the same thing, since $B_0 = A_0$ and $B_+ = A_+$.

Let us call $\varphi(x)$ an $\mathfrak{E}$-*function* if the sets

$$[y \in G] = \varphi^{-1}(G), \quad [y \in F] = \varphi^{-1}(F)$$

(G open in H, F closed in H) are always B-sets; then their complements are also B-sets. The B-functions belong to the $\mathfrak{E}$-functions, since, for a continuous function $\varphi(x)$, the sets $[y \in G]$ are open and the sets $[y \in F]$ closed, and, on passing to a limit, the property of being a B-set is preserved.

II. *Every* $\mathfrak{E}$*-function is a B-function.*

Let $\varphi(x)$ be an $\mathfrak{E}$-function and $G_1, G_2, \ldots$ the open sets of a countable basis of H. The B-sets

$$A_n = [y \in G_n] = \varphi^{-1}(G_n)$$

have a decomposition $E = C_n + D_n$ (D_n of the first category) for which $C_n A_n$ is open in C_n ($C_n = G_\delta$). Let $C = \mathfrak{D} C_n$, so that $E - C = \mathfrak{S}(E - C_n)$ is of the first category. For $x \in C$,

$$\varphi(x) \in G_n \quad \text{means} \quad x \in C \cdot A_n,$$

and this set is open in C, being the intersection of the open (in C_n) set $C_n A_n$ with $C \subseteq C_n$. Therefore $\varphi(x|C)$ is continuous, and $\varphi(x)$ is a B-function.

9.1.4 The restricted Baire condition

If M is a subset of E, then we call $\varphi(x)$ a B_M*-function* when $\varphi(x|M)$ is continuous up to sets of the first category (in M), or is a B-function in the space M, that is, when a decomposition $M = C + D$ is possible for which $D = M_{\mathrm{I}}$ and for which $\varphi(x|C)$ is continuous. Here it is possible to deduce M from the closed hull $\overline{M}$, as well as from the dense-in-itself kernel M_*; that is, we have:

If $F = \overline{M}$, then a B_M-function is also a B_F-function.

For if $F = C + D$, where D is of the first category in F and $\varphi(x|C)$ is continuous, then $\varphi(x|M \cdot C)$ is continuous, and MD is of the first category in F and so, by XIII of § 27, of the first category in M as well.

If $K = M_*$, then a B_K-function is at the same time a B_M-function.

For (compare p. 296 above),

$$M = H + (S - H) + M_*,$$

where H is the isolated part and S the separated part of M, then $M \cdot H_*$ is closed in $M \cdot H_*$ and is nowhere dense, since its complement $M \cdot H - M \cdot H_*$ is dense in $M \cdot H$, and, a fortiori, $M \cdot H_*$ is an M_{I}. By § 23, (15), S is contained in $\overline{H}_*$, and $S - H$ in $\overline{H}_*$. Furthermore, there exists a D, of the first category in K, and a fortiori in M, for which $\varphi(x|K - D)$ is continuous. Eliminating $M \cdot H_* + D$ from M leaves $H + C$, where $\varphi(x|C)$ is continuous and $C \cdot H_* = 0$; therefore $\varphi(x|H + C)$ is also continuous at the points of C and at the (isolated) points of H; $\varphi(x)$ is a B_M-function.

We say that $\varphi(x)$ *satisfies the restricted Baire condition* or is a *B^*-function* if it is a B_M-function for every M contained in E. For this to hold, it suffices that it be a B_P-function for every perfect P; since for $M_{\mathrm{I}} = F$ and $F_* = P$ it is possible to go from P to F and from F to M. The B_M-functions, for fixed M, form a Baire system to which the Baire functions of the space E belong, and the same holds for the intersection of all these systems, that is, for the system of the B*-functions.

The set A is called a *B_M-set* if MA is a B-set in the space M, that is, if a set MG open in M (G open in E) exists for which $[MA, MG]$ is of the first category in M. This is equivalent with the characteristic function $\varphi(x)$ of A being a B_M-function; for $\varphi(x|M)$ is the characteristic function of MA for the space M. *A satisfies the restricted Baire condition*, or is a *B^*-set*, if it is a B_M-set for every M; it suffices for this that it be a B_P-set for every perfect P. If PA is, in particular, of the first category in P for every perfect P, then A is always a B_P-set and therefore a B*-set (although MA is not always of the first category in M; for example, certainly not when M consists of a single point of A). This is the case of the Lusin set L described on p. 297.

§ 9.2. Half-schlicht Mappings

9.2.1 Borel separability

Two sets A and B of the space E are said to be *Borel separable* if they can be enclosed in disjoint Borel sets P and Q:

$$A \subseteq P, \quad B \subseteq Q, \quad PQ = 0.$$

This is equivalent to the statement that A and B are disjoint and are Borel sets in their sum. For from the Borel separability it follows that

$$AB = 0, \quad A = (A + B)P, \quad B = (A + B)Q;$$

conversely, $(A + B) \cdot PQ = 0$ follows from these equations, and

$$A \subseteq P - PQ, \quad B \subseteq Q - PQ.$$

In unsymmetric form the Borel separability of A and B can also be expressed thus: There exists a Borel set P which includes A and excludes B.

Before continuing further, we should like to show, following Lusin, how, by use of the concept of Borel separability, the two principal theorems, Theorems III and IV of § 34, concerning Suslin sets can be proved without recourse to transfinite ordinal numbers (§ 34,2). Let

$$A = \mathfrak{S}\mathfrak{D}C_{n_1 n_2 \cdots n_k \cdots}$$

be, in the — for the time being — arbitrary space E, a Suslin set formed with the closed sets $C_{n_1} \supseteq C_{n_1 n_2} \supseteq C_{n_1 n_2 n_3} \supseteq \cdots$; the indices range over the natural numbers. If we hold initial indices fixed, we obtain the sets

$$A_n = \mathfrak{S}\mathfrak{D}C_{n n_2 \cdots}, \quad A_{n_1 n_2} = \mathfrak{S}\mathfrak{D}C_{n_1 n_2 n_3 \cdots}$$

with $A = \mathfrak{S}A_n$, $A_n \supseteq \mathfrak{S}A_{n_1 n_2}$, $A_{n_1 n_2} \supseteq \mathfrak{S}A_{n_1 n_2 n_3}$, etc.

Similarly, we let $B = \mathfrak{S}\mathfrak{D}D_{n_1 n_2 \cdots}$ with the corresponding assumptions and notation. Let us assume that A and B are not Borel separable; by virtue of $A = \mathfrak{S} A_n$ and $B = \mathfrak{S} B_n$, there then exists a non-Borel-separable pair of sets A_n and B_p, next, a non-Borel-separable pair A_{np} and B_{pq}, etc. Because of $A_n \subseteq C_n$ and $B_n \subseteq D_n$ (C_n and D_n closed), we have $C_n D_p > 0$ and also

$$C_{np} D_{pq} > 0, \quad C_{npr} D_{pqs} > 0, \ldots$$

If now E is separable, then we may assume (p. 217) that the diameters of the C and D converge to 0 as the number of indices increases; if in addition, E is complete, then the sequence of the $C_n D_p$, $C_{np} D_{pq}$, $C_{npr} D_{pqs}$, ... has a point of intersection which obviously belongs to AB. That is, *if A and B are not Borel separable, then they are not even disjoint*, or:

In a separable complete space, two disjoint Suslin sets are Borel separable, that is, they are Borel sets in their sum (§ 34, III).

Furthermore, let

$$A = \mathfrak{S}\mathfrak{D} C_{n_1 n_2 \cdots}$$

be representable with disjoint summands, then

$$A = \mathfrak{S} A_n, \quad A_n = \mathfrak{S} A_{n_1 n_2}, \quad A_{n_1 n_2} = \mathfrak{S} A_{n_1 n_2 n_3}, \ldots$$

with disjoint summands. If E is separable and complete, then the disjoint Suslin sets A_n are pairwise Borel separable and therefore also simultaneously Borel separable: there exist disjoint Borel sets P_n containing A_n which can, in addition, be replaced by $C_n P_n$, so that they may be assumed to be contained in C_n. Similarly, there exist disjoint Borel sets $P_{n_1 n_2}$ containing $A_{n_1 n_2}$ and contained in $C_{n_1 n_2}$, where, in addition, $P_{n_1 n_2}$ can be replaced by $P_{n_1} P_{n_1 n_2}$, that is, where it may be assumed to be contained in P_{n_1}. In this way, we obtain

$$A_n \subseteq P_n \subseteq C_n, \quad A_{n_1 n_2} \subseteq P_{n_1 n_2} \subseteq C_{n_1 n_2}, \quad A_{n_1 n_2 n_3} \subseteq P_{n_1 n_2 n_3} \subseteq C_{n_1 n_2 n_3}, \ldots$$

$$P_n \supseteq P_{n_1 n_2} \supseteq P_{n_1 n_2 n_3} \supseteq \cdots$$

But since $A_n \cdot A_{n_1} \cdot A_{n_1 n_2} \cdots = C_n \cdot C_{n_1 n_2} \cdot C_{n_1 n_2 n_3} \cdots$, this is also $\subseteq P_n \cdot P_{n_1 n_2} \cdot P_{n_1 n_2 n_3} \cdots$,

$$A = \mathfrak{S} P_n \cdot P_{n_1 n_2} \cdot P_{n_1 n_2 n_3} \cdots,$$

and, since the sets P with equal numbers of indices are pairwise disjoint, this is equivalent with

$$A = \mathfrak{P}P_n + \mathfrak{S}P_{n_1n_2} + \mathfrak{S}P_{n_1n_2n_3} + \cdots ,$$

and A is a Borel set. *In a separable complete space, every Suslin set that can be represented by means of disjoint summands is a Borel set* (§ 34, IV).

9.2.2 Obtaining the B-sets

The (single-valued) mapping $y = \varphi(x)$ of A onto B was called one-to-one, or *schlicht*, if it has a single-valued inverse, so that the inverse image $\varphi^{-1}(y)$ of every point $y \in B$ consists of only one point; it will be called *half-schlicht* and B will be called the *half-schlicht image* of A if every inverse image $\varphi^{-1}(y)$ is at most countable.

We call the Borel and Suslin sets of a separable complete space *B-sets* and *S-sets*, for short. From Theorem XIII of § 43 we know that:

The Baire image of an S-set is an S-set; the one-to-one Baire image of a B-set is a B-set.

The special case of continuous images has already been treated in Theorem II of § 37. We wish to show that even a half-schlicht Baire image of a B-set is also a B-set and that such a mapping can be split into at most countably many simple Baire mappings of B-sets. We begin by stating a theorem on which the proofs of both statements will be based.

I. *Let the mapping $y = \varphi(x)$ of the complete space A be continuous. Let a sequence of bounded sets*[4]

$$U_n \supseteq U_{nq} \supseteq U_{nqr} \supseteq \cdots$$

open in A, with diameters > 0, and with the images

$$V_n = \varphi(U_n), \quad V_{nq} = \varphi(U_{nq}), \quad V_{nqr} = \varphi(U_{nqr}), \ldots$$

[4]For brevity, let $P \gg Q$ (or $Q \ll P$) mean $P_i \supseteq \overline{Q}_a$; the open kernel of P contains the closed hull of Q. For a sequence $P \gg Q \gg R \cdots$ we obviously have $P \cdot Q \cdot R \cdots = P_a \cdot Q_a \cdot R_a \cdots = P_i \cdot Q_i \cdot R_i \cdots$.

be assigned to every dyadic sequence $n, q, r, \ldots$ — that is, one formed from the numerals 1 and 2 — in such a way that the 2^n sets U with n indices are pairwise disjoint while their images V have points in common:

$$V_1 \cdot V_2 > 0, \quad V_{n1} \cdot V_{n2} \cdot V_{nq1} \cdot V_{nq2} > 0, \ldots \tag{9.11}$$

Then the uncountable set

$$D = \mathfrak{D} U_n \cdot U_{nq} \cdot U_{nqr} \cdots$$

has an image consisting of but one point, so that the mapping is not half-schlicht.

The proof is exceedingly simple. The intersection $\mathfrak{D} U_n \cdot U_{nq} \cdot U_{nqr} \cdots$ can be replaced by the intersection of the closed hulls and therefore contains only one point; to different dyadic sequences there correspond different points $x = U_n \cdot U_{nq} \cdot U_{nqr} \cdots$, and D is of the cardinality of the continuum. If $y = \varphi(x)$, then because of the continuity, the diameters of the images V_n, V_{nq}, V_{nqr}, ... also converge to 0. If $\xi = U_n \cdot U_{nq} \cdot U_{nqr} \cdots$, then $V_n \cdot V_{nq} > 0$, $V_{nq} \cdot V_{nqr} > 0$, $V_{nqr} \cdot V_{nqrs} > 0$, ..., and since the sum of two sets with non-empty intersection has a diameter which is at most equal to the sum of the two diameters taken separately, the diameters of

$$V_n + V_{nq}, \quad V_{nq} + V_{nqr}, \quad V_{nqr} + V_{nqrs}, \ldots$$

also converge to 0, that is, $\eta = \varphi(\xi)$ coincides with y. Therefore $\varphi(D) = \eta$ has only one point, and $\varphi^{-1}(\eta) \supseteq D$ is uncountable.

The difficulty consists primarily in the realization of the conditions (1). In the case we shall treat first (Theorem II), the non-vanishing of the sets (1) will be proved by their being non-Borel-separable from a certain other set; in the later case (Theorem IV), by using the fact that the sets U with n indices form "coherent" systems.

For the mapping $y = \varphi(x)$ of A onto B, we call the point y of B a *simple image* or a *multiple image* depending on whether its inverse image $\varphi^{-1}(y)$ consists of one point or of more than one point. The set M of the multiple images can obviously be represented in the form

$$M = \mathfrak{S}\varphi(U) \cdot \varphi(U'), \tag{9.12}$$

where U and U' range over all the pairs of disjoint sets of a basis (p. 320) of A. From this it follows that:

(A) *For a continuous or a Baire mapping of an S-set, the set of multiple images is an S-set.*

For the sets U and U' open in A and the images of these sets are S-sets, and the basis can be taken to be (at most) countable.

(B) *If $y = \varphi(x)$ is a continuous mapping of the B-set A into the separable complete space Y, if $B = \varphi(A)$ is the image of A, and if M is the set of the multiple images, then every S-set S contained in $B - M$ is Borel separable from $Y - B$.*

The disjoint S-sets S and M are Borel separable (in Y), and there exists a B-set Q which contains S and for which $Q \cdot M = 0$. Since $B \cdot Q$ is a Borel set in B, its inverse image $P = \varphi^{-1}(BQ)$ is a Borel set in A (§ 35, II) and is therefore a B-set. Inasmuch as $Q \cdot M = 0$, all the points of BQ are simple images, and therefore BQ, as the one-to-one continuous image of P, is itself a B-set (and thus is a Borel set in Y and not only in B). We have $S \subseteq BQ \subseteq B$; between S and $Y - B$ there lies the B-set BQ; that is, S is Borel separable from $Y - B$.

(C) *For $k = 1, 2, \ldots, n$, let $y_k = \varphi_k(x_k)$ be continuous mappings of arbitrary B-sets A_k into the separable complete space Y, let $B_k = \varphi_k(A_k)$ be the image of A_k, let M_k be the set of the multiple images for φ_k, and finally let B be any set containing the B_k and contained in Y. If then $M_1 \cdots M_n$ is Borel separable from $Y - B$, then $B_1 \cdots B_n$ is also Borel separable from $Y - B$.*

Let $D = B_1 \cdots B_n$. By hypothesis, it is possible to interpose a B-set between $D \cdot M_1 \cdots M_n$ $(= M_1 \cdots M_n)$ and B. If it has already been established for some k of the series $n, n-1, \ldots, 1$ that a B-set Q can be interposed between $D \cdot M_1 \cdots M_k$ and B, then this is also true, we claim, for $D \cdot M_1 \cdots M_{k-1}$ (which, for $k-1$, is to mean the set D). In fact, let

$$D \cdot M_1 \cdots M_k \subseteq Q \subseteq B;$$

then $D \cdot M_1 \cdots M_{k-1} \cdot (Y - Q)$ is an S-set contained in B_k (because D is contained in B_k), is disjoint to M_k (for $D \cdot M_1 \cdots M_{k-1} \cdot M_k$ is disjoint to $Y - Q$), is therefore contained in $B_k - M_k$, and consequently, by (B), is Borel separable from $Y - B$, and, *a fortiori*, from the smaller set $Y - B$. Therefore for some B-set Q', we have

$$D \cdot M_1 \cdots M_{k-1} \cdot (Y - Q) \subseteq Q' \subseteq B$$

as well as $D \cdot M_1 \cdots M_{k-1} \cdot Q \subseteq Q \subseteq B$, and, adding, $D \cdot M_1 \cdots M_{k-1} \subseteq Q + Q' \subseteq B$. This establishes the induction from k to $k-1$, and D is seen to be Borel separable from $Y - B$.

II. *The half-schlicht continuous image of a separable complete space A is a B-set.*

We preface the proof by the following remarks: For any $\delta > 0$, those sets of a basis for the space A that have diameters $< \delta$ themselves form a basis; a separable space thus has for every $n = 1, 2, \ldots$ a countable basis $\mathfrak{B}_n$ consisting of sets U of diameters $< 1/n$. Further: If the non-empty set G is open in A, then the sets $U \subseteq G$ (as we stated earlier, this means that U_a is contained in G) belonging to a basis of A form a basis for the space G.[5] Accordingly, we let $p, q, r, \ldots$ range over the natural numbers and let U_p stand for the sets of $\mathfrak{B}_1$, U_{pq} for the sets of $\mathfrak{B}_2(U_p)$, U_{pqr} for the sets of $\mathfrak{B}_3(U_{pq})$, and so on.[6] Let V_p, V_{pq}, V_{pqr}, $\ldots$ be the images of U_p, U_{pq}, U_{pqr}, $\ldots$ under the mapping $y = \varphi(x)$.

We prove II in this form: *If the continuous image $B = \varphi(A)$ of A is not a B-set, then the mapping is not half-schlicht.* Let B lie in the separable complete space Y.

The set of the multiple images in B is, by (2),

$$M = \mathfrak{S} V_p \cdot V_q;$$

where the sum is extended over all disjoint pairs of sets U_p and U_q of the basis $\mathfrak{B}_1$. Now B is not Borel separable from $Y - B$

[5] In the case that A is separable, G thus has a basis $\mathfrak{B}_n(G)$ which is at most countable and consists of sets $U \subseteq G$ with diameters $< 1/n$.

[6] We therefore have $U_p \supseteq U_{pq} \supseteq U_{pqr} \supseteq \cdots$, and the sets U with m indices have diameters $< 1/n$.

(this means just that B is not a Borel set in Y); by (C) (for $n = 1$) M is also not Borel separable from $Y - B$, so that one of the (at most countably many) summands of M is not Borel separable from $Y - B$; numbering the sets suitably, we therefore have

$$V_1 \cdot V_2 \text{ is not Borel separable from } Y - B.$$

Let the set of the multiple images under the mapping $\varphi(x|U_p)$ of U_p onto V_p be M_p $(p = 1, 2)$; then we have $M_p \subseteq V_p$ and

$$V_1 \cdot V_2 = M_1 \cdot M_2 + M_1 \cdot (V_2 - M_2) + (V_1 - M_1) \cdot M_2 + (V_1 - M_1) \cdot (V_2 - M_2).$$

The last three summands are Borel separable from $Y - B$ (they are contained in B and are one-to-one continuous images of B-sets, and are therefore themselves B-sets); therefore $M_1 \cdot M_2$ is not Borel separable from $Y - B$. Since

$$M_1 = \mathfrak{S} V_{1p} \cdot V_{1q}, \quad M_2 = \mathfrak{S} V_{2p} \cdot V_{2q},$$

one of the summands $V_{1p} \cdot V_{1q} \cdot V_{2r} \cdot V_{2s}$ is not Borel separable from $Y - B$, and in particular (after suitable numbering)

$$V_{11} \cdot V_{12} \cdot V_{21} \cdot V_{22} \text{ is not Borel separable from } Y - B.$$

Continuing in this way, we realize the hypotheses of I, and in particular the inequalities (1), in that the sets on the left are not Borel separable from $Y - B$. Hence $\varphi(x)$ is not half-schlicht, and II is proved.

III. *The half-schlicht Baire image of a B-set is itself a B-set.*

Let $y = \varphi(x)$ be a half-schlicht mapping of the B-set A onto B. If $\varphi(x)$ is, first, continuous, then we note (§ 37, III) that A is the image of a separable complete space T, under a one-to-one continuous mapping $x = \psi(t)$, e.g., of a set closed in the Baire null space (the trivial case that A and T are only finite can be excluded). Then $y = \varphi(\psi(t)) = \chi(t)$ is a half-schlicht continuous mapping of T onto B, and therefore, by II, B is a B-set.

In the general case that $y = \varphi(x)$ is a Baire mapping of A into Y (which contains B), we proceed as in the proof of

Theorem XIII of § 43. In the product space (A, Y), the set C of the points $(x, \varphi(x))$ is a Borel set and therefore a B-set, since Y can be taken as separable and complete; B is the projection of C onto Y and is therefore a continuous — and in our case half-schlicht — image of the B-set C and, by what has already been proved, is itself a B-set.

9.2.3 Cleavability

We call a set P contained in A *cleavable* under the mapping $y = \varphi(x)$ of the space A if P is the sum of a sequence of B-sets that are mapped one-to-one by φ, so that

$$P = \mathfrak{S} P_n, \quad P_n \text{ a B-set}, \quad \varphi(x|P_n) \text{ one-to-one.}$$

Then P itself is of course a B-set and $\varphi(x|P)$ is half-schlicht. Moreover, the summand P_n may, if we wish, be taken as disjoint by replacing P_n by $P_n - P_1 \cdot P_2 \cdots P_{n-1}$. (The essence of the condition is that the P_n are to be B-sets; without this provision every set P would be cleavable for half-schlicht $\varphi(x)$.)

A sum of a countable number of cleavable sets is cleavable. Every B-set contained in a cleavable set is itself cleavable.

(D) *Under a half-schlicht Baire mapping of a B-set, the set of the multiple images is a B-set.*

This follows from (2) where, by II, III, the U and their images are B-sets, and the basis can be chosen so as to be (at most) countable.

(E) *Let $y = \varphi(x)$ be a half-schlicht continuous mapping of the B-set A into the space Y; let $\psi(y)$ be the inverse image of y for $y \in Y$ and let $\psi_0(y)$ be the set of the isolated points of $\psi(y)$. Then the set $P_0 = \mathfrak{S}\psi_0(y)$ is cleavable.*

Let $U_1, U_2, \ldots$ be the sets of a basis of A, let $V_n = \varphi(U_n)$ be the image of U_n, and, under the mapping $\varphi(x|U_n)$, let M_n be the set of the multiple images and $Q_n = V_n - M_n$ the set of the simple images. The sets U_n, V_n, M_n, and Q_n are B-sets, and similarly, the inverse images $P_n = \varphi^{-1}_{U_n}(Q_n)$ of Q_n under the mapping $\varphi(x|U_n)$ (for Q_n is a Borel set in V_n, and thus P_n is a Borel set in U_n). Then $\varphi(x|P_n)$ is one-to-one, and accordingly, the set $\mathfrak{S}P_n$ is cleavable. But this is the set $P_0 = \mathfrak{S}\psi_0(y)$. For if $x \in \psi_0(y)$, then there exists some U_n for which $\psi(y) \cdot U_n = x$, so that $y = \varphi(x)$ is the simple image under $\varphi(x|U_n)$, and $x \in P_n$. And vice versa.

In order to continue the separation of a cleavable portion of A begun here, let us take $\psi_\xi(y)$ to mean, for the ordinal numbers $\xi < \Omega$ (compare pp. 190 and 194) the *coherences* of $\psi(y)$; that is, $\psi_0(y) = \psi(y)$, $\psi_{\xi+1}(y)$ is the set of the points of accumulation of $\psi_\xi(y)$ lying in $\psi_\xi(y)$, so that $\psi_\xi(y) - \psi_{\xi+1}(y)$ is the set of isolated points of $\psi_\xi(y)$; for a limit ordinal η we have $\psi_\eta(y) = \mathfrak{D}_{\xi<\eta}\psi_\xi(y)$. Correspondingly, we set

$$A_\xi = \mathfrak{S}\psi_\xi(y) \quad (A_0 = A);$$

then for $\xi < \eta$ we have

$$A_\xi - A_\eta = \mathfrak{S}[\psi_\xi(y) - \psi_\eta(y)],$$

and the equation

$$\psi(y) - \psi_\eta(y) = \mathfrak{S}_\xi[\psi_\xi(y) - \psi_{\xi+1}(y)]$$

summed over y, gives

$$A - A_\eta = \mathfrak{S}_\xi(A_\xi - A_{\xi+1}).$$

Under the assumptions of (E), we then have:

(F) *The sets $A_\xi - A_{\xi+1}$ are cleavable.*

By (E), this holds for 0. If it is already proved for $\xi < \eta$, then by (3) $A - A_\eta$ is cleavable and is thus a B-set, and A_η is also a B-set. Under the half-schlicht continuous mapping $\varphi(x|A_\eta)$, $A_\eta \cdot \psi(y) = \psi_\eta(y)$ is the inverse image of y and $\psi_\eta(y) - \psi_{\eta+1}(y)$ is the set of its isolated points; application of (E) yields the cleavability of $A_\eta - A_{\eta+1}$.

[7] By (3), every $A - A_\xi$ is now cleavable, and for the cleavability of the whole set A it is sufficient that there exist an index

[7]Here, and in what follows, the sum with respect to y is to extend over $y \in Y$; the summands may, in part, be empty; for example, $\psi(y) = 0$ when y does not belong to $B = \varphi(A)$.

γ for which $A_\gamma = 0$, so that $\psi_\gamma(y) = 0$ for every y. This can also be expressed by saying that the sets $\psi(y)$ are "uniformly" separated, that is, that not only are their smallest coherences (dense-in-itself kernels) empty but all of them are already reached for indices $\leq \gamma$.

In the case of an arbitrary mapping $\varphi(x)$ of the arbitrary space A, we shall say that the sets $U_1, \ldots, U_n$, open in A form a *coherent system* if the intersection

$$D_\xi = \varphi(A_\xi \cdot U_1) \cdots \varphi(A_\xi \cdot U_n)$$

of the images of $A_\xi \cdot U_1$, ..., $A_\xi \cdot U_n$ is non-empty for all $\xi < \Omega$; in particular, the intersection $D_0 = \varphi(U_1) \cdots \varphi(U_n)$ is then non-empty.

(G) *If under the arbitrary mapping $\varphi(x)$ of the separable space A the n sets U_k $(k = 1, \ldots, n)$ form a coherent system and $\mathfrak{B}(U_k)$ is an at most countable basis of U_k, then this basis contains two disjoint sets U_{k1} and U_{k2} such that the $2n$ sets U_{k1} and U_{k2} form a coherent system.*

Let $y_\xi \in D_{\xi+1}$, so that for every ξ an inverse image point of y_ξ occurs in $A_{\xi+1} \cdot U_k$: $\psi(y_\xi) \cdot A_{\xi+1} \cdot U_k = \psi_{\xi+1}(y_\xi) \cdot U_k > 0$; since $\psi_{\xi+1}(y_\xi)$ is the coherence of $\psi_\xi(y_\xi)$, $\psi_\xi(y_\xi) \cdot U_k$ is infinite.[8] There thus exist in $\mathfrak{B}(U_k)$ two disjoint sets $U_{k1}(\xi)$ and $U_{k2}(\xi)$ depending on ξ which have a non-empty intersection with $\psi_\xi(y_\xi)$. But there exist at most countably many such systems of 2 sets U_{k1} and U_{k2}; at least one of them must occur an uncountable number of times, and for an uncountable number of ξ we thus have

$$U_{k1}(\xi) = U_{k1}, \quad U_{k2}(\xi) = U_{k2},$$

$$\psi_\xi(y_\xi) \cdot U_{k1} > 0, \quad \psi_\xi(y_\xi) \cdot U_{k2} > 0,$$

$$\varphi(A_\xi \cdot U_{k1}) \cdot \varphi(A_\xi \cdot U_{k2}) \cdots \varphi(A_\xi \cdot U_{n1}) \cdot \varphi(A_\xi \cdot U_{n2}) > 0.$$

This last inequality, valid for uncountably many ξ, holds, however, for all ξ, since the A_ξ decrease with increasing ξ, so that the $2n$ sets $U_{11}, \ldots, U_{n2}$ form a coherent system.

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[8] If P_* is the derived set of P and $P_* = P \cdot P_d$ the coherence of P, and if U is open, then by § 23, (13) we have $(P \cdot U)_* \supseteq P_* \cdot U \supseteq P \cdot U$; if $P_* \cdot U$ is non-empty, then $P \cdot U$ cannot be finite.

IV. *Under a half-schlicht continuous mapping of the separable complete space A, there exists a γ for which $A_\gamma = 0$; A is therefore cleavable.*

We prove that if all the A_ξ are non-empty, then the continuous mapping $y = \varphi(x)$ is not half-schlicht. A itself forms a coherent system, so that there exist in $\mathfrak{B}_1$ (in the notation of the proof of II) two disjoint U_1 and U_2 forming a coherent system, and then in $\mathfrak{B}_2(U_p)$ two disjoint U_{pq1} and U_{pq2} such that U_1, U_2, U_{p1}, U_{p2}, U_{q1}, U_{q2} form a coherent system, and so forth. Theorem I is now applicable; conditions (1) are now realized because the images of the sets of a coherent system have a non-empty intersection.

V. *A B-set is cleavable under every half-schlicht Baire mapping; that is, it is the sum of a countable number of B-sets that are mapped one-to-one.*

This is reduced to IV as III was reduced to II. If $y = \varphi(x)$ is a half-schlicht map of the B-set A and, to begin with, is continuous, then under the half-schlicht continuous mapping $y = \varphi(\psi(t)) = \chi(t)$ (compare the proof of III), $T = \mathfrak{S}T_n$ is cleavable into B-sets T_n which have the one-to-one images $B_n = \chi(T_n)$; since T_n and $A_n = \psi(T_n)$ are also one-to-one maps of each other, so also are A_n and $B_n = \varphi(A_n)$; $A = \mathfrak{S}A_n$ is cleavable under the mapping $\varphi(x)$.

In the general case of the Baire half-schlicht mapping $y = \varphi(x)$ of A onto B, we consider the points $z = (x, y)$ of the product space (A, B) and the continuous functions $x = \xi(z)$ and $y = \eta(z)$ (projections of z on A and B). The B-set C of the points $(x, \varphi(x))$ is mapped one-to-one and continuously on A by $\xi(z)$, half-schlicht and continuously on B by $\eta(z)$; $C = \mathfrak{S}C_n$ is cleavable here into B-sets C_n having one-to-one images $B_n = \eta(C_n)$ and $A_n = \xi(C_n)$; the B-set A_n is mapped one-to-one onto B_n by $\varphi(x)$, and $A = \mathfrak{S}A_n$ is cleavable under the mapping $\varphi(x)$.

Appendixes

(A). Appendix to p. 165

The space E is called an F_{II}-set if every closed non-empty set F is of the second category in itself. We call it an F_{I}-set if it is not an F_{II}-set, so that there is some closed non-empty set F that is of the first category in itself. Some remarkable results on this subject have been obtained by Hurewicz:

The space E is an F_{II}-set if and only if it contains a perfect countable set P; therefore it is an F_{I}-set if and only if every non-empty perfect set P is uncountable.

In a separable space E, the Borel sets (and even the complements of Suslin sets) are all of them F_{II}-sets, with the possible exception of the G_δ. These latter can be F_{II}-sets; we already know that the G_δ of a complete space, and even of a non-separable space — the Young sets — are, in point of fact, F_{II}-sets (p. 165).

The two following theorems also belong to this order of ideas; in the case of a complete space E, the theorems on cardinality — Theorems VIII of § 26 and I of § 32 — follow from them.

Every set G_δ in the arbitrary space E whose dense-in-itself kernel does not vanish contains a non-empty perfect (in E) subset P. Every uncountable Suslin set in the separable space E contains a non-empty perfect (in E) subset.

(B). Appendix to p. 182

For the method of proof in the text it would be necessary for two points of $P(0)$ to have the separation 0 in $P(0)$ itself and not merely in A. The connectedness of $P(0)$ can be proved somewhat as follows. Let us assume that $P(0) = Q + R$ has been partitioned into the sets Q and R compact in themselves;

we join a point $q \in Q$ to a point $r \in R$ by means of a $(1/n)$-chain in A, this being possible for every natural number n because $qr = 0$. For the points x of the chain let us observe whether the lower distance $d(x, Q)$ is less than $d(x, R)$ or is at least equal to it; for the first point q, the former is the case, for the last point r, the latter. The transition from the one case to the other takes place somewhere in the chain: there exists a point x_n for which $d(x_n, Q) \geq d(x_n, R)$ while for the preceding point x'_n, $d(x'_n, Q) < d(x'_n, R)$. We choose a convergent subsequence $x_n \to x$, where $x \in A$; then we also have $x'_n \to x$ and, because of the continuity of the separations and the distances, $xq = 0$, and $d(x, Q) = d(x, R)$. Therefore x would have to belong to $P(0) = Q + R$, and this yields a contradiction, since, of the two numbers $d(x, Q)$ and $d(x, R)$, one is 0 and the other is positive.

(C). Appendix to § 39, 2

Let $t = \psi(x)$ be a continuous mapping of the compact continuum C, irreducible between the two points a and b, onto the real interval T $(0 \leq t \leq 1)$. Let us call the inverse images of the numbers t the *layers* of the mapping ψ; let $C(x, \psi)$ be the layer to which the point x belongs, that is, the set of all the y for which $\psi(y) = \psi(x)$. Let us say that the mapping is *monotone* if all the layers are continua (containing one or more points); for example, Hahn's Theorem, Theorem VIII, proves the existence, for a composite continuum C, of a monotone mapping whose layers are the prime parts. It now follows, from an investigation by Kuratowski, that if C admits of any monotone mappings at all and if $C(x)$ denotes the intersection of the $C(x, m)$ for all the monotone mappings m for which $m(a) = 0$ and $m(b) = 1$, then these $C(x)$ are themselves the layers of a monotone mapping, which thus realizes a splitting of C into the smallest possible layers. These *minimal layers* $C(x)$ of Kuratowski now are more deserving of the name prime parts than the sets given this name in § 39, 2, which possibly can be split further into minimal layers.

(D). Appendix to p. 304

The question has been discussed by Banach, the following being his principal result. Let the "modified" Baire functions $f^{*\xi}$ be defined for $0 \leq \xi < \Omega$ by the inductive rule:

The f^{*1} are the functions of the class $(M^1, N^1) = (F_\sigma, G_\delta)$.

For $\xi > 0$, the $f^{*\xi+1}$ are the limit functions of convergent sequences of functions $f^{*\eta}$ $(\eta < \xi)$.

Then for an at most separable image space Y the functions $f^{*\xi}$ are identical with those of the class (M^ξ, N^ξ).

Thus, by taking as point of departure in the formation of limit functions the f^{*1} (which need not, in general, be limits of sequences of continuous functions) rather than the continuous functions, Theorem XII has a converse for the modified Baire functions in the case of an at most separable Y. The question remains open as to how, for a limit ordinal λ, the functions of the class (M^λ, N^λ) are formed from convergent sequences of functions of a lower class, or what conditions correspond, in the general case, to the completeness of systems of real functions (§ 41, 1 and § 43, 1).

(E). Appendix to p. 39

The naive theory of sets, due to Cantor, is, as the author remarks, self-contradictory. We may point for instance to the contradiction discovered by Bertrand Russell: let C be the class of all those classes which are not members of themselves, i.e., the class whose members x are such that $x \notin x$, and consider whether C is a member of itself or not. If $C \notin C$ then C satisfies the condition for membership of C, and therefore $C \in C$; but if $C \in C$ then C is one of those classes which is not a member of itself and so not a member of C, i.e., $C \notin C$. Many attempts have been made to protect set theory from contradiction. Russell's theory of types places objects, classes of objects, classes of classes, and so on, in a hierarchy of types and bans the formation of classes of mixed types, thus preventing a class being a member of itself; type theory, however, produces many curious distortions of mathematics—for instance, a different class of natural numbers in each type. A system of Willard van Orman Quine's known as New Foundations replaces type theory by 'potential typing' (in which in place of absolute types an ad hoc assignment of types for each relation $x \in y$ suffices), and utilises a distinction, due to von Neumann, between class and element, only classes which satisfy a condition of elementhood being eligible as members of classes; this system was shown (by Barkley Rosser in 1942) to be inconsistent, but a later version was protected from this contradiction by restricting (bound as well as free) variables in the membership axioms to elements. A recent system of Bernays maintains a distinction between class and set, sets being those classes which satisfy an elementhood condition, but dispenses with stratification, following Zermelo in using set-construction axioms.

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(F). Appendix to p. 66

The proof that every set can be well-ordered makes tacit appeal to Zermelo's Axiom of Choice, that if M is a set, the elements of which are non-empty sets (pairwise disjoint), then there exists a set which contains one member for each set which is an element of M. There are many equivalent forms of this axiom; for instance, that every set may be well-ordered, that every cardinal is finite or an aleph, or that all cardinals are comparable. Other equivalent forms are that for any infinite cardinals m, n,

(i) $m + n = m \cdot n$;

(ii) $m^2 = m$;

(iii) if $m^2 = n^2$, then $m = n$.

Another equivalent of the Axiom of Choice is known as *Zorn's lemma*: a non-empty partially ordered set in which every chain has a least upper bound, has a maximal element. A maximal element of a partially ordered set is an element m such that if $m \leq x$ then $m = x$ (i.e., m is as great as any element with which it is comparable); a chain is a subset of a partially ordered set, any two elements of which are comparable. Another form of Zorn's lemma states that every partially ordered set contains a maximal chain, a maximal chain being a maximal element of the set of chains (partially ordered by the inclusion relation).

It is not known whether the Axiom of Choice is independent of the other axioms of set theory, but K. Gödel has proved (for a version of Bernay's set theory) that if set theory without the Axiom of Choice is free from contradiction then the addition of the Axiom of Choice to this theory will not introduce a contradiction. In the opposite direction, E. Specker has shown that the Axiom of Choice is false in Quine's New Foundations.

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References

BERNAYS, P. and A. A. FRAENKEL, Axiomatic Set Theory (Amsterdam, 1958).

FRAENKEL, A. A., and Y. BAR-HILLEL, Foundations of Set Theory (Amsterdam, 1958).

GÖDEL, K. The consistency of the axiom of choice and of the generalised continuum-hypothesis with the axioms of set theory (Princeton, 1940, 1951).

QUINE, W. VAN ORMAN, Mathematical Logic. (Revised Edition, Harvard, 1951).

Bibliography

ALEXANDROFF, P., Topologie, I (Berlin, 1935).

ALEXANDROFF, P. S. and KOLMOGOROFF, A. N., Introduction to the Theory of Sets and the Theory of Functions (Moscow-Leningrad, 1948).

AUMANN, G., Reelle Funktionen (Berlin, 1954).

BACHMANN, H., Transfinite Zahlen (Berlin, 1955).

BAIRE, R., Leçons sur les fonctions discontinues (Paris, 1905).

BANACH, S., Introduction to the Theory of Real Functions (Warsaw, 1951; in Polish).

BANACH, S., Théorie des opérations linéaires (Warsaw, 1932; repr., New York, 1955).

BOREL, E., Éléments de la théorie des ensembles (Paris, 1949).

BOREL, E., Les paradoxes de l'infini (Paris, 1946).

BOREL, E., Leçons sur la théorie des fonctions (Paris, 1898; 4th ed., 1950). Cited as Leçons (1898).

BOREL, E., Leçons sur les fonctions de variables réelles (Paris, 1905; 2nd ed., 1928). Cited as Leçons (1905).

BOREL, E., Méthodes et problèmes de la théorie des fonctions (Paris, 1922).

BOURBAKI, N., Éléments de mathématiques Livres I, III (Paris, 1940–55).

CARATHÉODORY, C., Reelle Funktionen, I (Leipzig, 1939; repr., New York, 1946).

CARATHÉODORY, C., Vorlesungen über Reelle Funktionen (Leipzig, 1918; 2nd ed., 1927; repr., New York, 1948).

DENJOY, A., L'énumération transfinie (Paris, 1954).

FRAENKEL, A. A., Abstract Set Theory (Amsterdam, 1953; in English).

FRAENKEL, A. S., Einleitung in die Mengenlehre (3rd ed., Berlin, 1928).

FRÉCHET, M., Les espaces abstraits (Paris, 1928).

GÖDEL, K., Consistency of the Continuum Hypothesis (Princeton, 1940).

GRAVES, L. M., The Theory of Functions of Real Variables (New York, 1946; 2nd ed., 1953).

HAHN, H., Theorie der reellen Funktionen, I (Berlin, 1921). Cited as Theorie.

HAHN, H., Reelle Funktionen, I (Leipzig, 1932; repr., New York, 1948).

HAHN, H. and ROSENTHAL, A., Set Functions (Albuquerque, 1948).

HAUSDORFF, F., Grundzüge der Mengenlehre (Leipzig, 1914; repr., New York, 1949).

HESSENBERG, G., Grundbegriffe der Mengenlehre (Göttingen, 1906).

HOBSON, E. W., The Theory of Functions of a Real Variable (Cambridge, 1907; I, 3rd ed., 1927; II, 2nd ed., 1926).

HUREWICZ, W. and WALLMAN, H., Dimension Theory (Princeton, 1941).

KAMKE, E., Theory of Sets (New York, 1950).

KELLEY, J. L., General Topology (New York, 1955).

KURATOWSKI, C., Topologie, I (Warsaw, 1933; 2nd ed., 1948).

KURATOWSKI, C., Topologie, II (Warsaw, 1950).

LANDAU, E., Foundations of Analysis (New York, 1957).

LUSIN, N., Leçons sur les ensembles analytiques (Paris, 1930).

LUZIN (LUSIN), N., Theory of Functions of a Real Variable (2nd ed., Moscow, 1948; in Russian).

MENGER, K., Dimensionstheorie (Leipzig, 1928).

Moore, R. L., Foundations of Point Set Theory (New York, 1932).

Natanson, I. P., Theory of Functions of a Real Variable (New York, 1953).

Newman, M. H. A., Topology of Plane Sets of Points (2nd ed., Cambridge, 1950).

Pierpont, J., Lectures on the Theory of Functions of Real Variables, I and II (Boston, 1905 and 1912).

Rosenthal, A., Neuere Untersuchungen über Funktionen reeller Veränderlichen (Math. Enzykl. II C 9).

Schoenflies, A., Die Entwickelung der Lehre von den Punktmannigfaltigkeiten, II and III (Leipzig, 1900 and 1908).

Schoenflies, A., Entwickelung der Mengenlehre und ihrer Anwendungen (Leipzig, 1913).

Sierpiński, W., Hypothèse du continu (2nd ed., New York, 1956).

Sierpiński, W., General Topology (Toronto, 1934; 2nd ed., 1952).

Sierpiński, W., Leçons sur les nombres transfinis (Paris, 1928).

Sierpiński, W., Cardinal and Ordinal Numbers (Warsaw, 1958).

Souslin (Suslin), M., The continuum problem (Moscow, 195?).

Further References

The following references are intended only to give an account of the origins of the more important concepts and theorems of the subject; we refer the reader for further information to the reports by A. Schoenflies and the review by A. Rosenthal. Most of the theorems about point sets, incidentally, are stated only for one-dimensional space, or at most Euclidean space, and have been considerably transformed.

P. 10. Kowalewski, G., *Einführung in die Infinitesimalrechnung* (Leipzig, 1908), p. 14.

§ 1. In general, we have dispensed with references to Cantor; the contents of the first four chapters and the fundamental concepts of the theory of point sets derive almost exclusively from him. Of his numerous papers we refer the reader primarily to the following:

Über unendliche lineare Punktmannigfaltigkeiten I–VI: Math. Ann., Vol. 15 (1879), p. 1; Vol. 17 (1880), p. 355; Vol. 20 (1882), p. 113; Vol. 21 (1883), pp. 51, 945; Vol. 23 (1884), p. 453. *Beiträge zur Begründung der transfiniten Mengenlehre*, I, II: Math. Ann., Vol. 46 (1895), p. 481; Vol. 49 (1897), p. 207.

P. 11. Genocchi, A. and Peano, G., *Differentialrechnung und Grundzüge der Integralrechnung* (Leipzig, 1899), p. 340.

§ 3. P. 18. Carathéodory, C. *Vorlesungen*, p. 23.

P. 23. Upper and lower limit as *ensemble limite complet* and *ensemble limite restreint*, in: Borel, E., *Leçons* (1905), p. 18.

P. 22. de la Vallée Poussin, C., *Intégrales*, p. 7.

§ 5. Theorem III. Bernstein, F., in Borel, E., *Leçons* (1898), p. 103.

§ 7. P. 39. Zermelo, E., Math. Ann., Vol. 65 (1908), p. 261. The reader will find more on the subject of the antinomies in Fraenkel, A., *Einleitung in die Mengenlehre* (1928).

Theorem III. König, J., Math. Ann., Vol. 60 (1904), p. 177.

§ 11. Theorem I. Bernstein, F., *Untersuchungen aus der Mengenlehre*, Thesis (Halle, 1901), p. 34.

P. 61. Minkowski, H., Verh. Heidelb. Kongr. (Leipzig, 1905), p. 171.

P. 62. Dedekind, R., *Stetigkeit und irrationale Zahlen* (Braunschweig, 1872).

§ 12. Zermelo, E., Math. Ann., Vol. 59 (1904), p. 514, and Vol. 65 (1908), p. 107.

§ 13. Important contributions to well-ordering theory have been made by Hessenberg, *Grundbegriffe der Mengenlehre* (1906).

§ 14. P. 80. Hessenberg, G., *Grundbegriffe*, § 75.

§ 15. P. 82. For the question as to $2^{\aleph_0} = 2^{\aleph_1}$ compare Hilbert, D., Math. Ann., Vol. 95 (1925), p. 181.

§ 16. Hausdorff, F., Leipz. Ber., Vol. 58 (1906), p. 108.

§ 18. Borel, E., *Leçons* (1898), p. 46.

§ 19. Suslin,[1] M., C. R. Acad. Sci. Paris, Vol. 164 (1917), p. 88. Lusin, N., ibid., p. 91. Cf. § 32.

§ 20, 1. In Fréchet, Rend. Circ. Mat. Palermo, Vol. 22 (1906), pp. 17 and 30, a metric space is called class (E) (distance = écart); in later papers it is called class (D) (distance).

§ 20, 3. P. 114. Hilbert, D., Götting. Nachr., Vol. 8 (1906), p. 157.

P. 115. Minkowski, H., *Geometrie der Zahlen* (Leipzig and Berlin, 1910; repr., New York, 1956).

§ 20, 4. Baire, R., Acta Math., Vol. 32 (1909), p. 105.

§ 20, 5. Hensel, K. *Theorie der algebraischen Zahlen*, Vol. 1 (Leipzig and Berlin, 1908). Kürschák, J., J. für Math. Vol. 142 (1913), p. 212.

§ 21, 1. Complete space, essentially according to Fréchet, M., Rend. Circ. Mat. Palermo, Vol. 22 (1906), p. 23.

§ 21, 3. Cantor, G., Math. Ann., Vol. 5 (1872), p. 123; Méray, C., *Nouveau précis d'analyse infinitésimale* (Paris, 1872).

§ 21, 4. Fréchet, M., Rend. Circ. Mat. Palermo, Vol. 22 (1906) p. 6.

§ 22. Open set: Lebesgue, H., Ann. Mat. Pura Appl. (3), Vol. 7 (1902), p. 242; Carathéodory, C., *Vorlesungen*, p. 40. In the first edition the open sets were called domains (Gebiete), a term now used to denote connected open sets.

§ 23. Most of the concepts of this section derive from Cantor. Condensation point: Lindelöf, E., Acta Math., Vol. 29 (1905), p. 184. Dense-in-itself kernel: Hahn, H., *Theorie*, p. 76.

§ 25. The theorems of this section derive mostly from Cantor.

P. 144. Separable (in a slightly different sense): Fréchet, M., Rend. Circ. Mat. Palermo, Vol. 22 (1906), p. 23.

Theorem IV. Bernstein, F., Thesis (Halle, 1901), p. 44.

§ 26, 1. Theorem II. Borel, E., Ann. Sci. Ecole Norm. Sup., (3) Vol. 12 (1895), p. 51.

Theorem III. Lindelöf, E., C. R. Acad. Sci. Paris, Vol. 137 (1903), p. 697; Young, W. H., Proc. Lond. Math. Soc., Vol. 35 (1903), p. 384.

Theorem IV. Sierpiński, W., Bull. Acad. Sci. Cracovie (1918), p. 49.

§ 26, 3. Theorem VIII. Young, W. H., Leipz. Ber., Vol. 55 (1903), p. 287; *Theory*, p. 64.

§ 27. P. 163. Baire, R., Ann. Mat. Pura Appl. (3), Vol. 3 (1899), p. 67.

§ 28, 1. Distance AB: Pompéju, D., Ann. Fac. Toulouse (2), Vol. 7 (1905), p. 281.

§ 28, 2. The sets F, F_i (in substance) in Painlevé, P.; see Zoretti, L., J. Math. Pures Appl. (6), Vol. 1 (1905), p. 8; Bull. Soc. Math. France, Vol. 37 (1909), p. 116.

§ 29, 1. Connectedness: Lennes, N. J., Amer. J. Math., Vol. 33 (1911) p. 303.

Continuum: Jordan, C., *Cours d'analyse*, Vol. I (2nd ed., Paris, 1893), p. 25.

Domain (going back to Weierstrass): Carathéodory, C., *Vorlesungen*, p. 208, p. 222.

THEOREM III. Janiszewski, Z. and Kuratowski, C., Fund. Math., Vol. 1 (1920), p. 211.

§ 29, 2. P. 177. Hahn, H., Jber. Deutsch Math. Verein., Vol. 23 (1914), p. 319; Wiener Ber., Vol. 123 (1914), p. 2,433.

THEOREM X. Hahn, H., Fund. Math., Vol. 2 (1921), p. 189. Kuratowski, C., Fund. Math., Vol. 1 (1920), p. 43.

P. 179. Mazurkiewicz, S., Fund. Math., Vol. 1 (1920), p. 167.

§ 29, 3. P. 181. Cantor, G., Math. Ann., Vol. 21 (1883), p. 576.

THEOREM XV. Brouwer, L. E. J., Amst. Ak. Proc., Vol. 12 (1910), p. 785.

THEOREM XVII. Janiszewski, Z., Thesis (Paris, 1911) = Journ. éc. Polyt. (2), Vol. 16 (1912), p. 100.

P. 185. Sierpiński, W., Tohoku Math. J., Vol. 13 (1918), p. 300.

§ 29, 4. THEOREM XIX. Zoretti, L., J. Math. Pures Appl. (6), Vol. 1 (1905), p. 8.

§ 30, 4. THEOREM VII. Zalcwasser, Fund. Math., Vol. 3 (1922), p. 44.

§ 31, 1. P. 199. Hamel, G., Math. Ann., Vol. 60 (1905), p. 459.

P. 200. Zoretti, L., Ann. Sci. Ecole Norm. Sup. (3), Vol. 26 (1909), p. 487.

P. 201. Janiszewski, C. R. Acad. Sci. Paris, Vol. 151 (1910), p. 198.

§ 31, 2. P. 201. Bernstein, F., Leipz. Ber., Vol. 60 (1908), p. 325.

P. 202. Sierpiński, W., Fund. Math., Vol. 1 (1920), p. 7.

§ 32. THEOREM I. For Borel sets: Alexandroff, P., C. R. Acad. Sci. Paris, Vol. 162 (1916), p. 323; Hausdorff, F., Math. Ann., Vol. 77 (1916), p. 430.

For the theory of Suslin sets compare: Suslin, M., C. R. Acad. Sci. Paris, Vol. 164 (1917), p. 88. Lusin, N., *Ensembles analytiques* (1930). Lusin, N. and Sierpiński, W., Bull. Acad. Cracovie (1918), p. 35; J. Math. Pures Appl. (7), Vol. 2 (1923), p. 53. Sierpiński, W., Bull. Acad. Cracovie (1919), pp. 161, 179.

§ 33. Theorem I. Sierpiński, W., C. R. Acad. Sci. Paris, Vol. 175 (1922), p. 859.

§ 34, 2. Lusin, N. and Sierpiński, W., J. Math. Pures Appl. (7), Vol. 2 (1923), p. 53.

§ 35, 2. Theorem III. Jordan, C., *Cours d'Analyse*, Vol. I (2nd ed., Paris, 1893), p. 53.

Theorem V. Brouwer, L. E. J., Amst. Akad. Proc., Vol. 12 (1910), p. 785.

§ 35, 3. P. 227. Lebesgue, H., *Leçons sur l'intégration* (Paris, 1904), p. 105.

§ 36, 1. Peano curves: Peano, G., Math. Ann., Vol. 36 (1890), p. 157. Hilbert, D., Math. Ann., Vol. 38 (1891), p. 459. Knopp, K, Archiv. d. Math. u. Ph. (3), Vol. 26 (1918), p. 103. Lebesgue, H., *Leçons sur l'intégration* (Paris, 1904), p. 44.

P. 232. Brouwer, L. E. J., Math. Ann., Vol. 70 (1911), p. 161. On the subject of the Brouwer-Urysohn-Menger dimension theory compare Menger, K., *Dimensionstheorie* (1928).

§ 36, 2. Theorem II. Sierpiński, W., Fund. Math., Vol. 1 (1920), p. 44.

Theorem III. Hahn, H., Jahresb. D. Math., Vol. 23 (1914), p. 319; Wiener Ber., Vol. 123 (1914), p. 2,433; Mazurkiewicz, S., Compt. R. Varsovie (III), Vol. 6 (1913), p. 305 (in Polish); Fund. Math., Vol. 1 (1920), p. 191.

§ 37. Theorems II and III. Sierpiński, W., Bull. Ac. Crac. (1918), p. 29 and (1919), p. 161; Fund. Math., Vol. 5 (1924), p. 155.

Theorem IV. Suslin, M., Compt. Rend., Vol. 164 (1917), p. 88.

§ 38. Theorems I and II. Sierpiński, W., Compt. Rend., Vol. 171 (1920), p. 24.

Theorem III. Mazurkiewicz, S., Bull. Acad. Cracovie (1916), p. 490. Alexandroff, P., C. R. Acad. Sci. Paris, Vol. 178 (1924), p. 185. Hausdorff, F., Fund. Math., Vol. 6 (1924), p. 146.

Theorem IV. Lavrentieff, M., C. R. Acad. Sci. Paris, Vol. 178 (1924), p. 187; Fund. Math., Vol. 6 (1924), p. 149.

§ 39, 1. Theorem II. Vietoris, L., Monatshefte Math. Phys., Vol. 31 (1921), p. 179.

Theorem III. Lennes, N. J., Amer. J. Math., Vol. 33 (1911), p. 308; Sierpiński, W., Ann. Mat. Pura Appl. (3), Vol. 26 (1916), p. 131.

§ 39, 2. Hahn, H., Wiener Ber., Vol. 130 (1921), p. 217.

Theorem VII. Moore, R. L., Math. Zeit., Vol. 22 (1925), p. 307.

§ 40. For more precise information refer to: Rosenthal, A. (Enzycl. II C 9) No. 26, Tietze, H. and Vietoris, L. (Enzykl. III, A, B, 13), Fréchet, M., *Les espaces abstraits* (1928); Kuratowski, C., *Topologie*, Vol. I (1933); Sierpiński, W., *General Topology* (1934).

P. 258. Kuratowski, C., Fund. Math., Vol. 3 (1922), p. 182.

P. 260. Separation axioms: Tietze, H., Math. Ann., Vol. 88 (1923), p. 290.

P. 261. Urysohn, P., Math. Ann., Vol. 94 (1925), p. 309. Tychonoff, A., Math. Ann., Vol. 95 (1925), p. 139.

P. 261. Fréchet, M., Rend. Circ. Mat. Palermo, Vol. 22 (1906).

§ 41. This theory is due primarily to Lebesgue, H., J. Math. Pures Appl. (6) Vol. 1 (1905), pp. 139–216. Cf. Young, W. H., Proc. Lond. Math. Soc. (2), Vol. 12 (1912), p. 260. Hausdorff, F., Math. Zeit., Vol. 5 (1919), p. 292.

§ 41, 4. Theorems XIII and XIV. Hahn, H., Wien. Ber., Vol. 126 (1917), p. 103. Tietze, H., J. f. Math., Vol. 145 (1914), p. 9. Hausdorff, F., Math. Zeit. Vol. 5 (1919), p. 295.

§ 41, 5. Sierpiński, W., Fund. Math., Vol. 2 (1921), pp. 15, 37; Mazurkiewicz, S., C. R. Varsovie, Vol. 6 (1913), pp. 106, 161 (in Polish).

P. 278. Baire, R., Ann. Sci. Ecole Norm. Sup. (3), Vol. 27 (1910), pp. 402, 407. Volterra, V., Giornale Mat., Vol. 19 (1881), p. 111.

§ 42. P. 281. Baire, R., Ann. Sci. Ecole Norm. Sup. (3), Vol. 27 (1910), p. 381.

P. 282. Ascoli, G., Ann. Mat. Pura Appl. (2), Vol. 9 (1883), p. 81.

P. 282. Baire, R., ibid., pp. 65, 81.

P. 284. Borel, E., *Leçons* (1905), p. 163.

P. 285. Baire, R., ibid., p. 102.

P. 287. Borel, E., ibid., pp. 161, 170; Baire, R., ibid., p. 100.

P. 289. Hahn, H., *Theorie*, p. 308; Sierpiński, W., Fund. Math., Vol. 1 (1920), p. 124.

Theorem III. For g^1 and h^1: Baire, R., ibid., pp. 76, 81.

Theorem IV. Sierpiński, W., Fund. Math., Vol. 1 (1920), p. 124.

Theorem VI. Sierpiński, W., ibid., p. 131.

§ 43. P. 296. Baire, R., ibid., pp. 76, 81. Borel, E., *Leçons* (1905), p. 163.

P. 297. Lusin, N., C. R. Acad. Sci. Paris, Vol. 158 (1914), p. 1259.

Theorem I. Baire, R., ibid., pp. 65, 81.

Theorems III, IV. Sierpiński, W., Fund. Math., Vol. 3 (1922), pp. 120, 129; *Hypothèse du continu*, pp. 66, 68. Banach, S., Fund. Math., Vol. 1 (1920), p. 123; ibid., Vol. 3 (1922), p. 86.

Theorem V, VI. Kuratowski, C., Fund. Math., Vol. 1 (1920), pp. 105, 112.

Theorem VII. Nikodym, O., Fund. Math., Vol. 10 (1927), p. 201; Kuratowski, C., ibid., Vol. 11 (1928), p. 22.

Theorem XI. For the classes of Baire, Lebesgue, H., J. Math. Pures Appl. (6), Vol. 1 (1905), p. 173.

§ 44. Hahn, H., *Reelle Funktionen*, p. 221. Sierpiński, W., Fund. Math., Vol. 9 (1927), pp. 172, 177.

Theorem I. Hahn, H., *Theorie*, p. 221; *Reelle Funktionen*, p. 214.

§ 45. Hausdorff, F., Math. Ann., Vol. 76 (1915), pp. 422, 428. Nikodym, O., Fund. Math., Vol. 10 (1927), pp. 201, 219.

Theorem I. Banach, S., *Oeuvres* (in the press).

Theorem II. Nikodym, O., Fund. Math., Vol. 10 (1927), p. 201.

Theorem III. Sierpiński, W., Fund. Math., Vol. 11 (1928), p. 120.

Theorem IV. Kuratowski, C., Fund. Math., Vol. 1 (1920), p. 112.

Theorems V, VI. Kuratowski, C., ibid., pp. 105, 112.

Theorem VII. Nikodym, O., Fund. Math., Vol. 10 (1927), p. 219; Kuratowski, C., ibid., Vol. 11 (1928), p. 22.

§ 46. Lusin, N. and Sierpiński, W., Bull. Acad. Cracovie (1918), pp. 35, 48. Sierpiński, W., ibid. (1919), pp. 161, 179.

Theorem I. Sierpiński, W., Fund. Math., Vol. 3 (1922), p. 50; Vol. 5 (1924), pp. 160, 170.

Theorem II. Sierpiński, W., ibid., Vol. 5 (1924), pp. 160, 170.[9]

[9]Michael Suslin (1894–1919) himself published only this one paper.

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